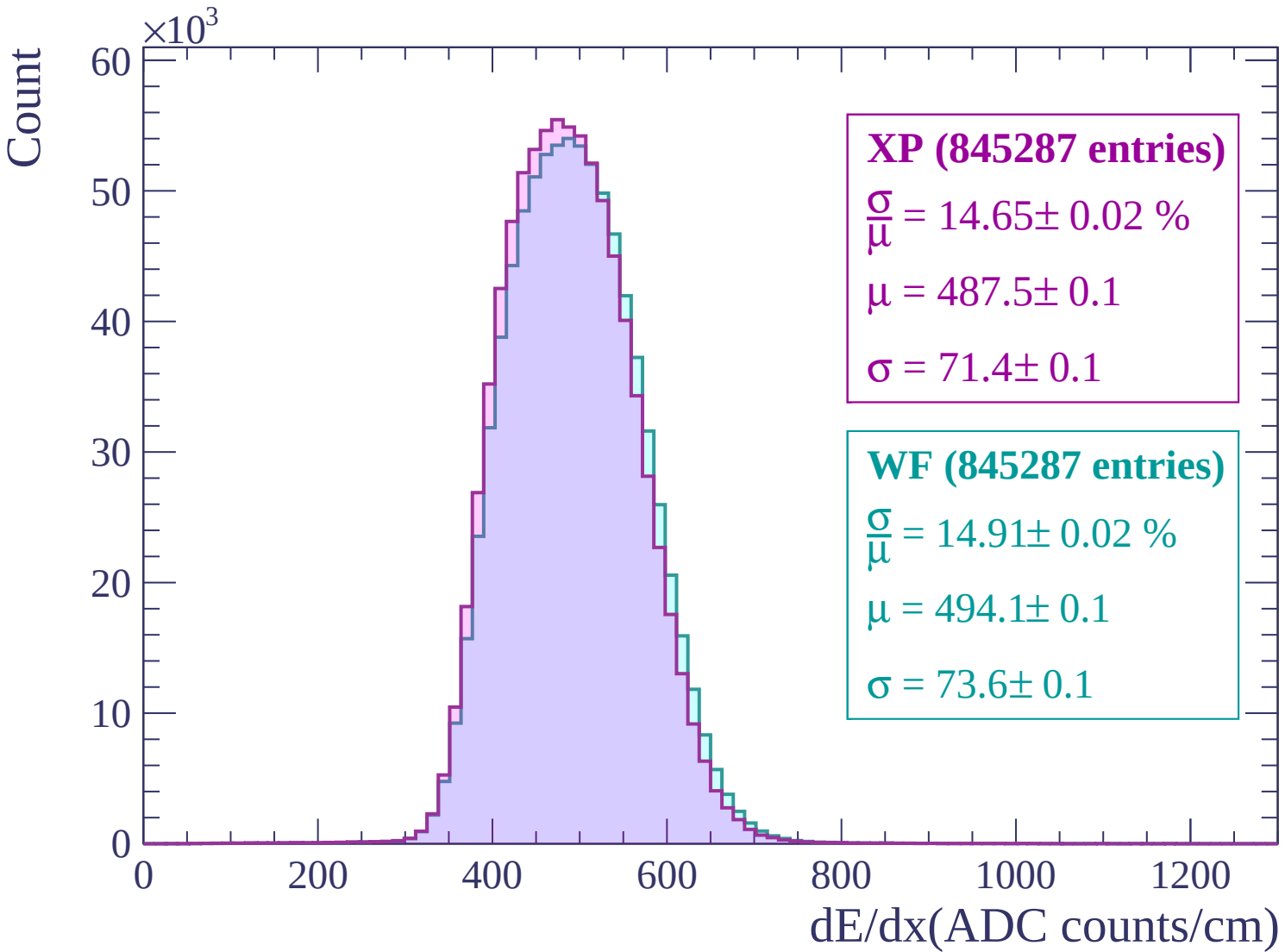
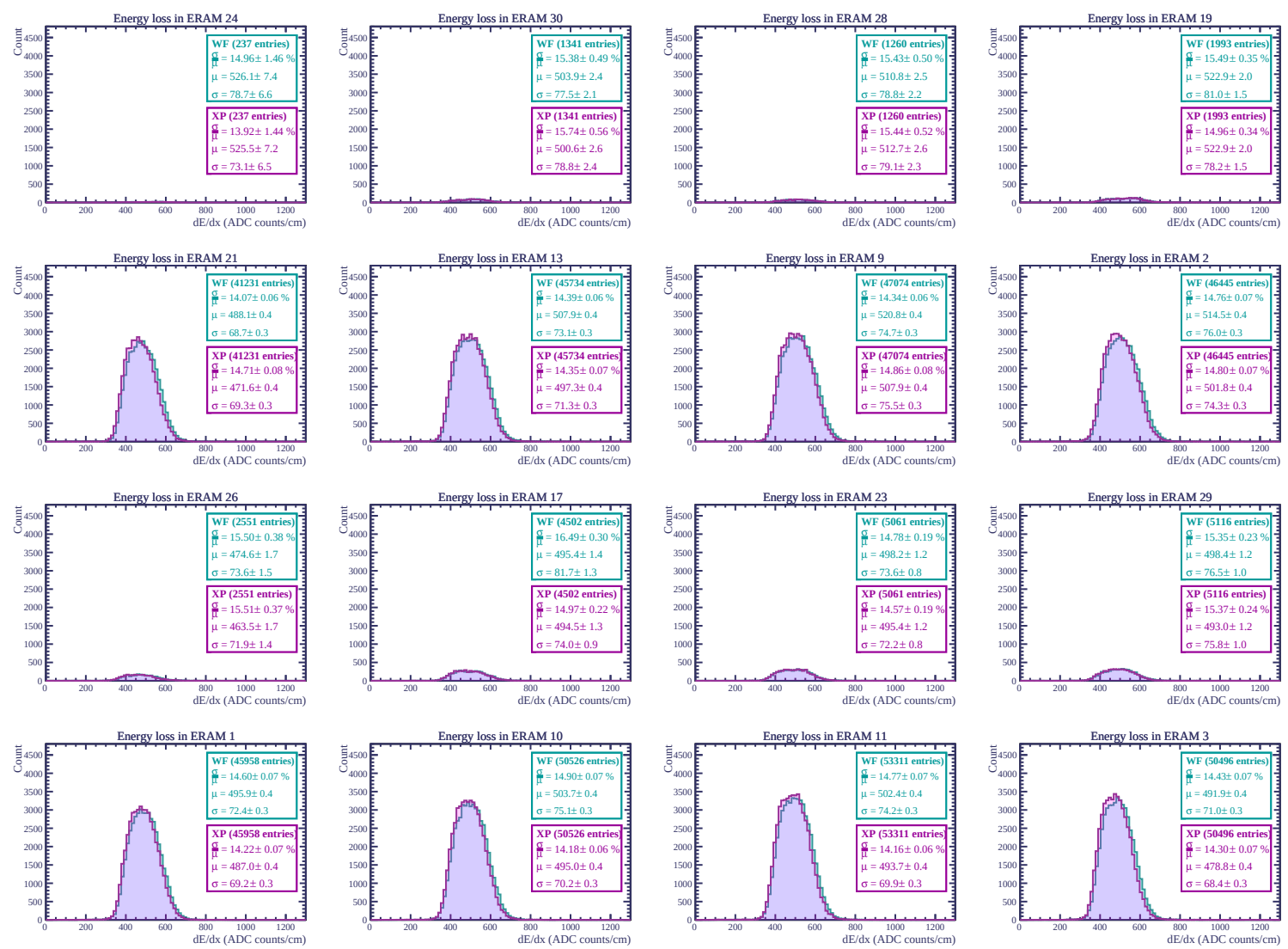
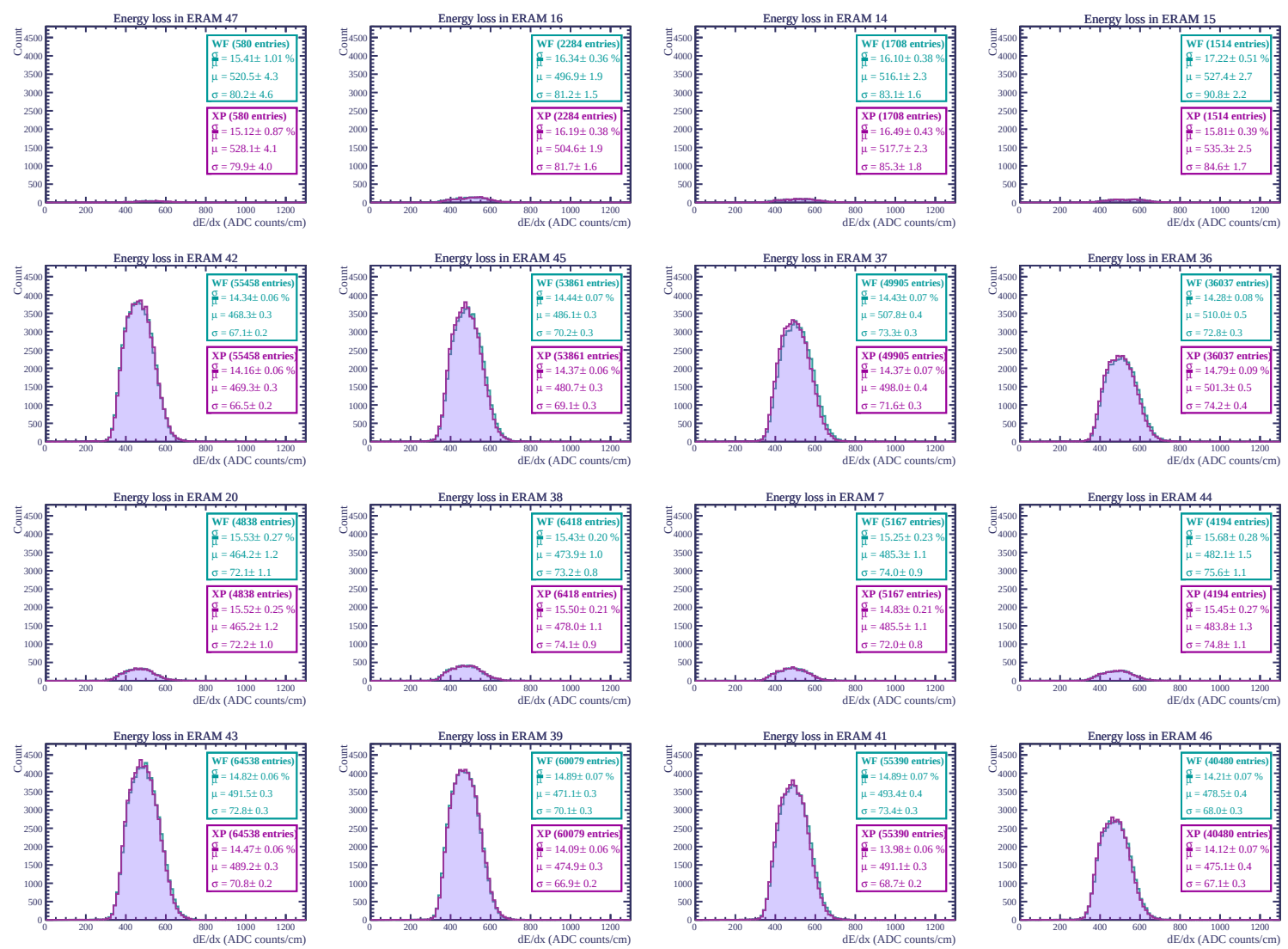
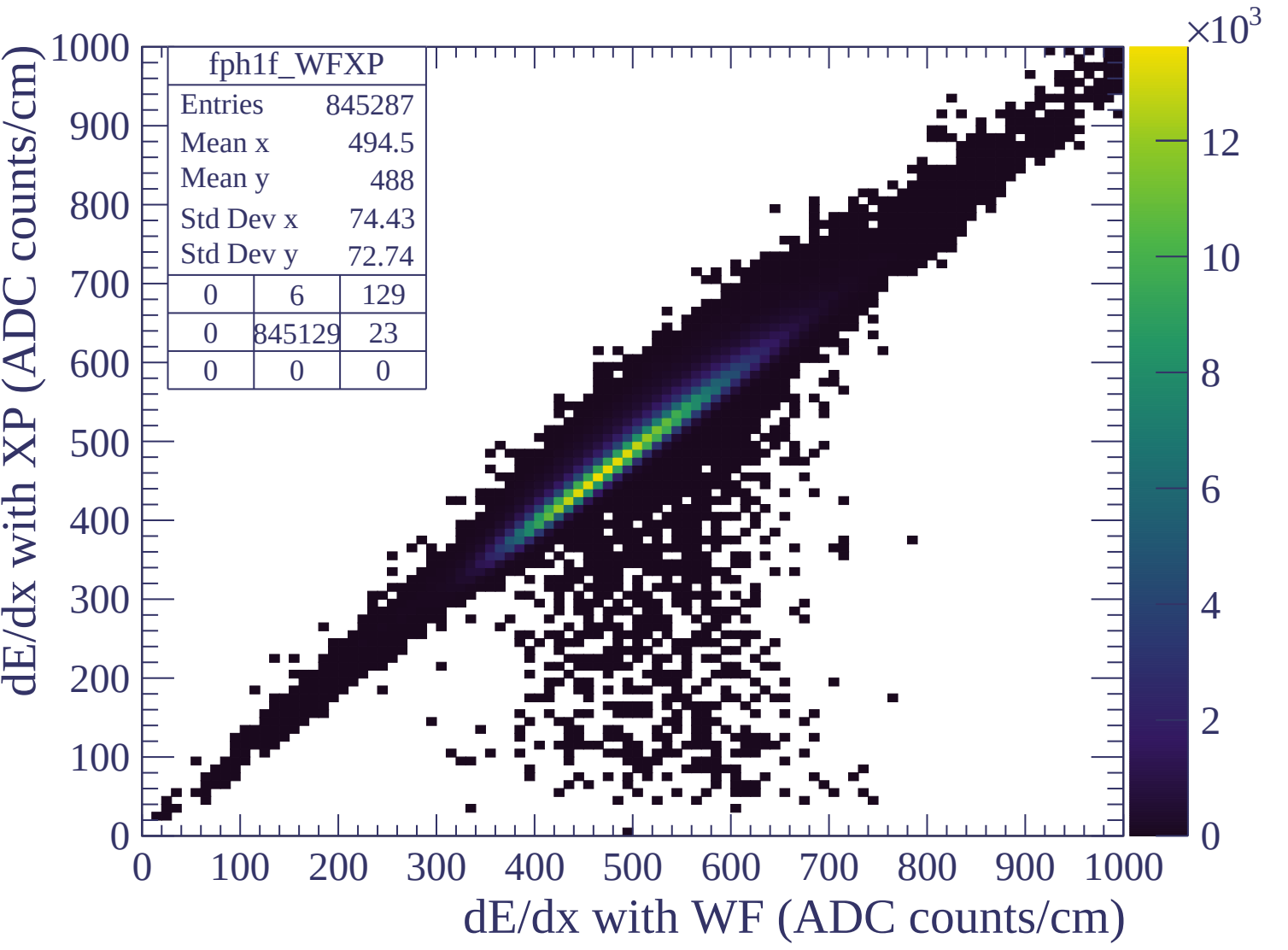
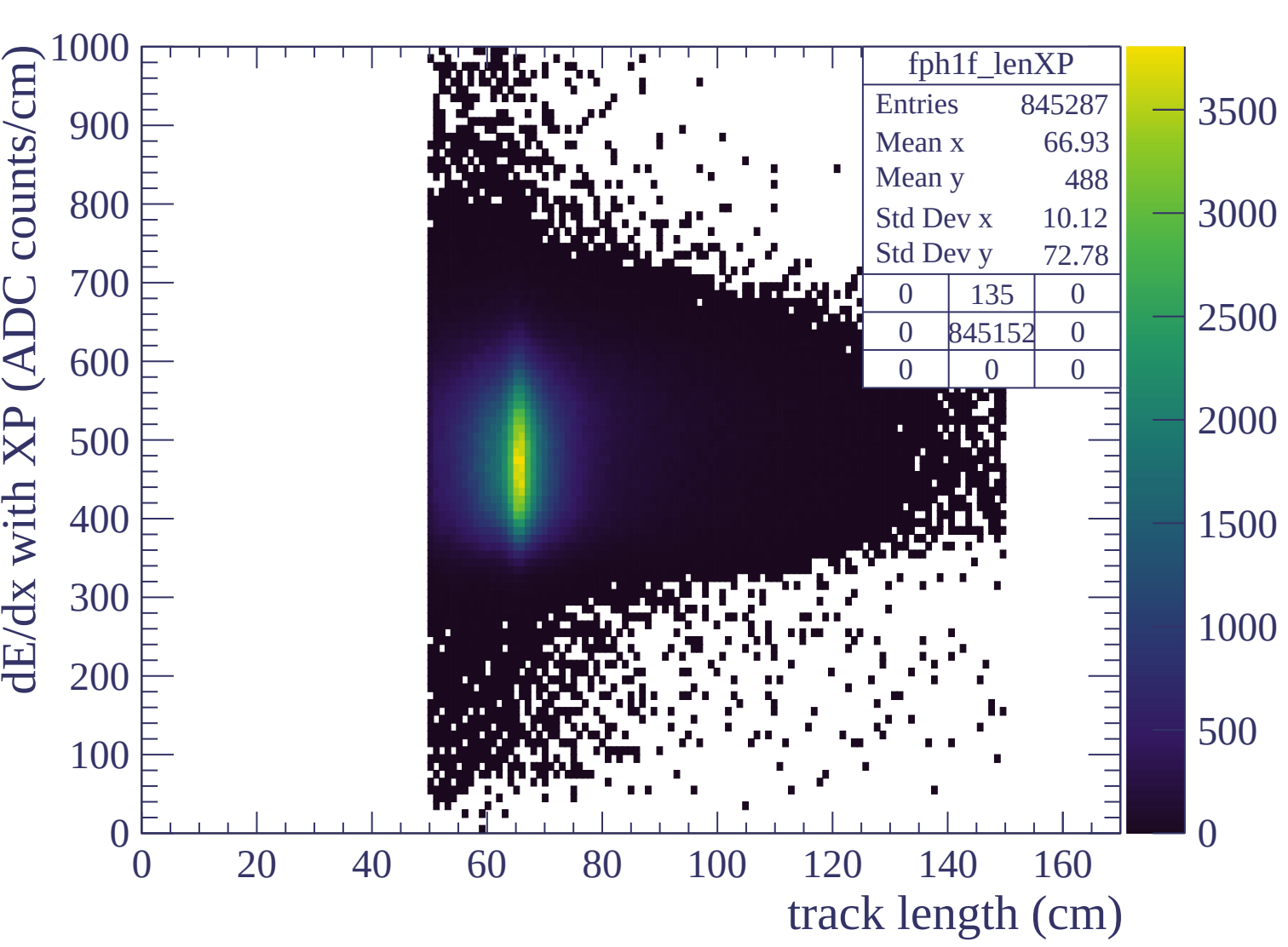


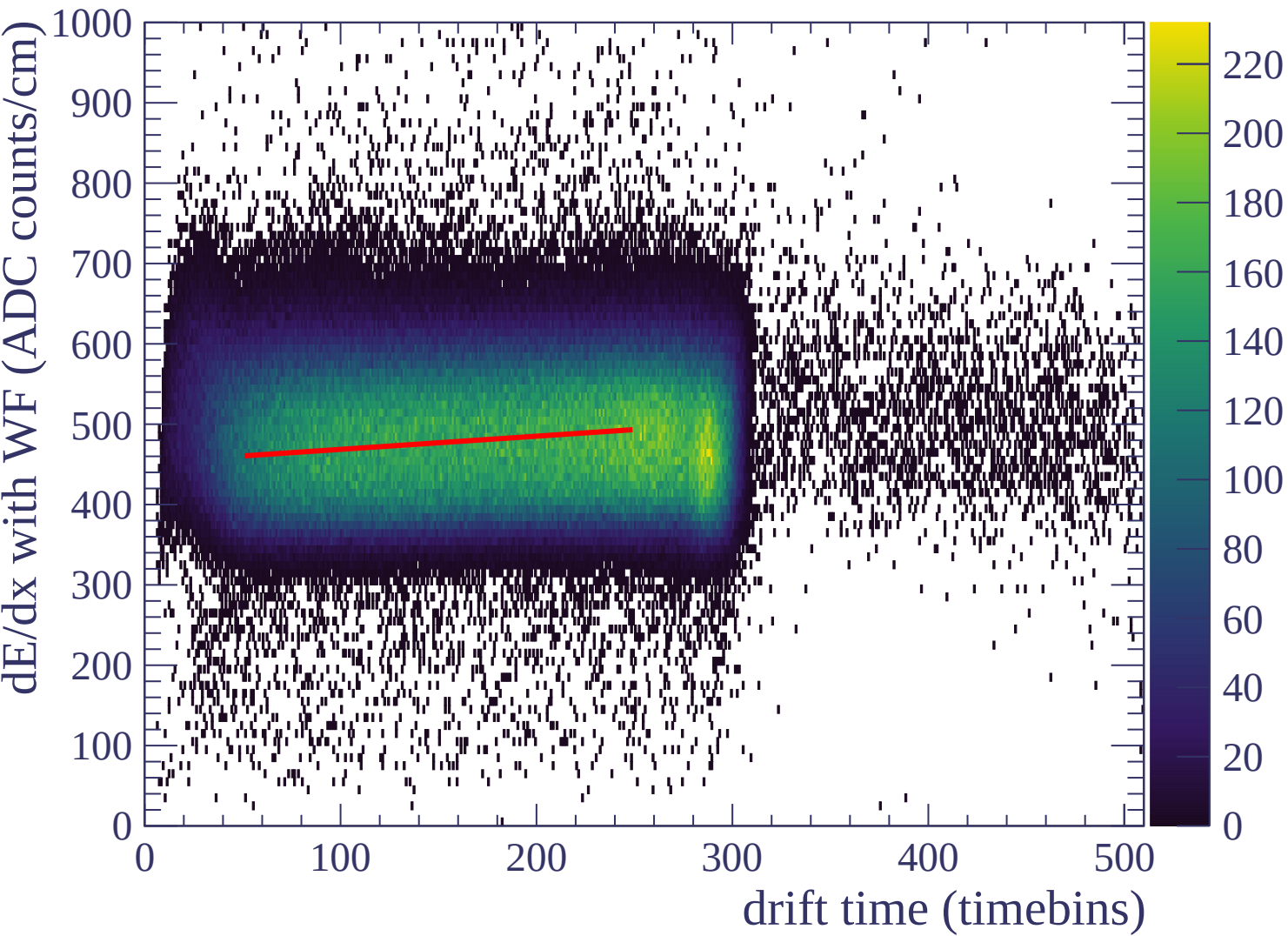


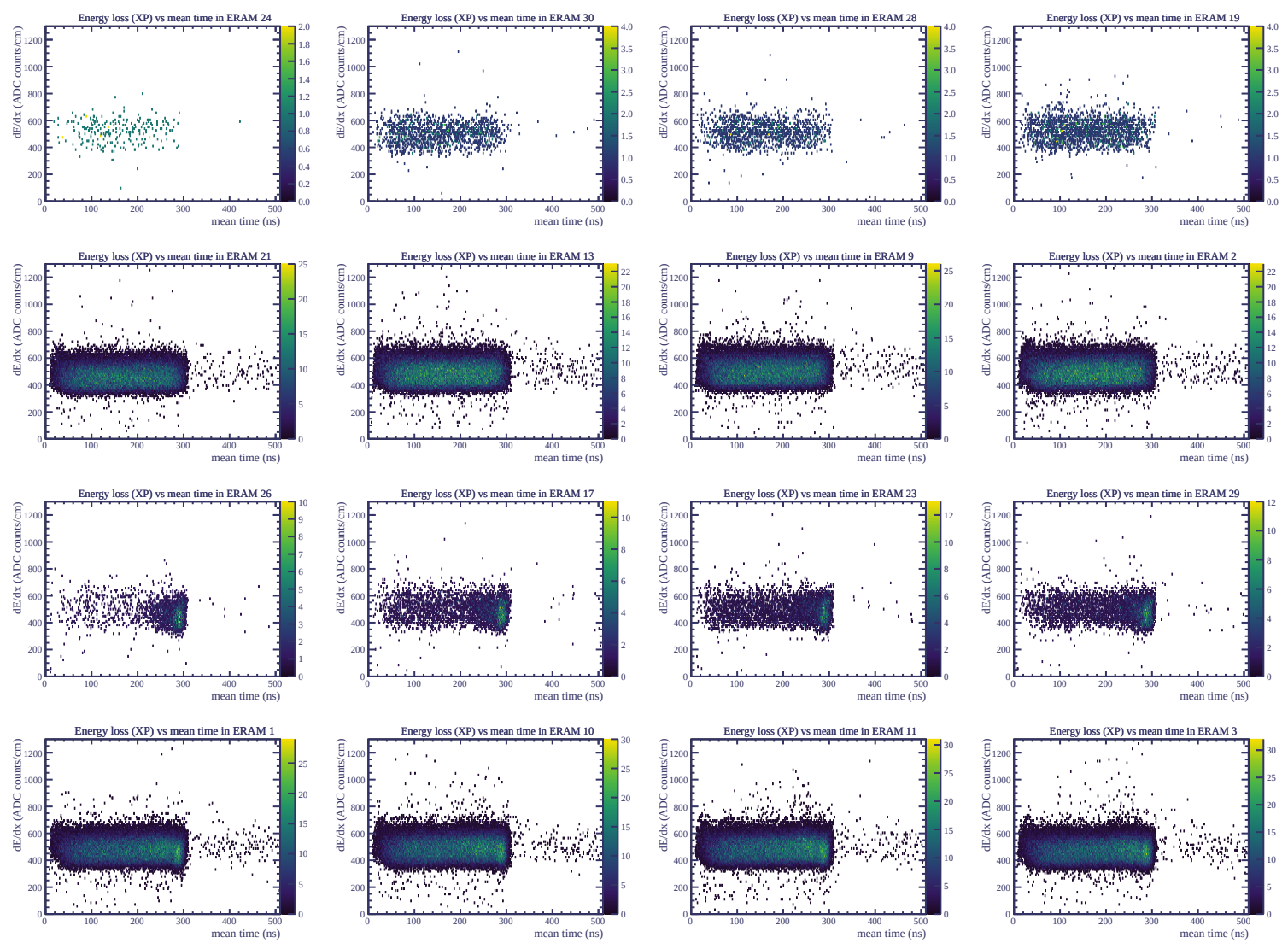


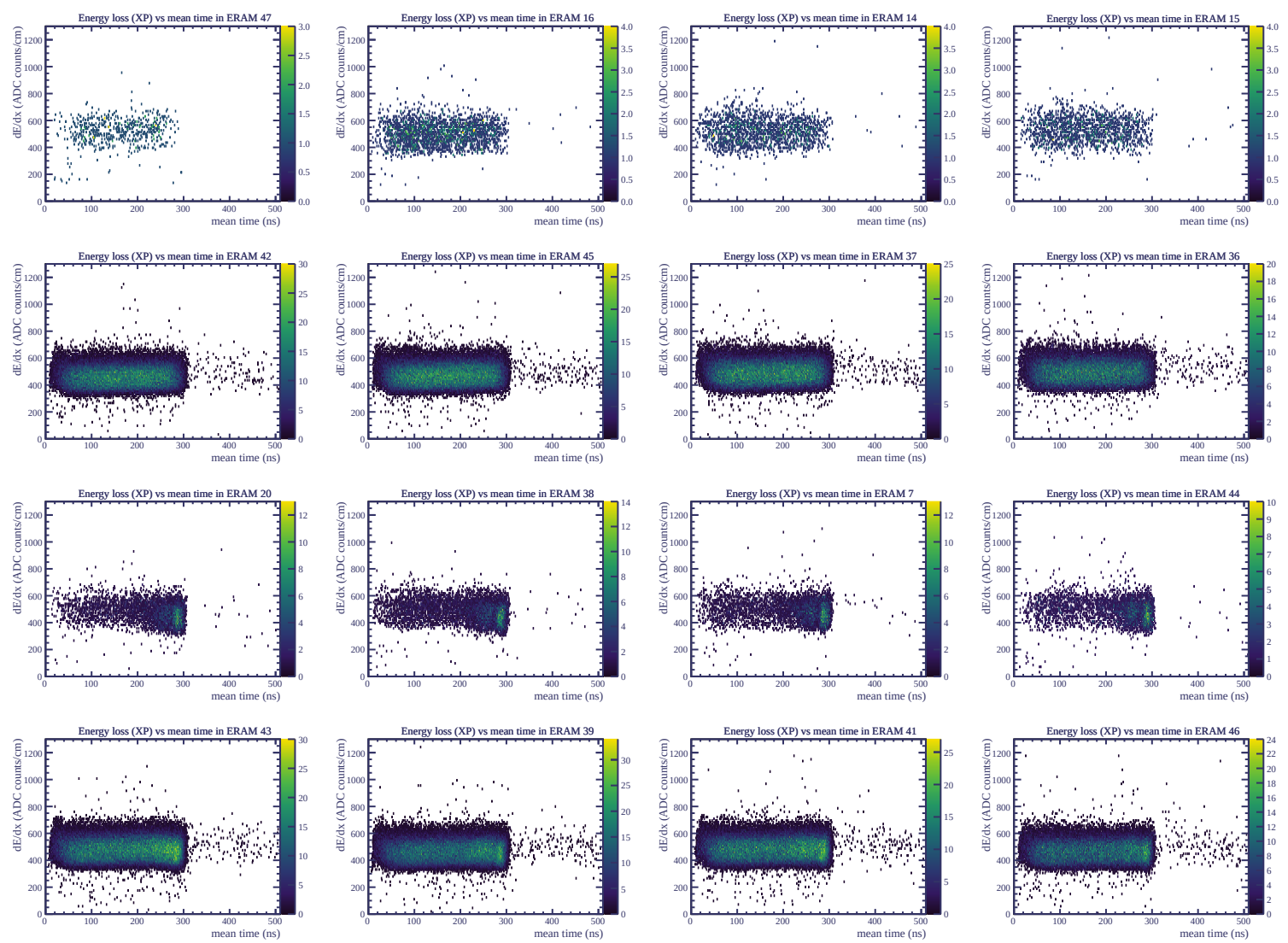




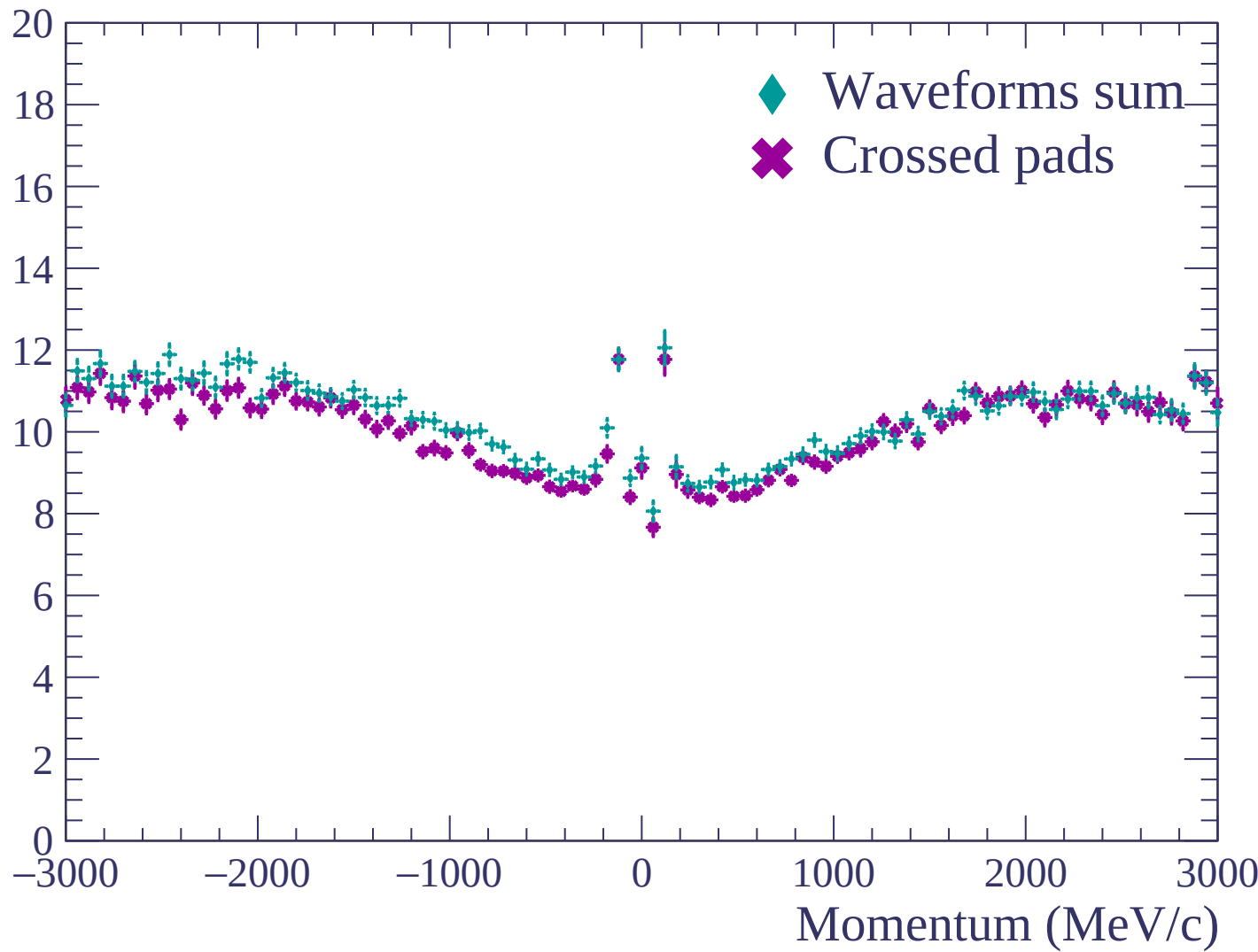


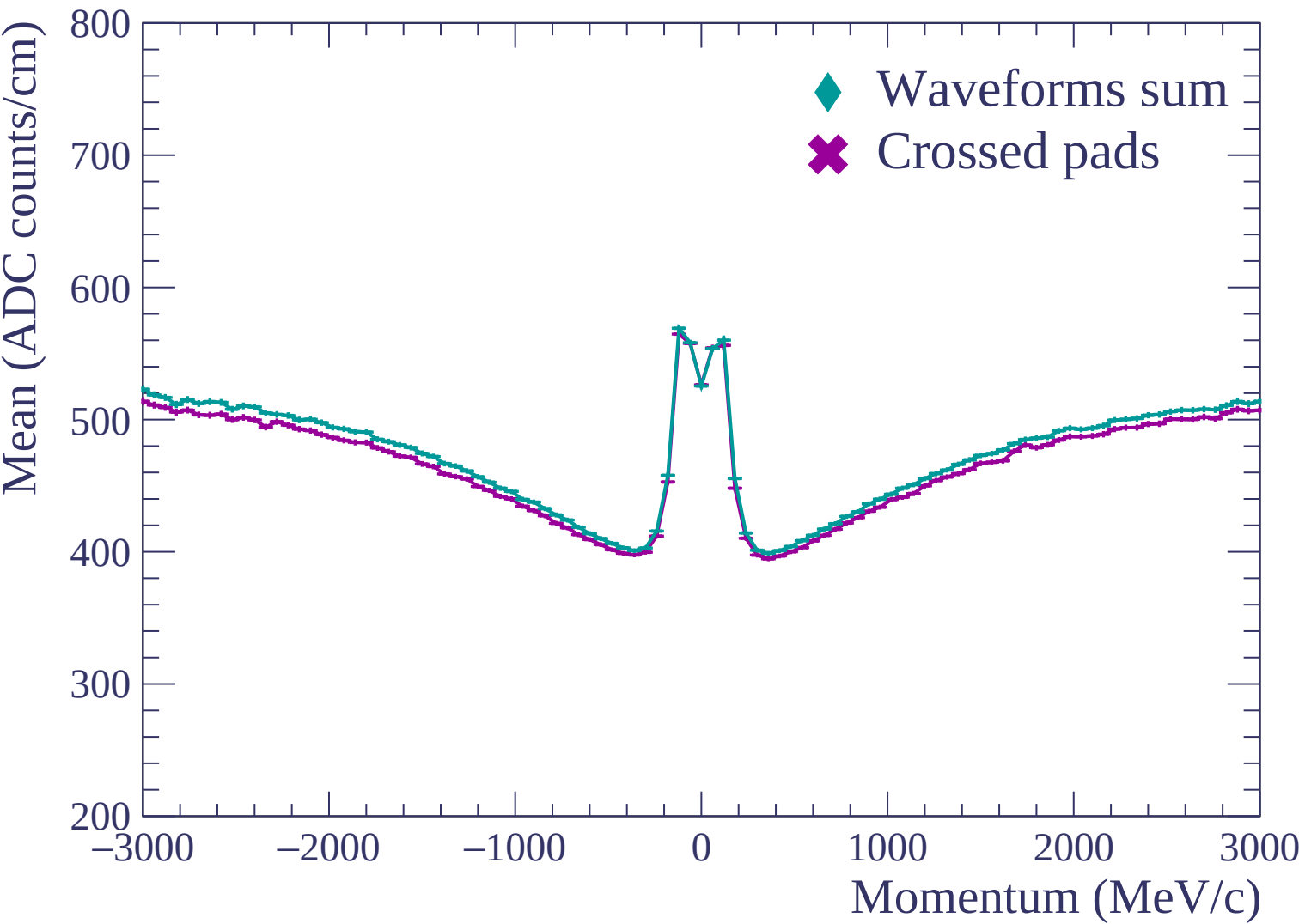


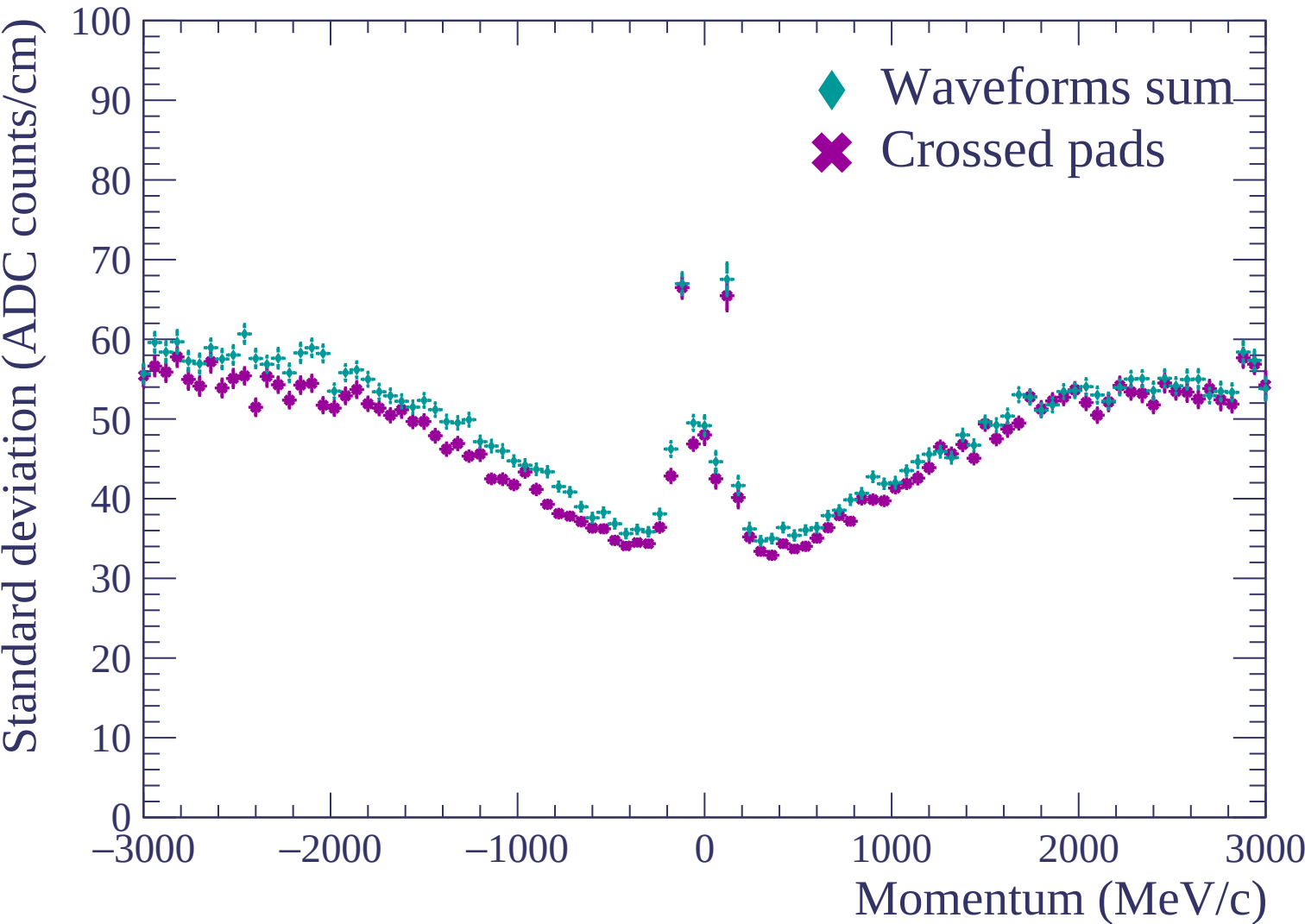


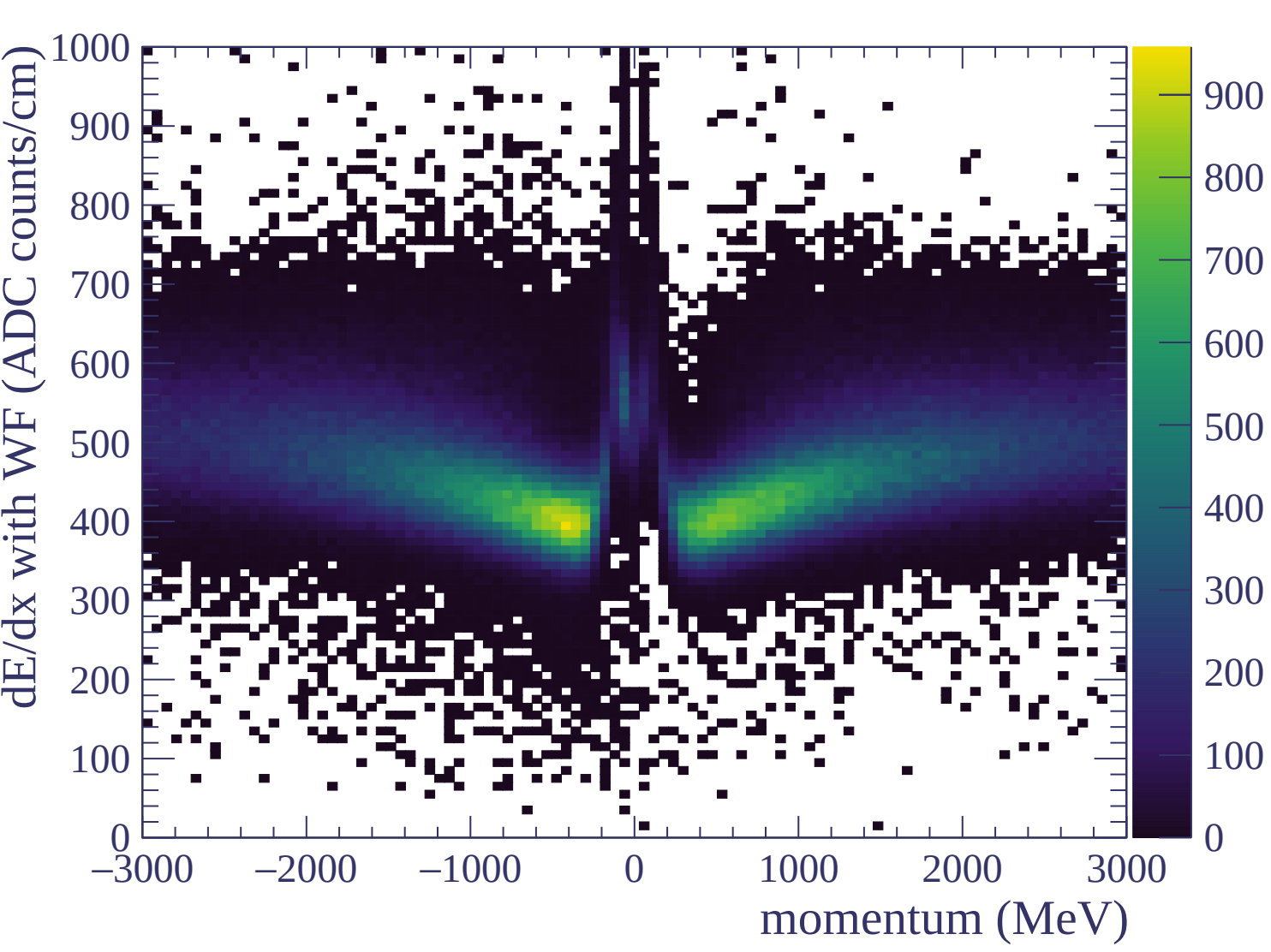


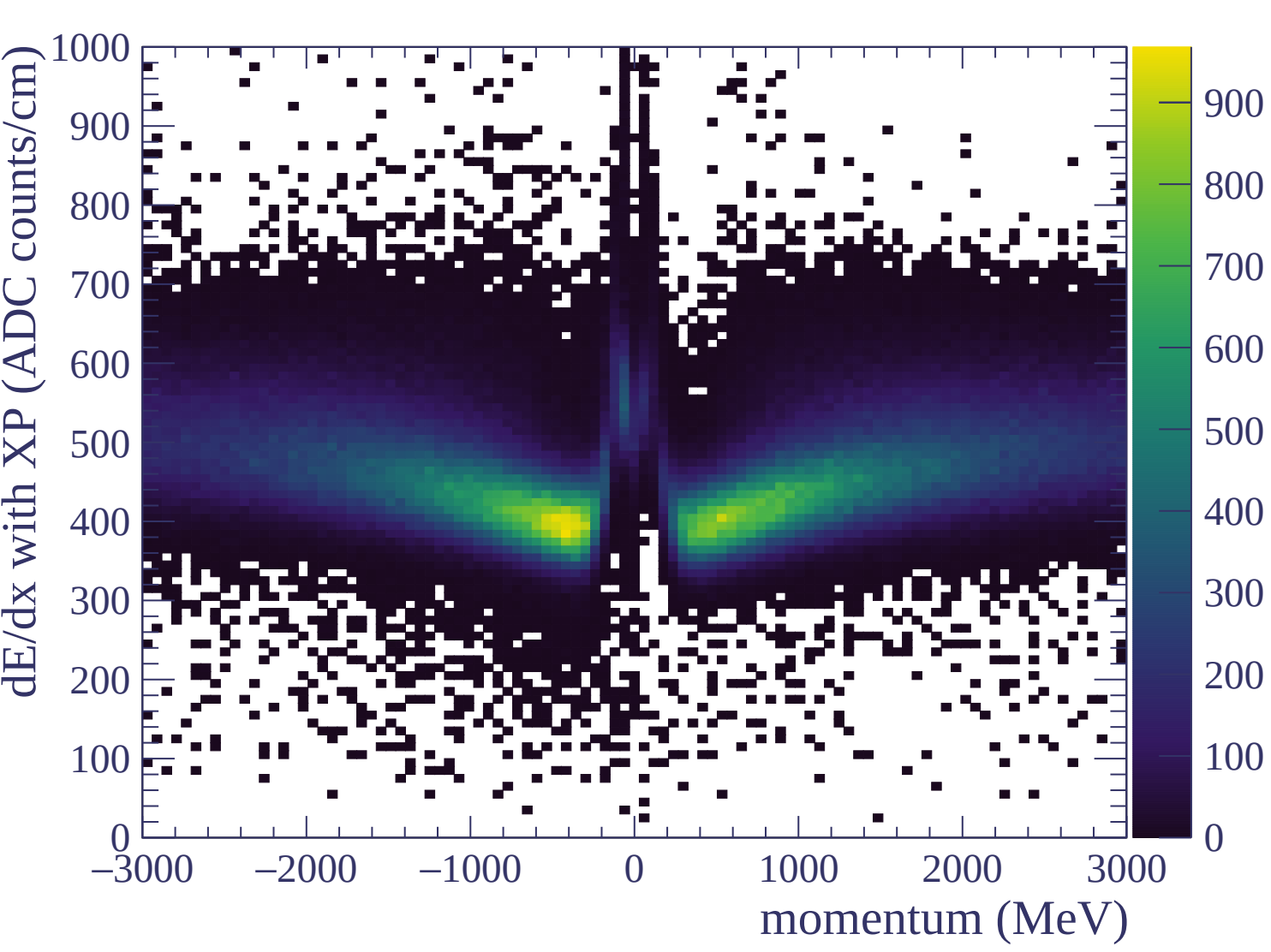
Resolution (%)

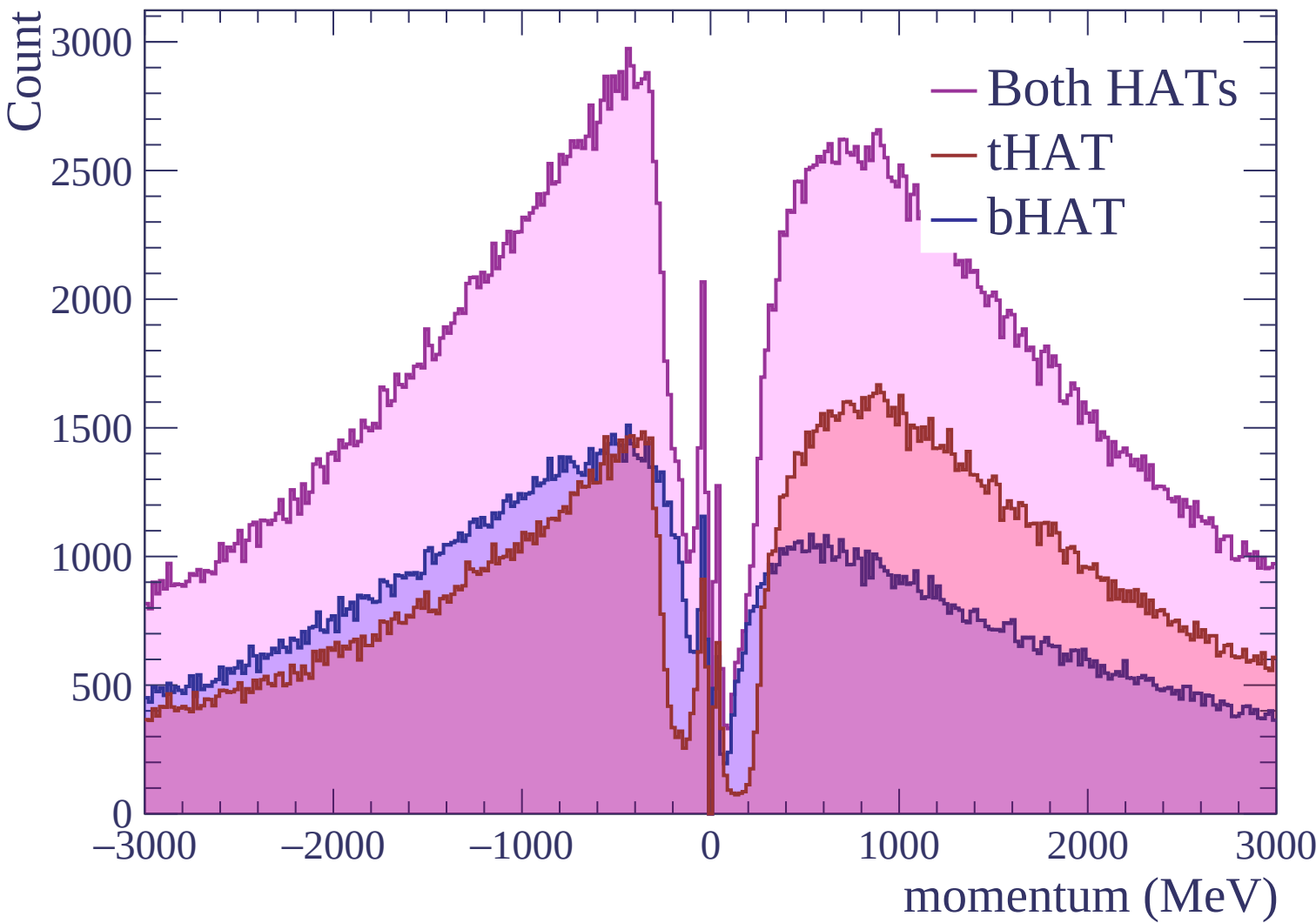


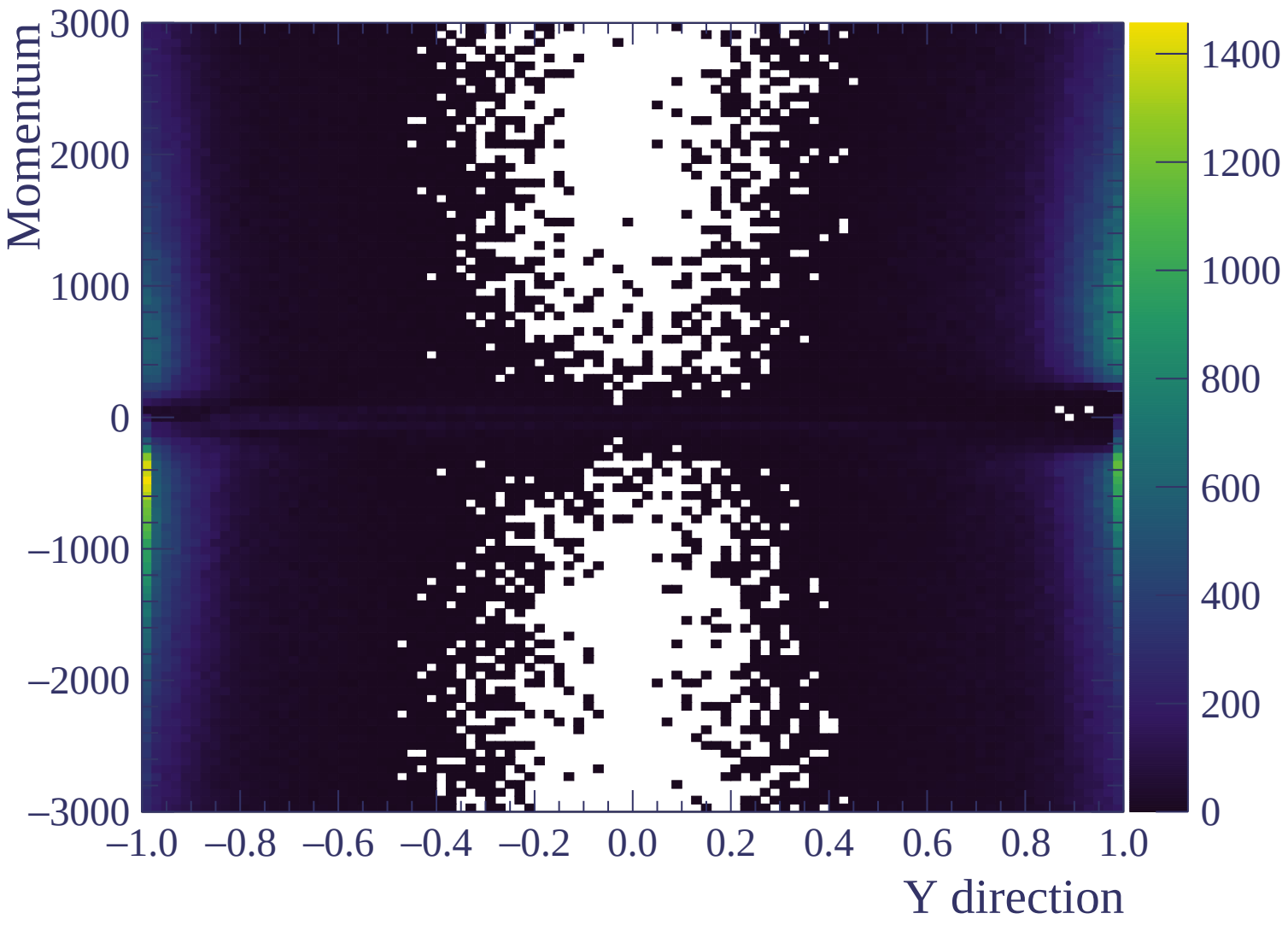


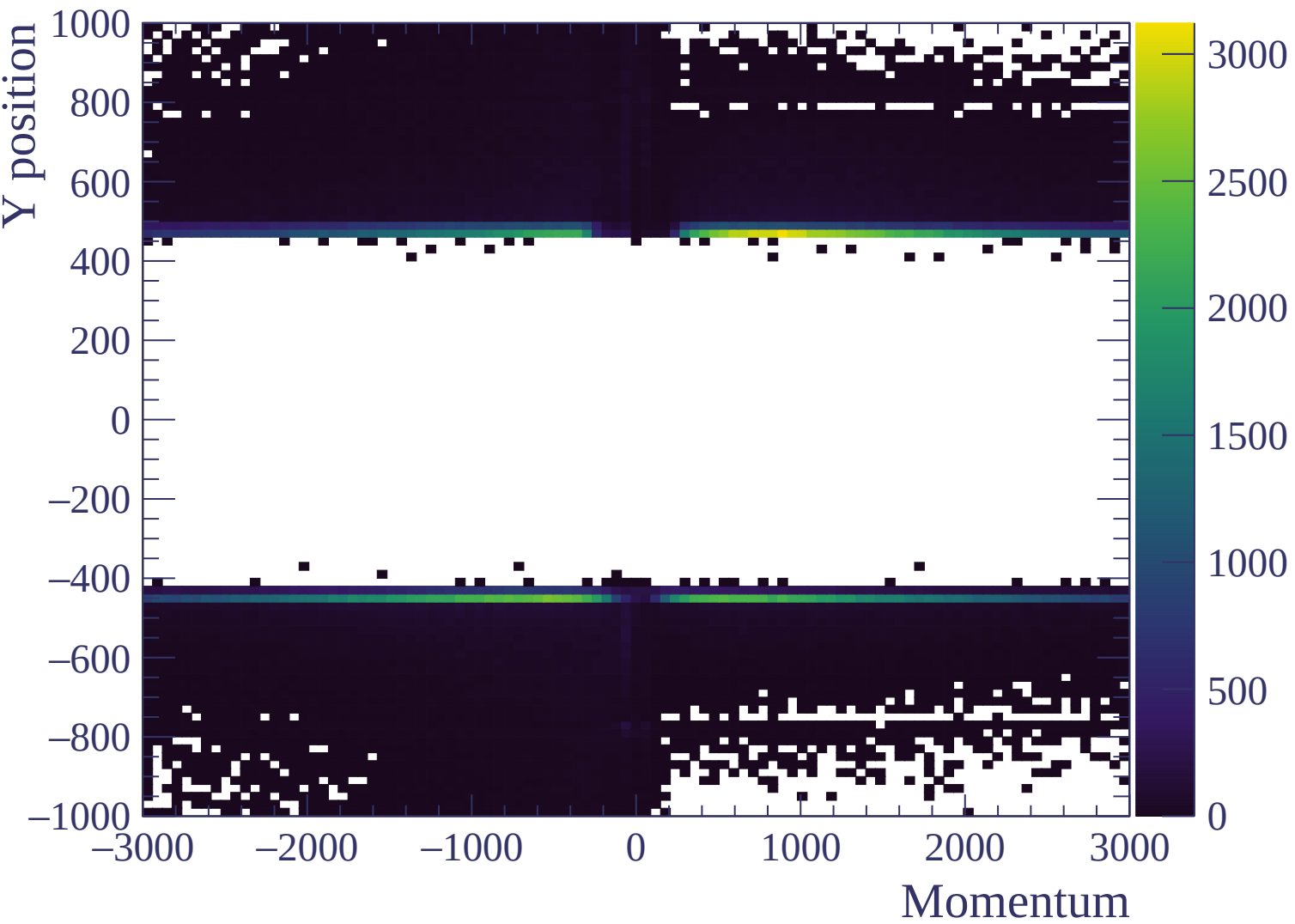


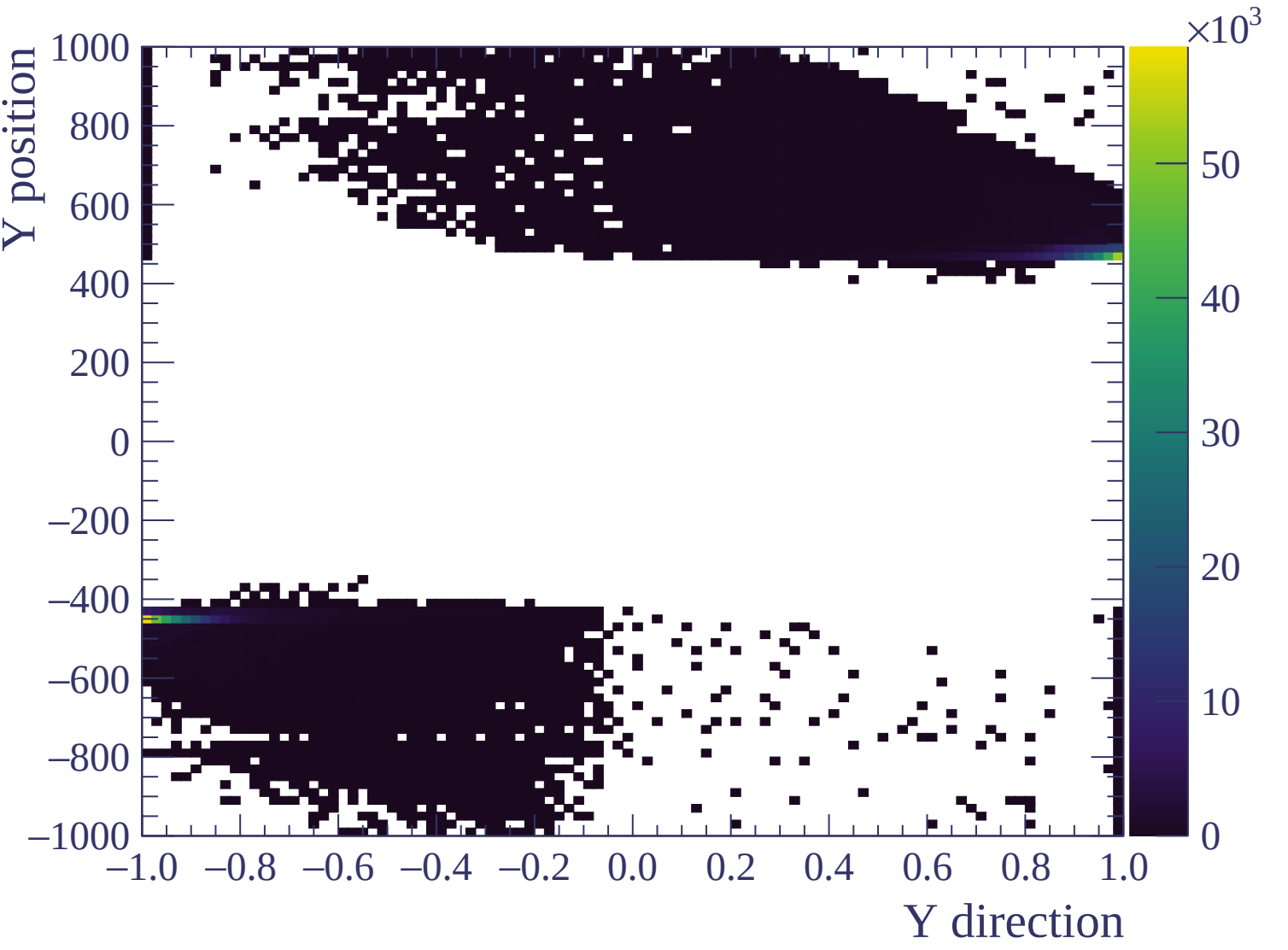






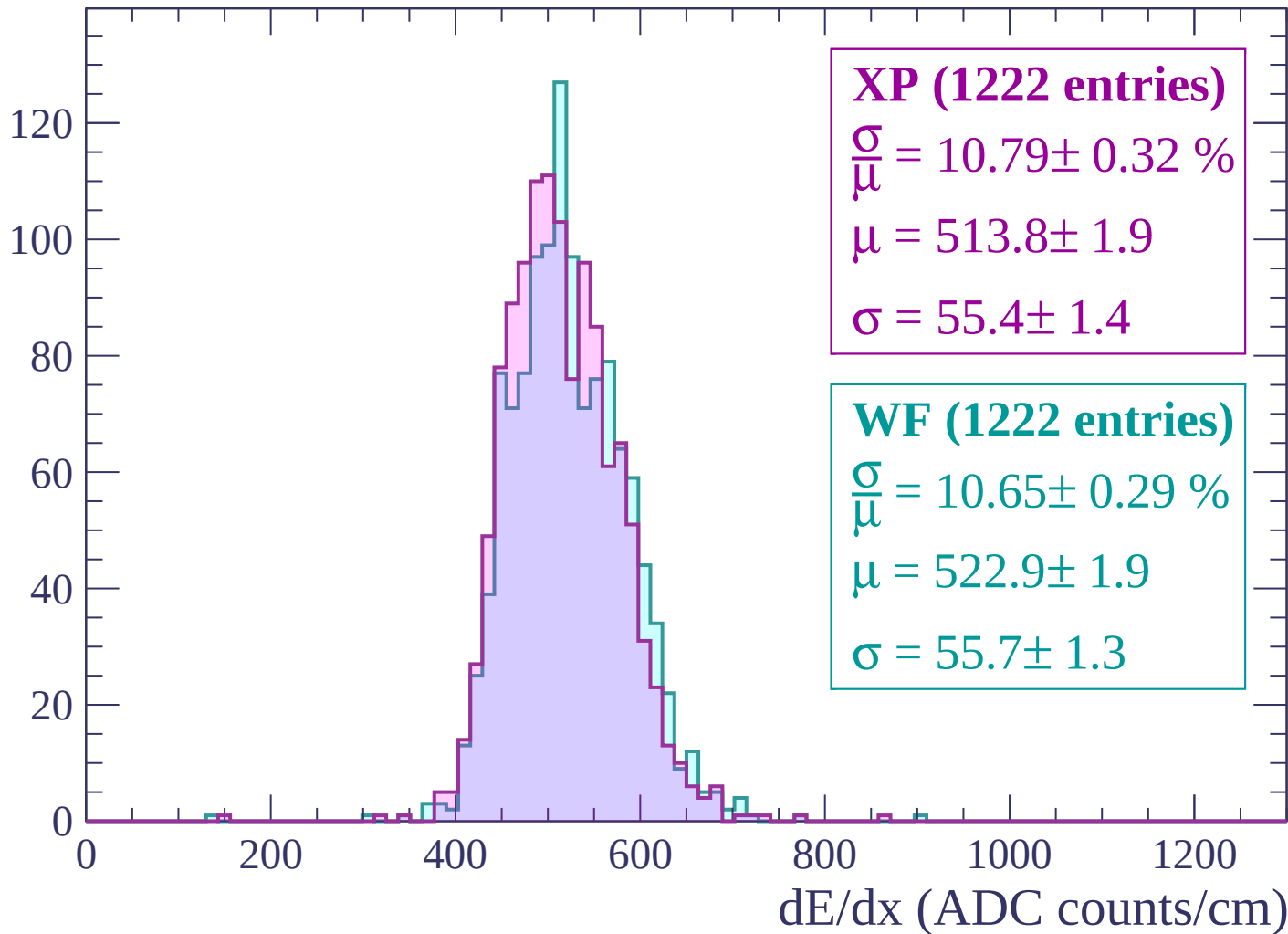






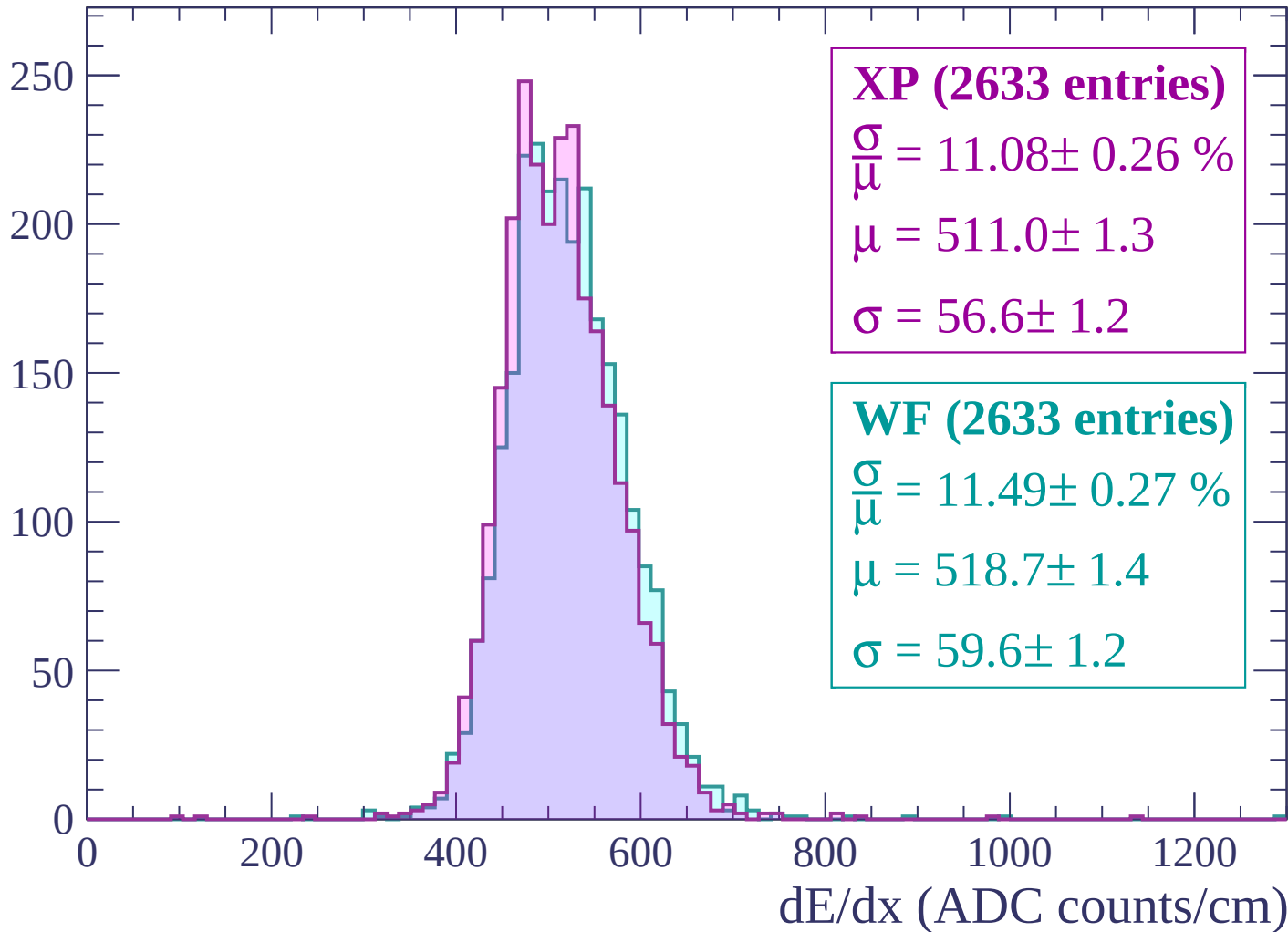
Energy loss | $-3000 < p < -2940$

Count



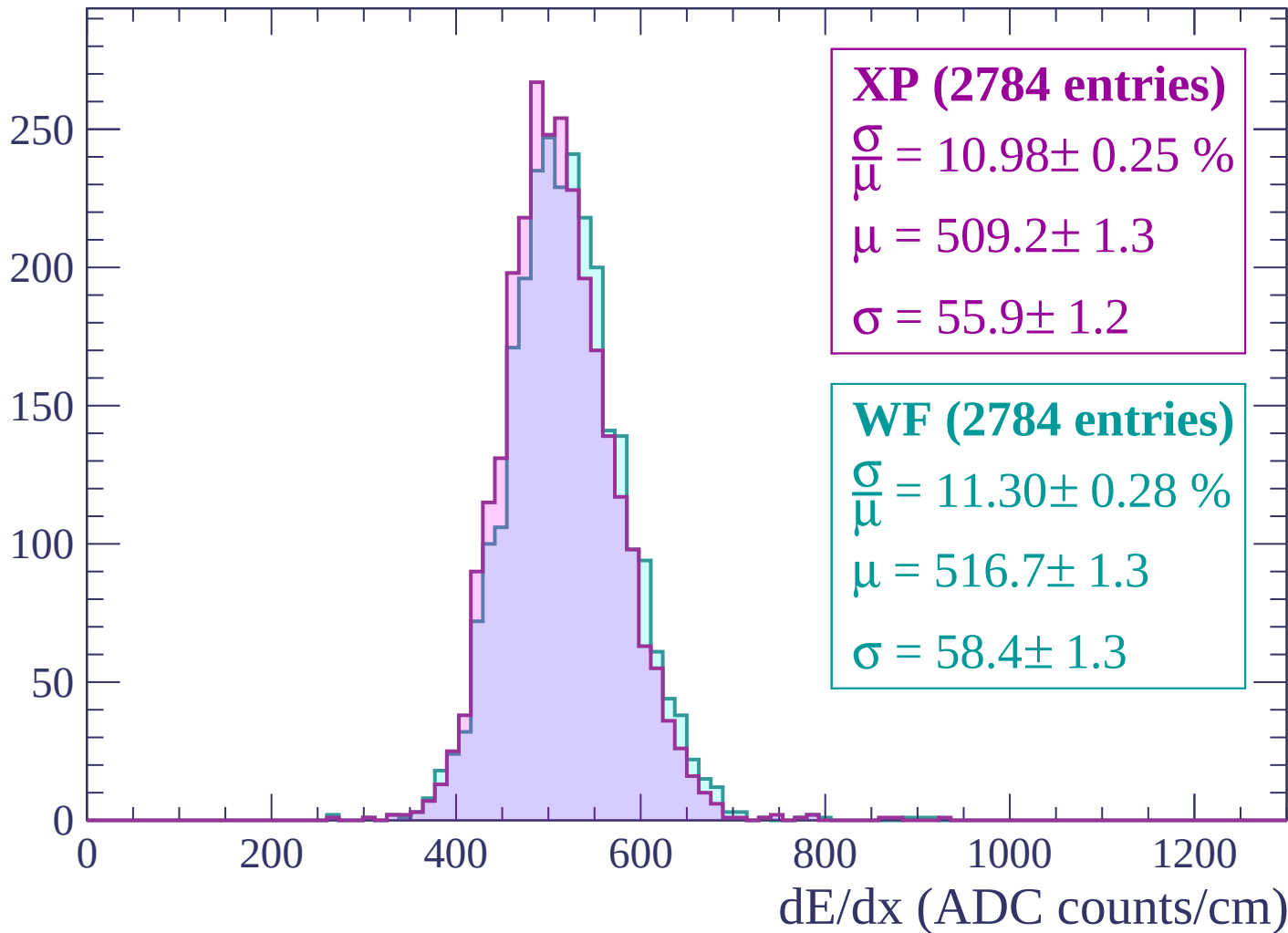
Energy loss | $-2940 < p < -2880$

Count



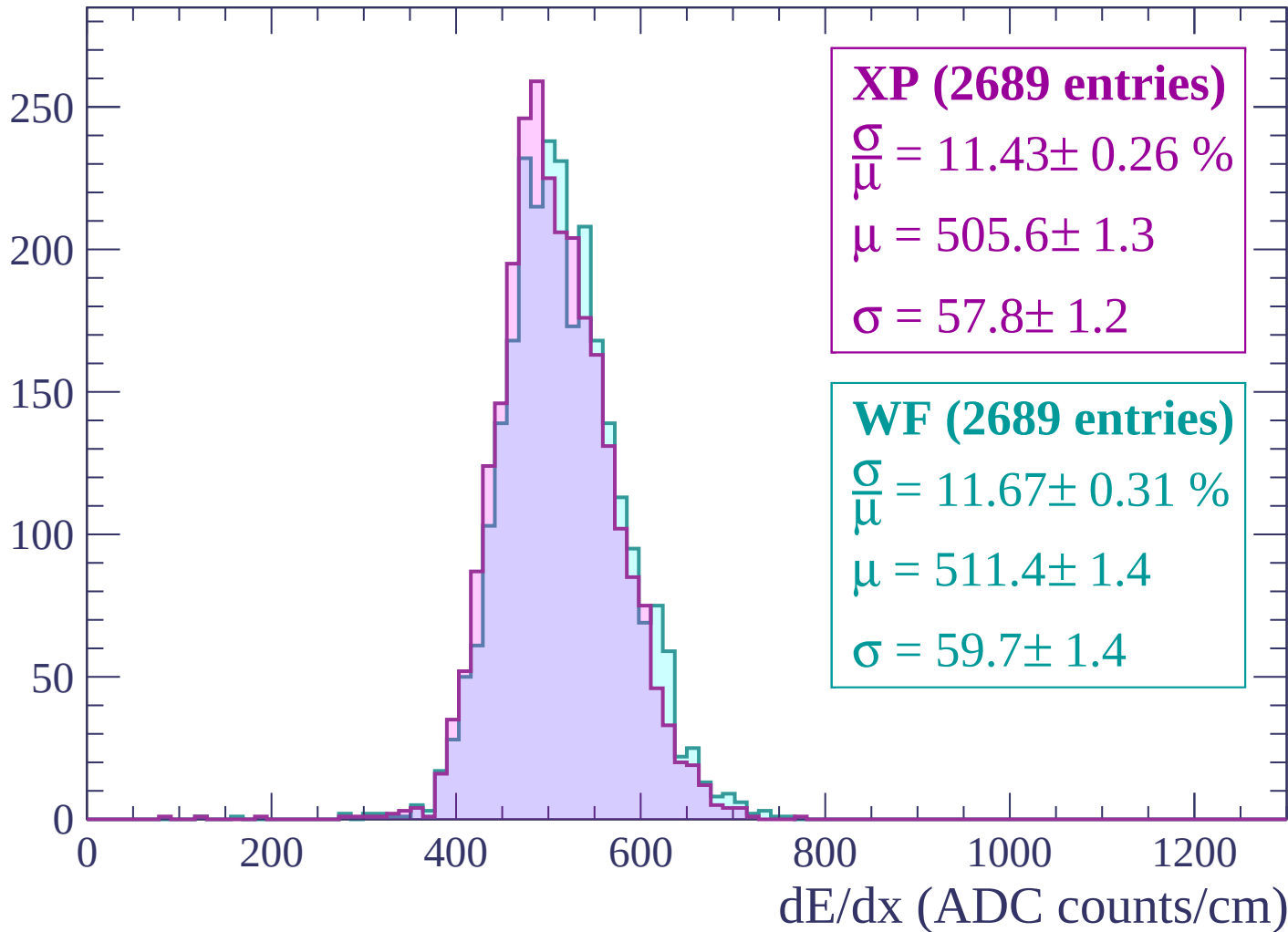
Energy loss | $-2880 < p < -2820$

Count



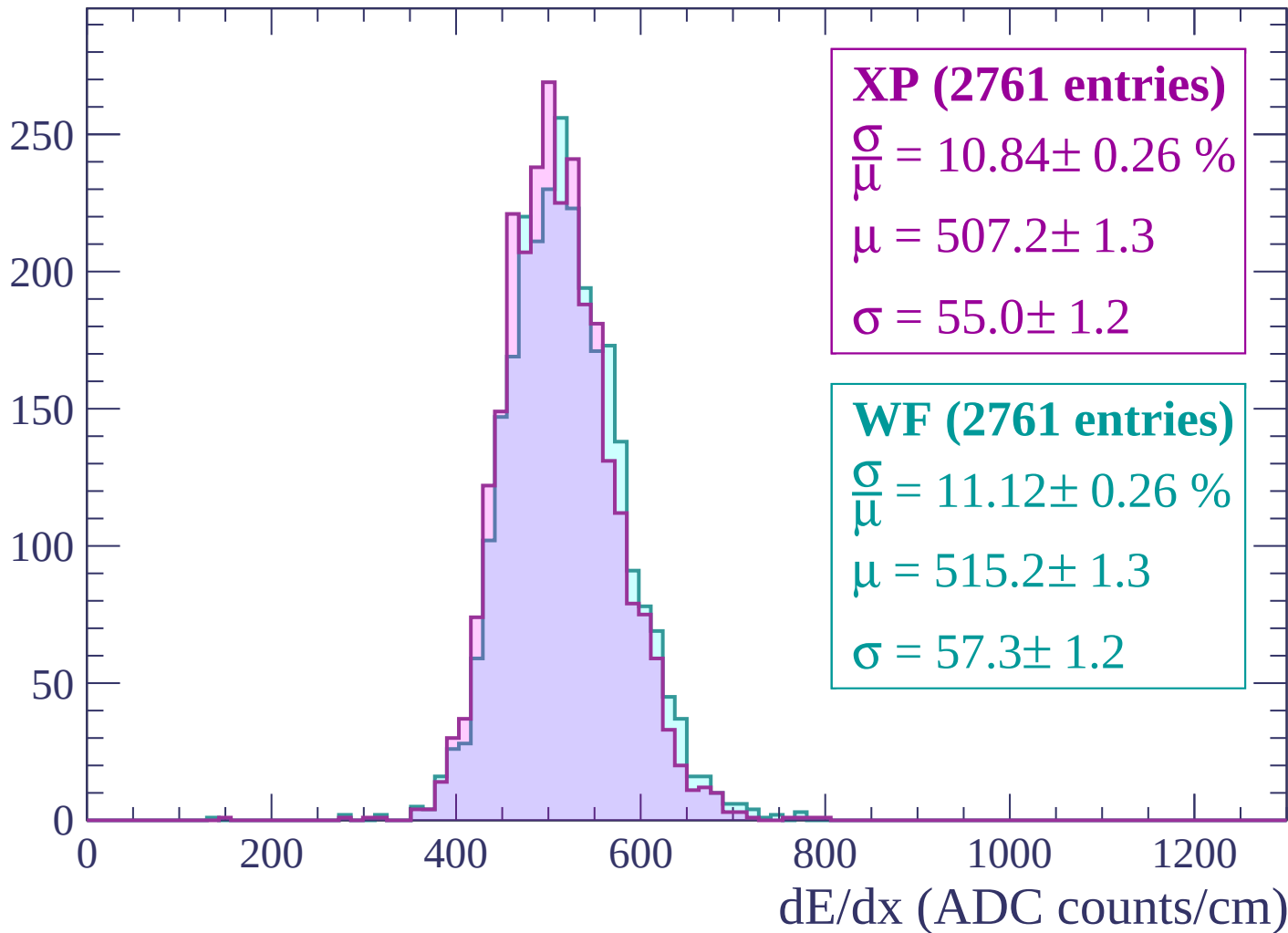
Energy loss | $-2820 < p < -2760$

Count



Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (2834 entries)

$\frac{\sigma}{\mu} = 10.75 \pm 0.25 \%$

$\mu = 503.6 \pm 1.2$

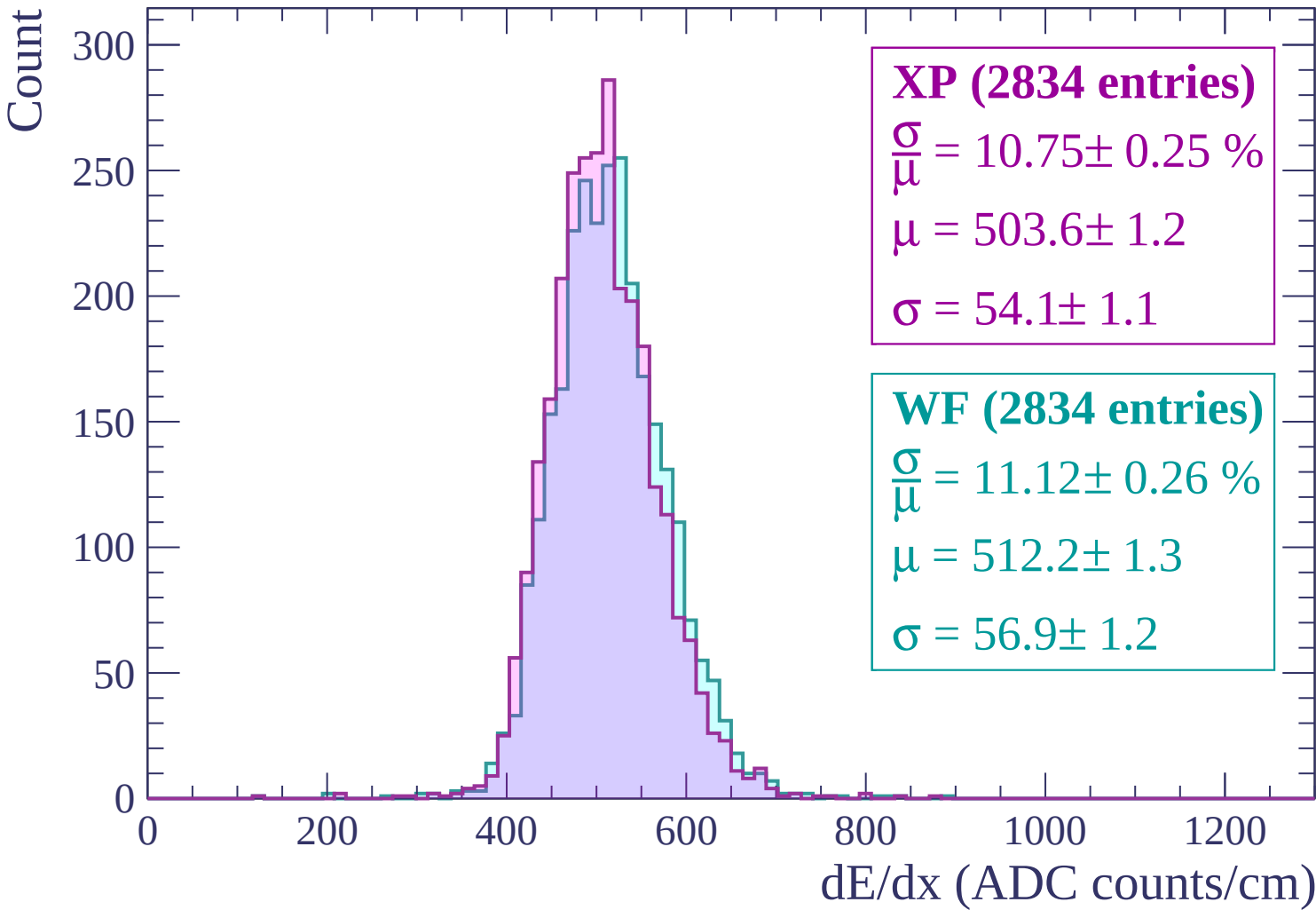
$\sigma = 54.1 \pm 1.1$

WF (2834 entries)

$\frac{\sigma}{\mu} = 11.12 \pm 0.26 \%$

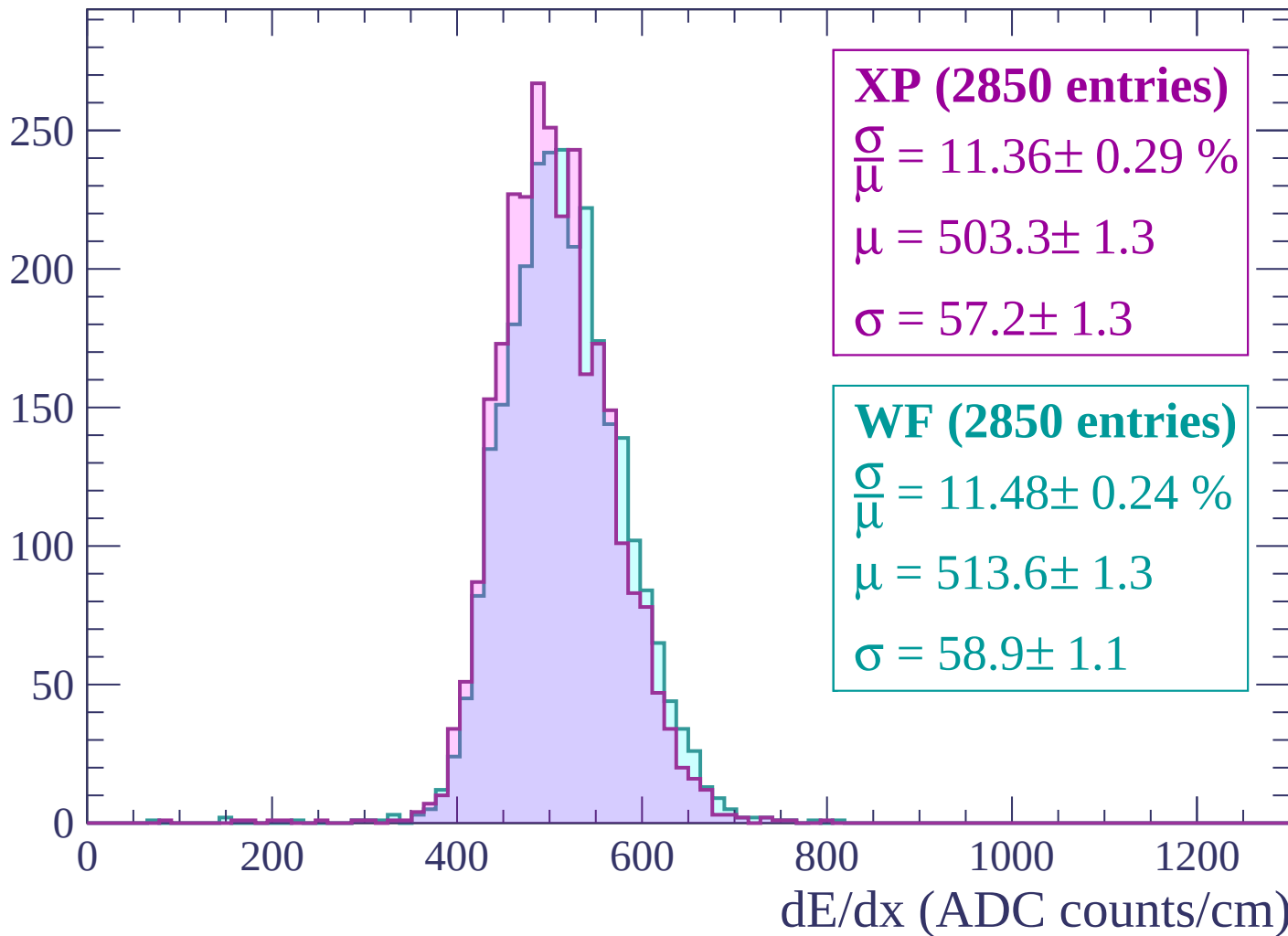
$\mu = 512.2 \pm 1.3$

$\sigma = 56.9 \pm 1.2$



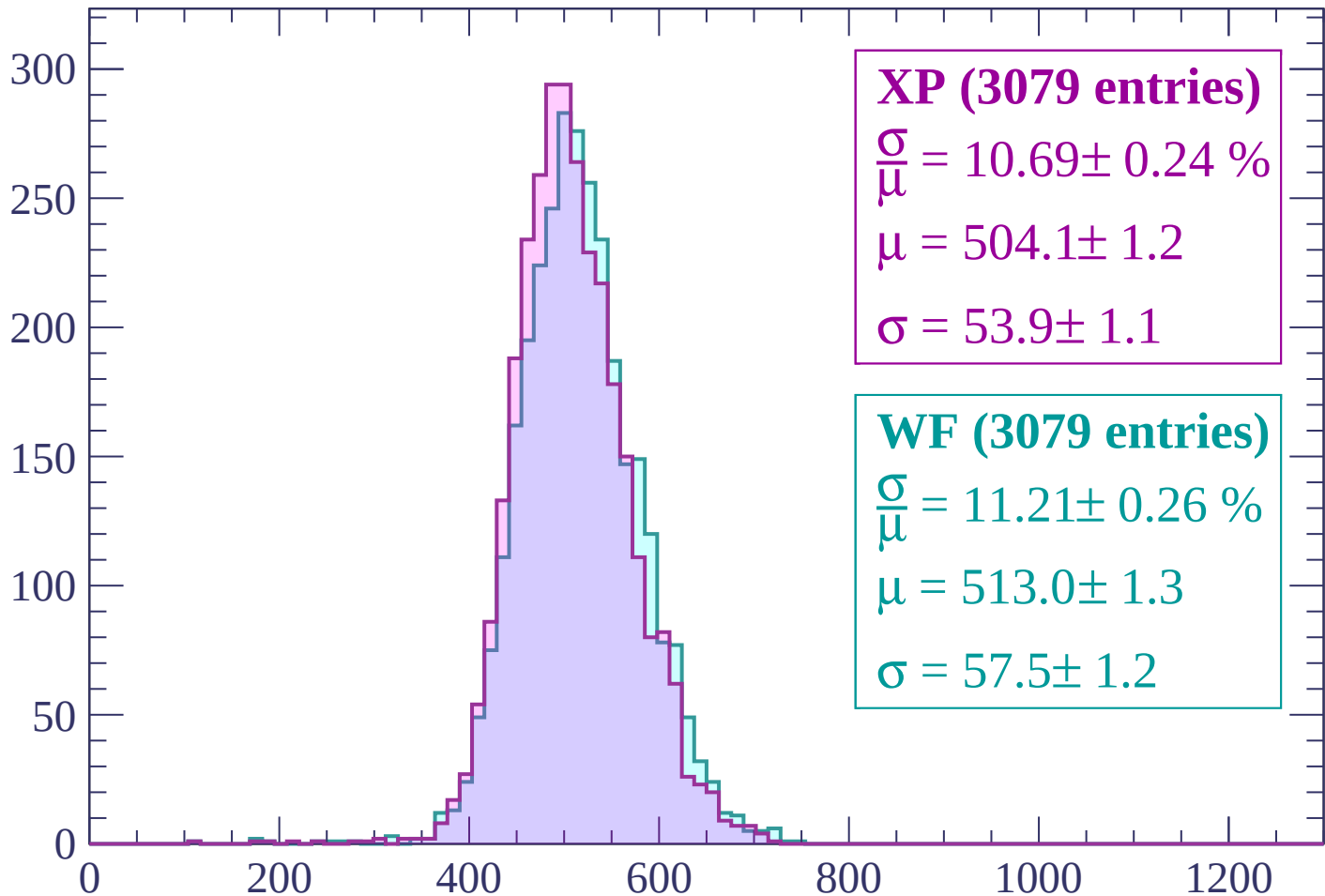
Energy loss | $-2640 < p < -2580$

Count



Energy loss | -2580 < p < -2520

Count



XP (3079 entries)

$$\frac{\sigma}{\mu} = 10.69 \pm 0.24 \%$$

$$\mu = 504.1 \pm 1.2$$

$$\sigma = 53.9 \pm 1.1$$

WF (3079 entries)

$$\frac{\sigma}{\mu} = 11.21 \pm 0.26 \%$$

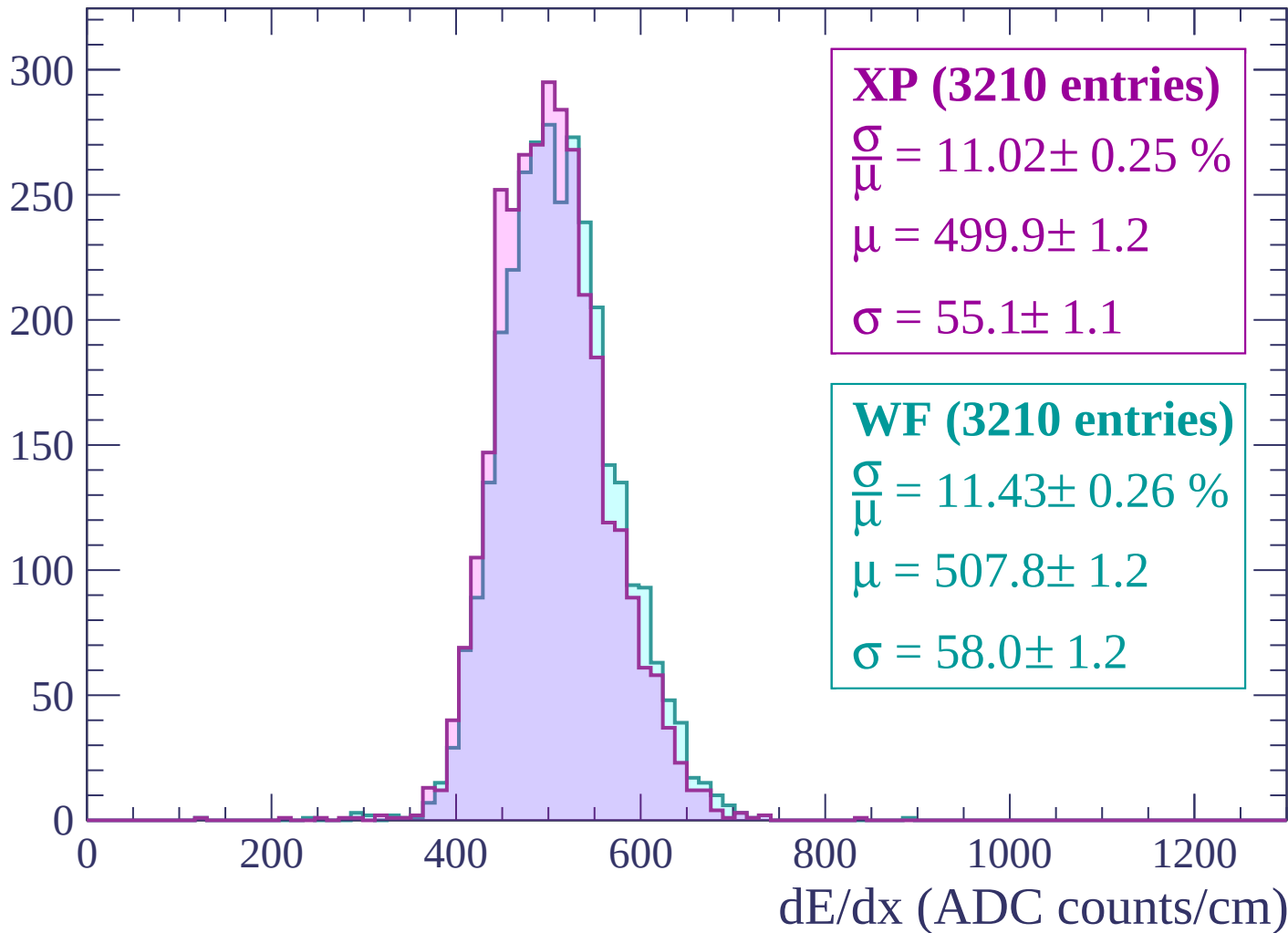
$$\mu = 513.0 \pm 1.3$$

$$\sigma = 57.5 \pm 1.2$$

dE/dx (ADC counts/cm)

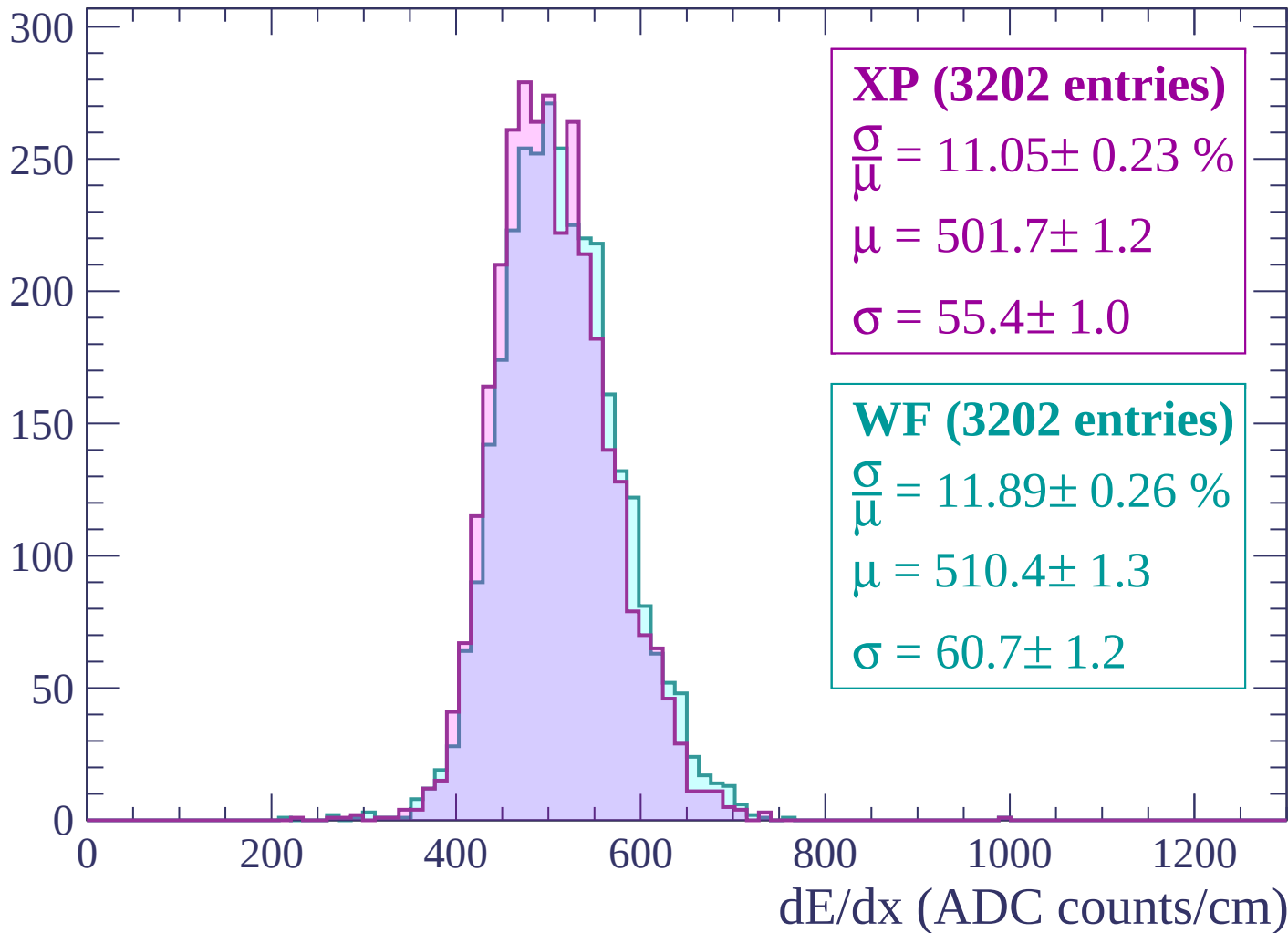
Energy loss | -2520 < p < -2460

Count



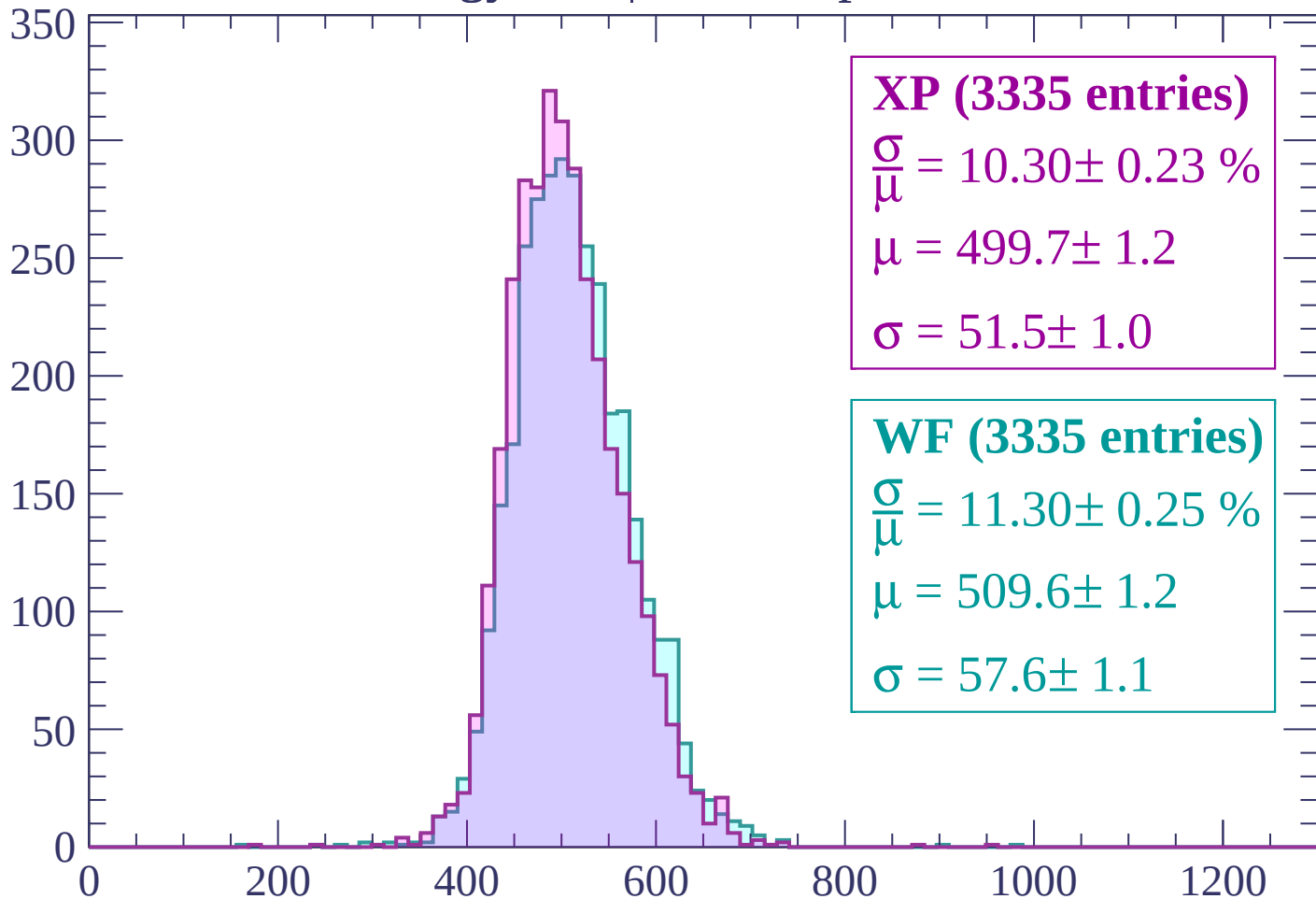
Energy loss | $-2460 < p < -2400$

Count



Energy loss | -2400 < p < -2340

Count



XP (3335 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.23 \%$$

$$\mu = 499.7 \pm 1.2$$

$$\sigma = 51.5 \pm 1.0$$

WF (3335 entries)

$$\frac{\sigma}{\mu} = 11.30 \pm 0.25 \%$$

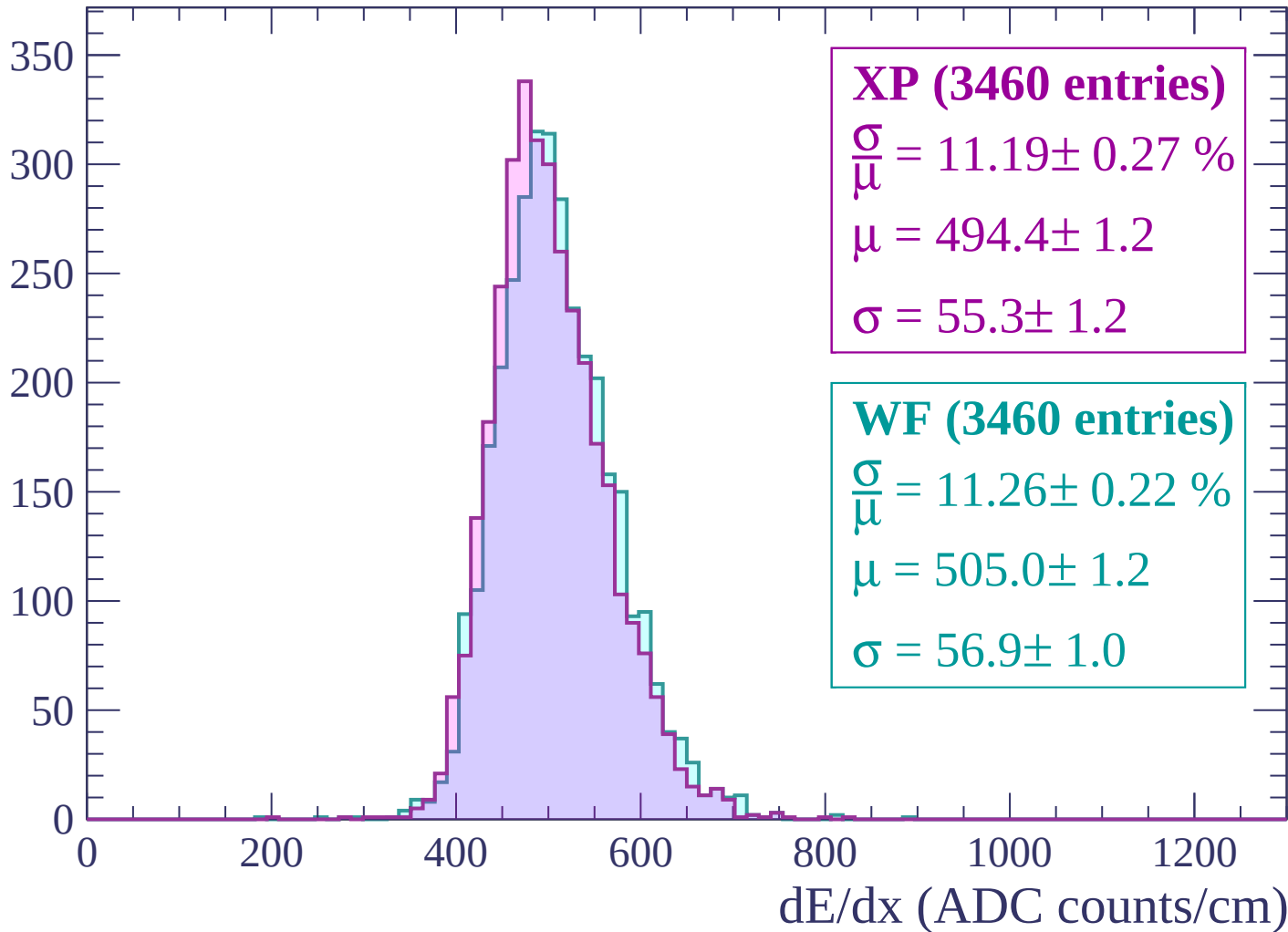
$$\mu = 509.6 \pm 1.2$$

$$\sigma = 57.6 \pm 1.1$$

dE/dx (ADC counts/cm)

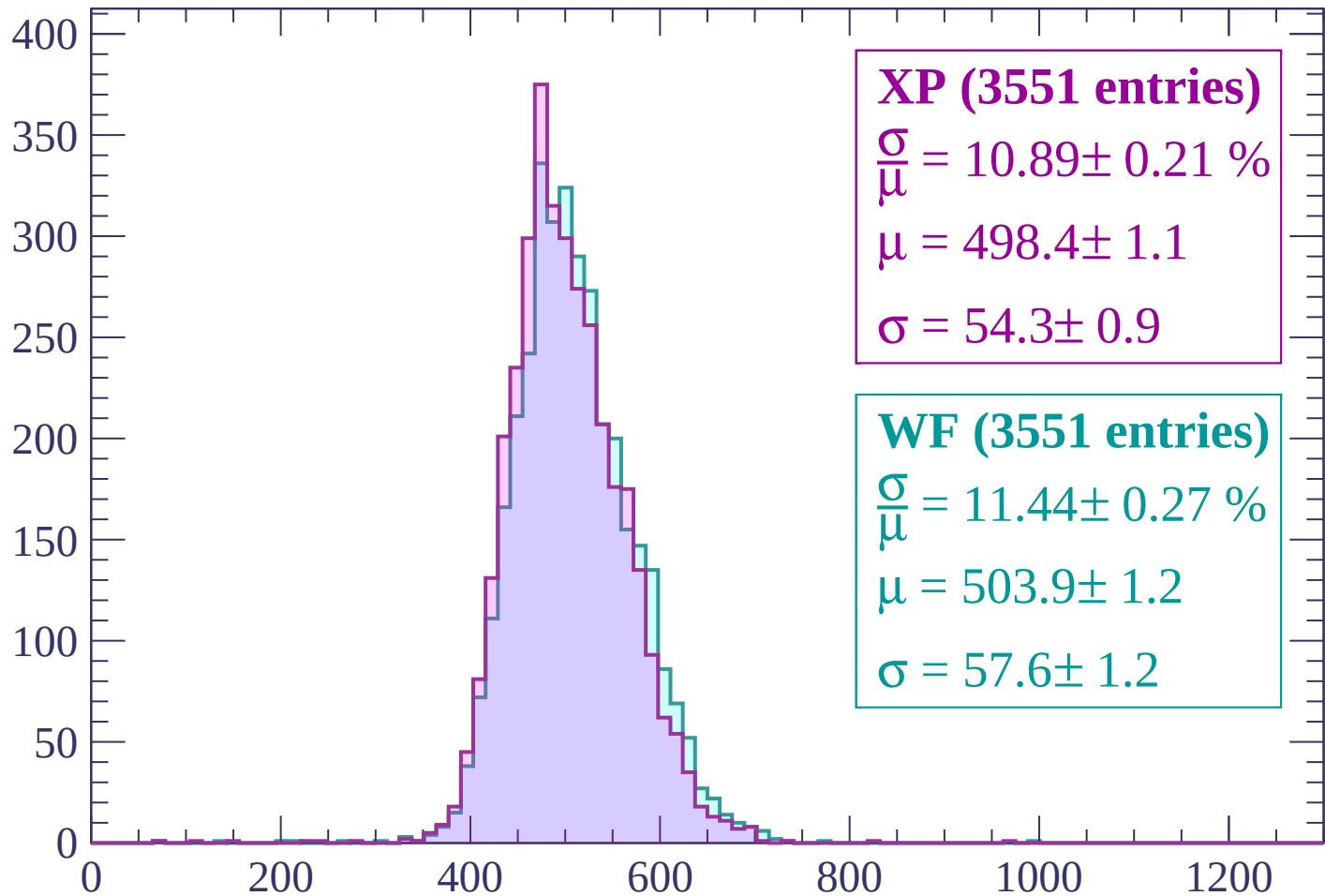
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (3551 entries)

$$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$$

$$\mu = 498.4 \pm 1.1$$

$$\sigma = 54.3 \pm 0.9$$

WF (3551 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.27 \%$$

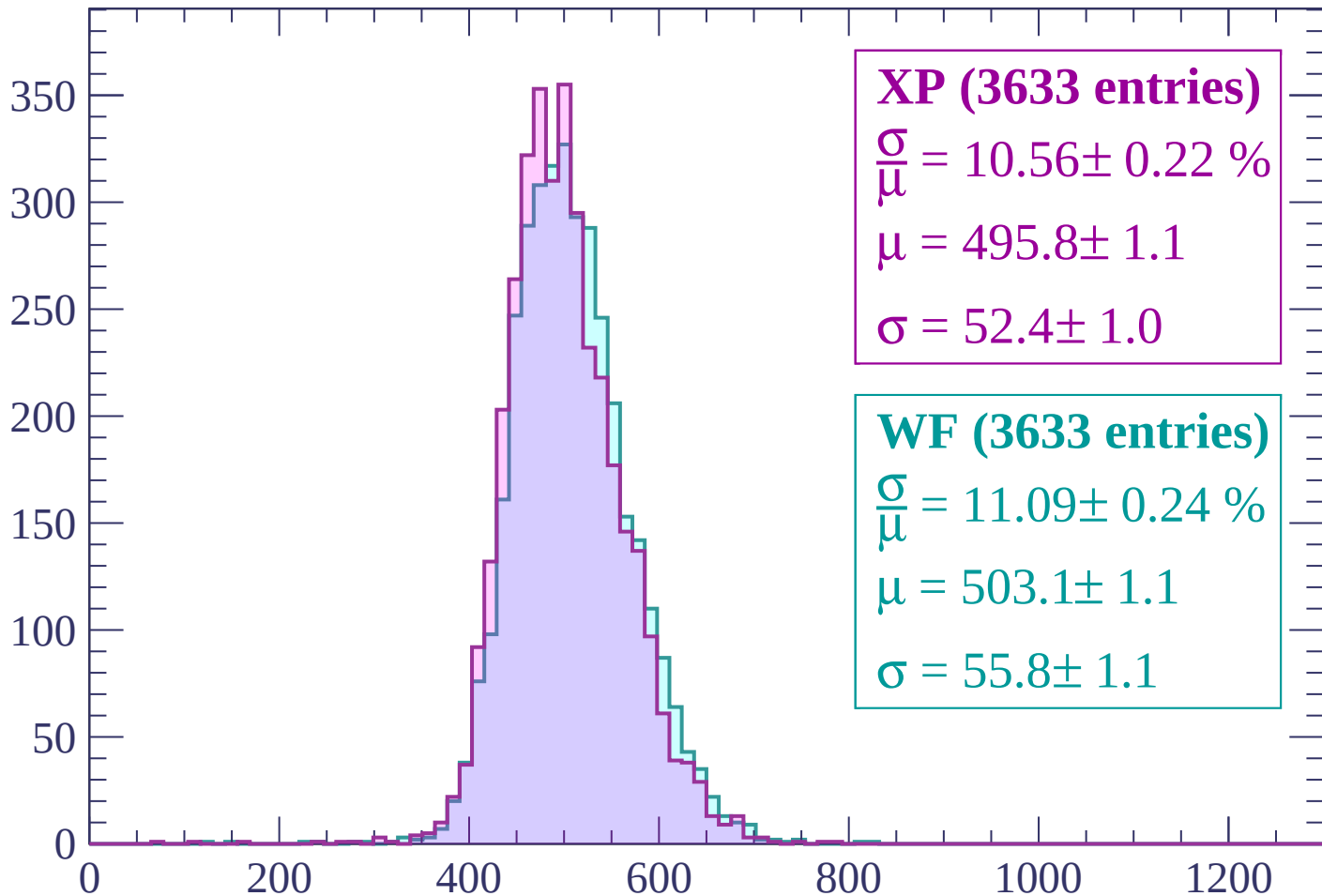
$$\mu = 503.9 \pm 1.2$$

$$\sigma = 57.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count



XP (3633 entries)

$$\frac{\sigma}{\mu} = 10.56 \pm 0.22 \%$$

$$\mu = 495.8 \pm 1.1$$

$$\sigma = 52.4 \pm 1.0$$

WF (3633 entries)

$$\frac{\sigma}{\mu} = 11.09 \pm 0.24 \%$$

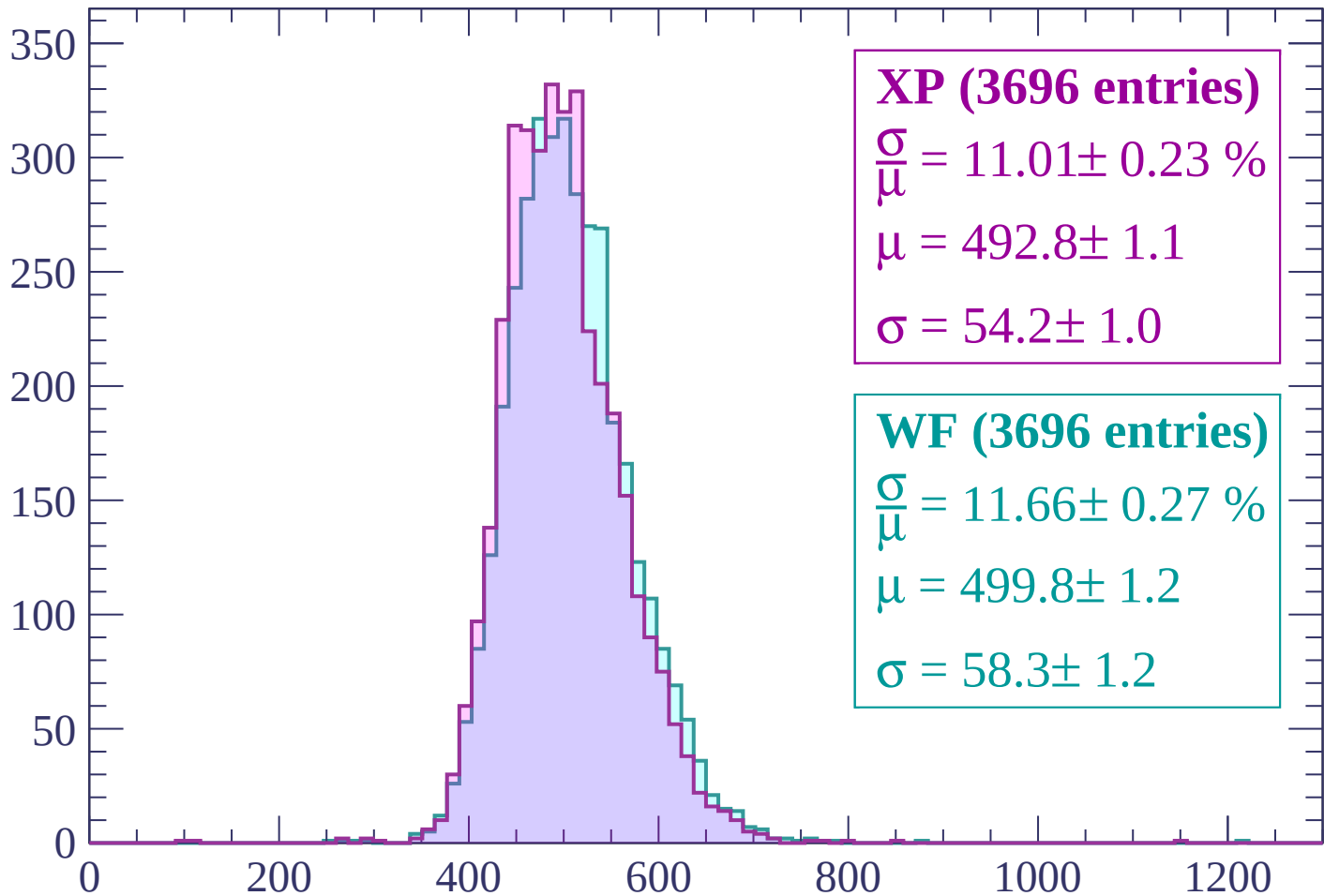
$$\mu = 503.1 \pm 1.1$$

$$\sigma = 55.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2160 < p < -2100

Count



XP (3696 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.23 \%$$

$$\mu = 492.8 \pm 1.1$$

$$\sigma = 54.2 \pm 1.0$$

WF (3696 entries)

$$\frac{\sigma}{\mu} = 11.66 \pm 0.27 \%$$

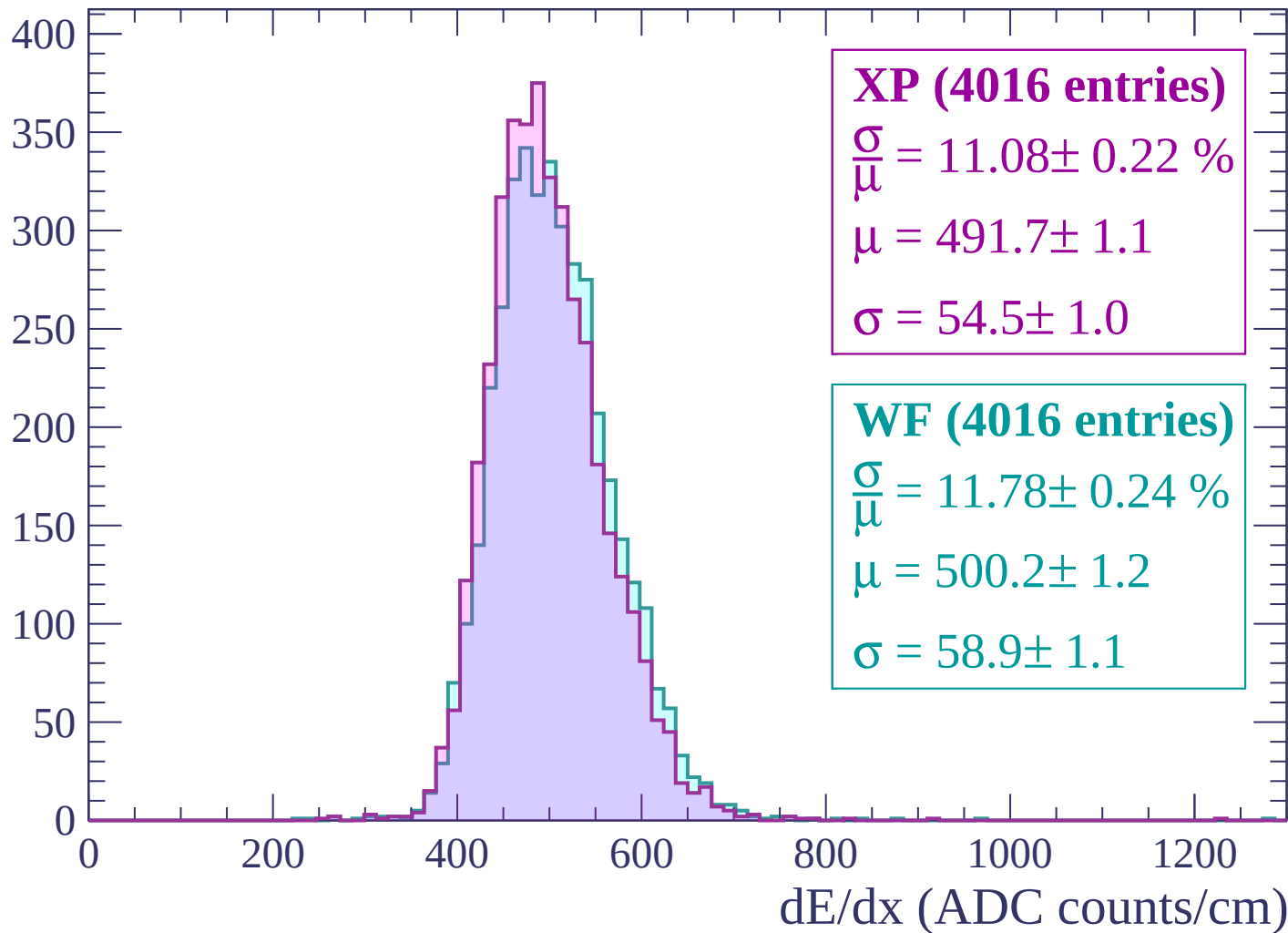
$$\mu = 499.8 \pm 1.2$$

$$\sigma = 58.3 \pm 1.2$$

dE/dx (ADC counts/cm)

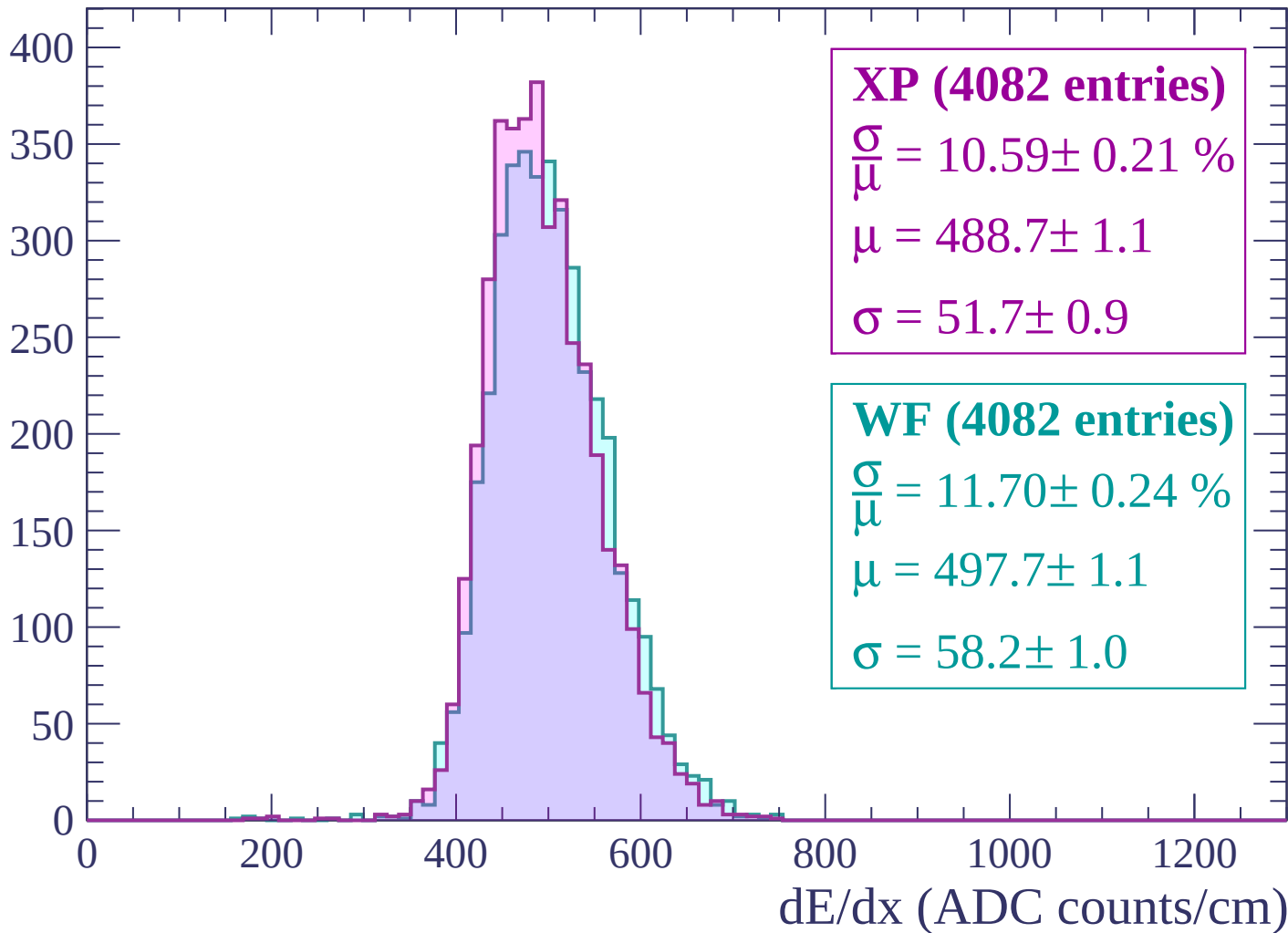
Energy loss | -2100 < p < -2040

Count



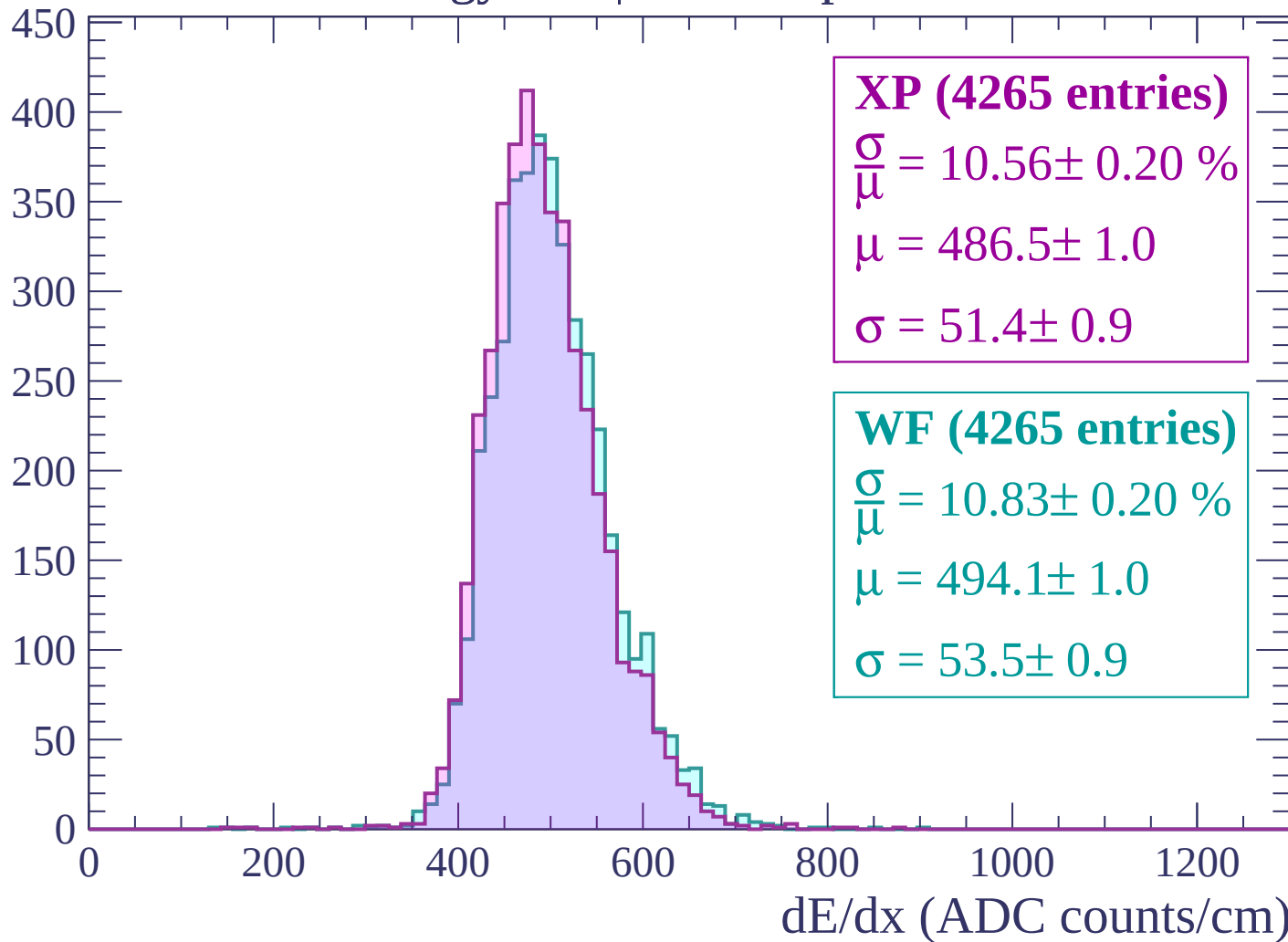
Energy loss | -2040 < p < -1980

Count

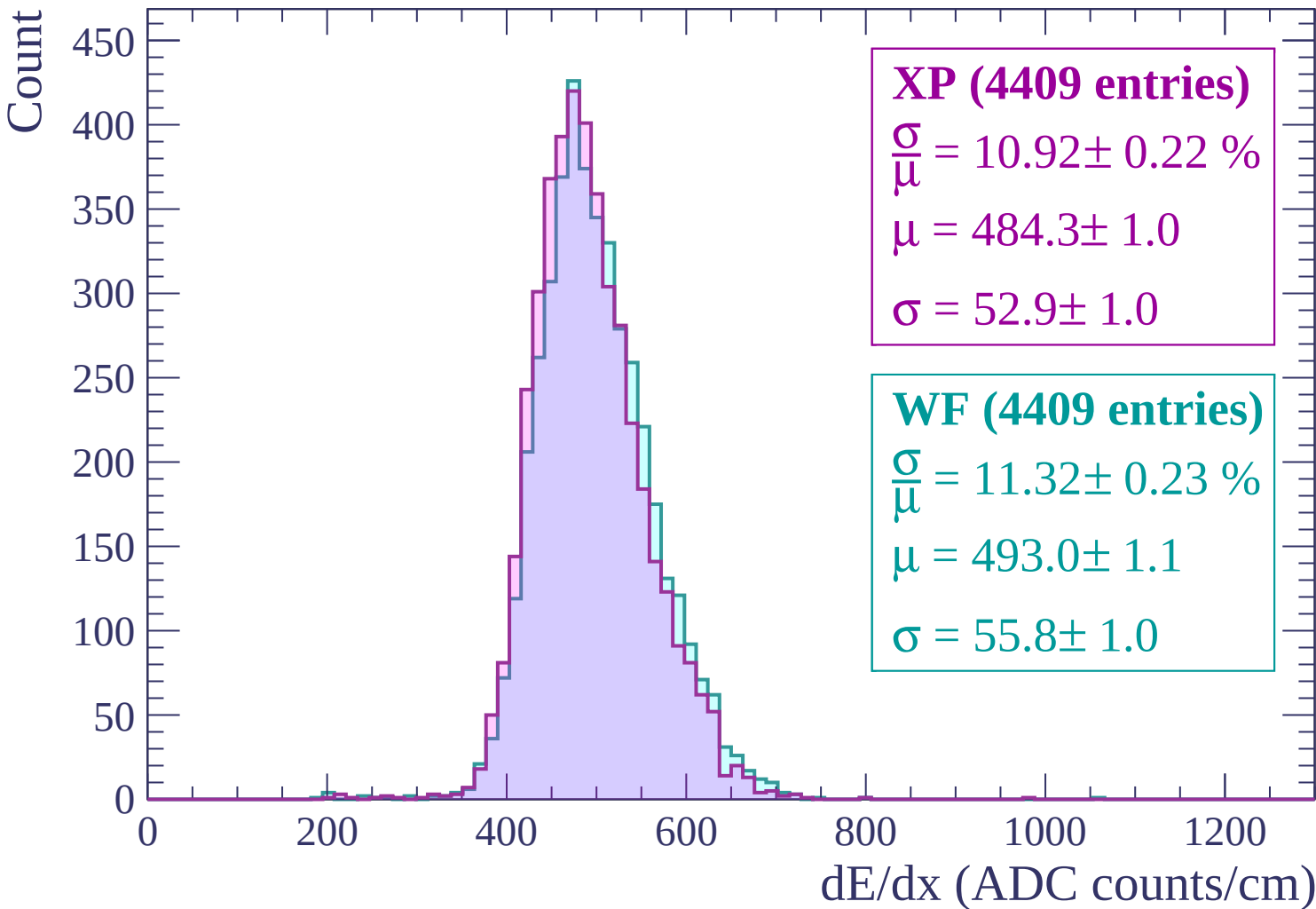


Energy loss | -1980 < p < -1920

Count

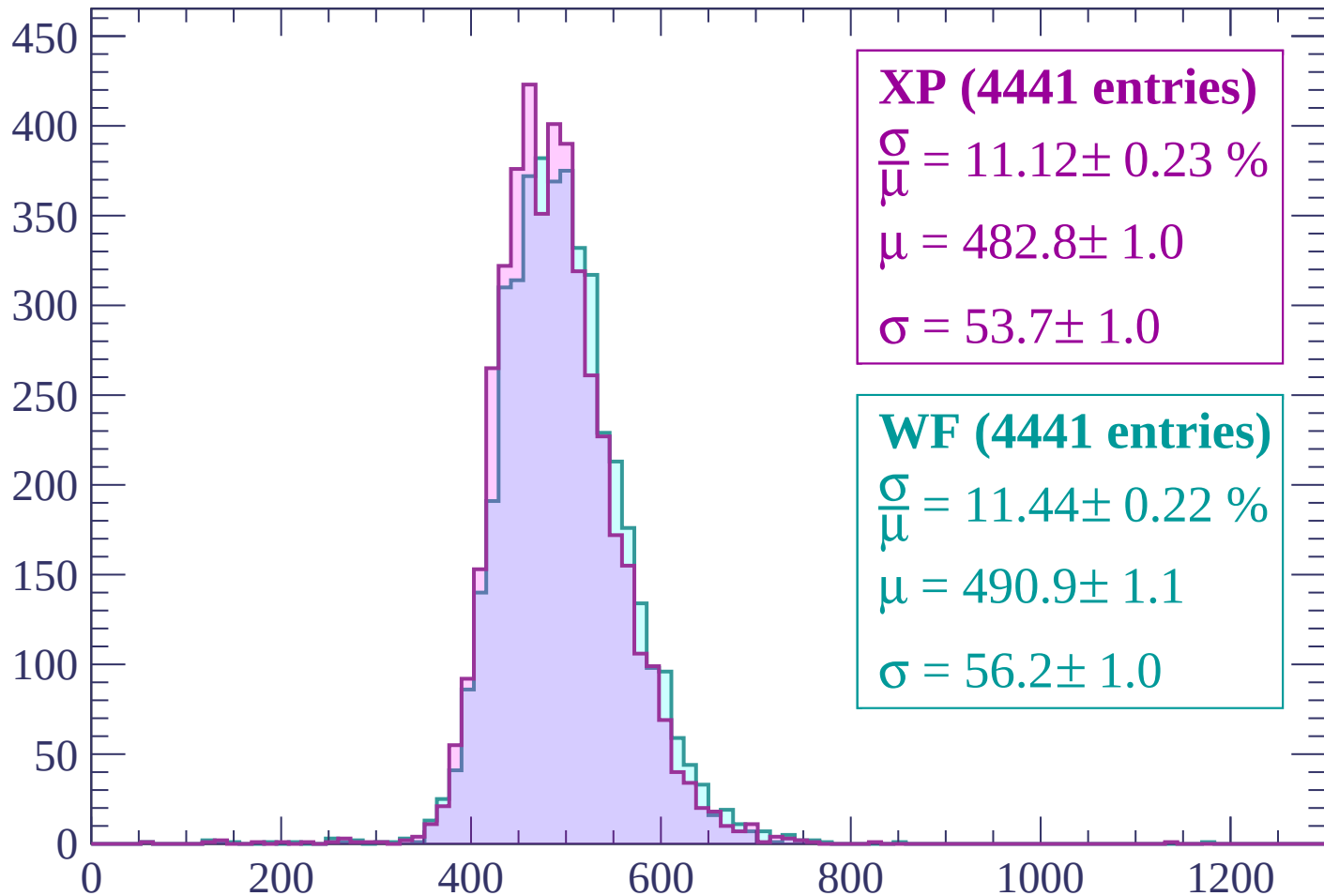


Energy loss | -1920 < p < -1860



Energy loss | -1860 < p < -1800

Count



XP (4441 entries)

$$\frac{\sigma}{\mu} = 11.12 \pm 0.23 \%$$

$$\mu = 482.8 \pm 1.0$$

$$\sigma = 53.7 \pm 1.0$$

WF (4441 entries)

$$\frac{\sigma}{\mu} = 11.44 \pm 0.22 \%$$

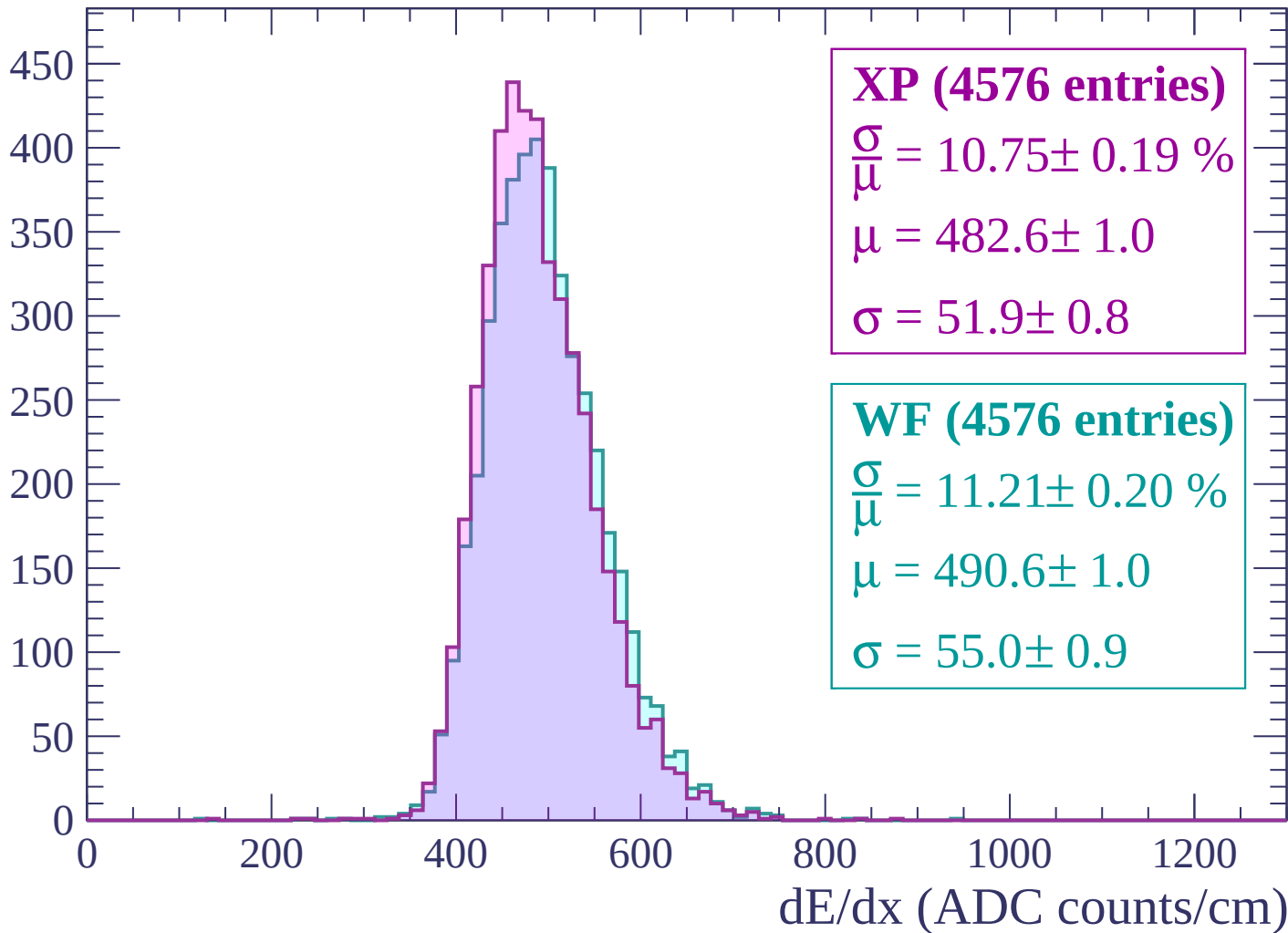
$$\mu = 490.9 \pm 1.1$$

$$\sigma = 56.2 \pm 1.0$$

dE/dx (ADC counts/cm)

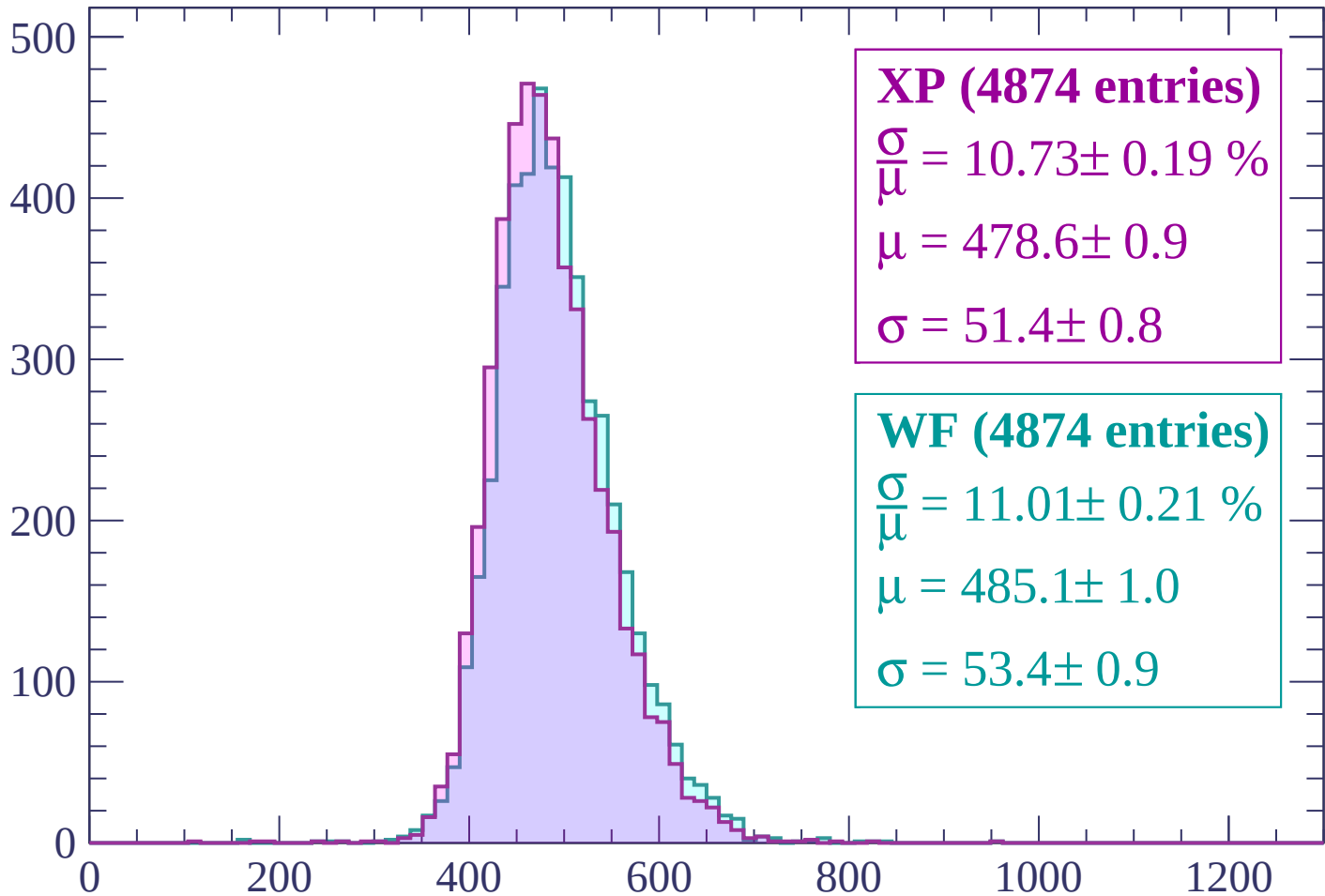
Energy loss | $-1800 < p < -1740$

Count



Energy loss | -1740 < p < -1680

Count



XP (4874 entries)

$\frac{\sigma}{\mu} = 10.73 \pm 0.19 \%$

$\mu = 478.6 \pm 0.9$

$\sigma = 51.4 \pm 0.8$

WF (4874 entries)

$\frac{\sigma}{\mu} = 11.01 \pm 0.21 \%$

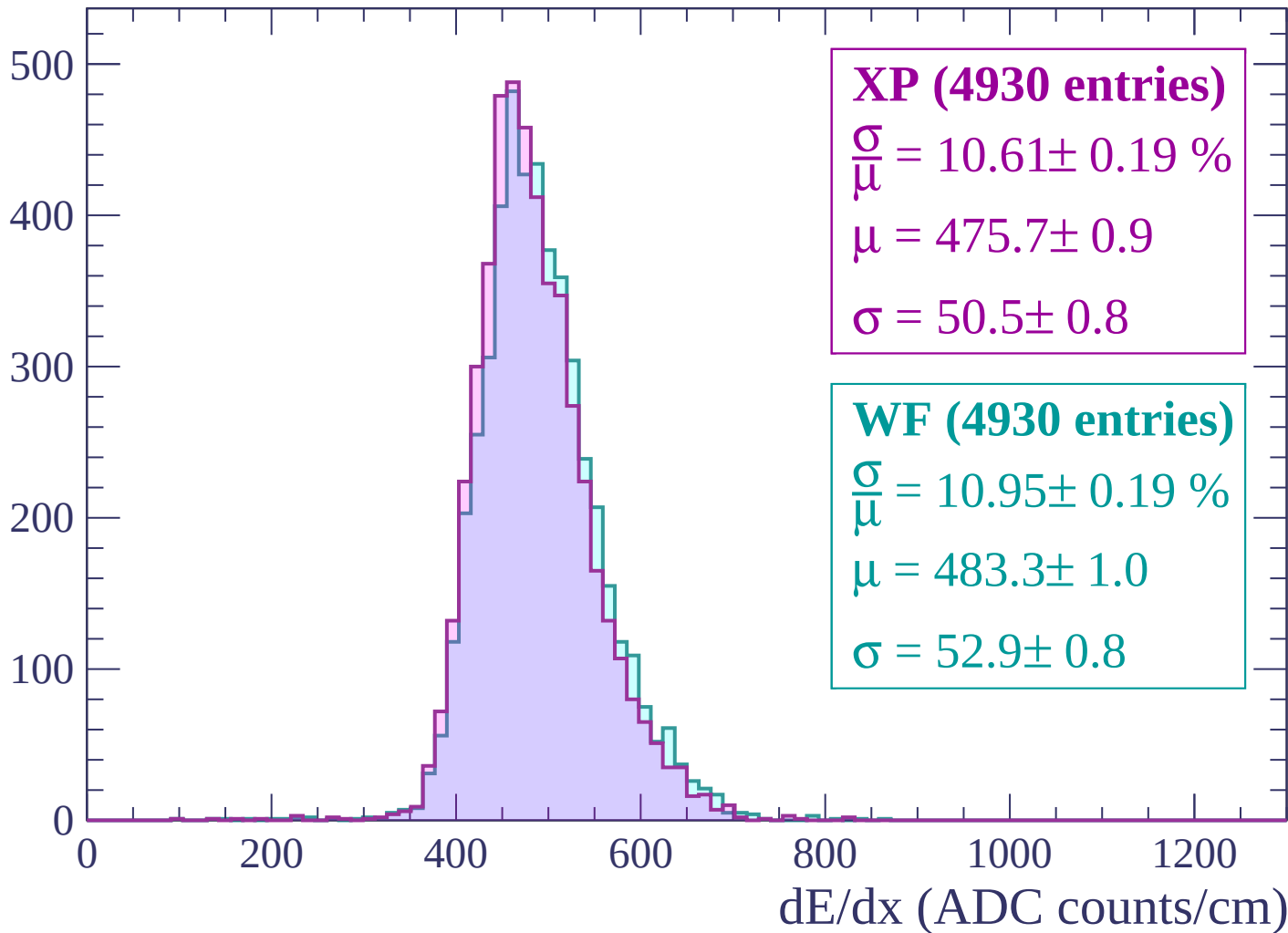
$\mu = 485.1 \pm 1.0$

$\sigma = 53.4 \pm 0.9$

dE/dx (ADC counts/cm)

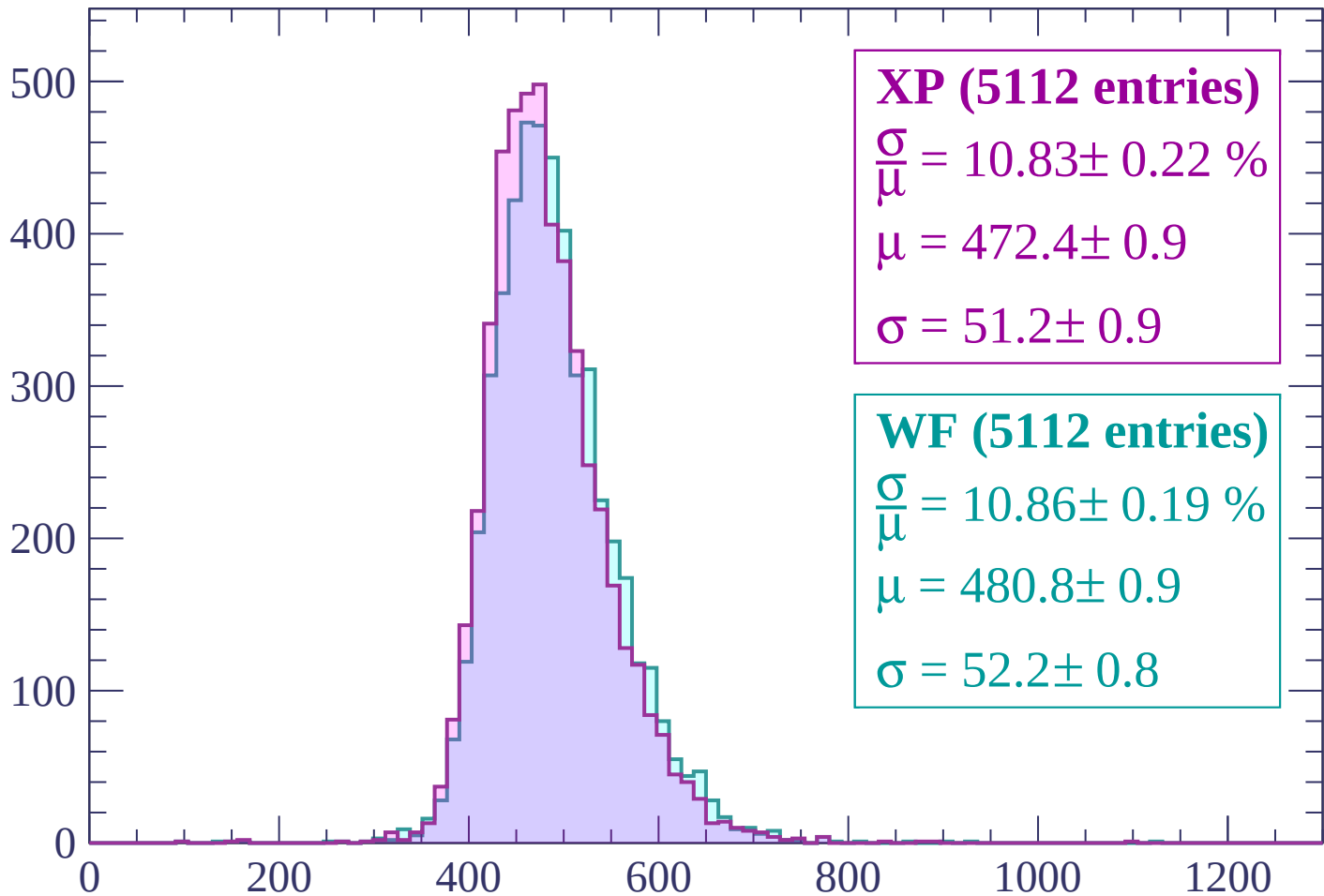
Energy loss | $-1680 < p < -1620$

Count



Energy loss | -1620 < p < -1560

Count



XP (5112 entries)

$\frac{\sigma}{\mu} = 10.83 \pm 0.22 \%$

$\mu = 472.4 \pm 0.9$

$\sigma = 51.2 \pm 0.9$

WF (5112 entries)

$\frac{\sigma}{\mu} = 10.86 \pm 0.19 \%$

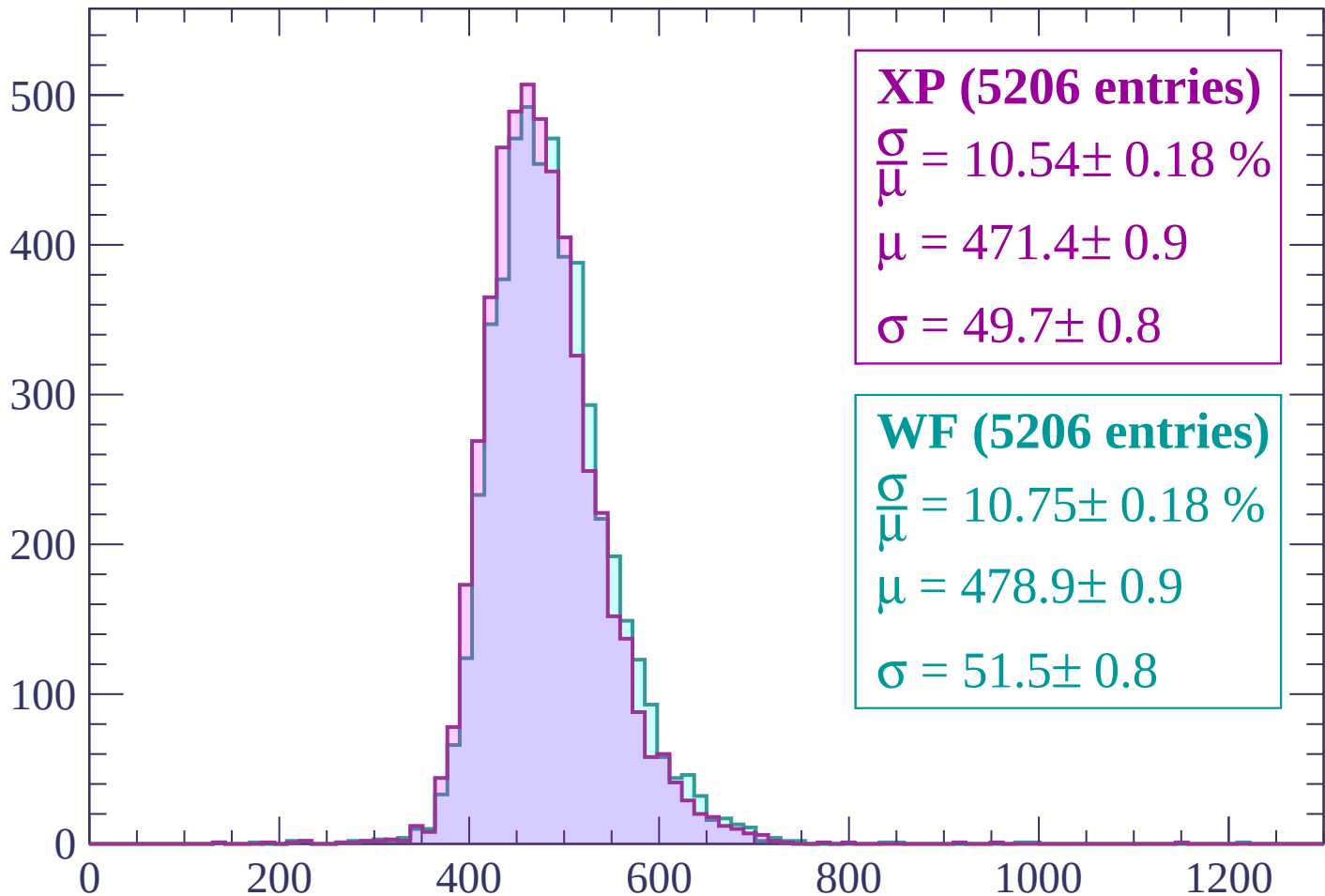
$\mu = 480.8 \pm 0.9$

$\sigma = 52.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (5206 entries)

$$\frac{\sigma}{\mu} = 10.54 \pm 0.18 \%$$

$$\mu = 471.4 \pm 0.9$$

$$\sigma = 49.7 \pm 0.8$$

WF (5206 entries)

$$\frac{\sigma}{\mu} = 10.75 \pm 0.18 \%$$

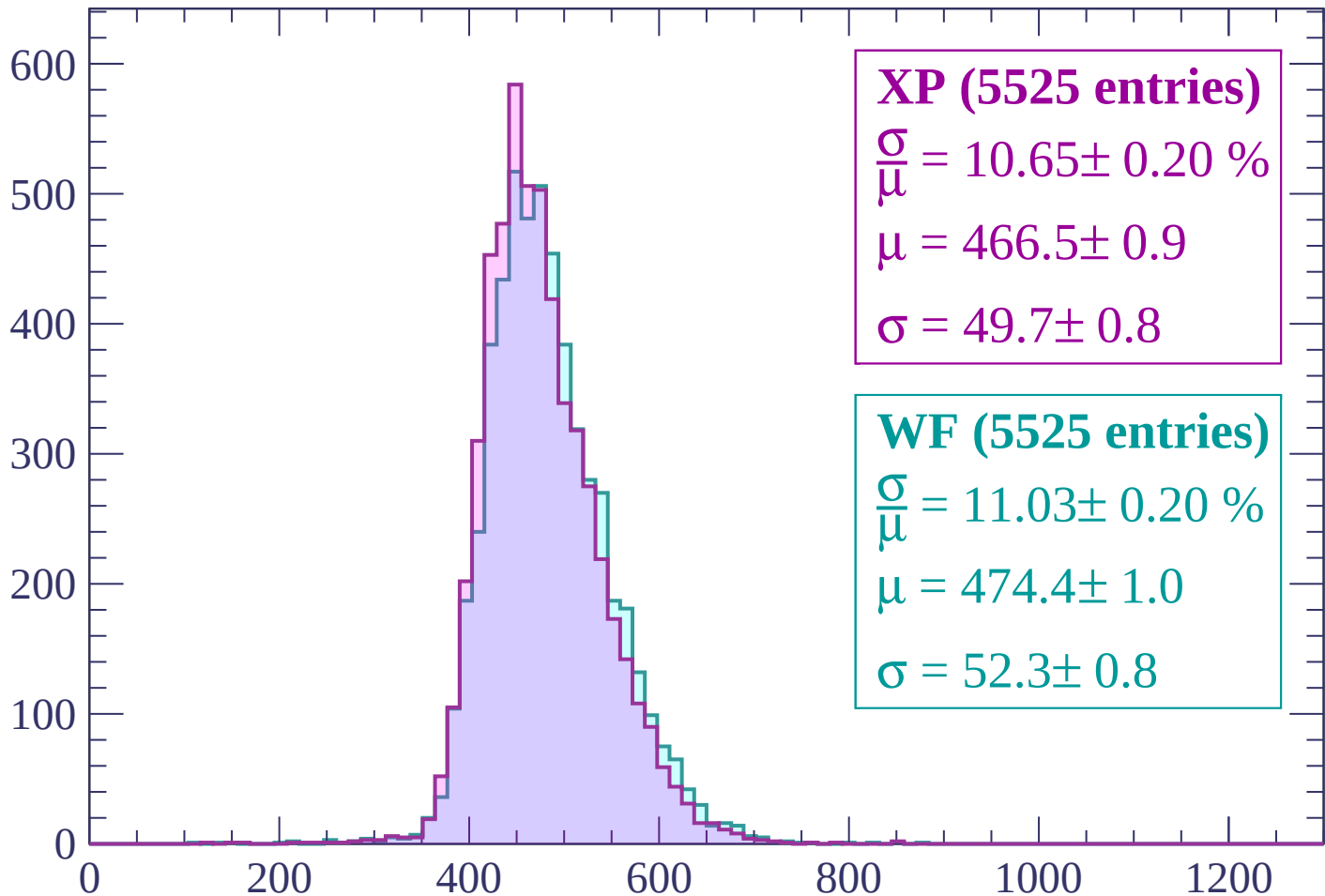
$$\mu = 478.9 \pm 0.9$$

$$\sigma = 51.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (5525 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.20 \%$

$\mu = 466.5 \pm 0.9$

$\sigma = 49.7 \pm 0.8$

WF (5525 entries)

$\frac{\sigma}{\mu} = 11.03 \pm 0.20 \%$

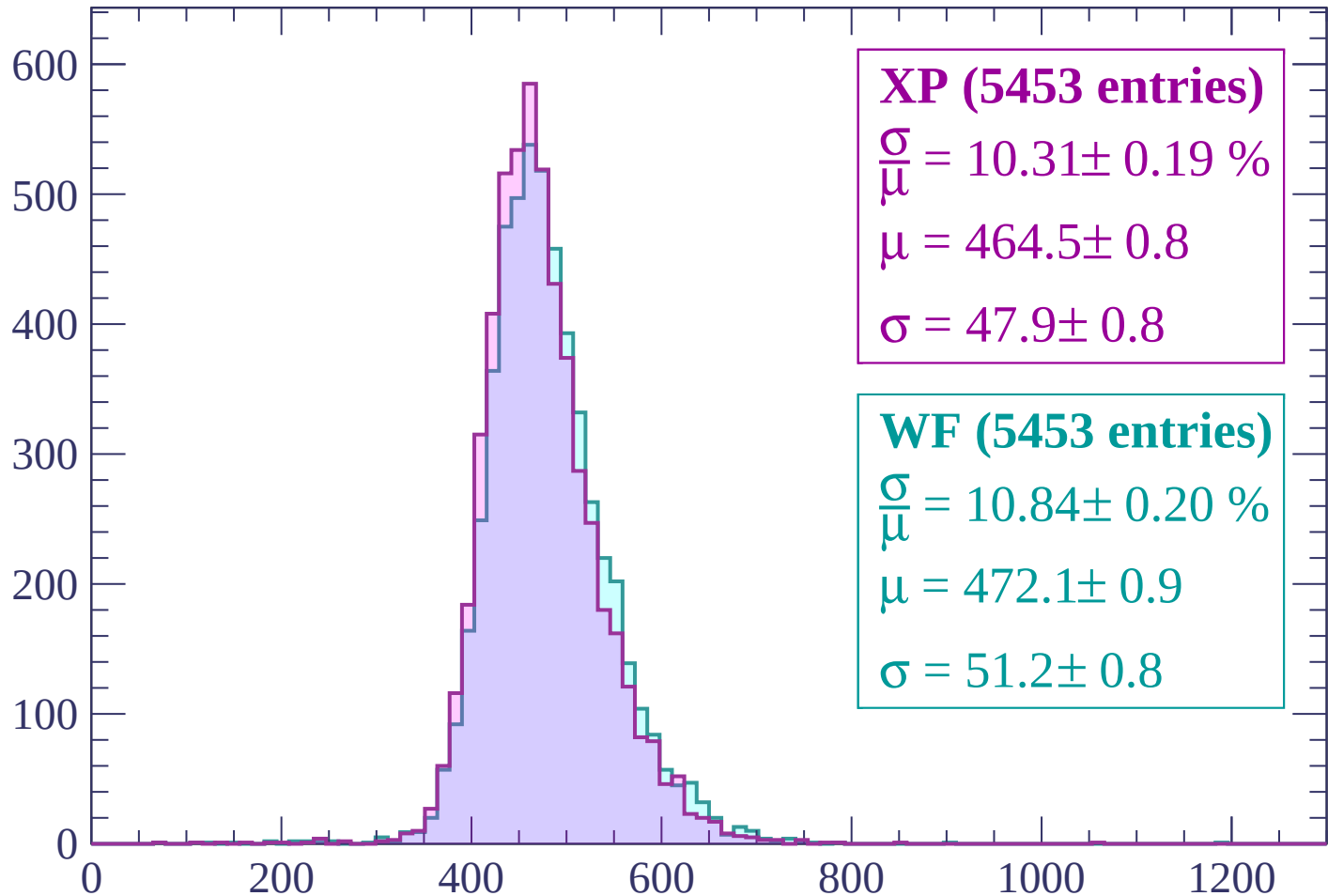
$\mu = 474.4 \pm 1.0$

$\sigma = 52.3 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1380$

Count



XP (5453 entries)

$\frac{\sigma}{\mu} = 10.31 \pm 0.19 \%$

$\mu = 464.5 \pm 0.8$

$\sigma = 47.9 \pm 0.8$

WF (5453 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.20 \%$

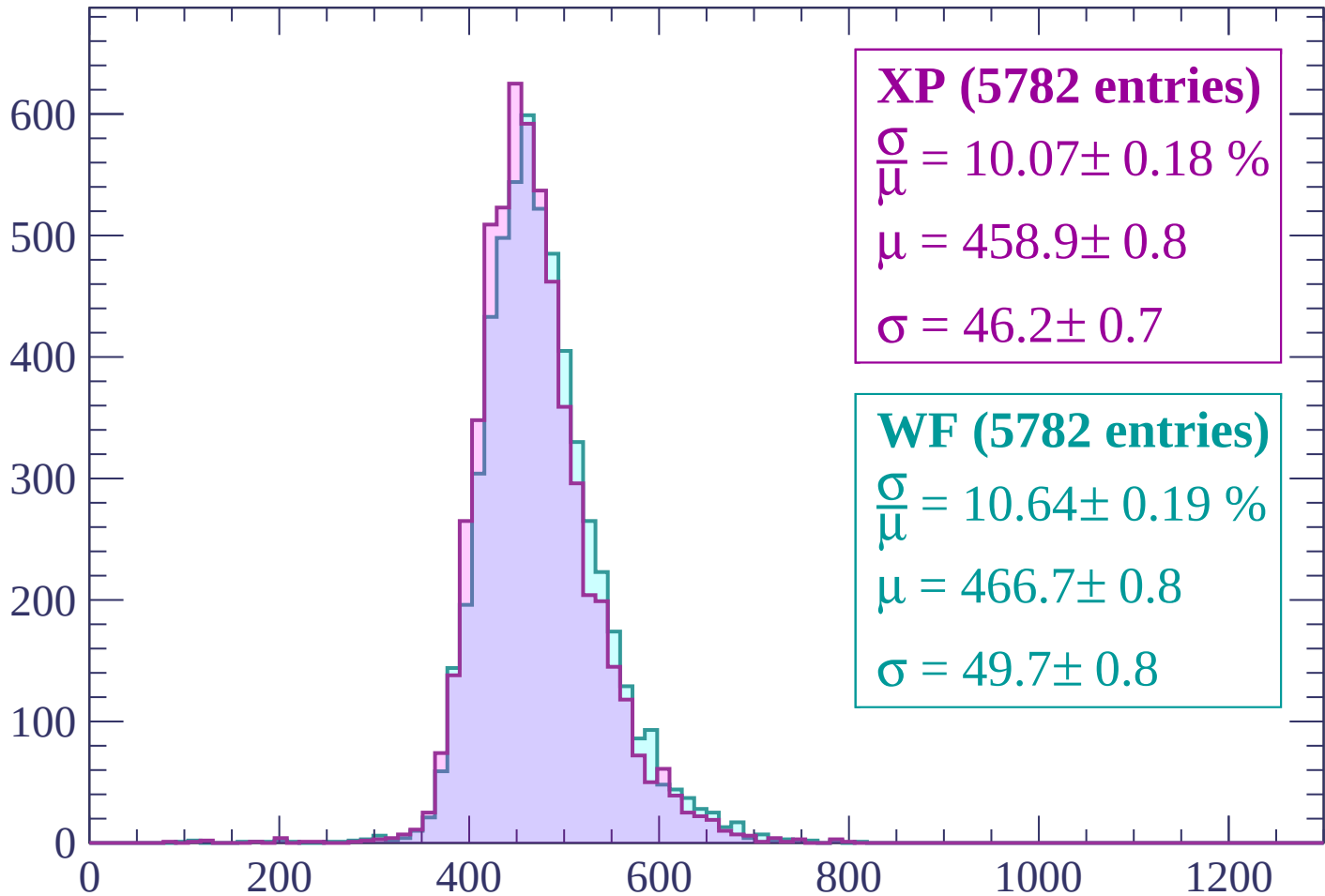
$\mu = 472.1 \pm 0.9$

$\sigma = 51.2 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count



XP (5782 entries)

$$\frac{\sigma}{\mu} = 10.07 \pm 0.18 \%$$

$$\mu = 458.9 \pm 0.8$$

$$\sigma = 46.2 \pm 0.7$$

WF (5782 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.19 \%$$

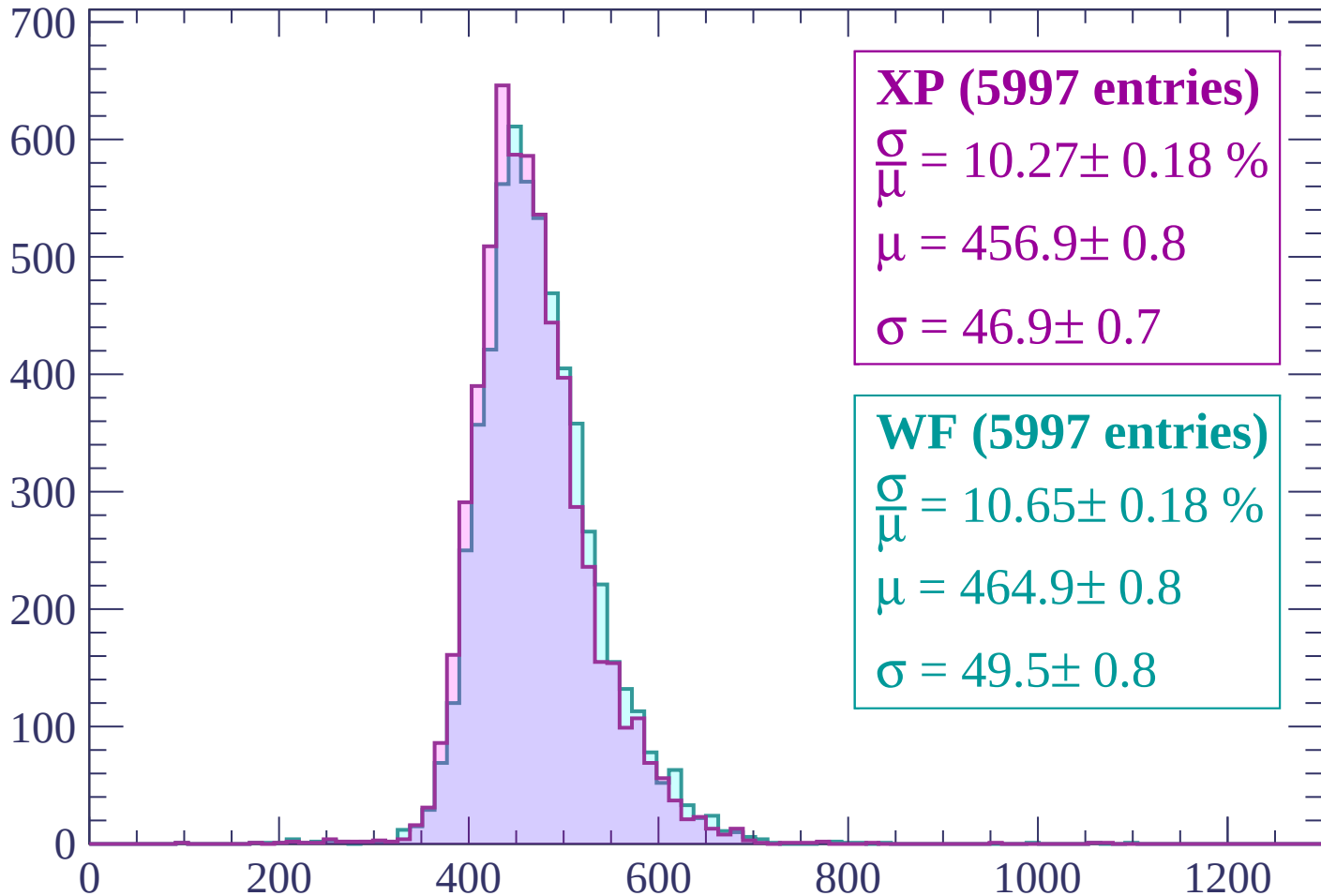
$$\mu = 466.7 \pm 0.8$$

$$\sigma = 49.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



XP (5997 entries)

$$\frac{\sigma}{\mu} = 10.27 \pm 0.18 \%$$

$$\mu = 456.9 \pm 0.8$$

$$\sigma = 46.9 \pm 0.7$$

WF (5997 entries)

$$\frac{\sigma}{\mu} = 10.65 \pm 0.18 \%$$

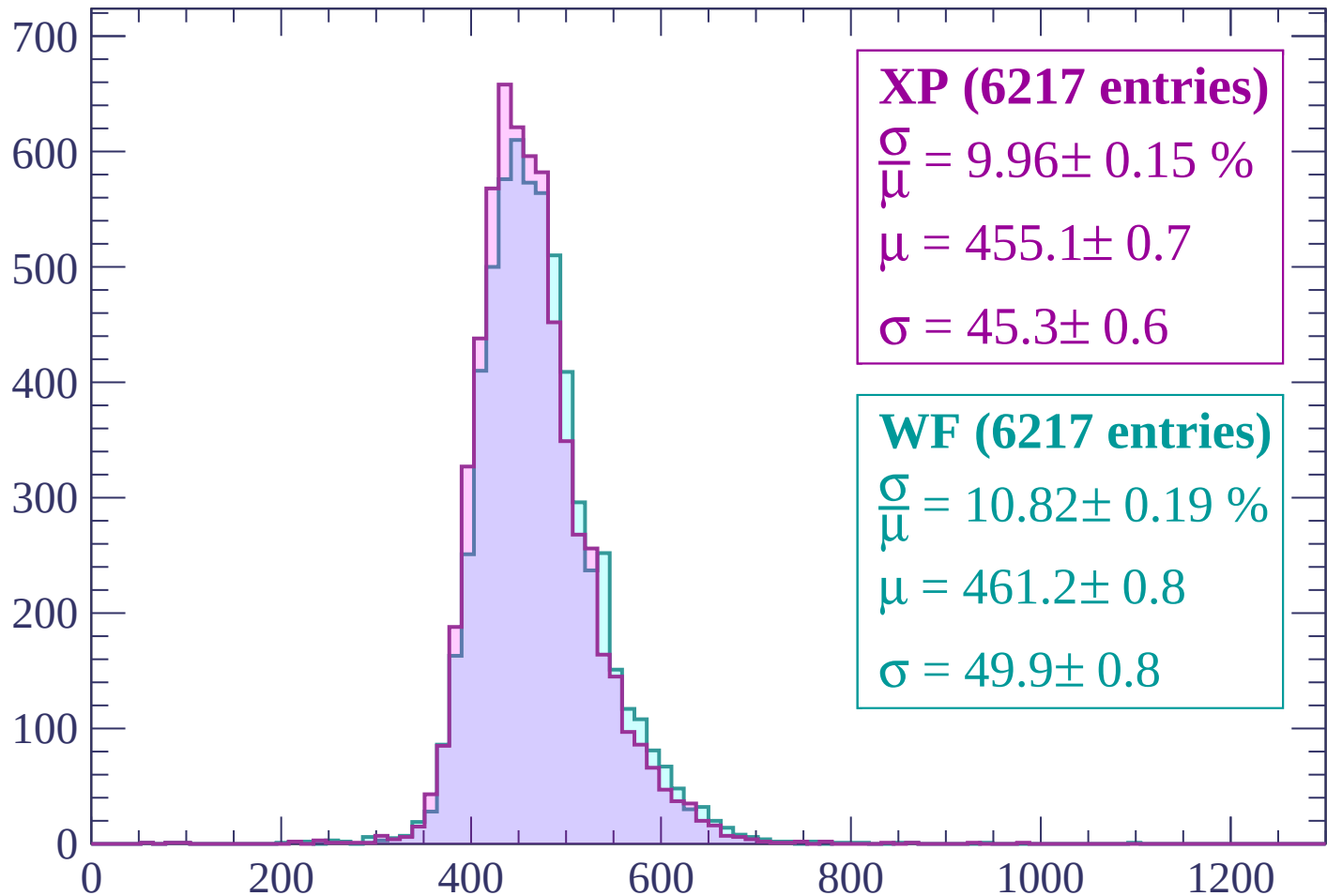
$$\mu = 464.9 \pm 0.8$$

$$\sigma = 49.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1260 < p < -1200$

Count



XP (6217 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.15 \%$$

$$\mu = 455.1 \pm 0.7$$

$$\sigma = 45.3 \pm 0.6$$

WF (6217 entries)

$$\frac{\sigma}{\mu} = 10.82 \pm 0.19 \%$$

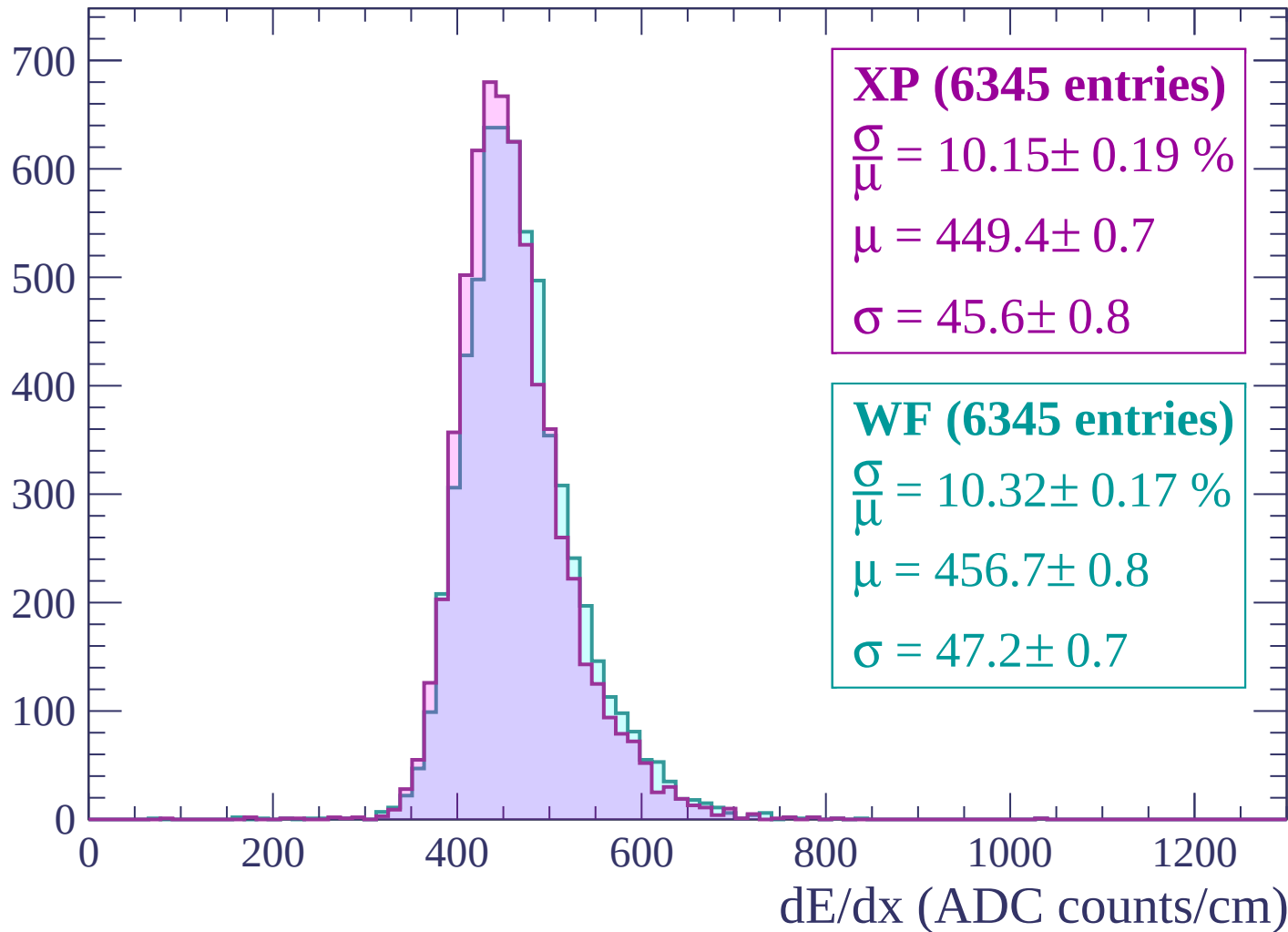
$$\mu = 461.2 \pm 0.8$$

$$\sigma = 49.9 \pm 0.8$$

dE/dx (ADC counts/cm)

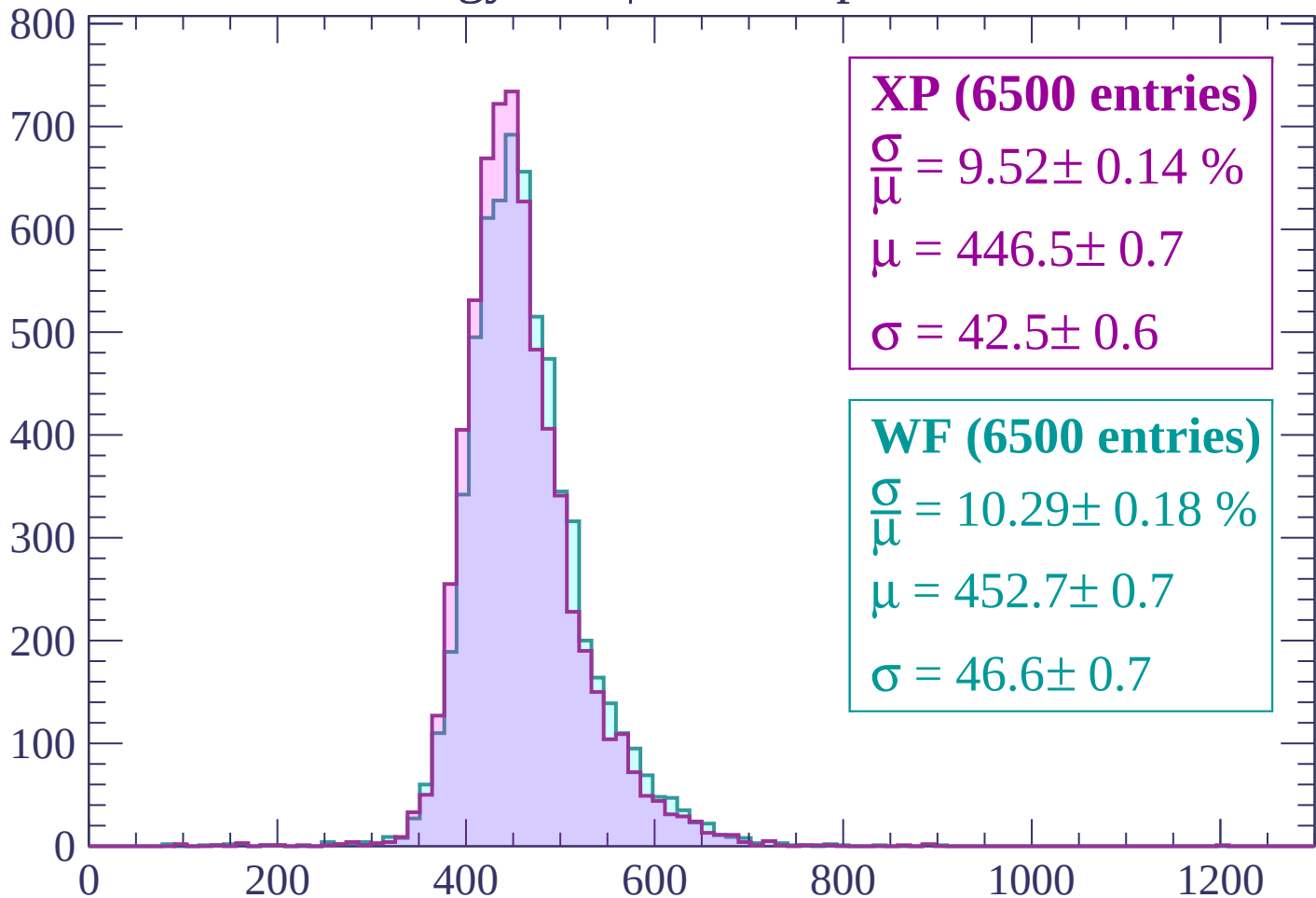
Energy loss | $-1200 < p < -1140$

Count



Energy loss | -1140 < p < -1080

Count



XP (6500 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.14 \%$$

$$\mu = 446.5 \pm 0.7$$

$$\sigma = 42.5 \pm 0.6$$

WF (6500 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.18 \%$$

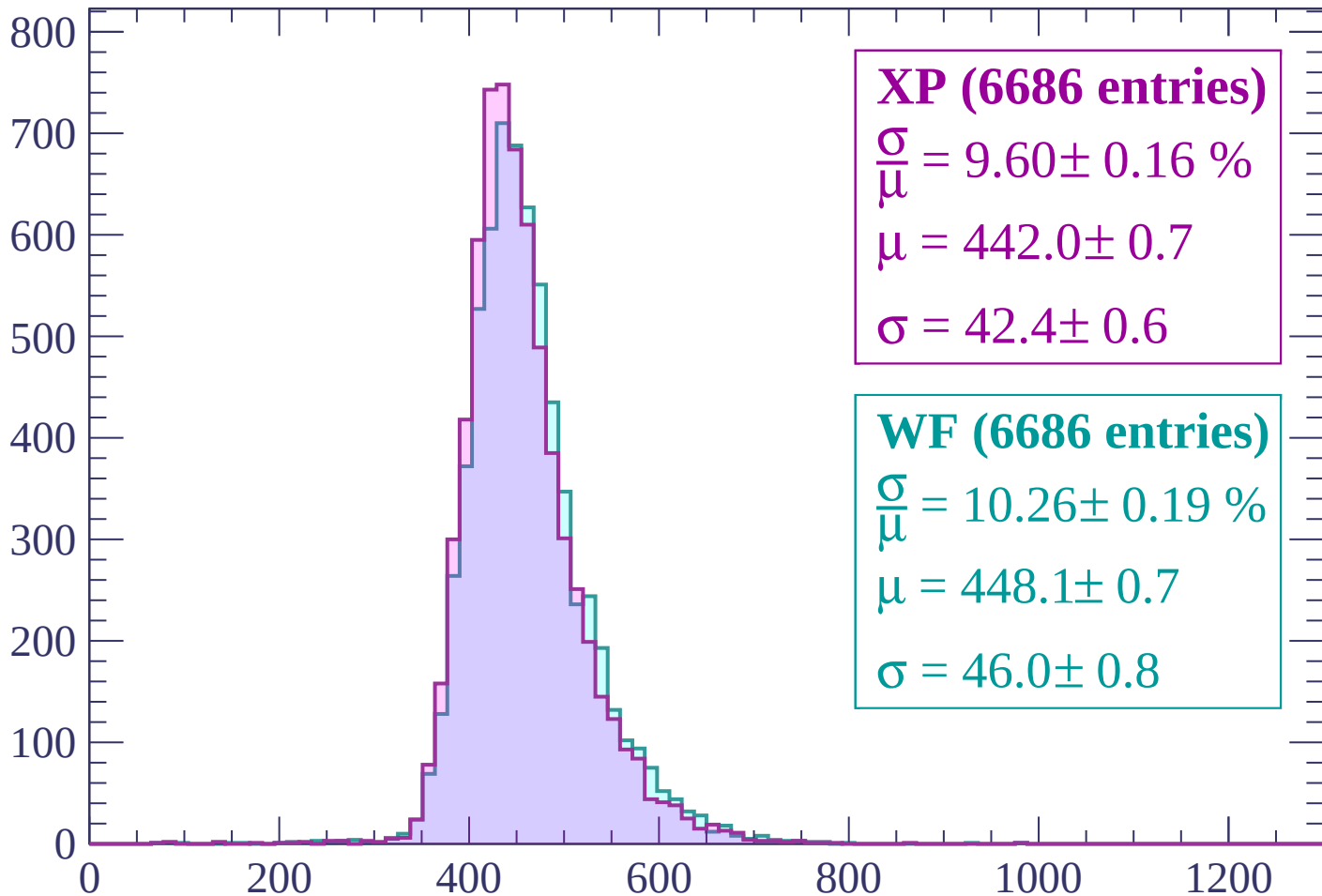
$$\mu = 452.7 \pm 0.7$$

$$\sigma = 46.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count



XP (6686 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

$$\mu = 442.0 \pm 0.7$$

$$\sigma = 42.4 \pm 0.6$$

WF (6686 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 0.19 \%$$

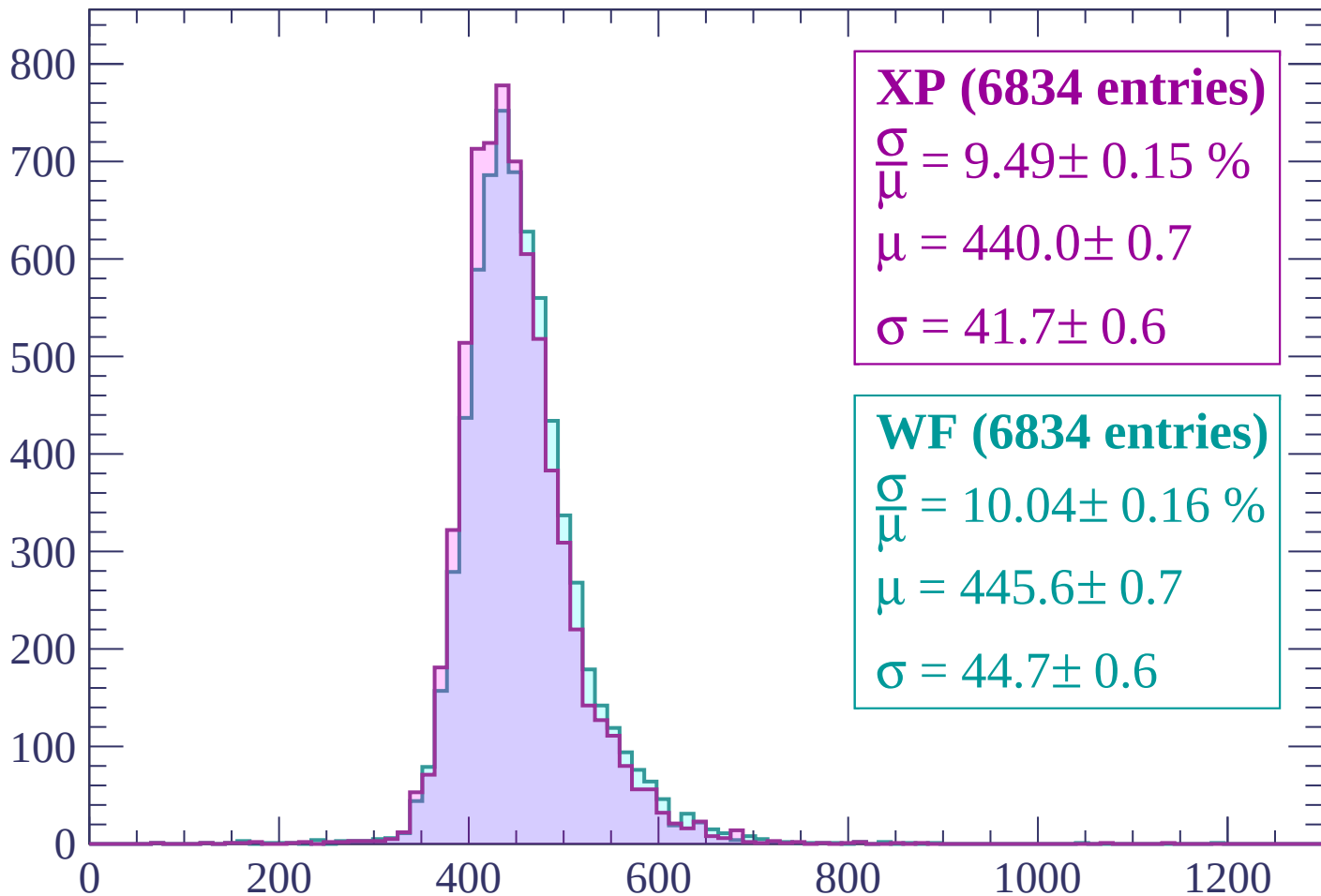
$$\mu = 448.1 \pm 0.7$$

$$\sigma = 46.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1020 < p < -960$

Count



XP (6834 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.15 \%$$

$$\mu = 440.0 \pm 0.7$$

$$\sigma = 41.7 \pm 0.6$$

WF (6834 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

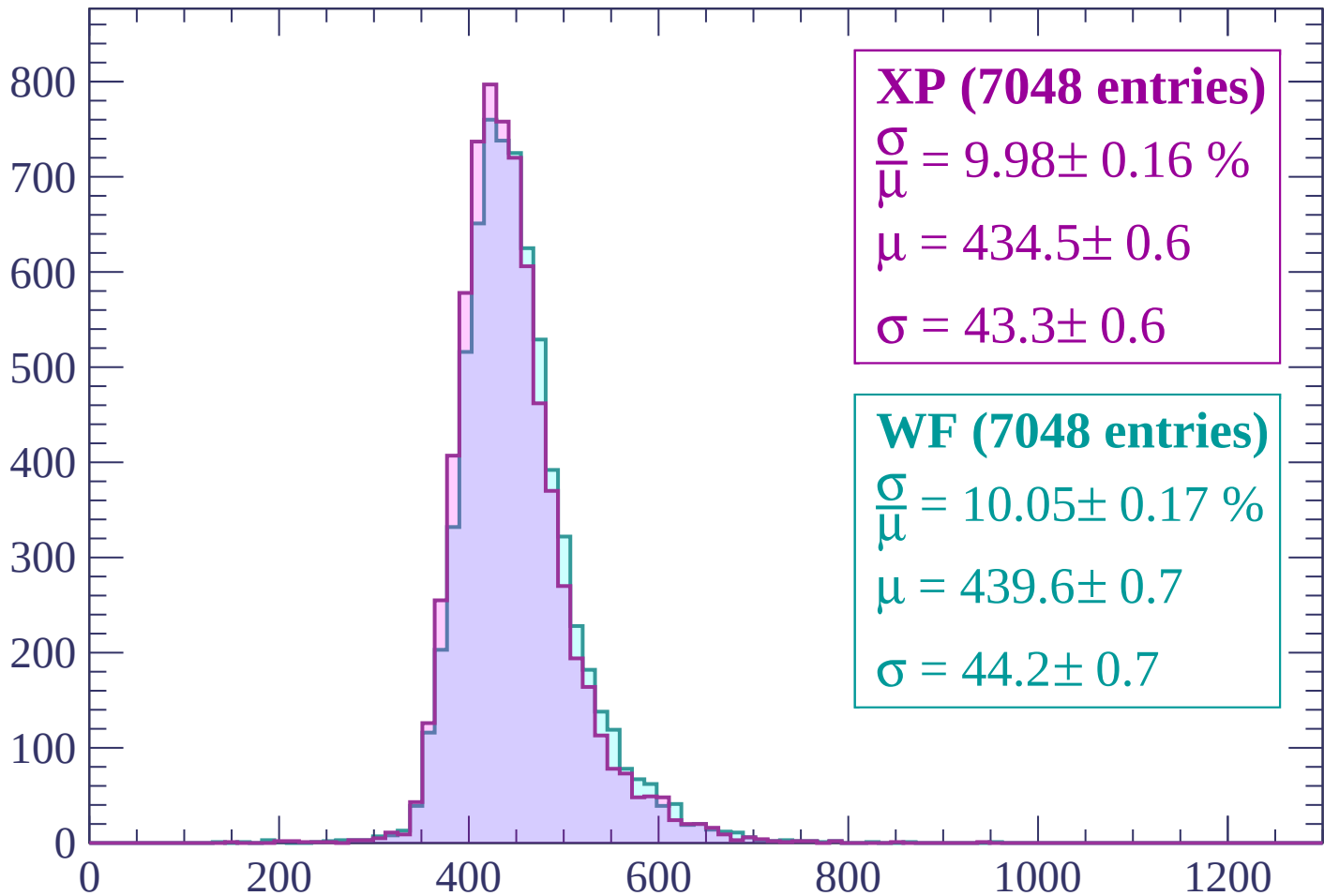
$$\mu = 445.6 \pm 0.7$$

$$\sigma = 44.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (7048 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$$

$$\mu = 434.5 \pm 0.6$$

$$\sigma = 43.3 \pm 0.6$$

WF (7048 entries)

$$\frac{\sigma}{\mu} = 10.05 \pm 0.17 \%$$

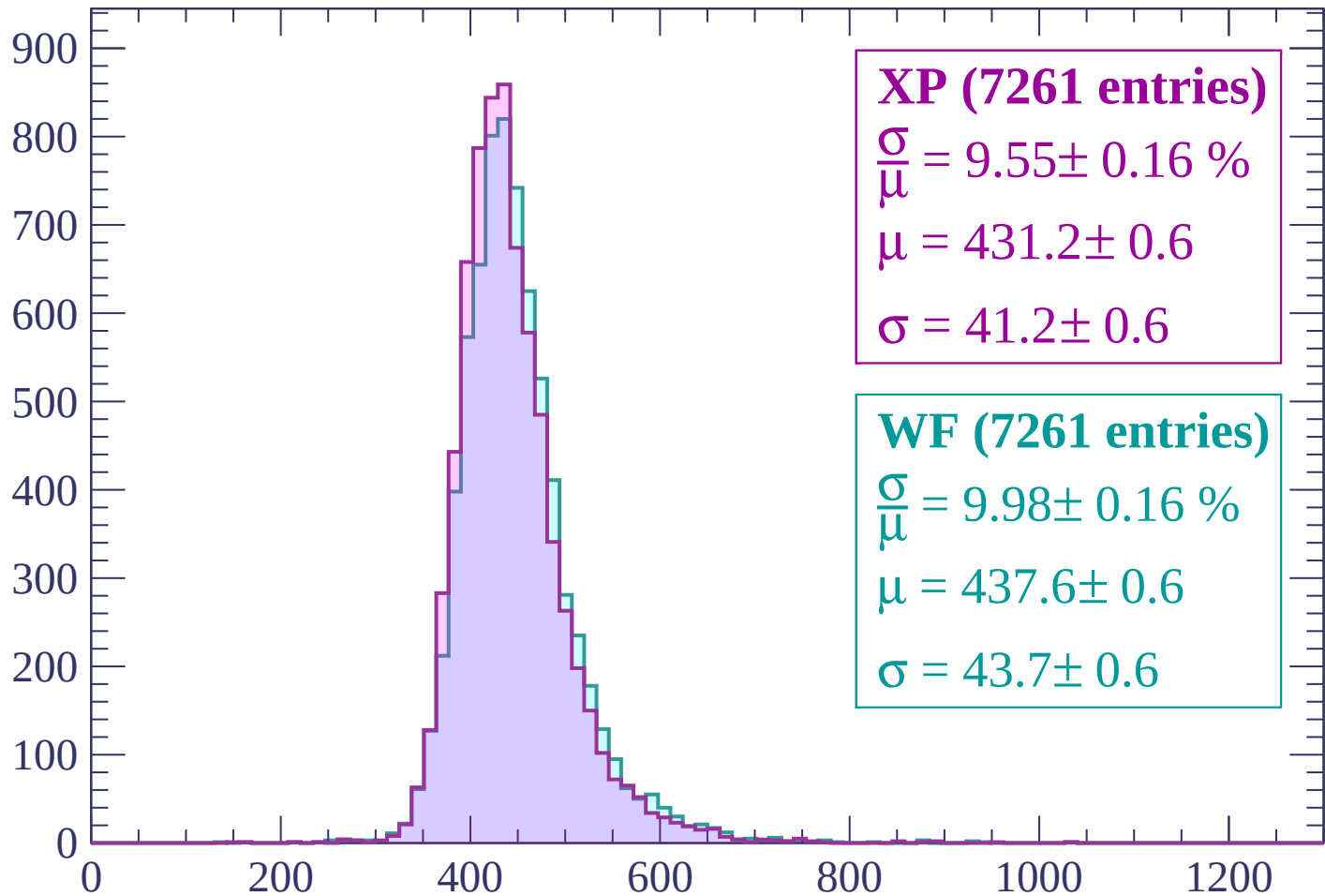
$$\mu = 439.6 \pm 0.7$$

$$\sigma = 44.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (7261 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.16 \%$$

$$\mu = 431.2 \pm 0.6$$

$$\sigma = 41.2 \pm 0.6$$

WF (7261 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$$

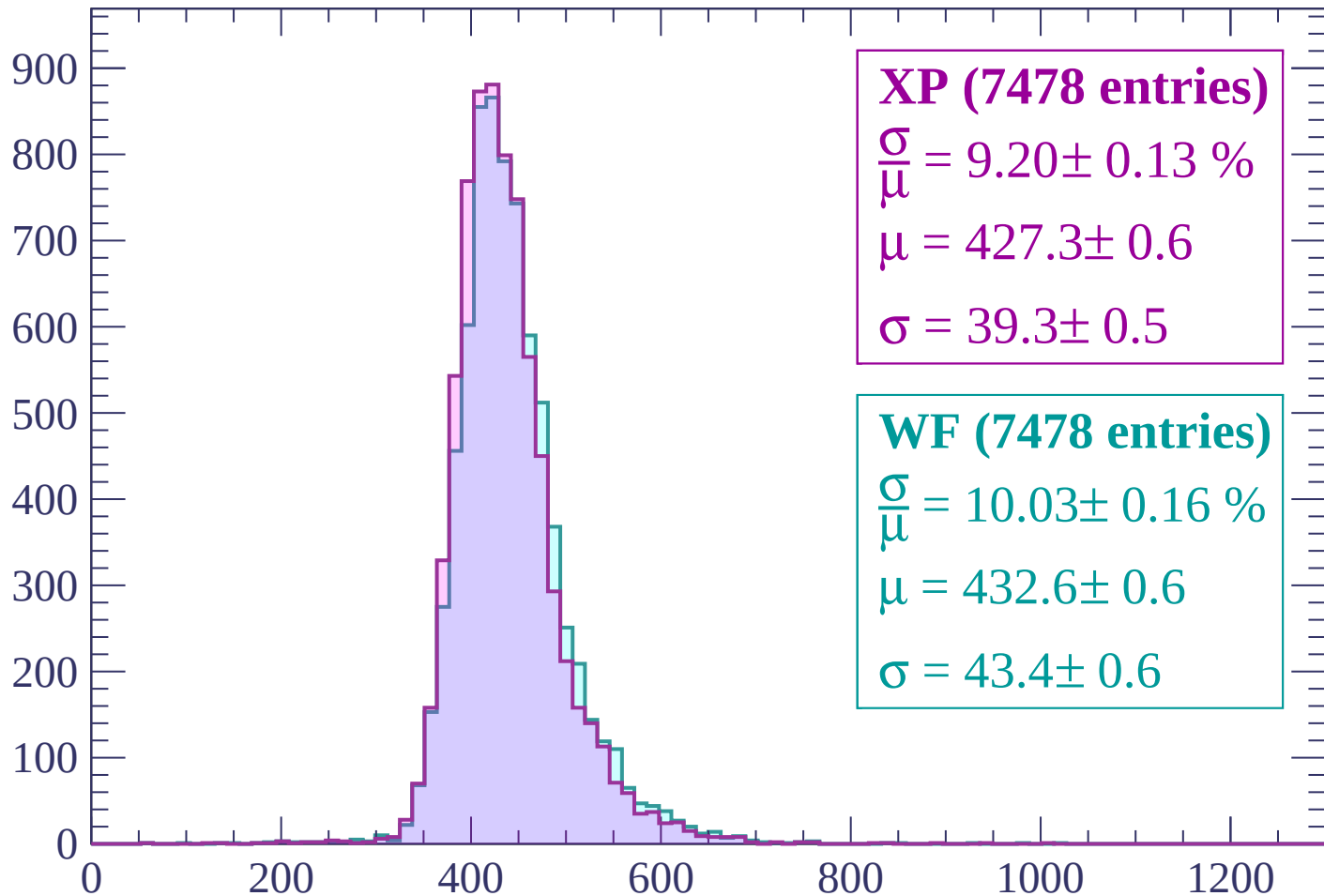
$$\mu = 437.6 \pm 0.6$$

$$\sigma = 43.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (7478 entries)

$$\frac{\sigma}{\mu} = 9.20 \pm 0.13 \%$$

$$\mu = 427.3 \pm 0.6$$

$$\sigma = 39.3 \pm 0.5$$

WF (7478 entries)

$$\frac{\sigma}{\mu} = 10.03 \pm 0.16 \%$$

$$\mu = 432.6 \pm 0.6$$

$$\sigma = 43.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7681 entries)

$\frac{\sigma}{\mu} = 9.04 \pm 0.13 \%$

$\mu = 421.5 \pm 0.6$

$\sigma = 38.1 \pm 0.5$

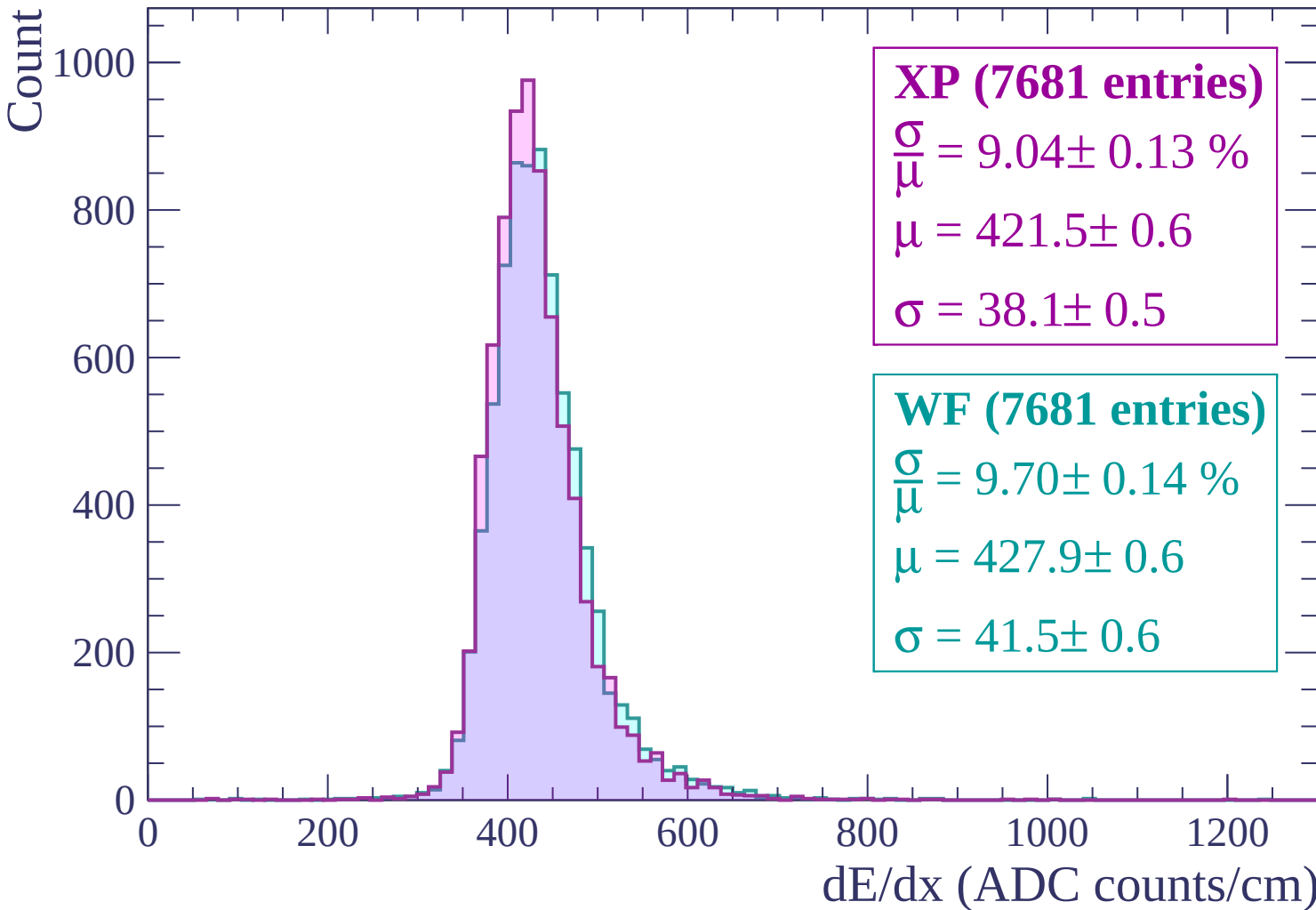
WF (7681 entries)

$\frac{\sigma}{\mu} = 9.70 \pm 0.14 \%$

$\mu = 427.9 \pm 0.6$

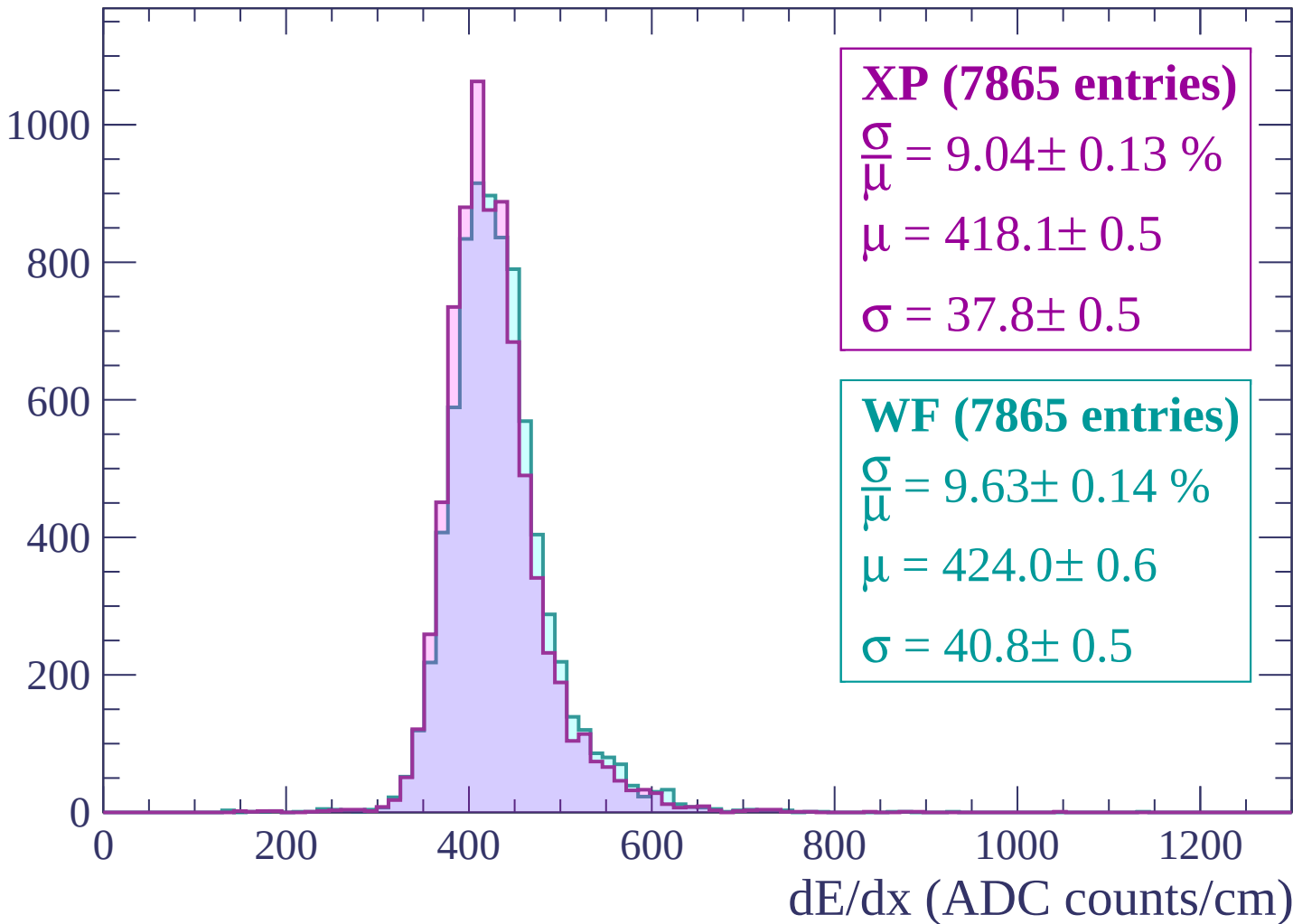
$\sigma = 41.5 \pm 0.6$

dE/dx (ADC counts/cm)



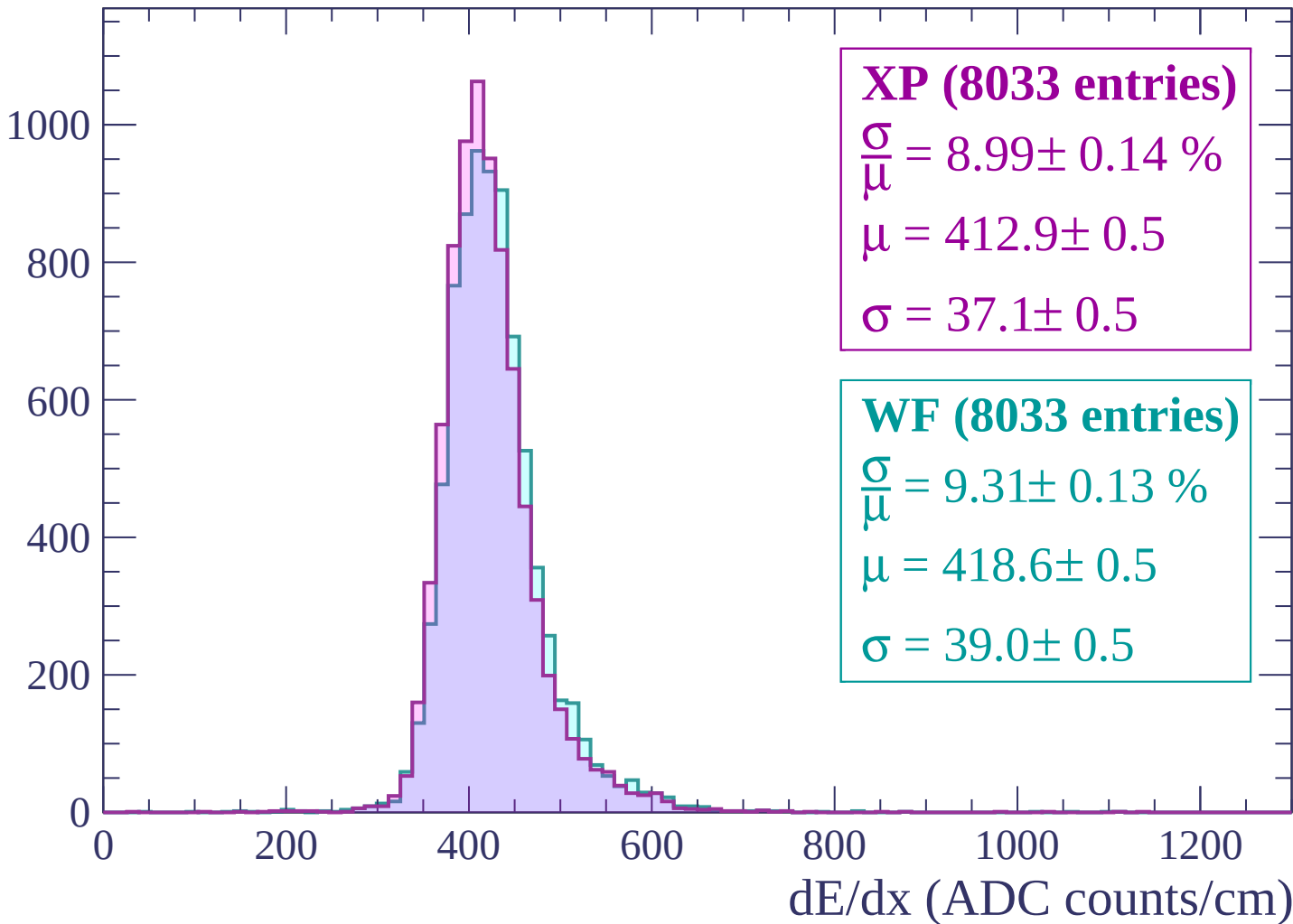
Energy loss | $-720 < p < -660$

Count



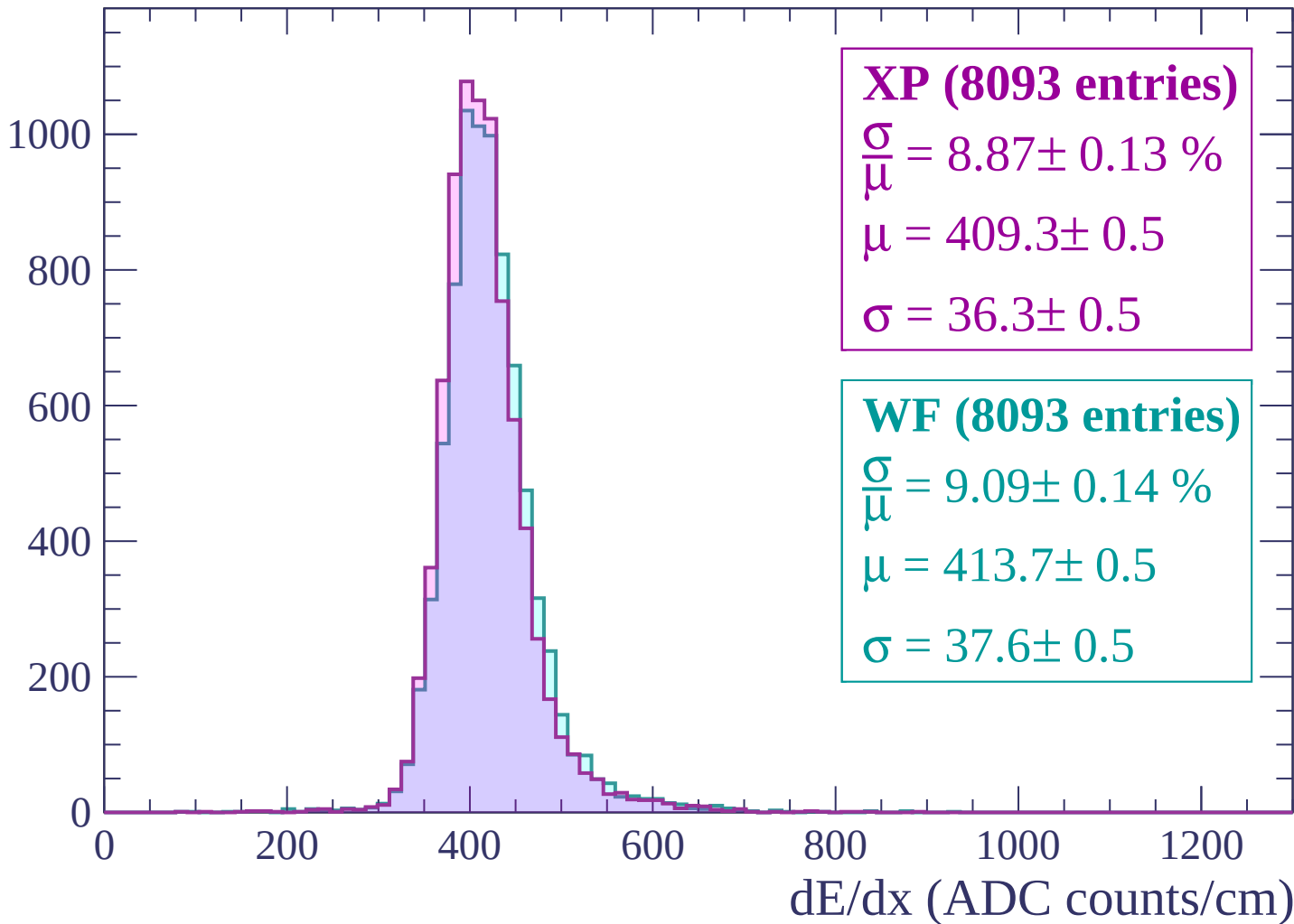
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count

1200

1000

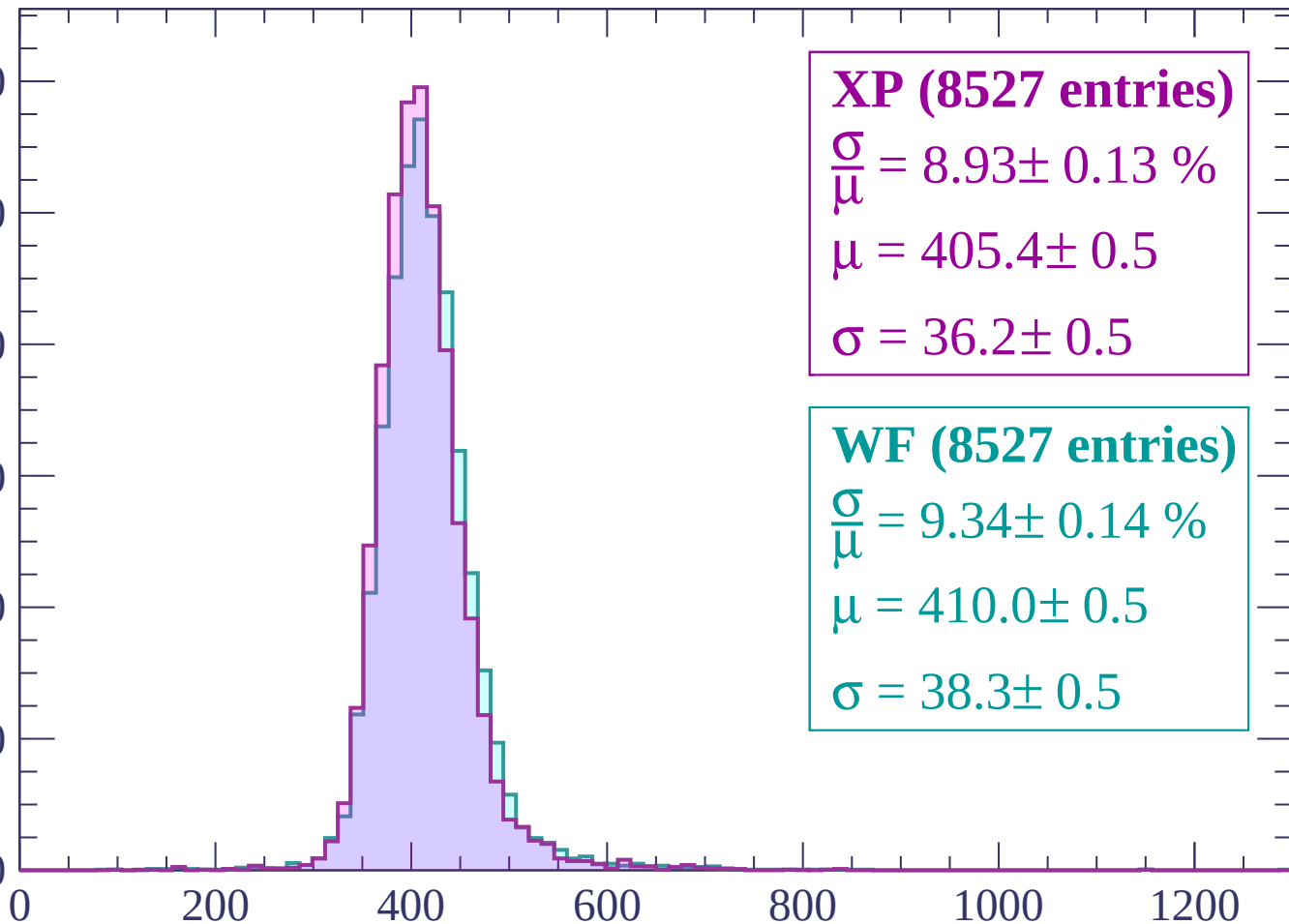
800

600

400

200

0



XP (8527 entries)

$\frac{\sigma}{\mu} = 8.93 \pm 0.13 \%$

$\mu = 405.4 \pm 0.5$

$\sigma = 36.2 \pm 0.5$

WF (8527 entries)

$\frac{\sigma}{\mu} = 9.34 \pm 0.14 \%$

$\mu = 410.0 \pm 0.5$

$\sigma = 38.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count

1200

1000

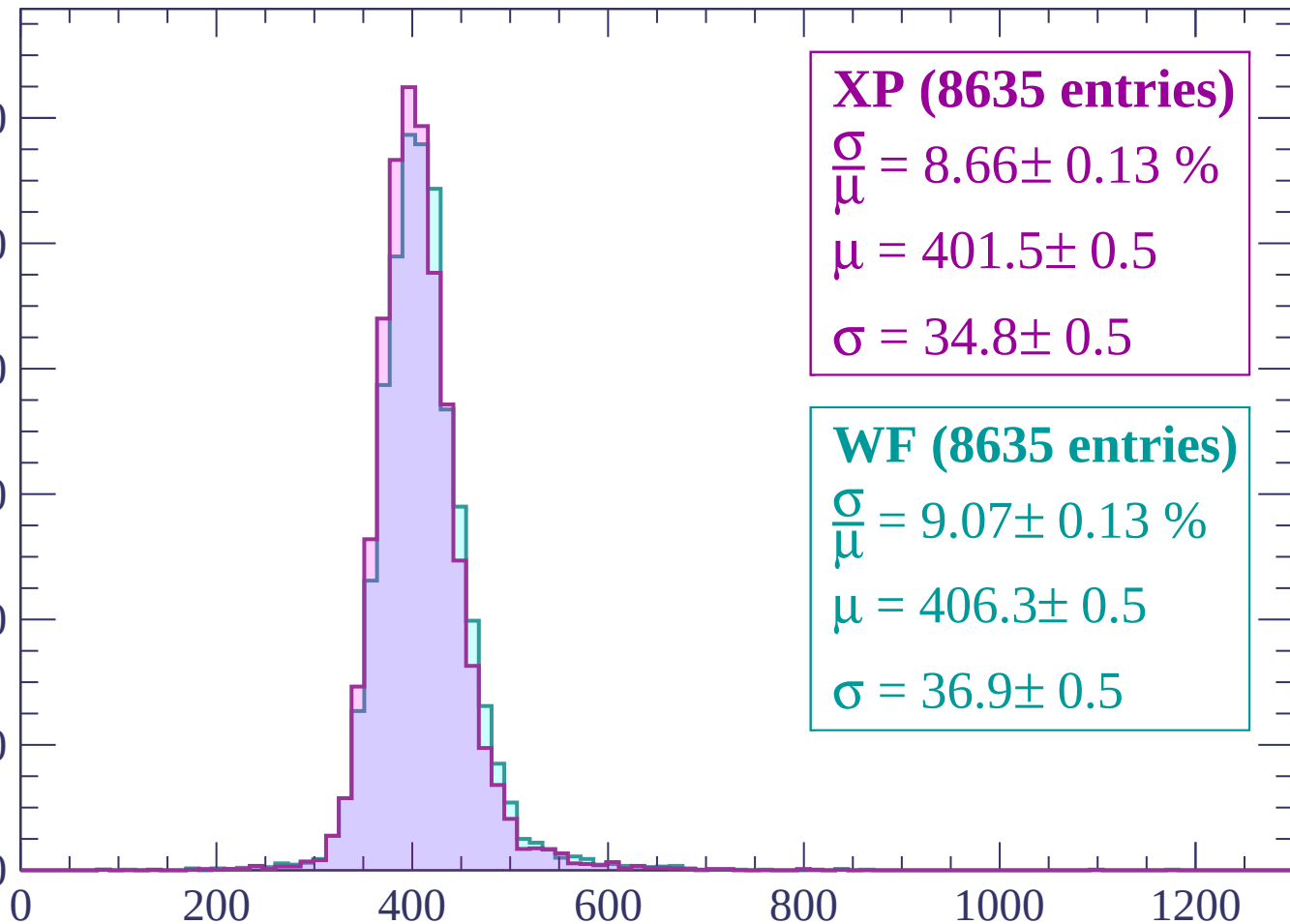
800

600

400

200

0



XP (8635 entries)

$\frac{\sigma}{\mu} = 8.66 \pm 0.13 \%$

$\mu = 401.5 \pm 0.5$

$\sigma = 34.8 \pm 0.5$

WF (8635 entries)

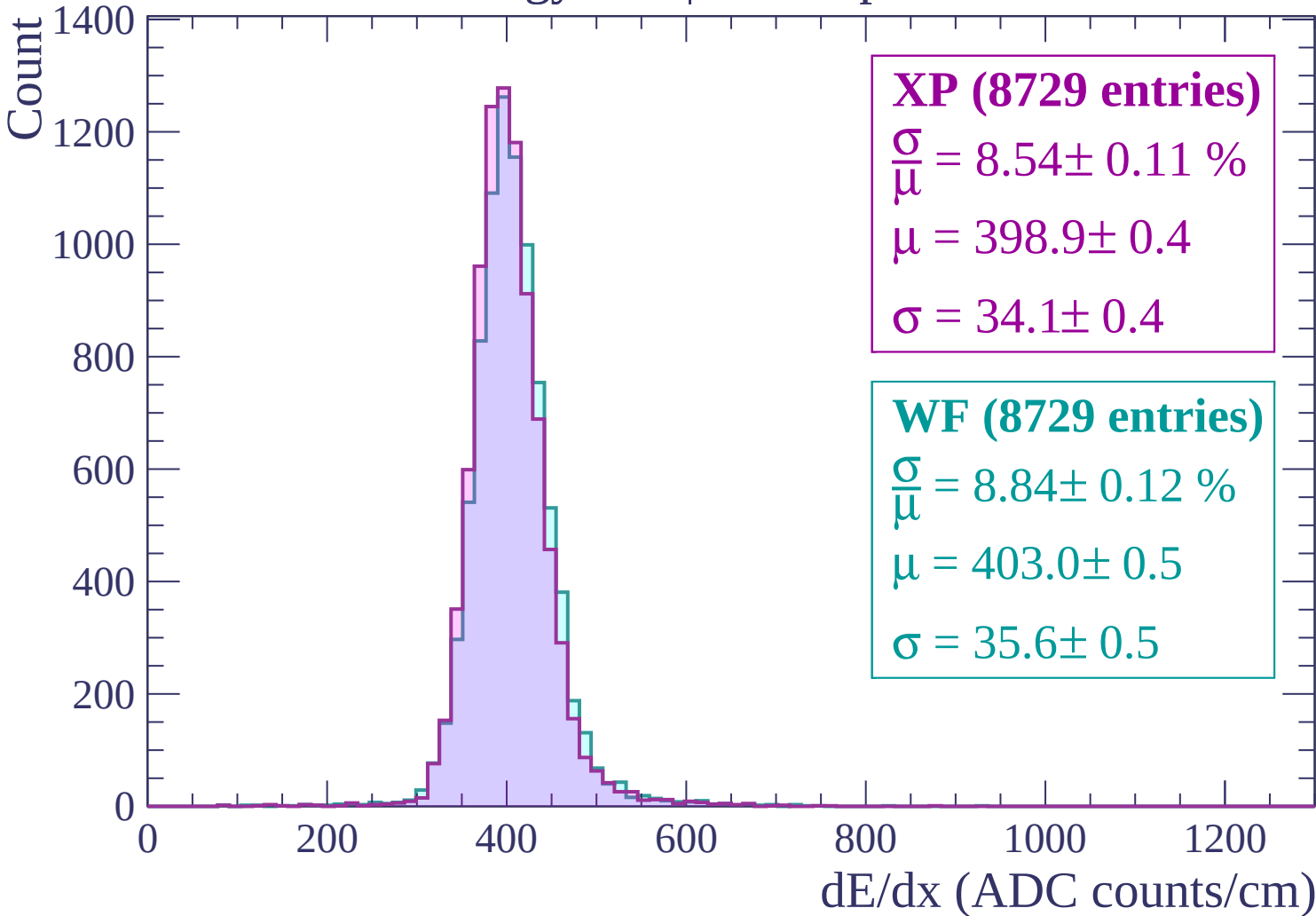
$\frac{\sigma}{\mu} = 9.07 \pm 0.13 \%$

$\mu = 406.3 \pm 0.5$

$\sigma = 36.9 \pm 0.5$

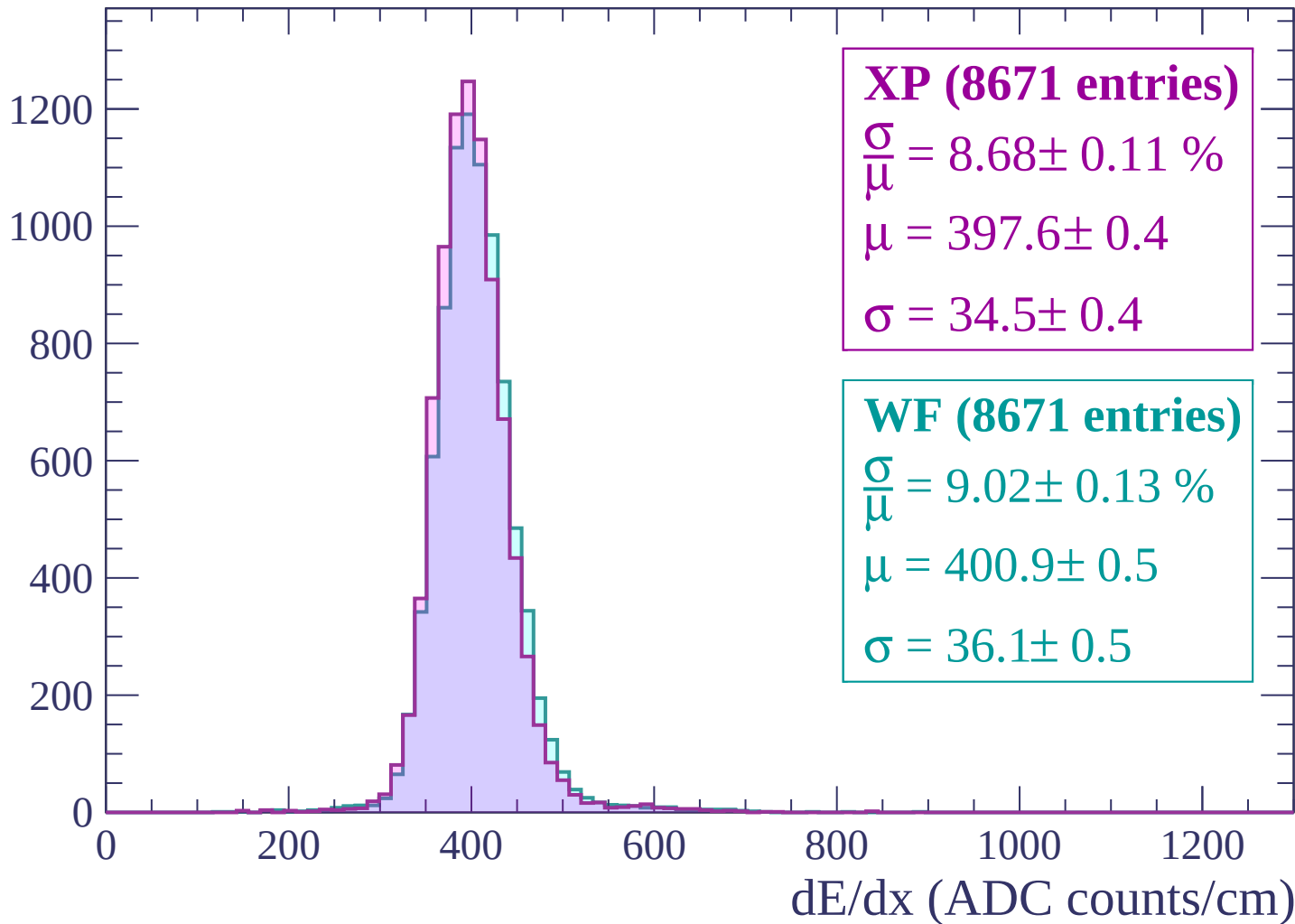
dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count

1200

1000

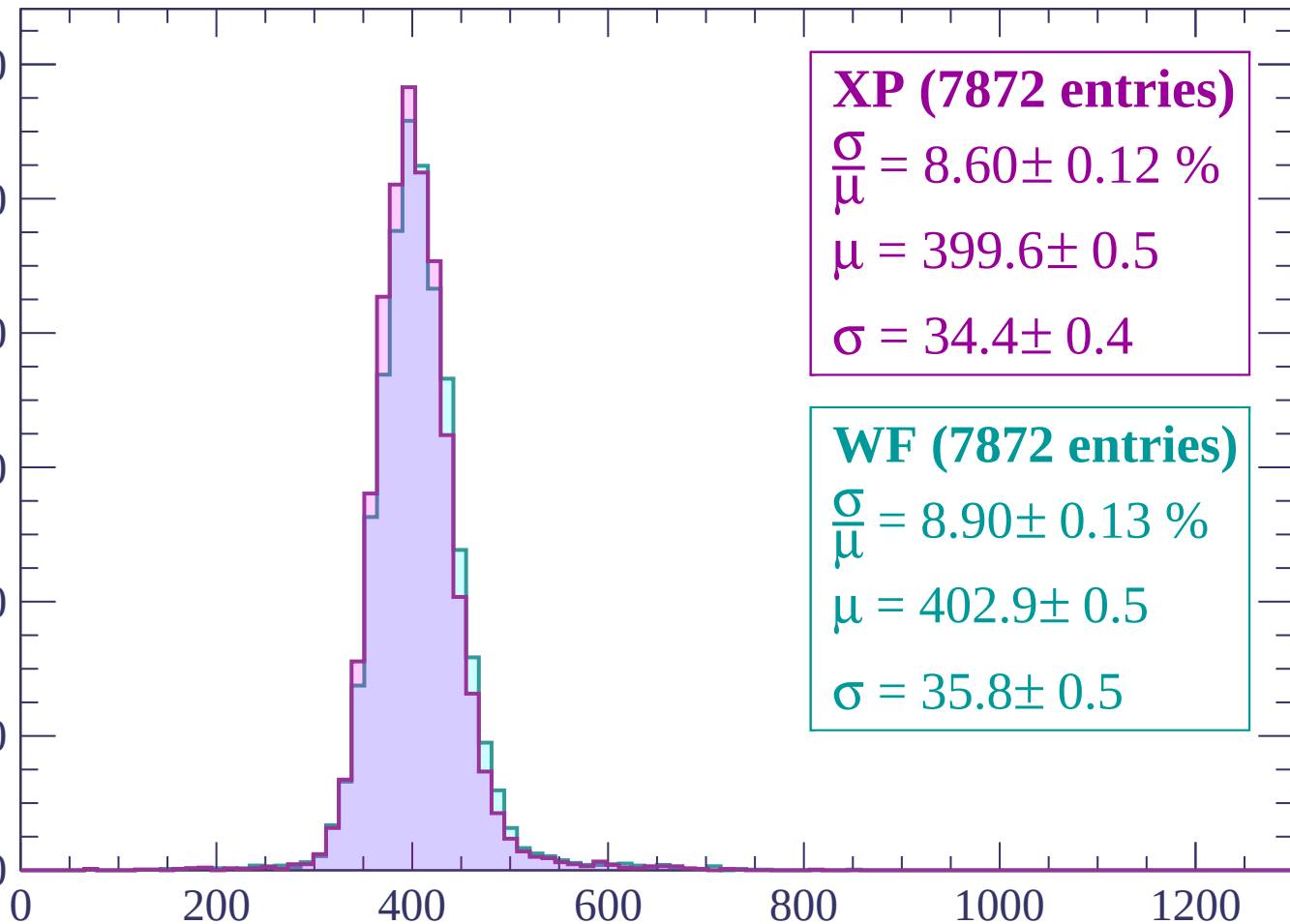
800

600

400

200

0



XP (7872 entries)

$\frac{\sigma}{\mu} = 8.60 \pm 0.12 \%$

$\mu = 399.6 \pm 0.5$

$\sigma = 34.4 \pm 0.4$

WF (7872 entries)

$\frac{\sigma}{\mu} = 8.90 \pm 0.13 \%$

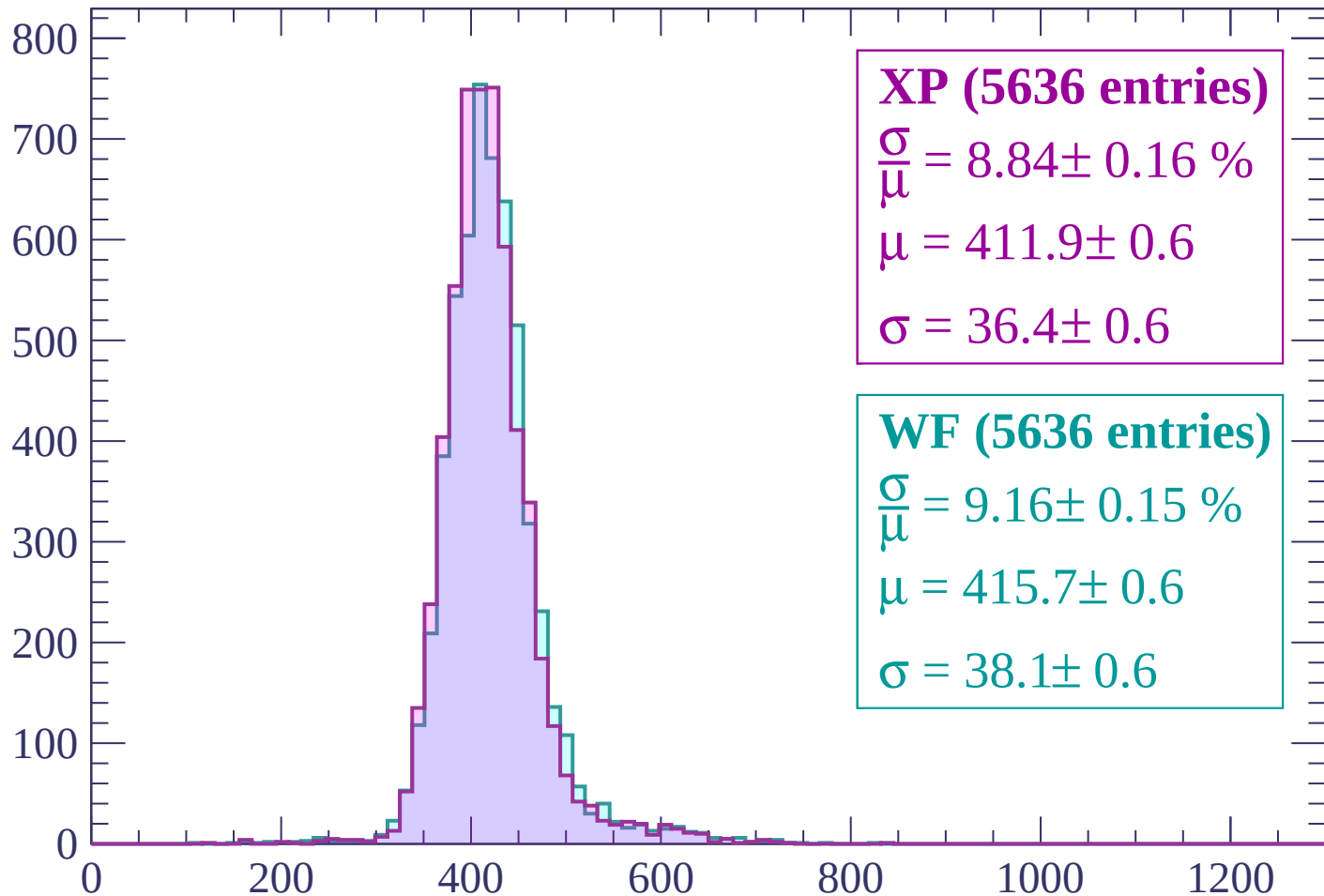
$\mu = 402.9 \pm 0.5$

$\sigma = 35.8 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (5636 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

WF (5636 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.15 \%$$

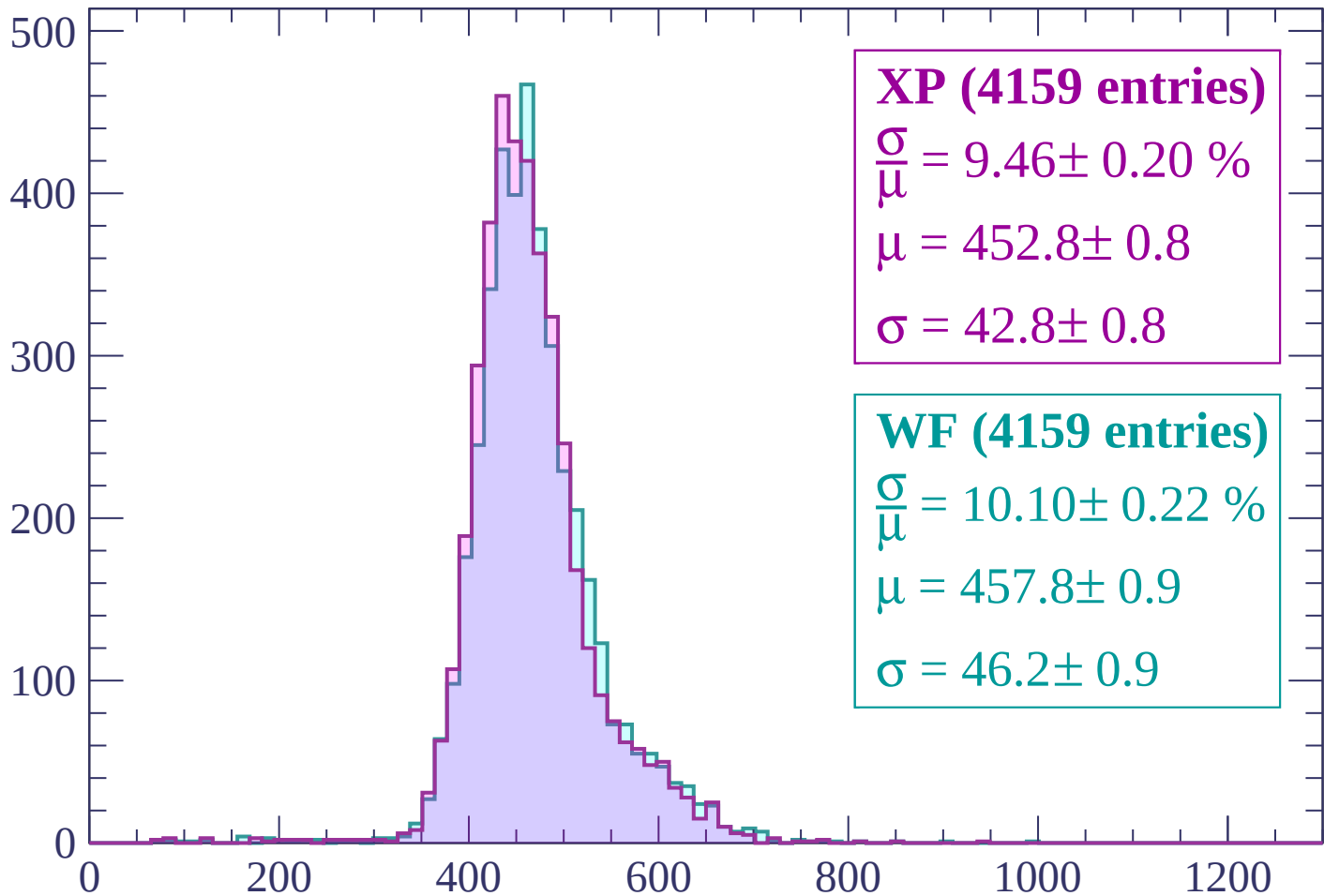
$$\mu = 415.7 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (4159 entries)

$$\frac{\sigma}{\mu} = 9.46 \pm 0.20 \%$$

$$\mu = 452.8 \pm 0.8$$

$$\sigma = 42.8 \pm 0.8$$

WF (4159 entries)

$$\frac{\sigma}{\mu} = 10.10 \pm 0.22 \%$$

$$\mu = 457.8 \pm 0.9$$

$$\sigma = 46.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count

250

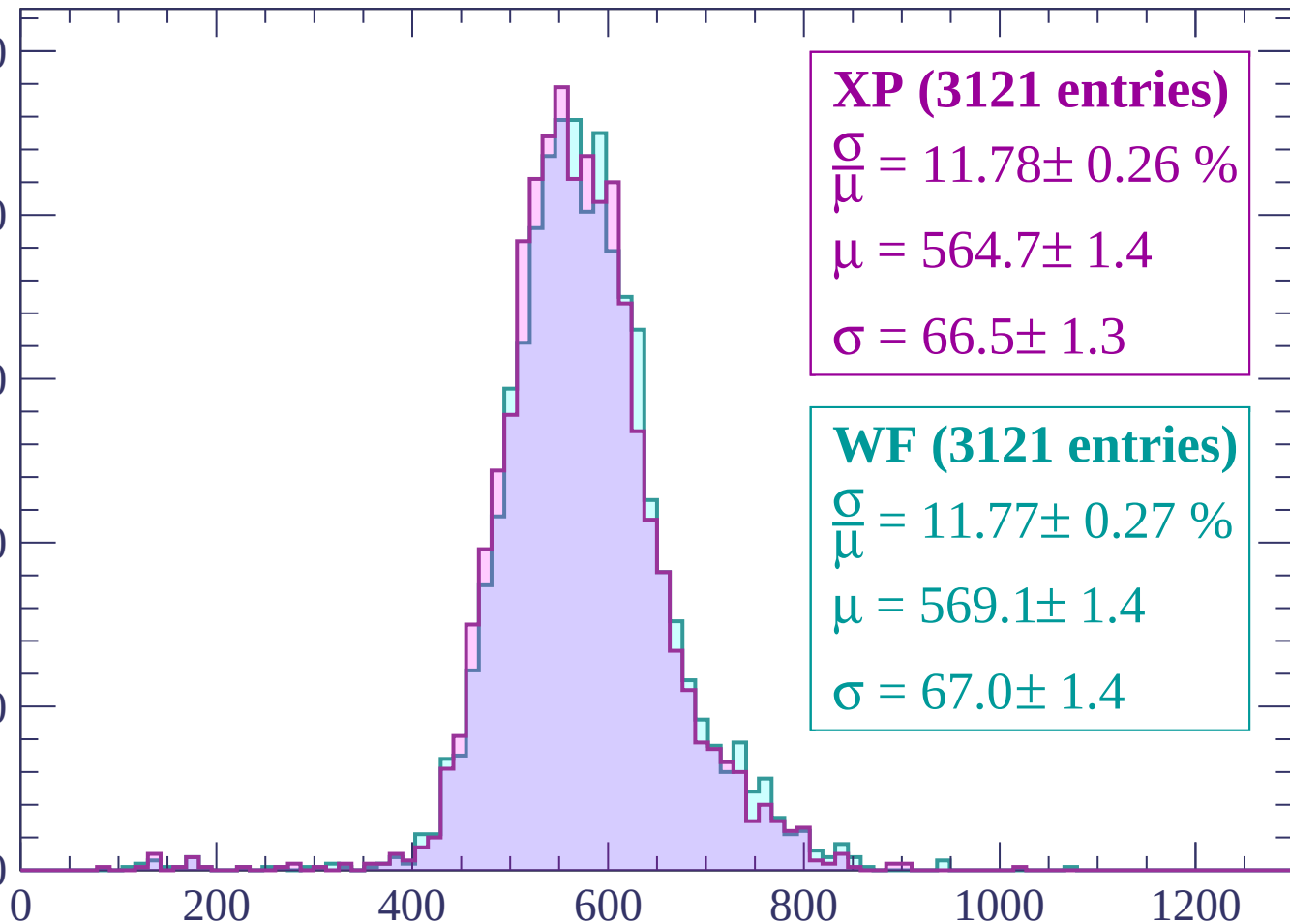
200

150

100

50

0



XP (3121 entries)

$\frac{\sigma}{\mu} = 11.78 \pm 0.26 \%$

$\mu = 564.7 \pm 1.4$

$\sigma = 66.5 \pm 1.3$

WF (3121 entries)

$\frac{\sigma}{\mu} = 11.77 \pm 0.27 \%$

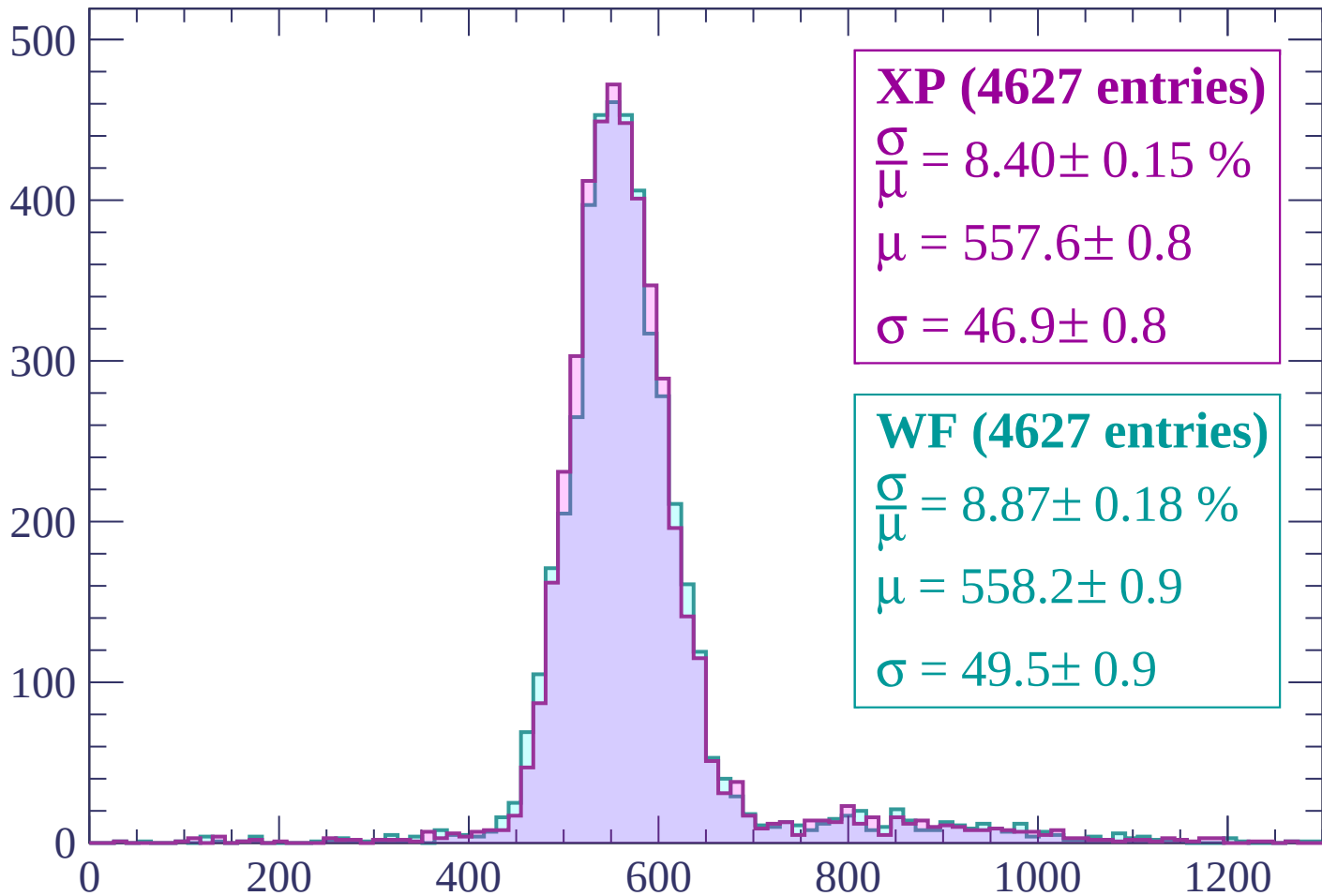
$\mu = 569.1 \pm 1.4$

$\sigma = 67.0 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $-60 < p < 0$

Count



XP (4627 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.15 \%$$

$$\mu = 557.6 \pm 0.8$$

$$\sigma = 46.9 \pm 0.8$$

WF (4627 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.18 \%$$

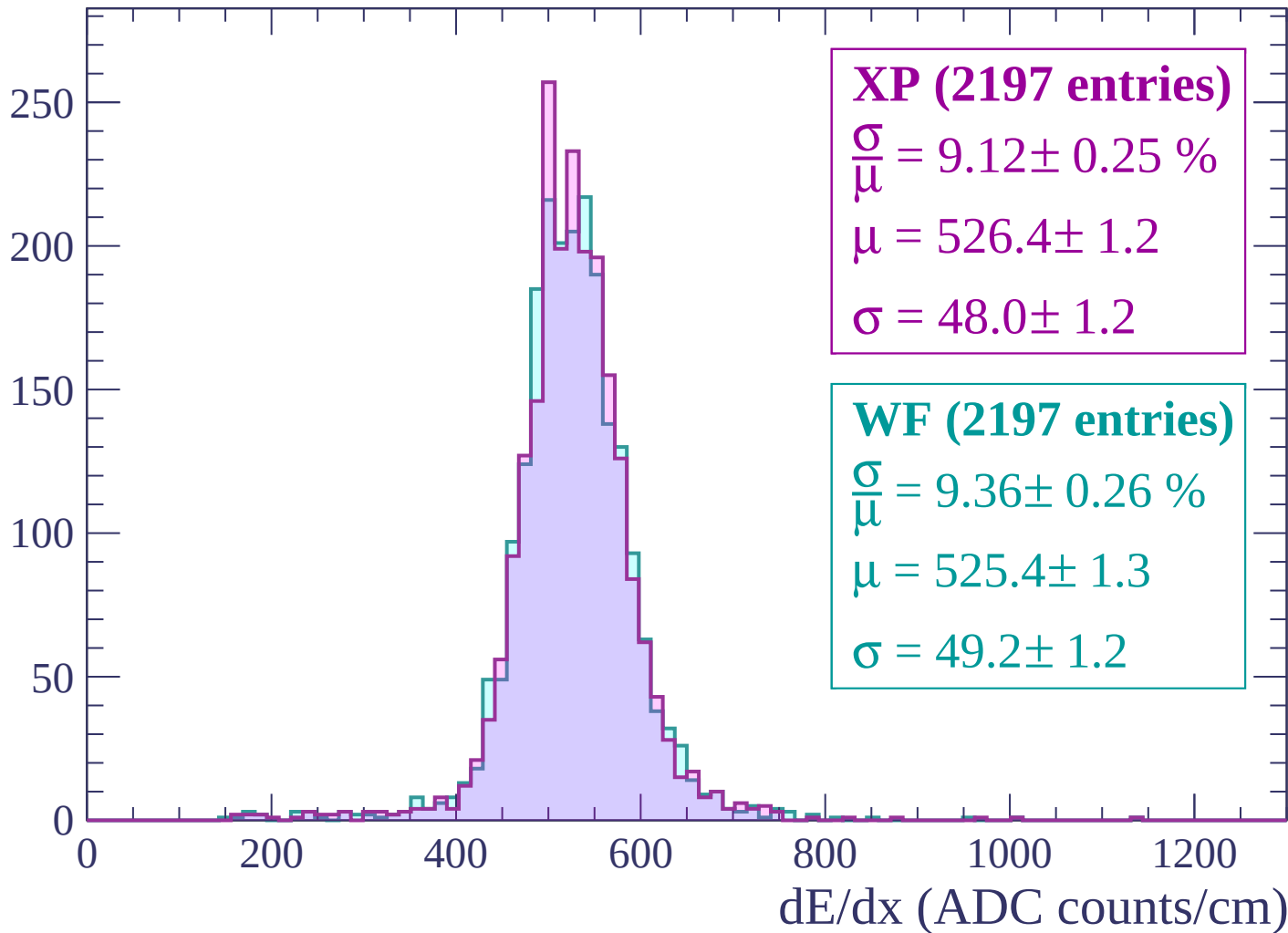
$$\mu = 558.2 \pm 0.9$$

$$\sigma = 49.5 \pm 0.9$$

dE/dx (ADC counts/cm)

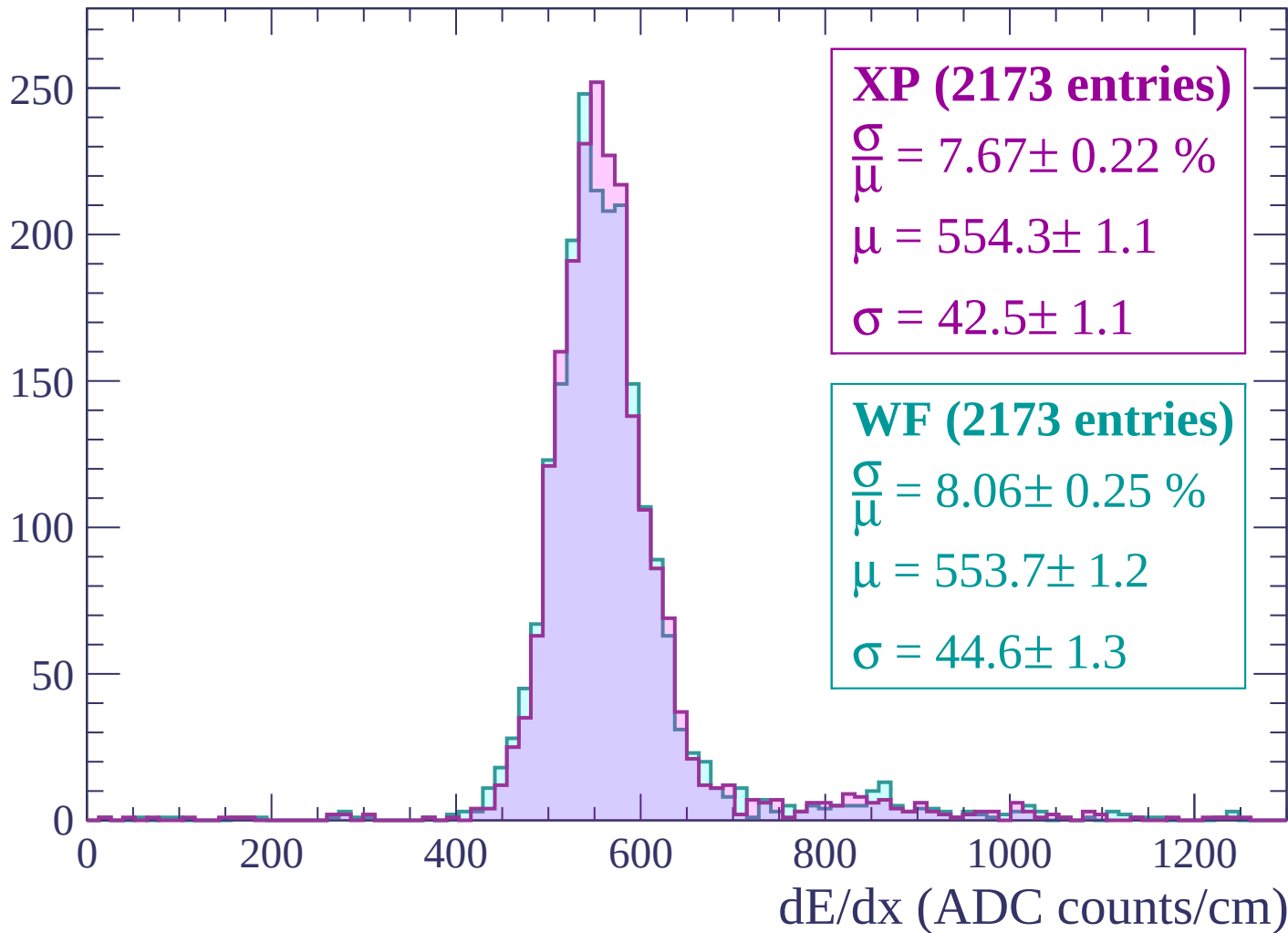
Energy loss | $0 < p < 60$

Count



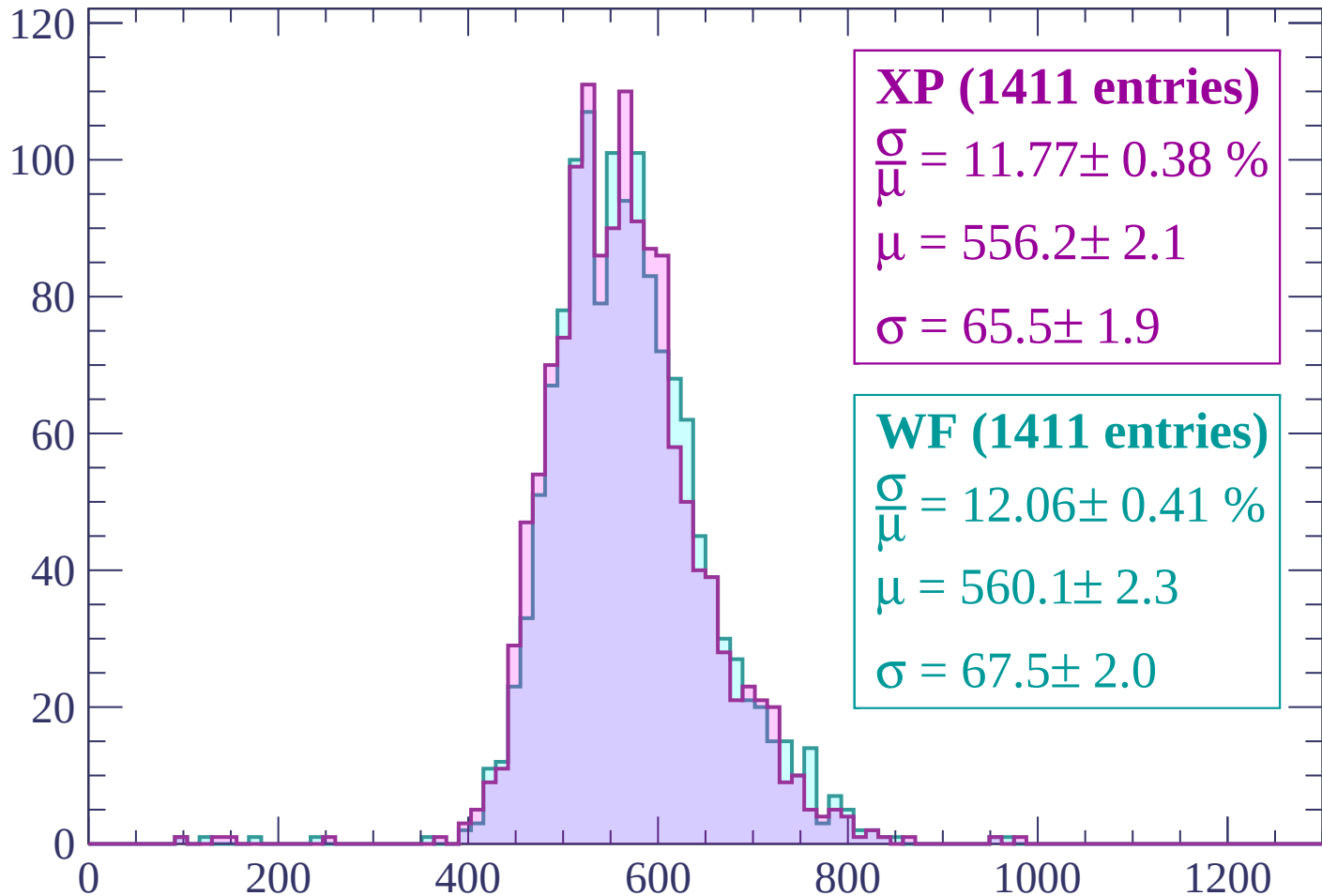
Energy loss | $60 < p < 120$

Count



Energy loss | $120 < p < 180$

Count



XP (1411 entries)

$\frac{\sigma}{\mu} = 11.77 \pm 0.38 \%$

$\mu = 556.2 \pm 2.1$

$\sigma = 65.5 \pm 1.9$

WF (1411 entries)

$\frac{\sigma}{\mu} = 12.06 \pm 0.41 \%$

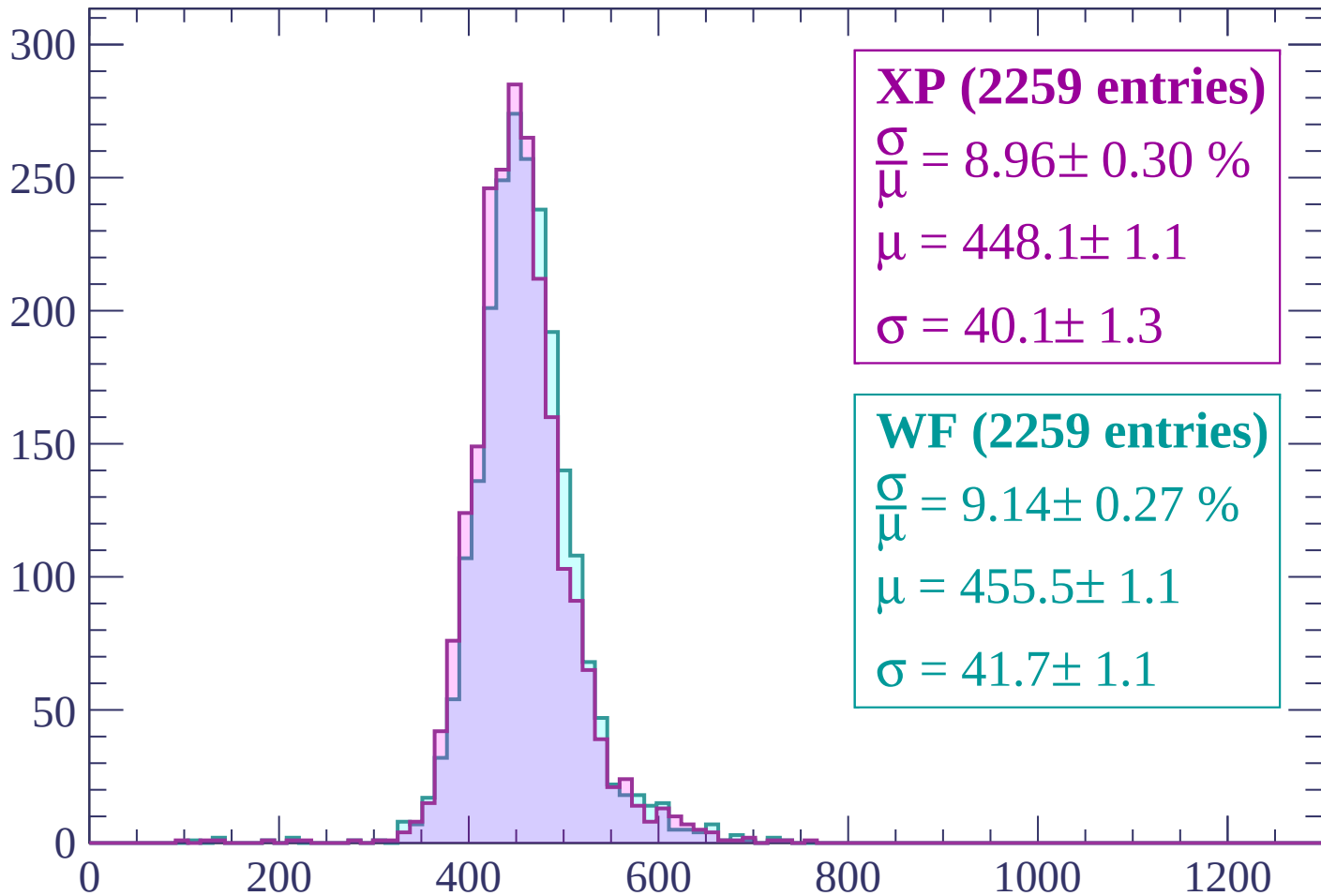
$\mu = 560.1 \pm 2.3$

$\sigma = 67.5 \pm 2.0$

dE/dx (ADC counts/cm)

Energy loss | $180 < p < 240$

Count



XP (2259 entries)

$$\frac{\sigma}{\mu} = 8.96 \pm 0.30 \%$$

$$\mu = 448.1 \pm 1.1$$

$$\sigma = 40.1 \pm 1.3$$

WF (2259 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.27 \%$$

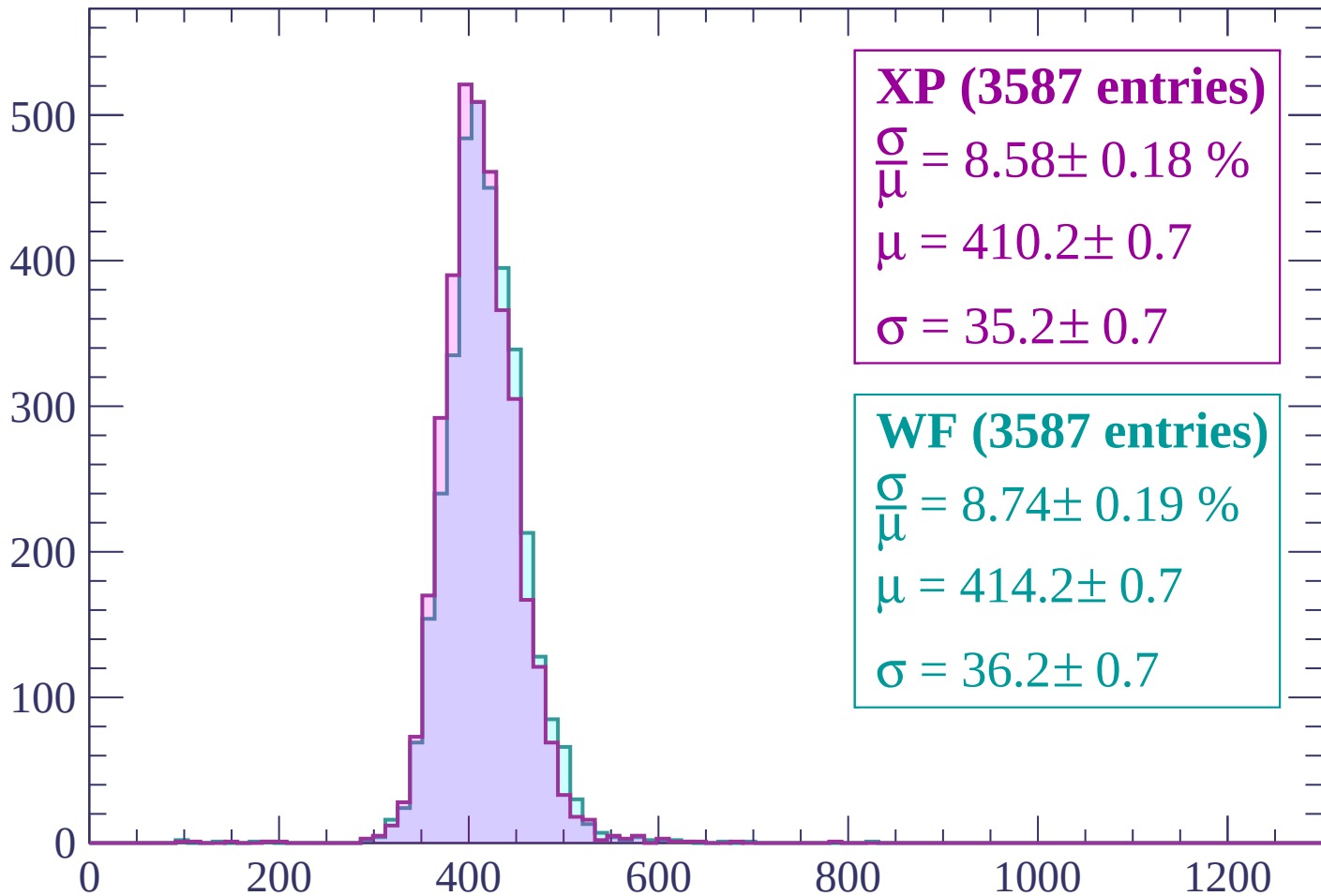
$$\mu = 455.5 \pm 1.1$$

$$\sigma = 41.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (3587 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.18 \%$$

$$\mu = 410.2 \pm 0.7$$

$$\sigma = 35.2 \pm 0.7$$

WF (3587 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.19 \%$$

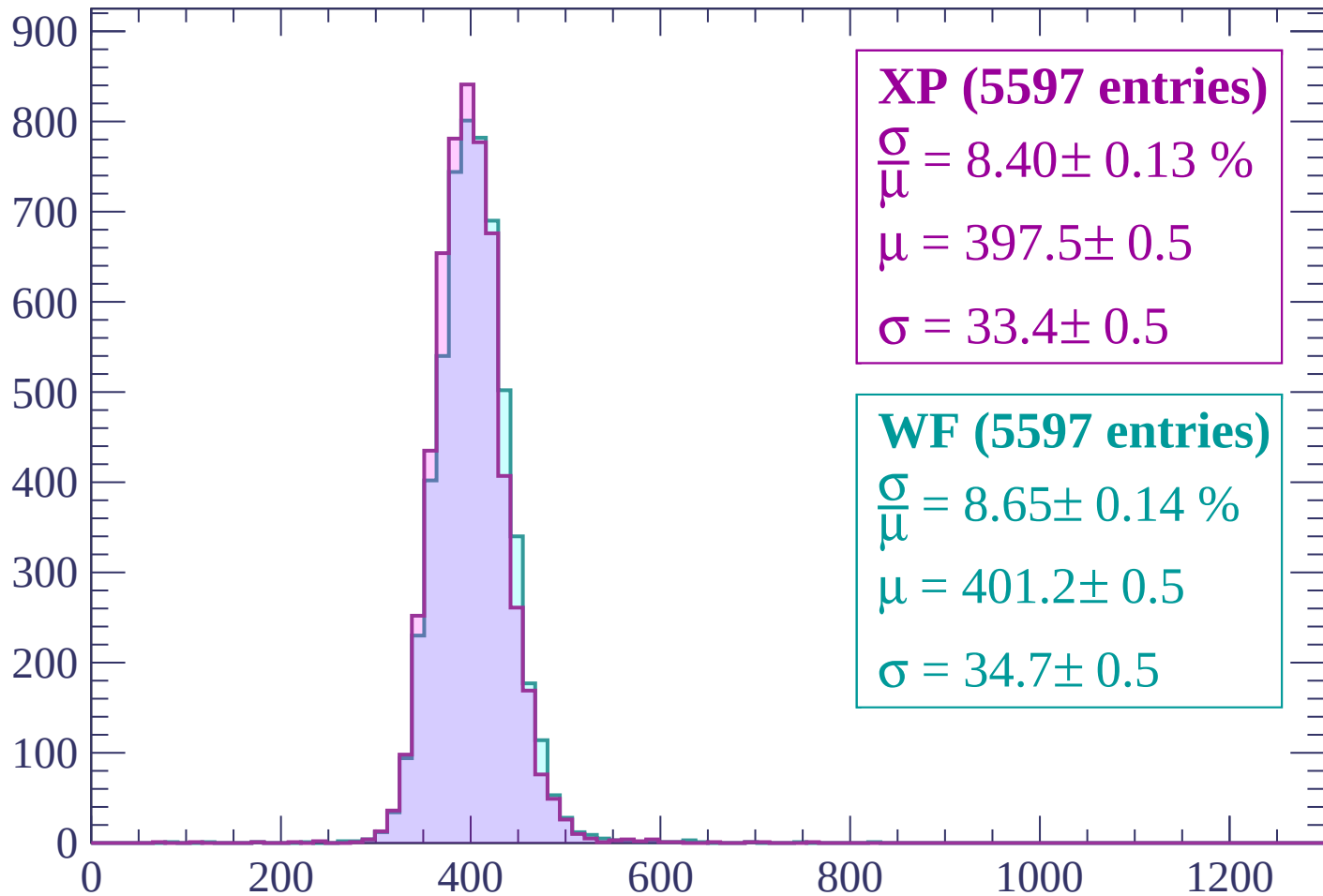
$$\mu = 414.2 \pm 0.7$$

$$\sigma = 36.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5597 entries)

$$\frac{\sigma}{\mu} = 8.40 \pm 0.13 \%$$

$$\mu = 397.5 \pm 0.5$$

$$\sigma = 33.4 \pm 0.5$$

WF (5597 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.14 \%$$

$$\mu = 401.2 \pm 0.5$$

$$\sigma = 34.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count

1000

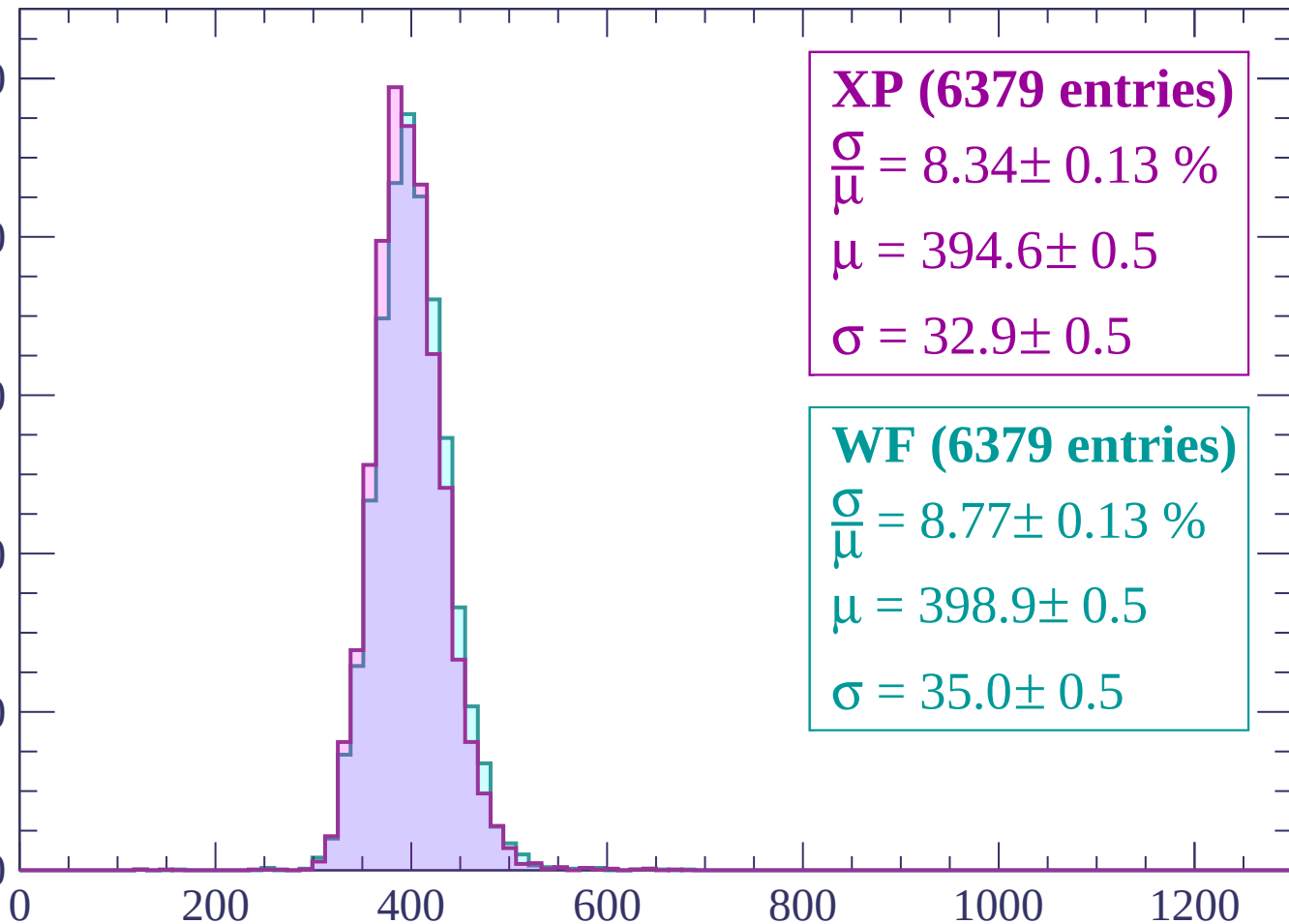
800

600

400

200

0



XP (6379 entries)

$\frac{\sigma}{\mu} = 8.34 \pm 0.13 \%$

$\mu = 394.6 \pm 0.5$

$\sigma = 32.9 \pm 0.5$

WF (6379 entries)

$\frac{\sigma}{\mu} = 8.77 \pm 0.13 \%$

$\mu = 398.9 \pm 0.5$

$\sigma = 35.0 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count

1000

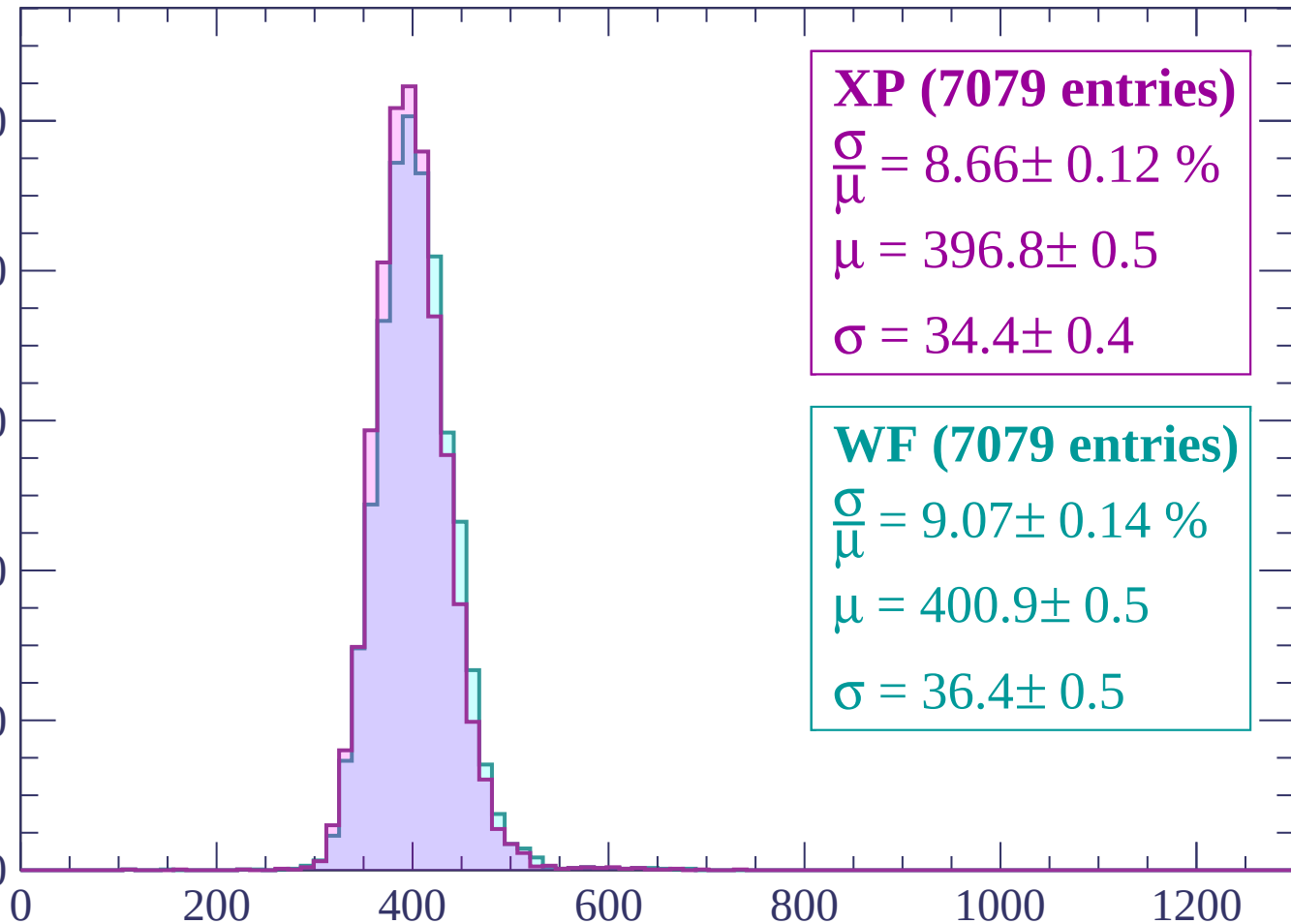
800

600

400

200

0



XP (7079 entries)

$\frac{\sigma}{\mu} = 8.66 \pm 0.12 \%$

$\mu = 396.8 \pm 0.5$

$\sigma = 34.4 \pm 0.4$

WF (7079 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.14 \%$

$\mu = 400.9 \pm 0.5$

$\sigma = 36.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count

1000

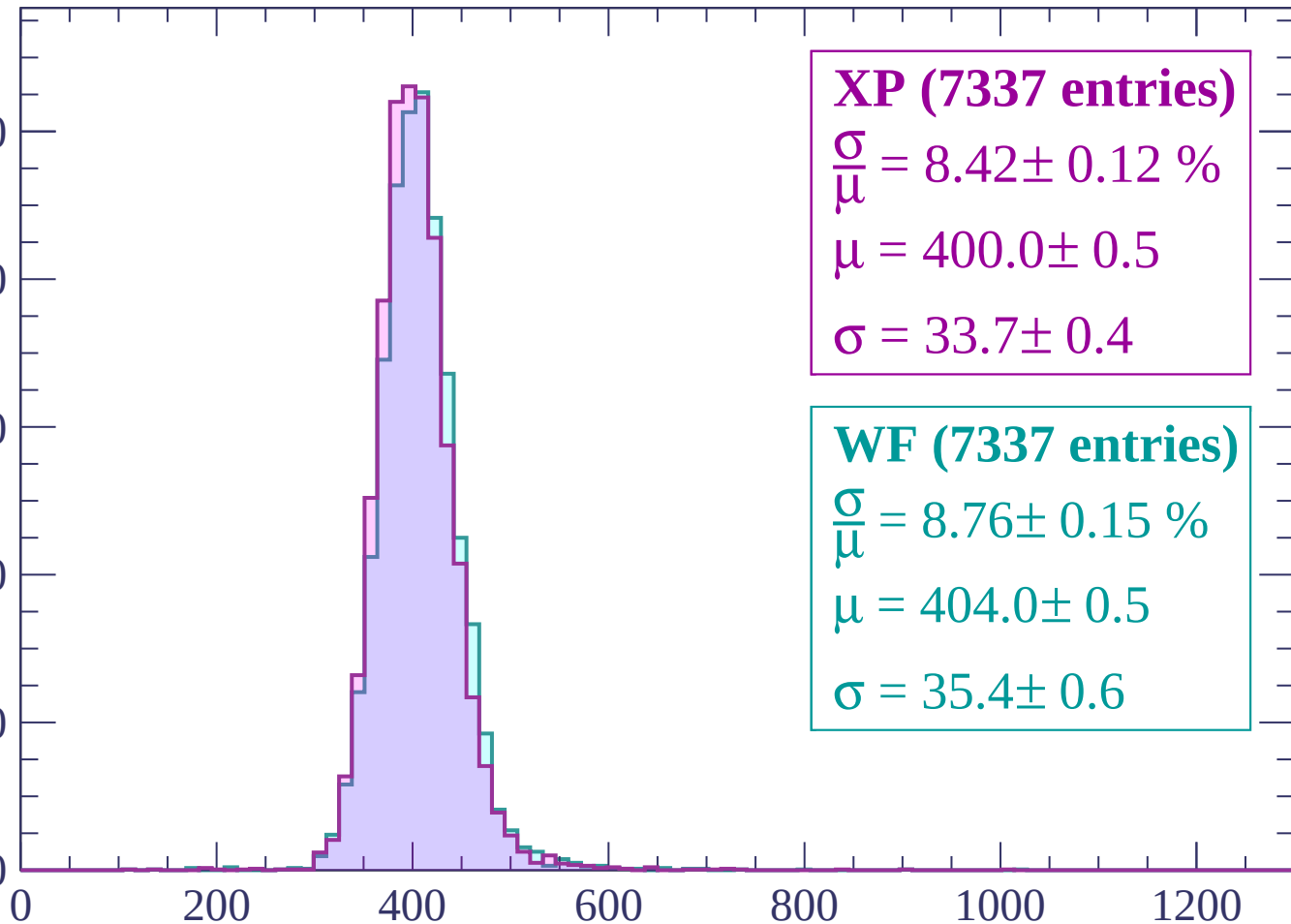
800

600

400

200

0



XP (7337 entries)

$\frac{\sigma}{\mu} = 8.42 \pm 0.12 \%$

$\mu = 400.0 \pm 0.5$

$\sigma = 33.7 \pm 0.4$

WF (7337 entries)

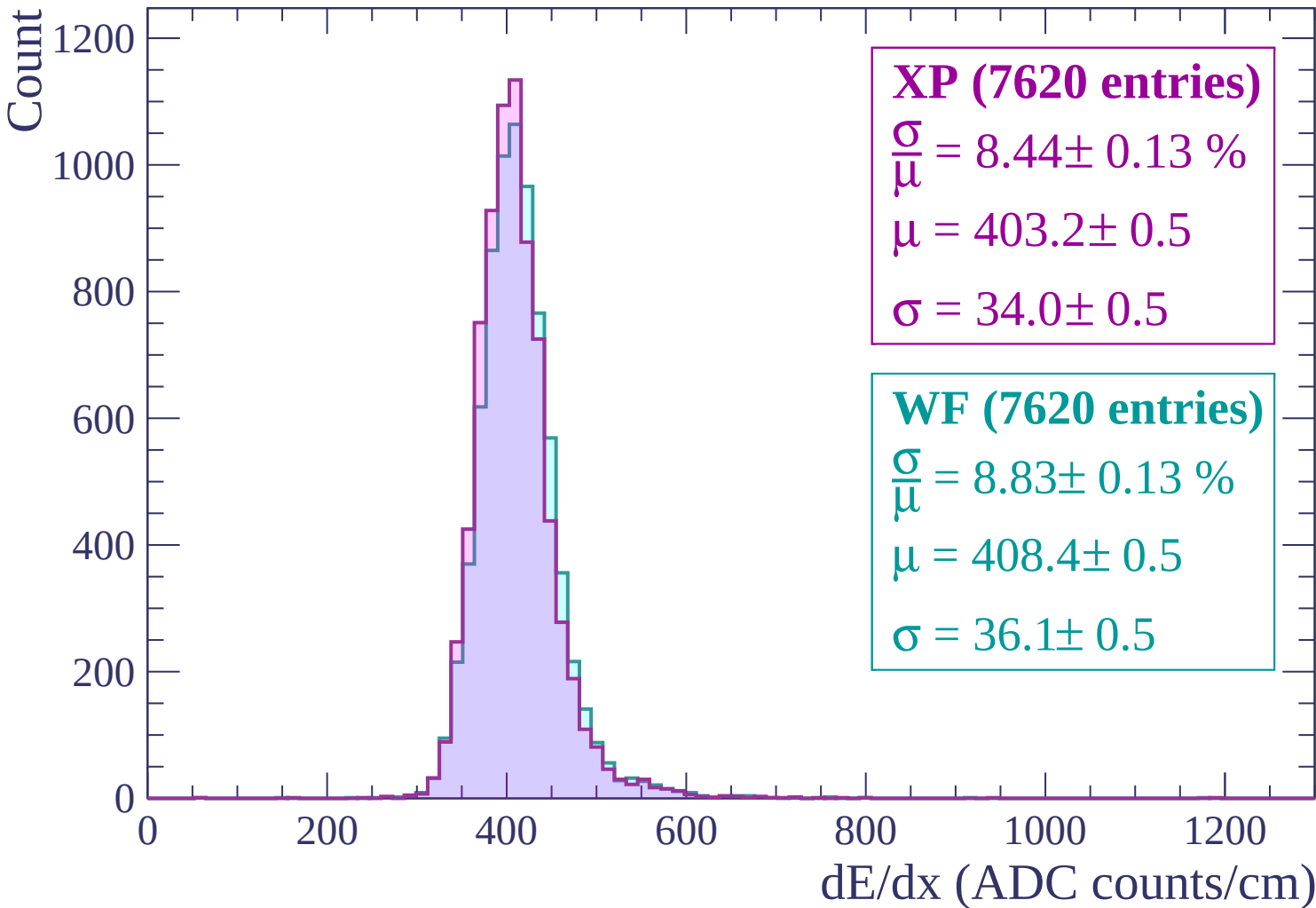
$\frac{\sigma}{\mu} = 8.76 \pm 0.15 \%$

$\mu = 404.0 \pm 0.5$

$\sigma = 35.4 \pm 0.6$

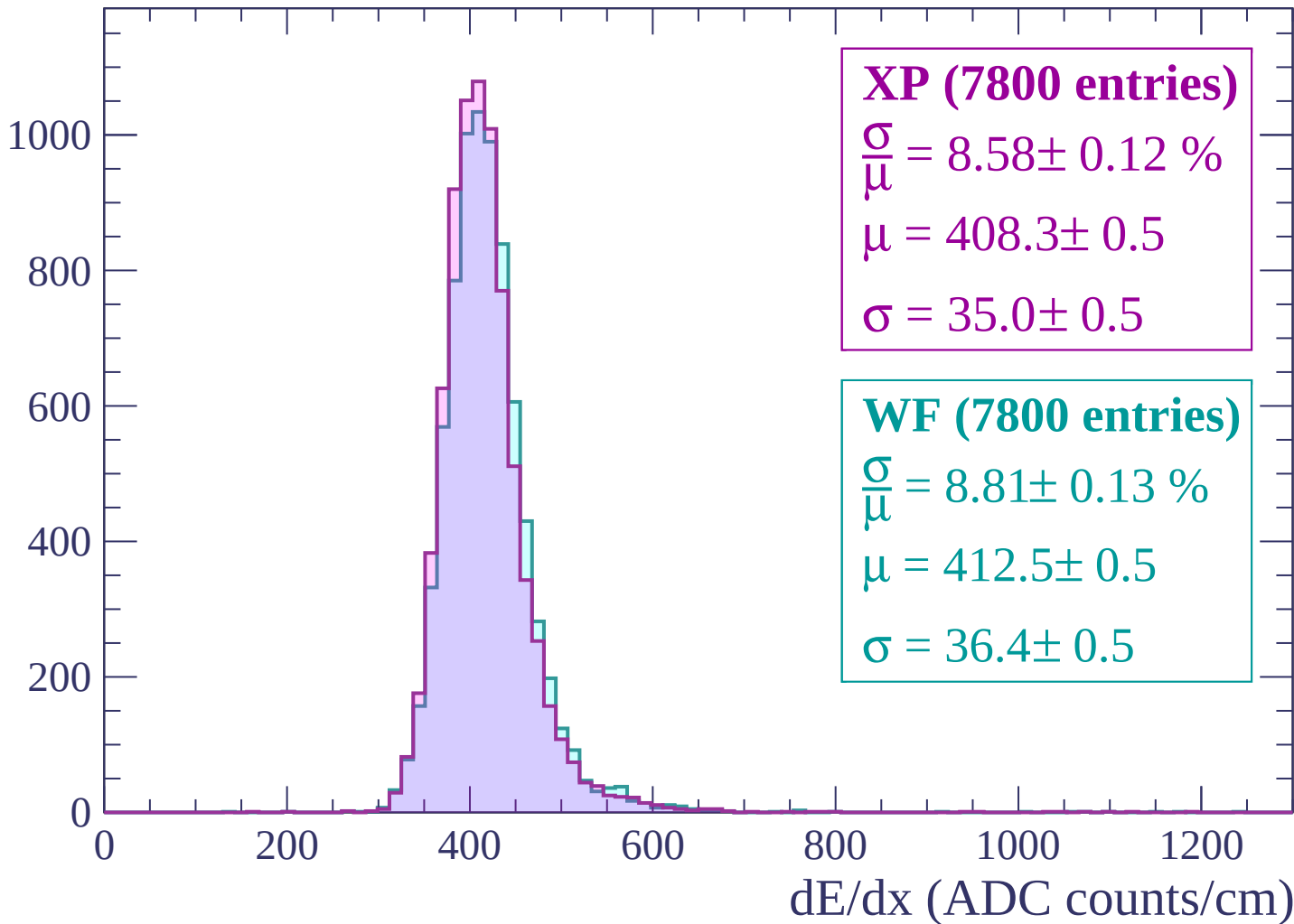
dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$



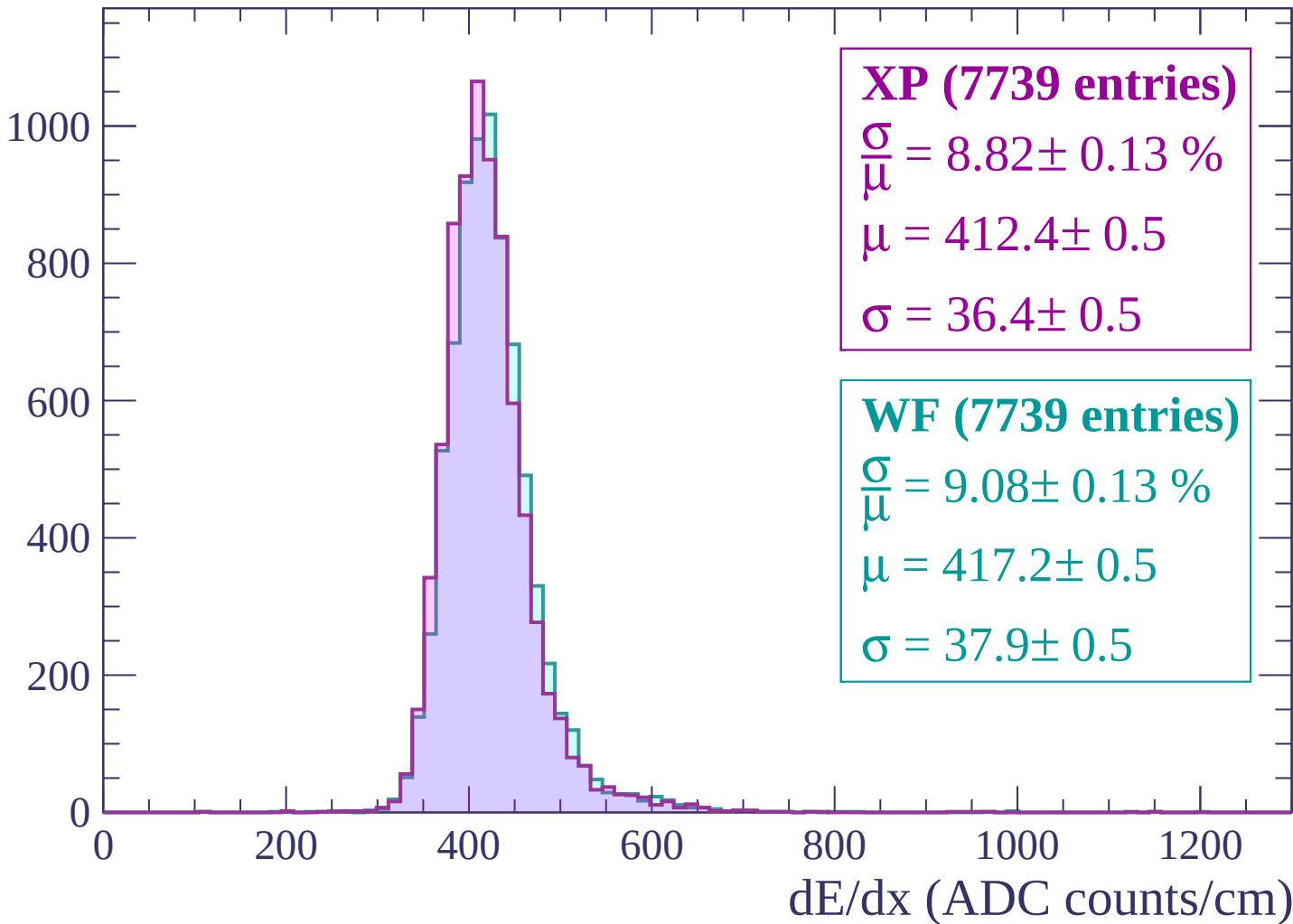
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



Energy loss | $720 < p < 780$

Count

1000

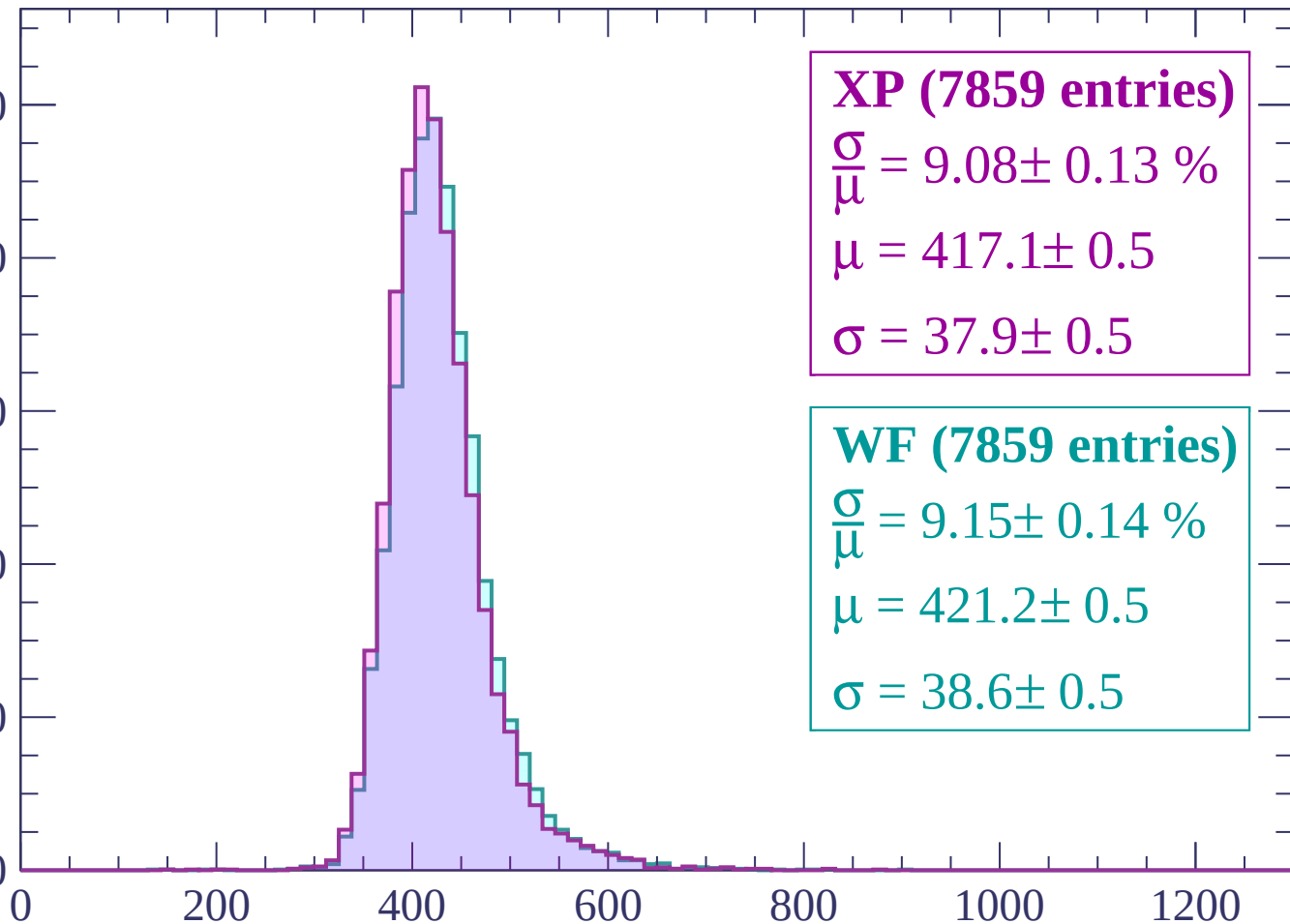
800

600

400

200

0



XP (7859 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.13 \%$

$\mu = 417.1 \pm 0.5$

$\sigma = 37.9 \pm 0.5$

WF (7859 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.14 \%$

$\mu = 421.2 \pm 0.5$

$\sigma = 38.6 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (7701 entries)

$$\frac{\sigma}{\mu} = 8.81 \pm 0.12 \%$$

$$\mu = 421.8 \pm 0.5$$

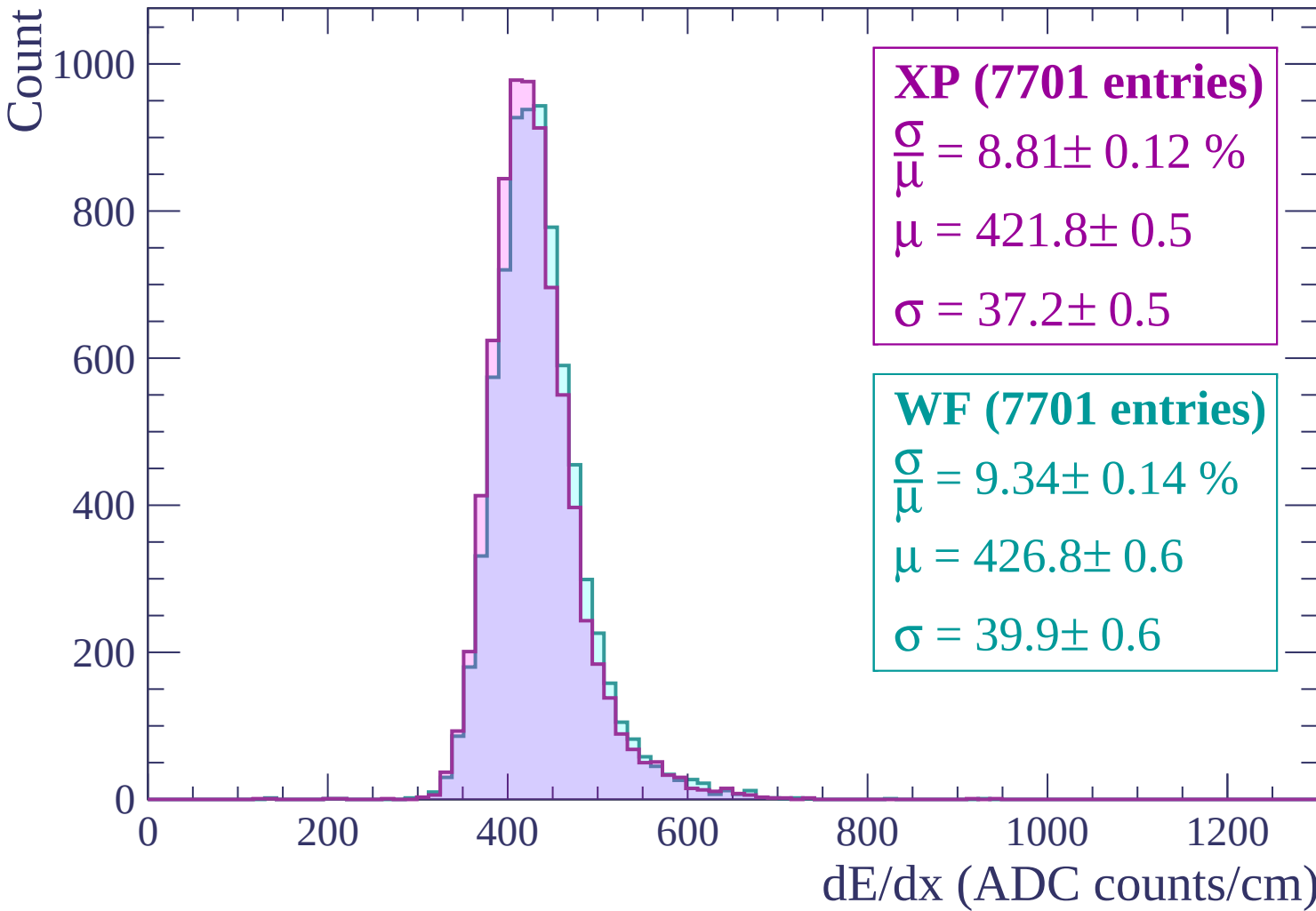
$$\sigma = 37.2 \pm 0.5$$

WF (7701 entries)

$$\frac{\sigma}{\mu} = 9.34 \pm 0.14 \%$$

$$\mu = 426.8 \pm 0.6$$

$$\sigma = 39.9 \pm 0.6$$



Energy loss | $840 < p < 900$

Count

1000

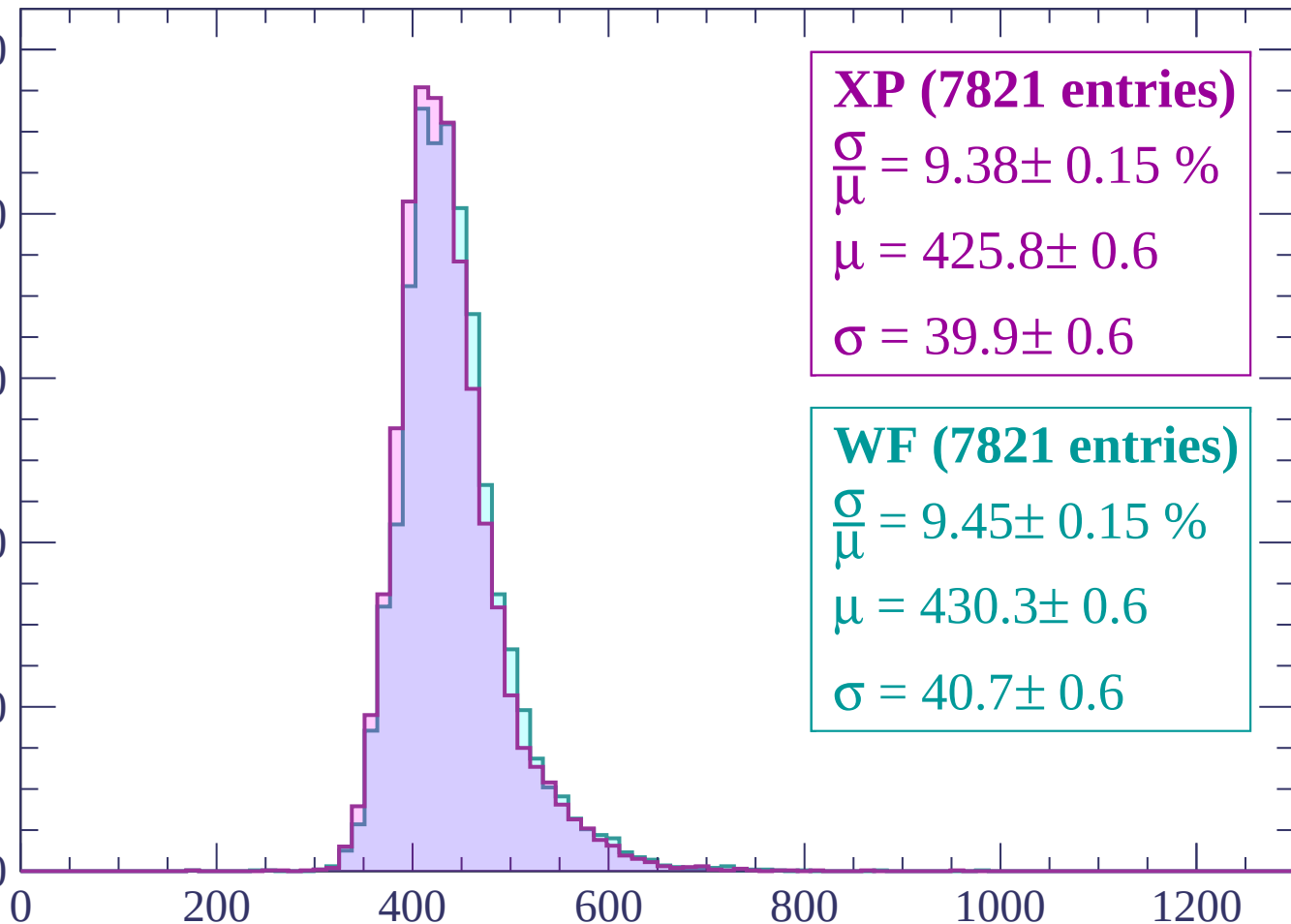
800

600

400

200

0



XP (7821 entries)

$\frac{\sigma}{\mu} = 9.38 \pm 0.15 \%$

$\mu = 425.8 \pm 0.6$

$\sigma = 39.9 \pm 0.6$

WF (7821 entries)

$\frac{\sigma}{\mu} = 9.45 \pm 0.15 \%$

$\mu = 430.3 \pm 0.6$

$\sigma = 40.7 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (7919 entries)

$\frac{\sigma}{\mu} = 9.26 \pm 0.14 \%$

$\mu = 430.7 \pm 0.6$

$\sigma = 39.9 \pm 0.6$

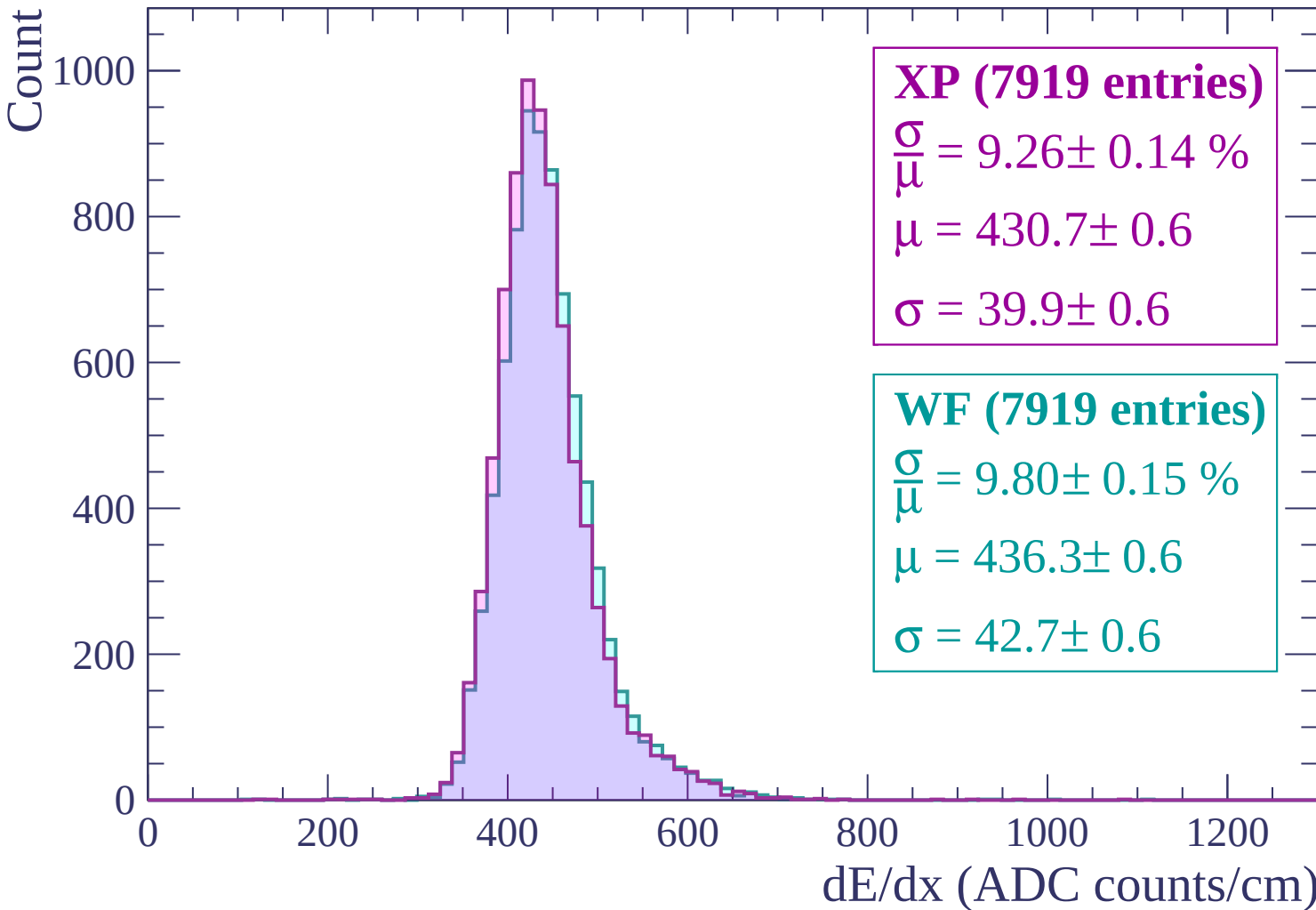
WF (7919 entries)

$\frac{\sigma}{\mu} = 9.80 \pm 0.15 \%$

$\mu = 436.3 \pm 0.6$

$\sigma = 42.7 \pm 0.6$

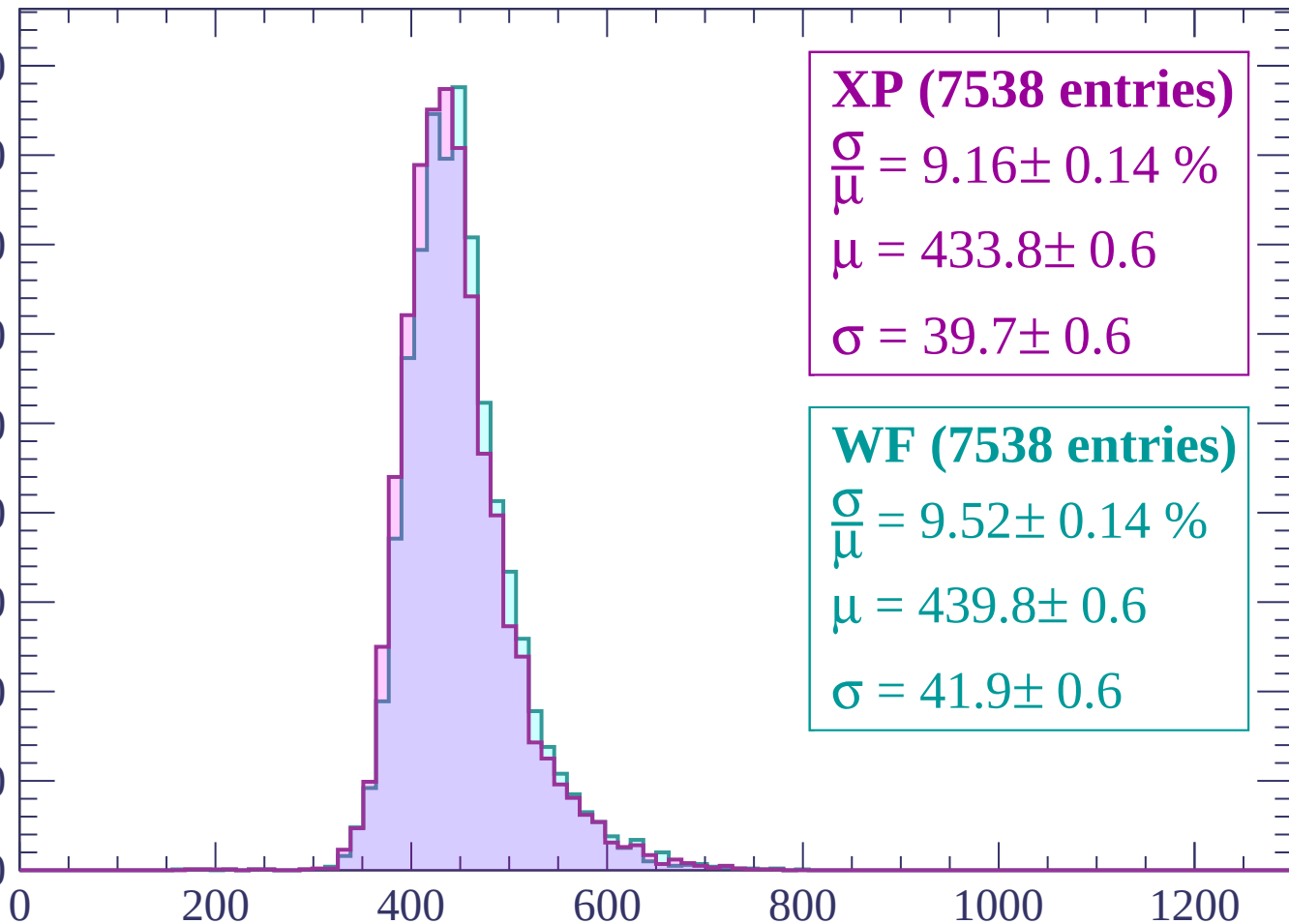
dE/dx (ADC counts/cm)



Energy loss | $960 < p < 1020$

Count

900
800
700
600
500
400
300
200
100
0



XP (7538 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.14 \%$$

$$\mu = 433.8 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

WF (7538 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.14 \%$$

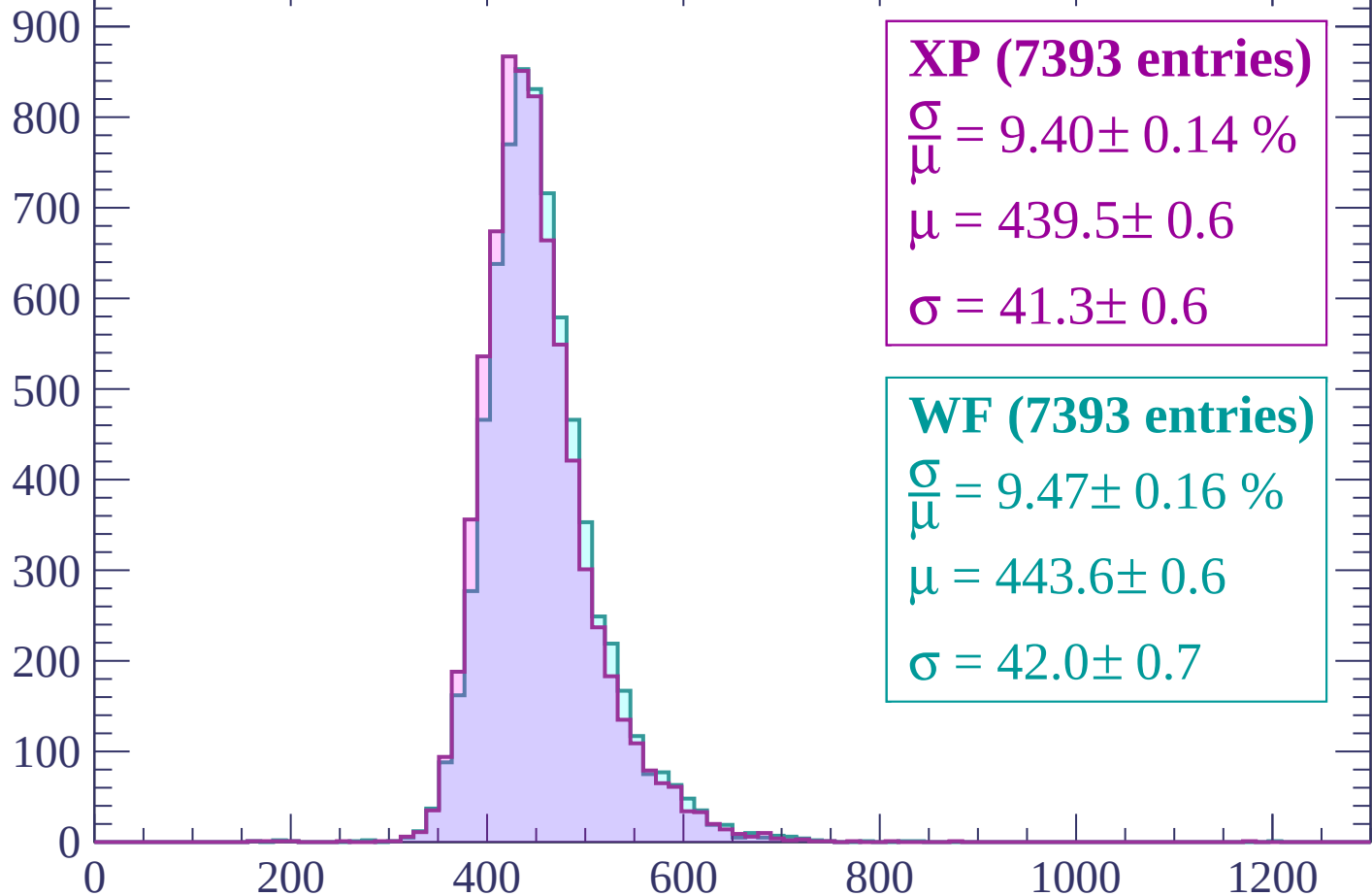
$$\mu = 439.8 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (7393 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.14 \%$$

$$\mu = 439.5 \pm 0.6$$

$$\sigma = 41.3 \pm 0.6$$

WF (7393 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

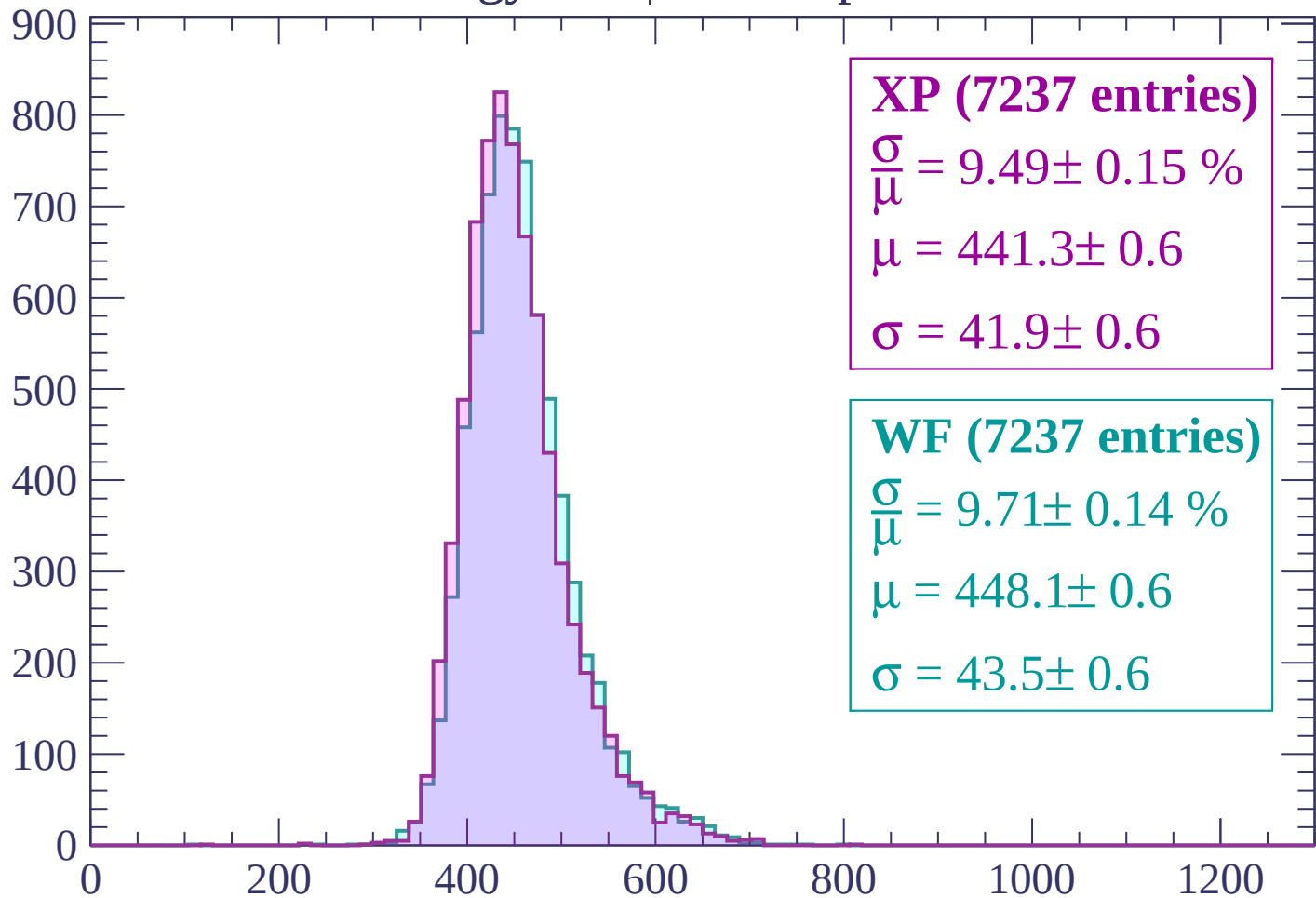
$$\mu = 443.6 \pm 0.6$$

$$\sigma = 42.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (7237 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.15 \%$$

$$\mu = 441.3 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

WF (7237 entries)

$$\frac{\sigma}{\mu} = 9.71 \pm 0.14 \%$$

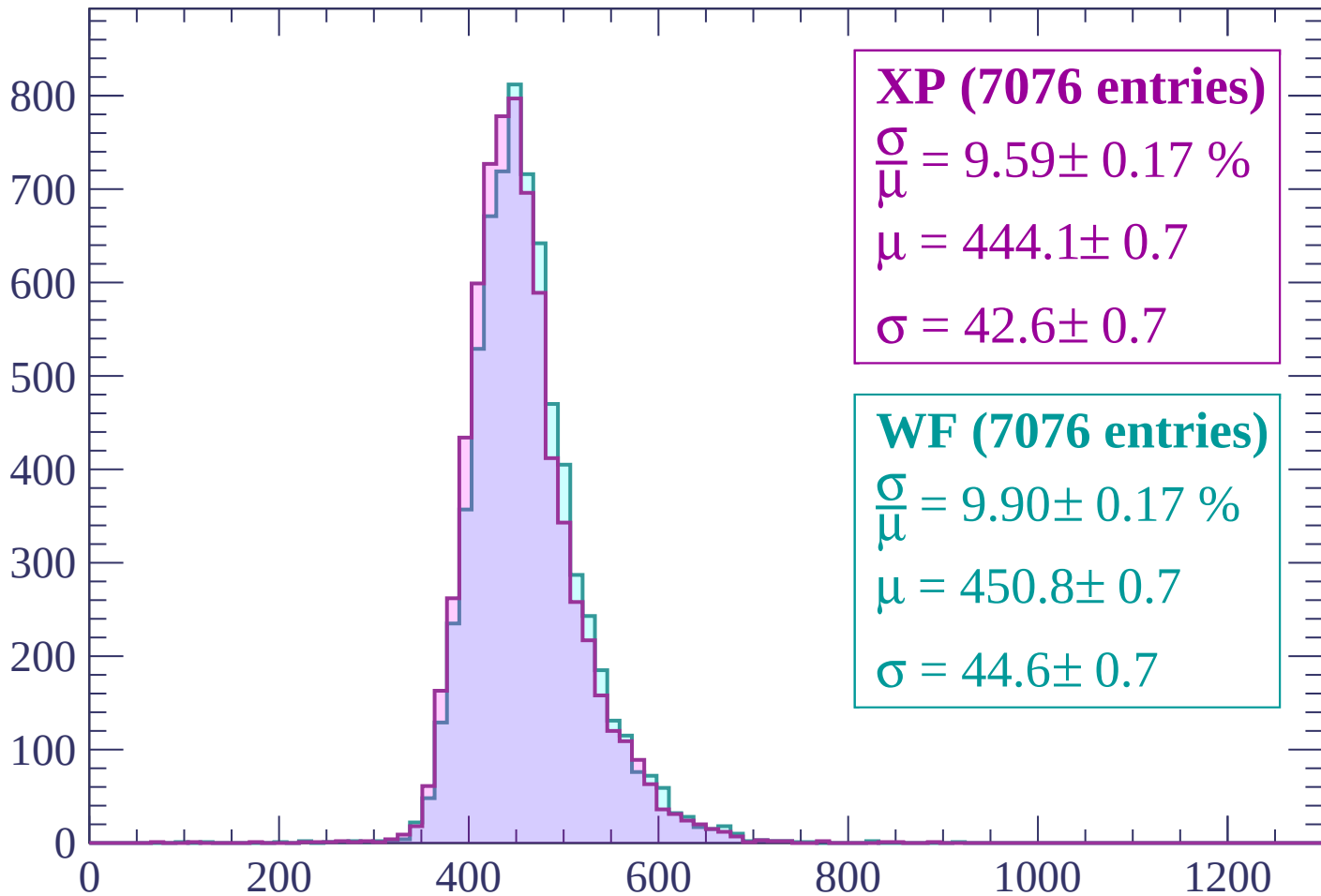
$$\mu = 448.1 \pm 0.6$$

$$\sigma = 43.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (7076 entries)

$$\frac{\sigma}{\mu} = 9.59 \pm 0.17 \%$$

$$\mu = 444.1 \pm 0.7$$

$$\sigma = 42.6 \pm 0.7$$

WF (7076 entries)

$$\frac{\sigma}{\mu} = 9.90 \pm 0.17 \%$$

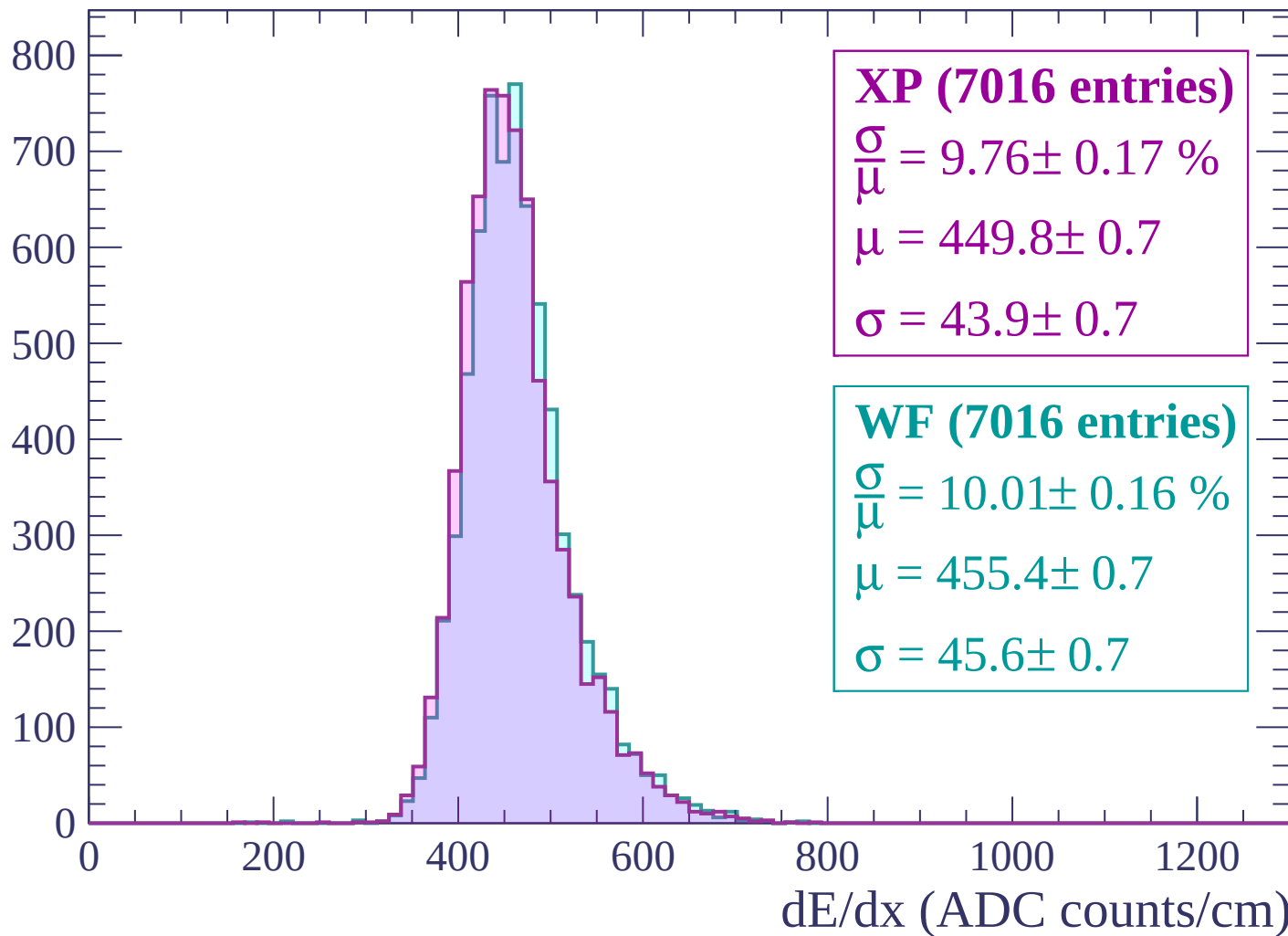
$$\mu = 450.8 \pm 0.7$$

$$\sigma = 44.6 \pm 0.7$$

dE/dx (ADC counts/cm)

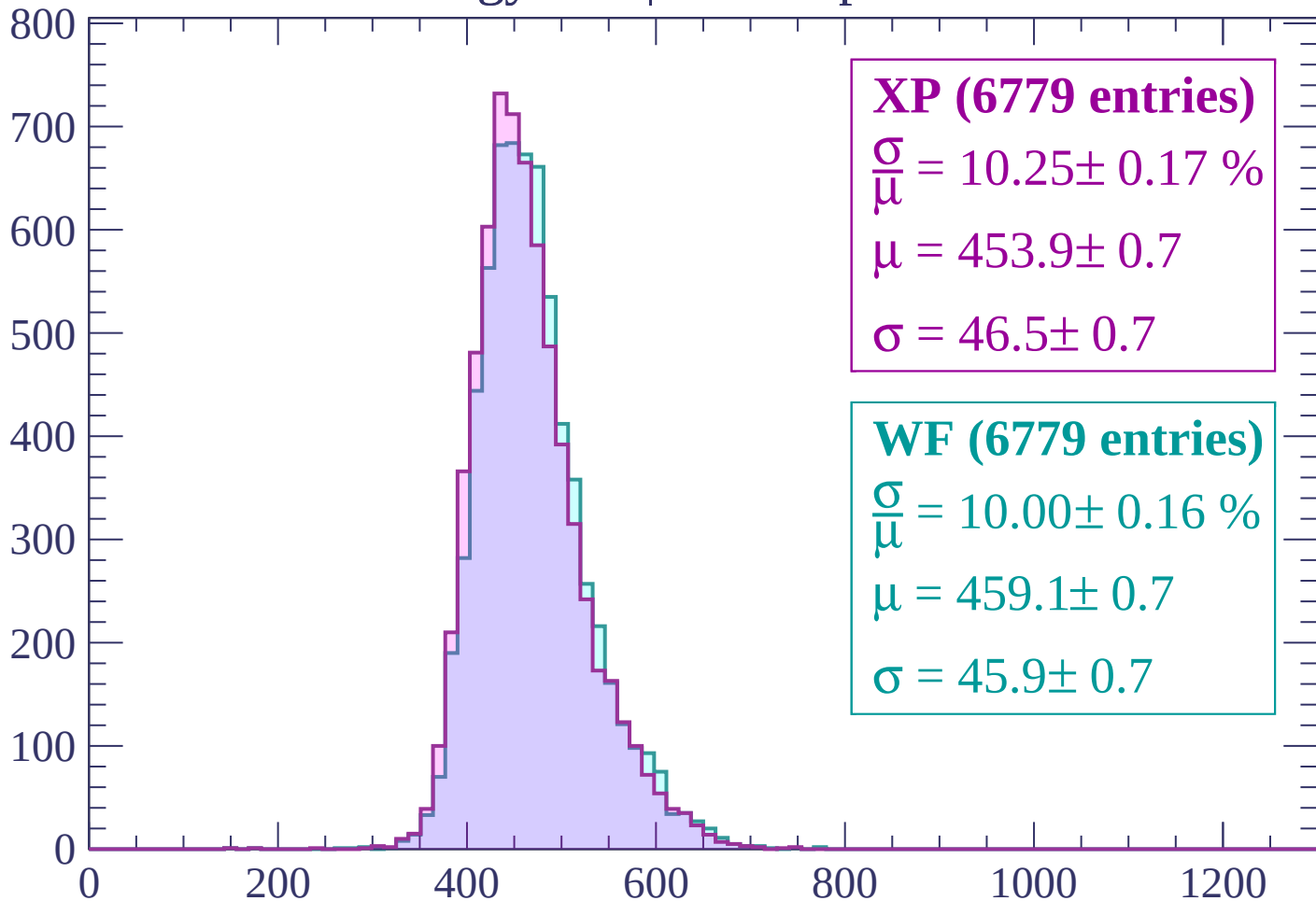
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (6779 entries)

$\frac{\sigma}{\mu} = 10.25 \pm 0.17 \%$

$\mu = 453.9 \pm 0.7$

$\sigma = 46.5 \pm 0.7$

WF (6779 entries)

$\frac{\sigma}{\mu} = 10.00 \pm 0.16 \%$

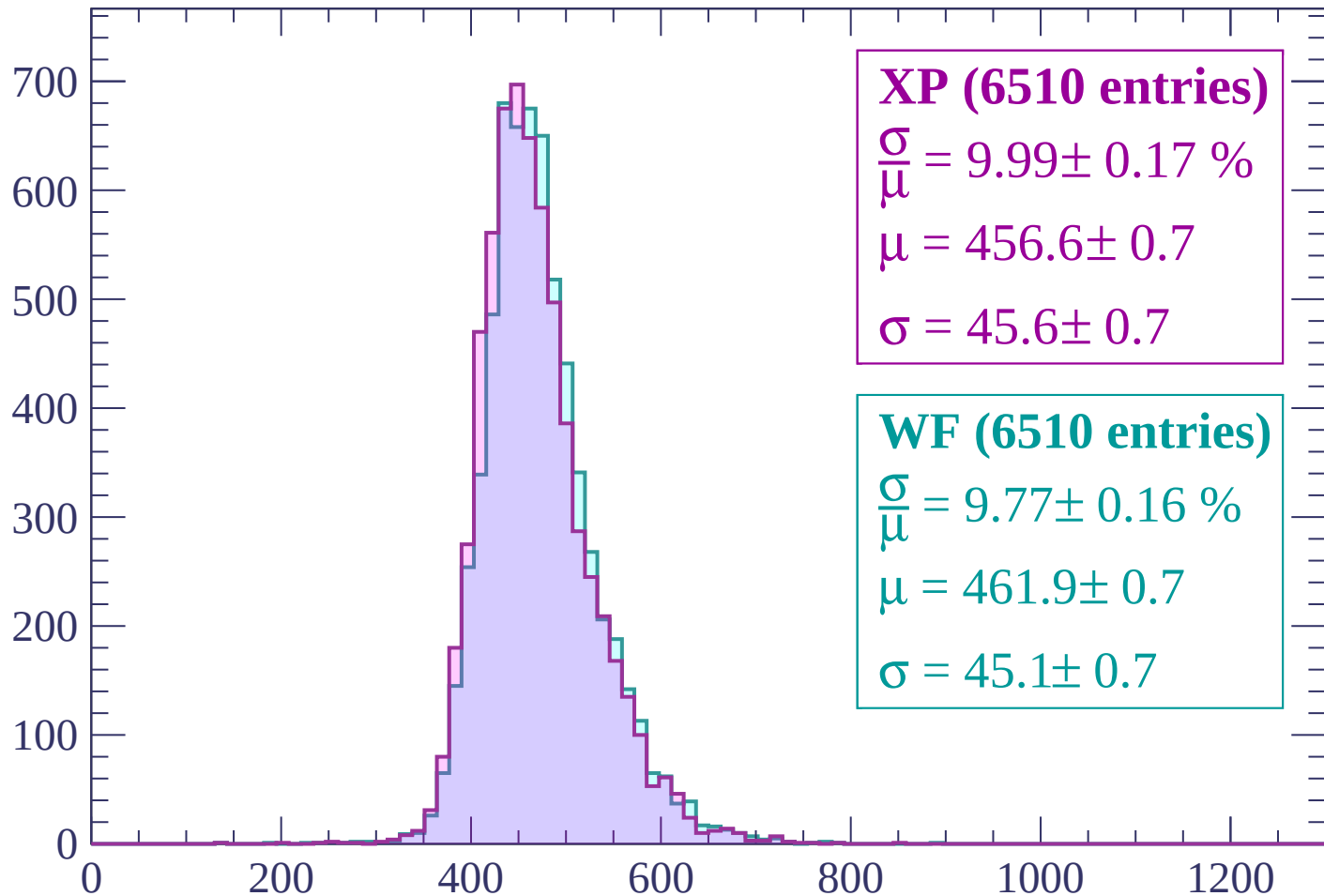
$\mu = 459.1 \pm 0.7$

$\sigma = 45.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (6510 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.17 \%$$

$$\mu = 456.6 \pm 0.7$$

$$\sigma = 45.6 \pm 0.7$$

WF (6510 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.16 \%$$

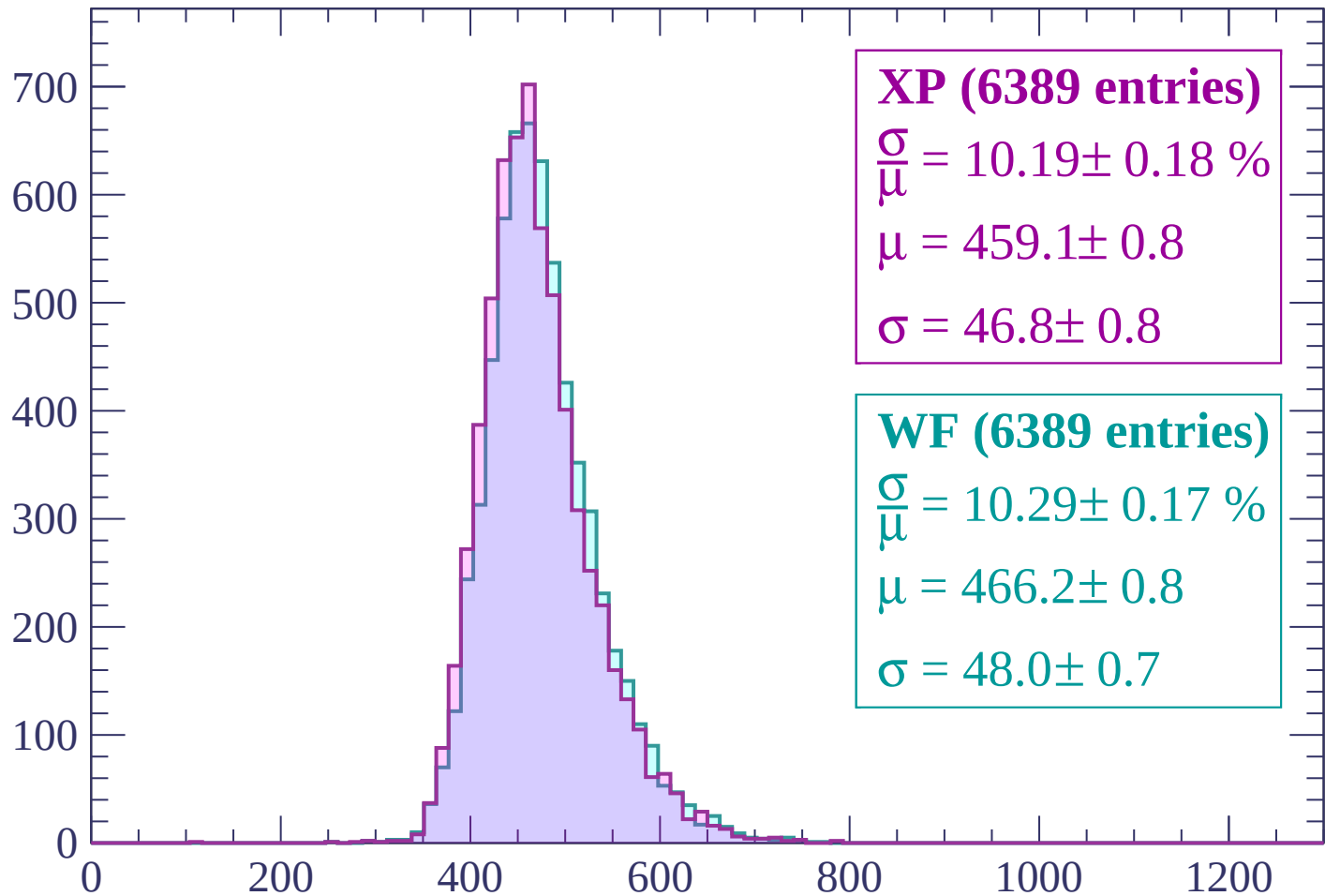
$$\mu = 461.9 \pm 0.7$$

$$\sigma = 45.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (6389 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.18 \%$

$\mu = 459.1 \pm 0.8$

$\sigma = 46.8 \pm 0.8$

WF (6389 entries)

$\frac{\sigma}{\mu} = 10.29 \pm 0.17 \%$

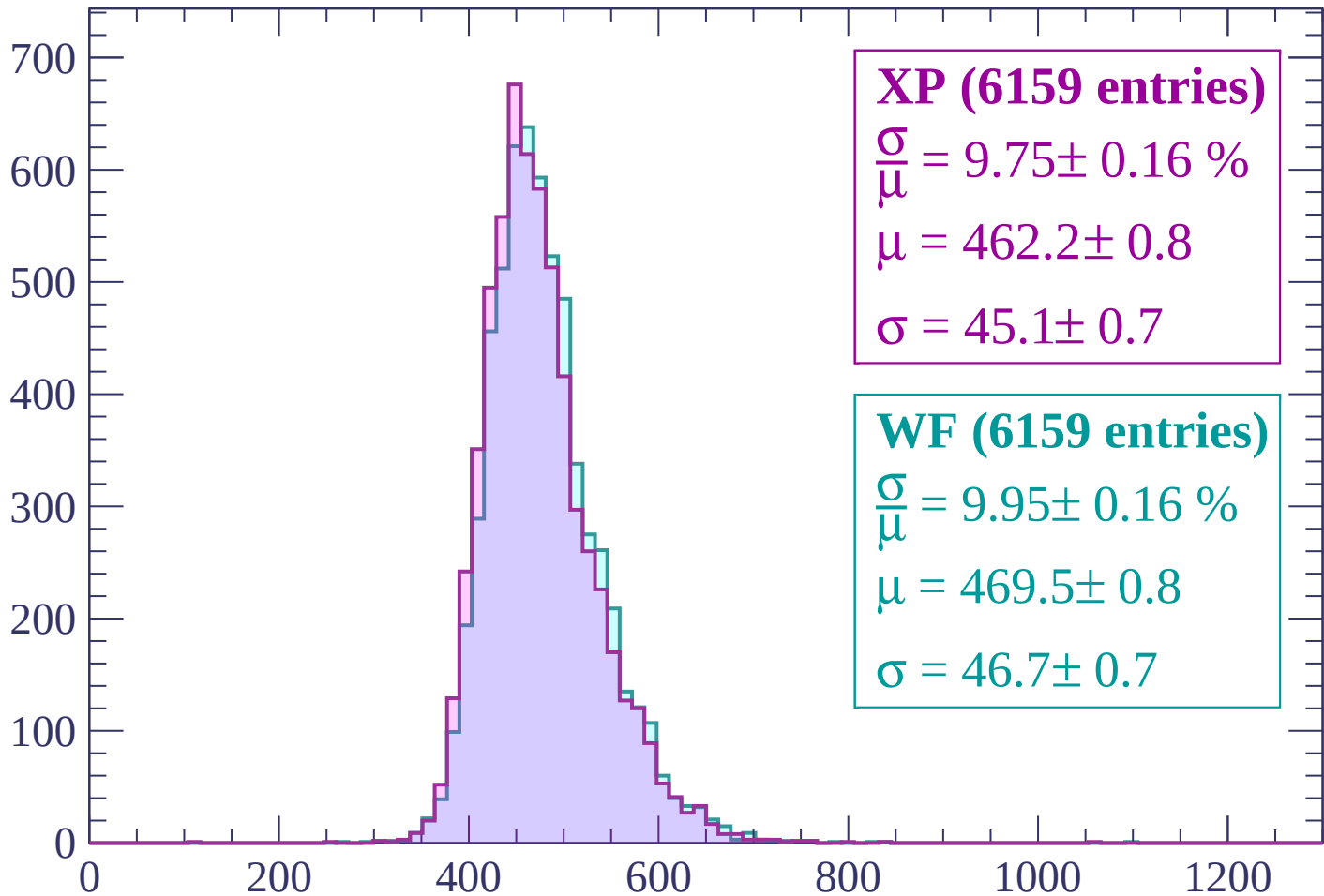
$\mu = 466.2 \pm 0.8$

$\sigma = 48.0 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (6159 entries)

$$\frac{\sigma}{\mu} = 9.75 \pm 0.16 \%$$

$$\mu = 462.2 \pm 0.8$$

$$\sigma = 45.1 \pm 0.7$$

WF (6159 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.16 \%$$

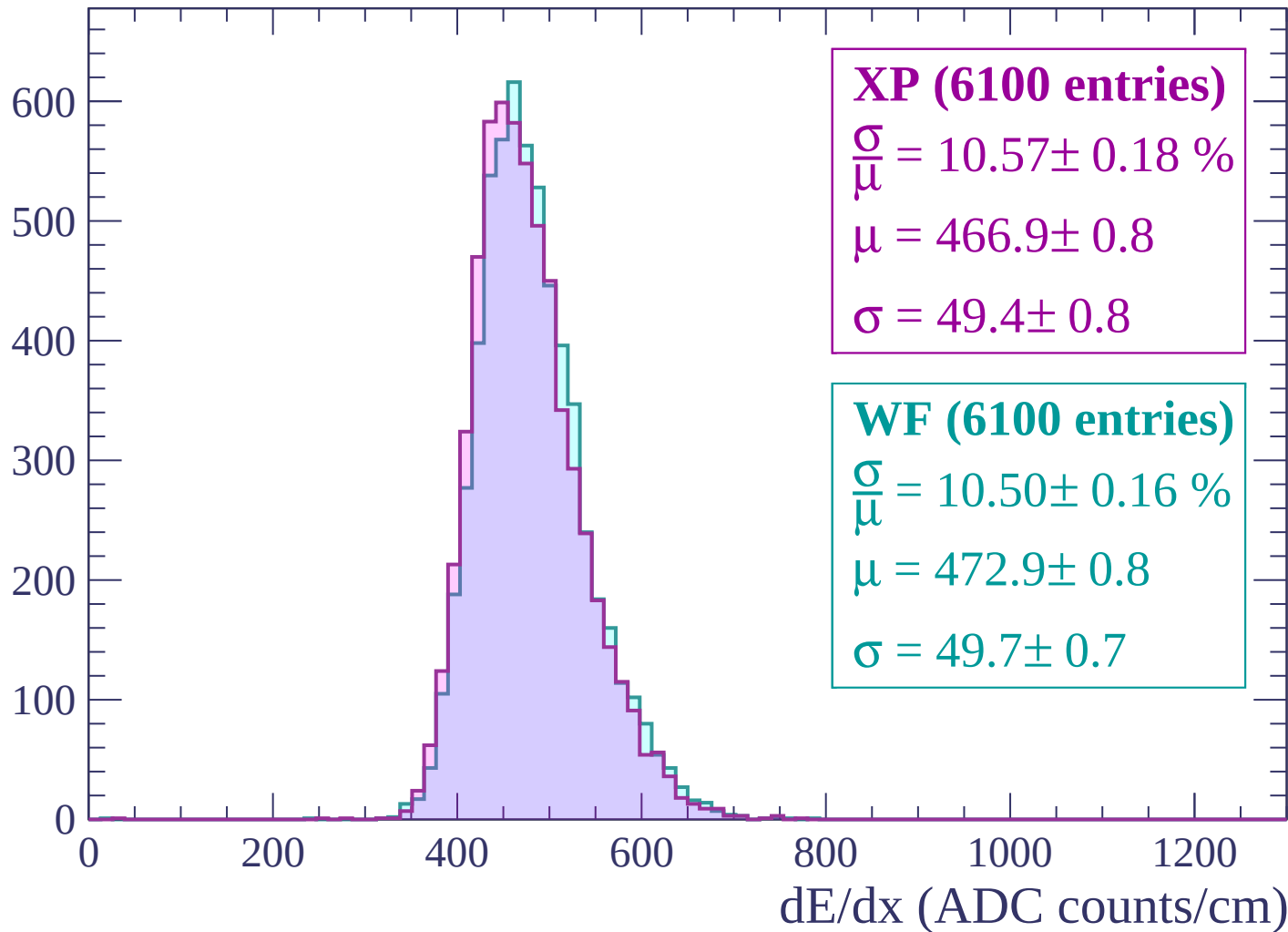
$$\mu = 469.5 \pm 0.8$$

$$\sigma = 46.7 \pm 0.7$$

dE/dx (ADC counts/cm)

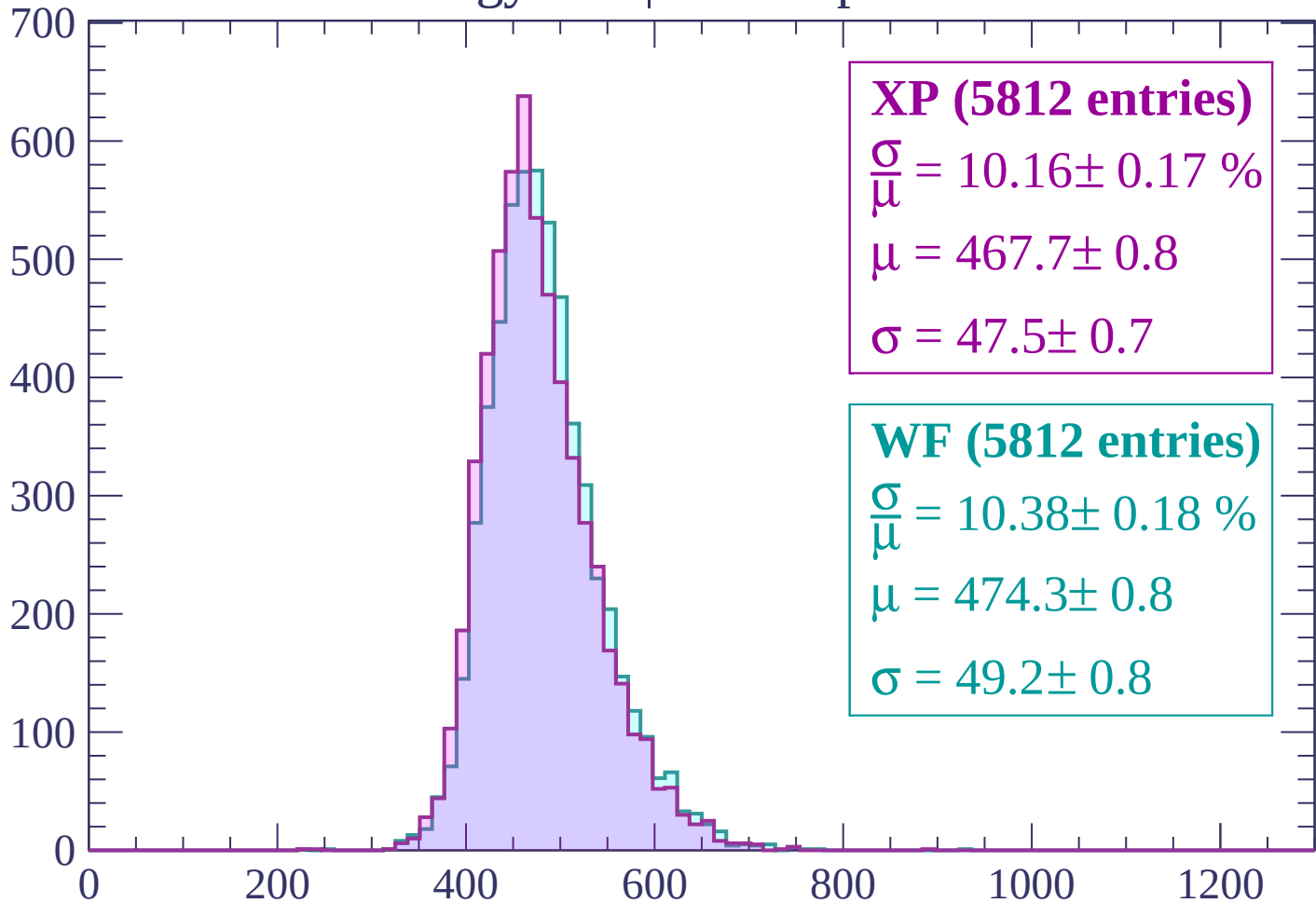
Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count



XP (5812 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.17 \%$$

$$\mu = 467.7 \pm 0.8$$

$$\sigma = 47.5 \pm 0.7$$

WF (5812 entries)

$$\frac{\sigma}{\mu} = 10.38 \pm 0.18 \%$$

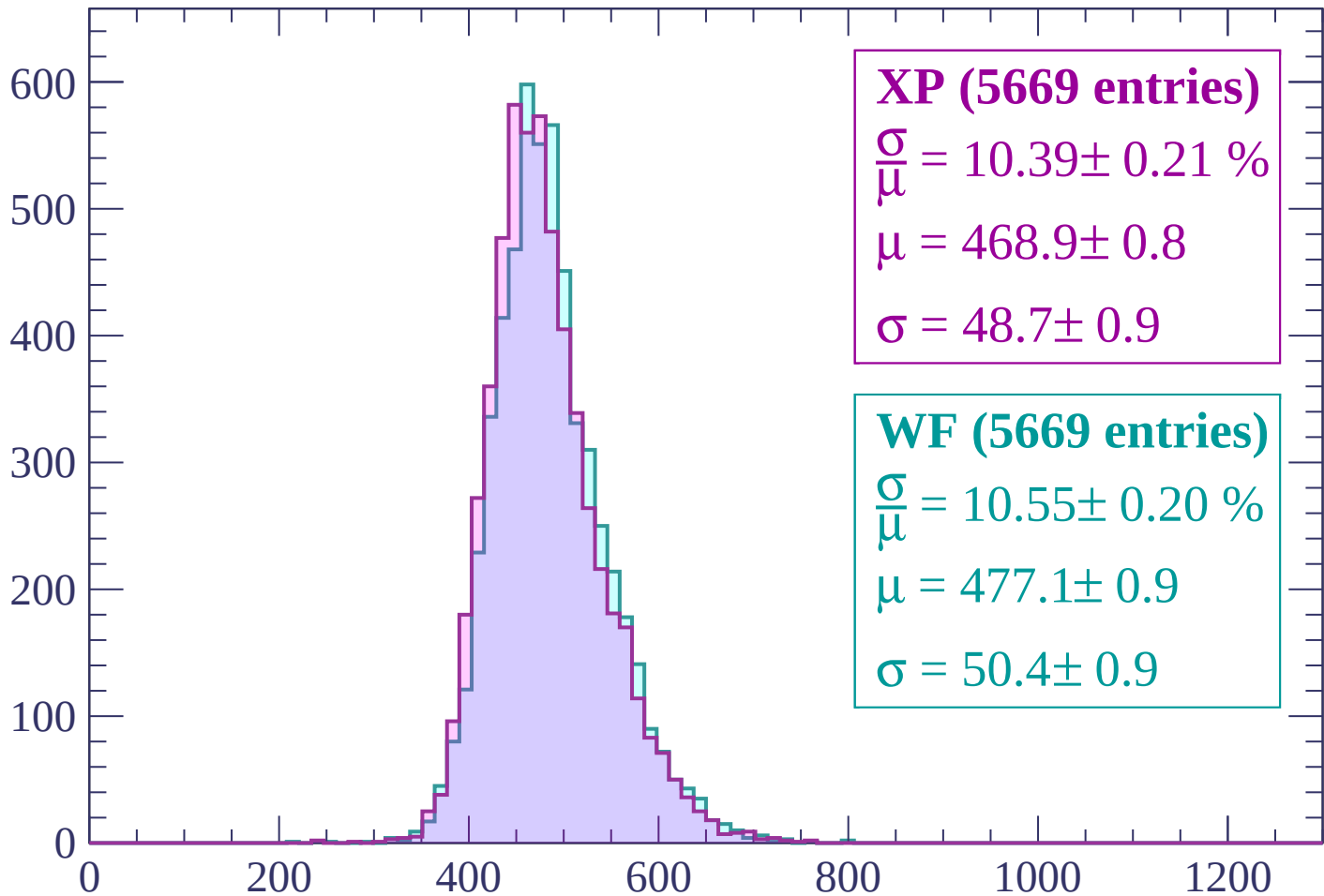
$$\mu = 474.3 \pm 0.8$$

$$\sigma = 49.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (5669 entries)

$$\frac{\sigma}{\mu} = 10.39 \pm 0.21 \%$$

$$\mu = 468.9 \pm 0.8$$

$$\sigma = 48.7 \pm 0.9$$

WF (5669 entries)

$$\frac{\sigma}{\mu} = 10.55 \pm 0.20 \%$$

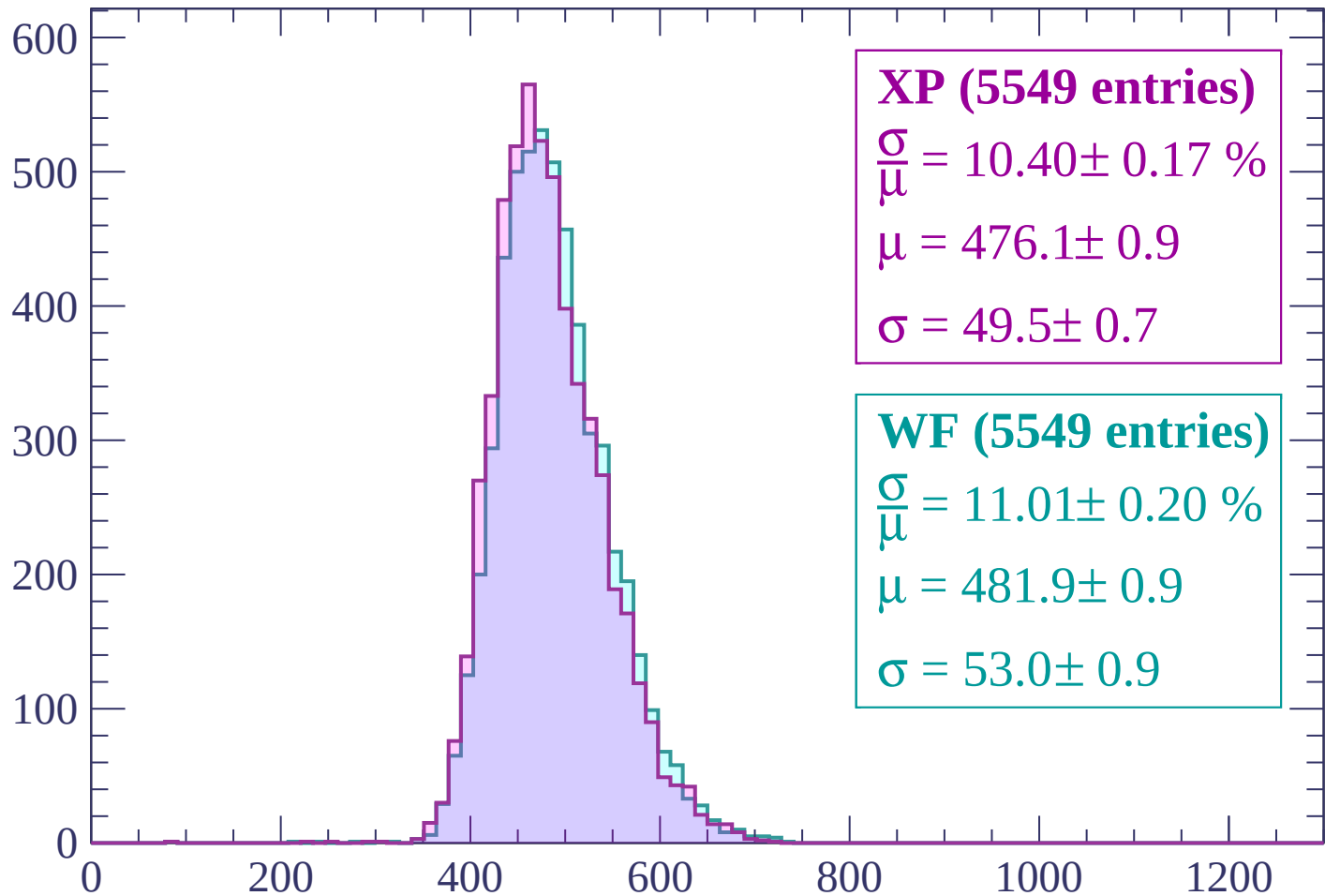
$$\mu = 477.1 \pm 0.9$$

$$\sigma = 50.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (5549 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.17 \%$$

$$\mu = 476.1 \pm 0.9$$

$$\sigma = 49.5 \pm 0.7$$

WF (5549 entries)

$$\frac{\sigma}{\mu} = 11.01 \pm 0.20 \%$$

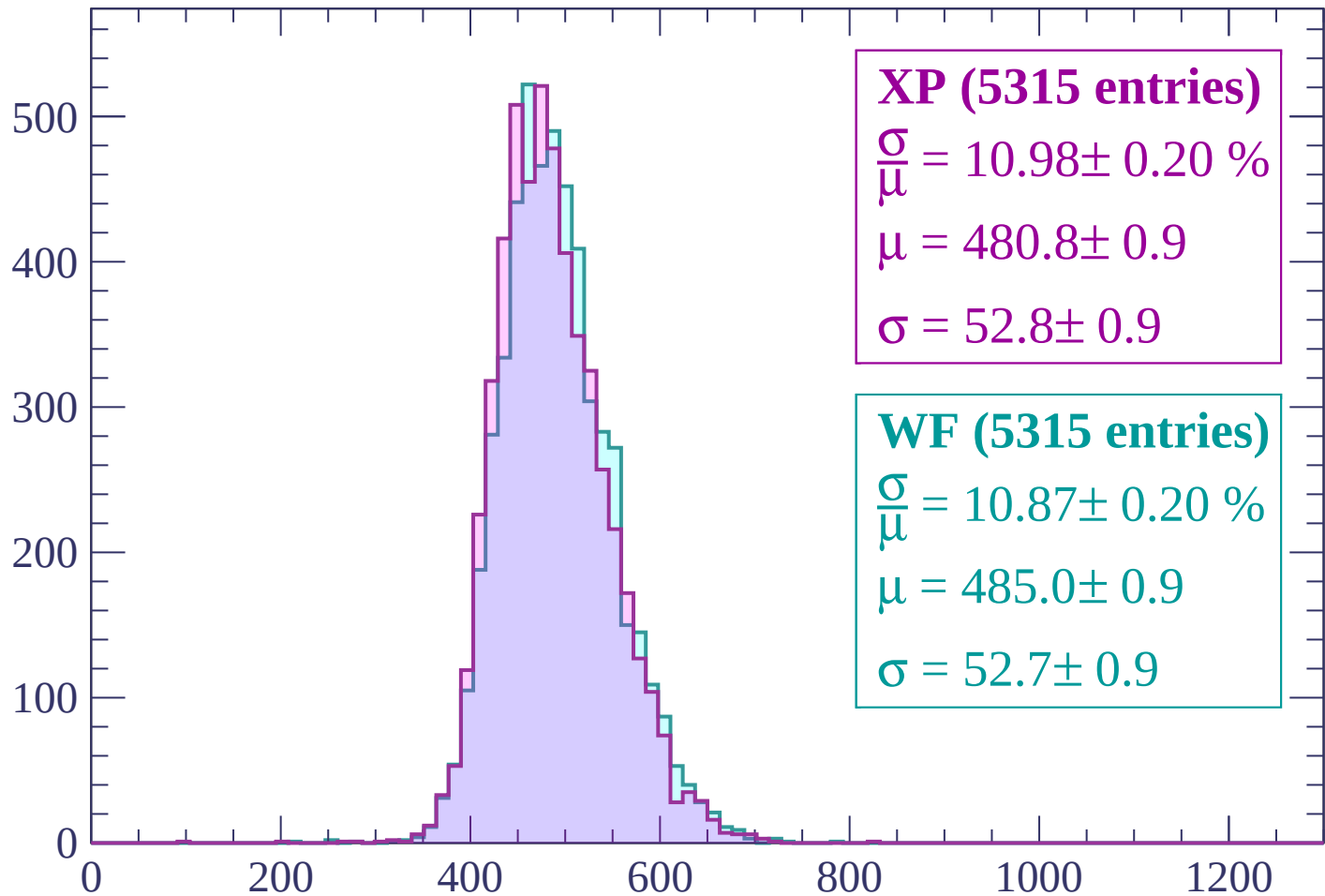
$$\mu = 481.9 \pm 0.9$$

$$\sigma = 53.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (5315 entries)

$\frac{\sigma}{\mu} = 10.98 \pm 0.20 \%$

$\mu = 480.8 \pm 0.9$

$\sigma = 52.8 \pm 0.9$

WF (5315 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.20 \%$

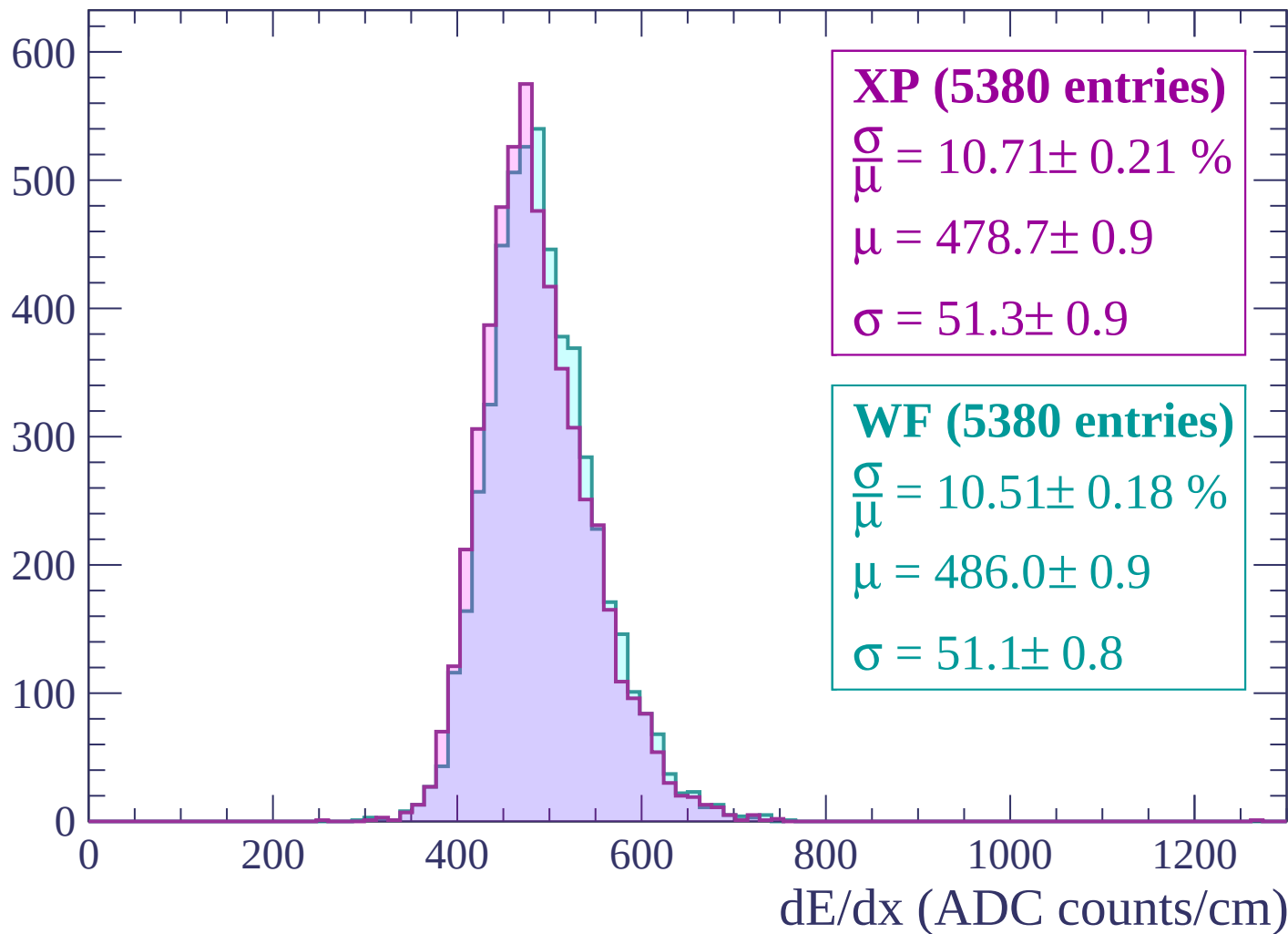
$\mu = 485.0 \pm 0.9$

$\sigma = 52.7 \pm 0.9$

dE/dx (ADC counts/cm)

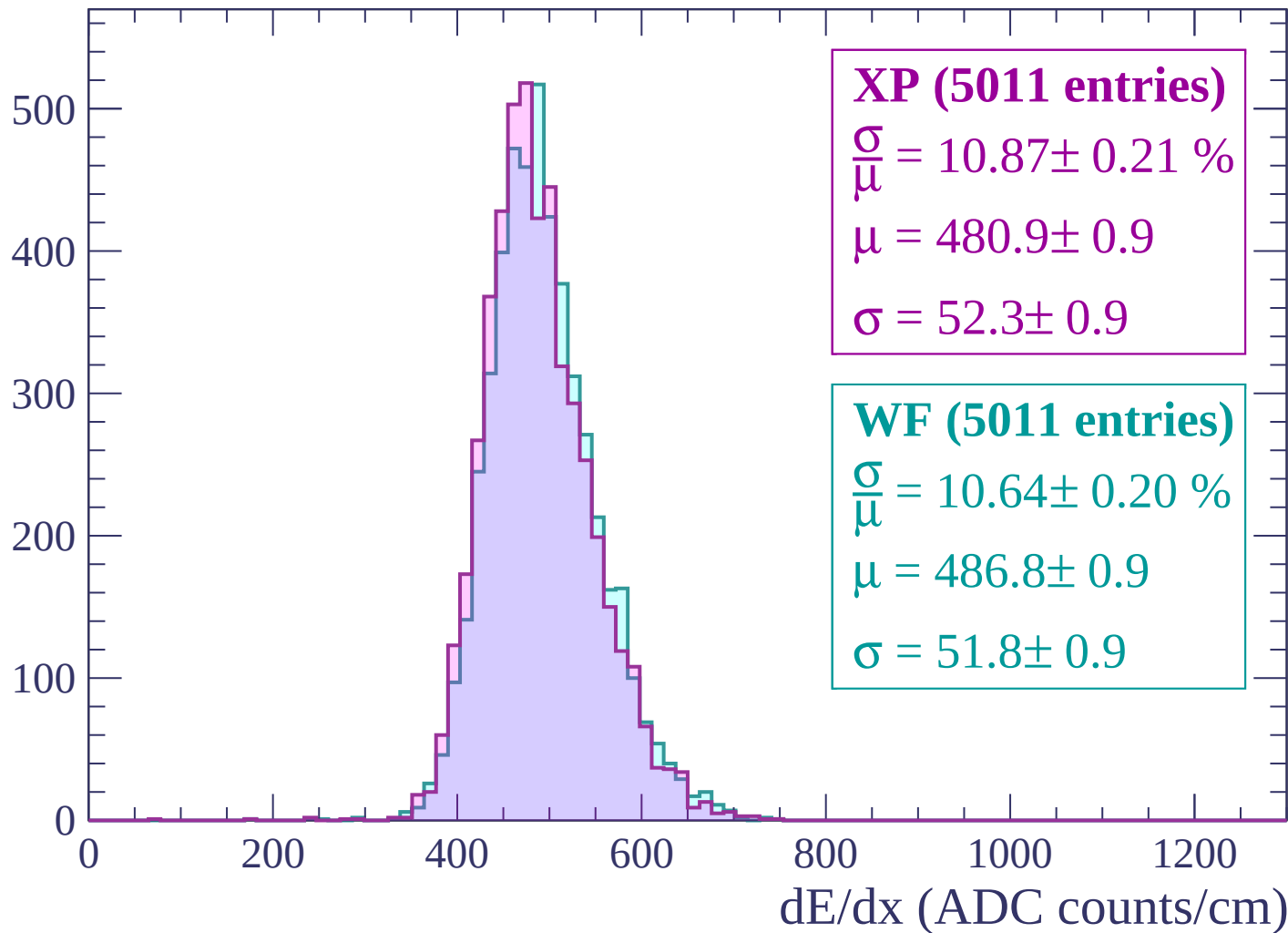
Energy loss | $1800 < p < 1860$

Count



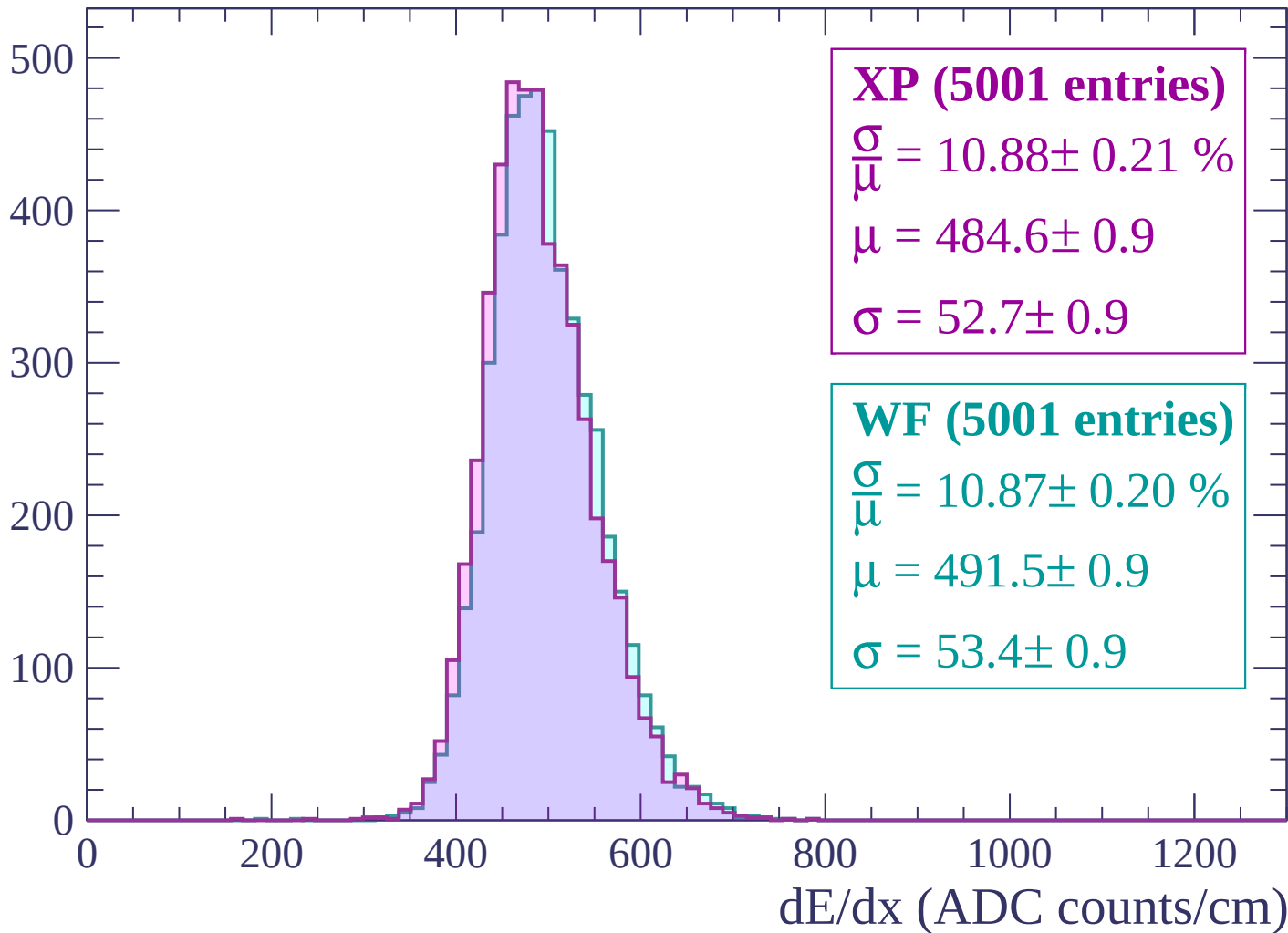
Energy loss | $1860 < p < 1920$

Count



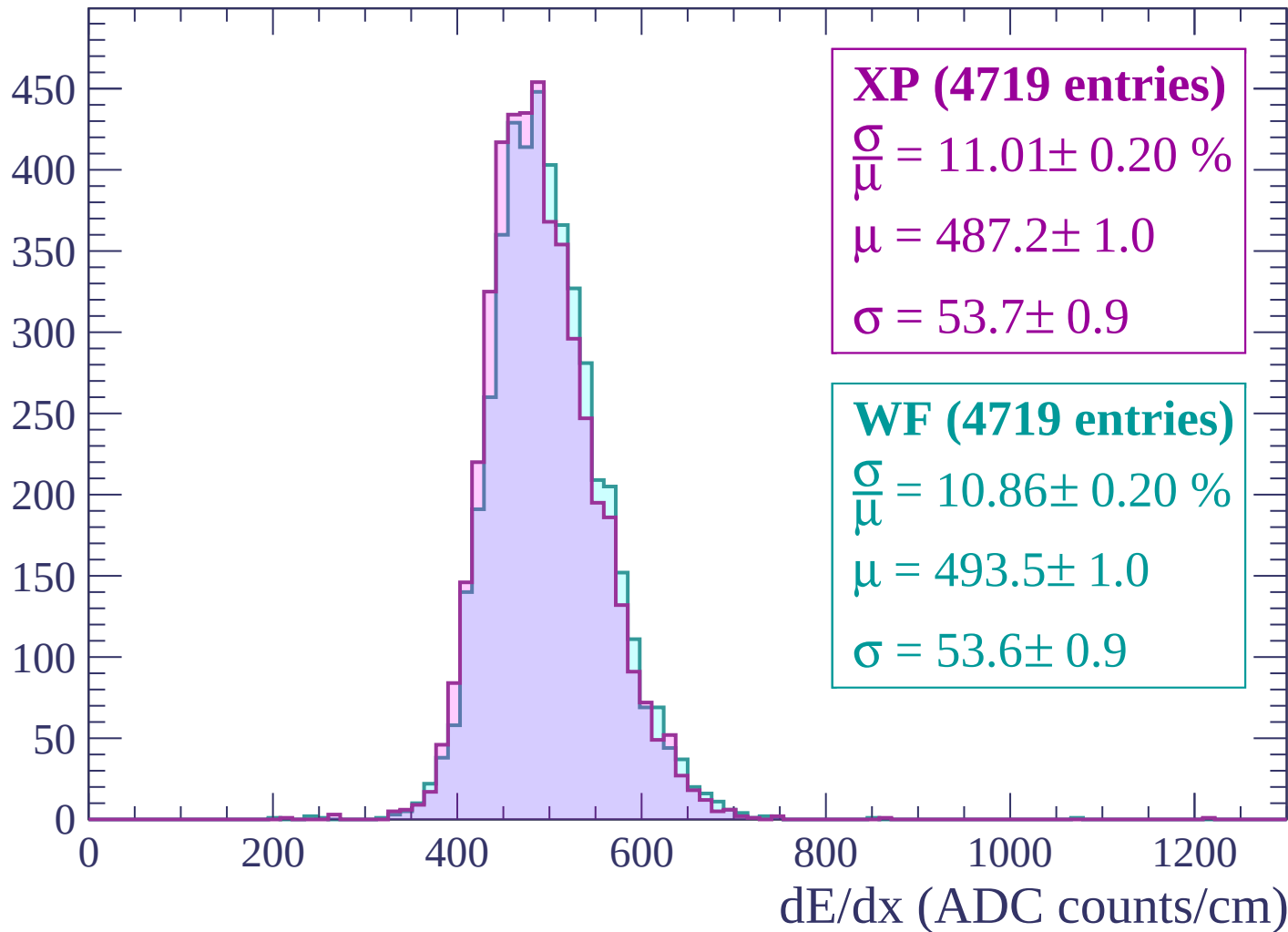
Energy loss | $1920 < p < 1980$

Count



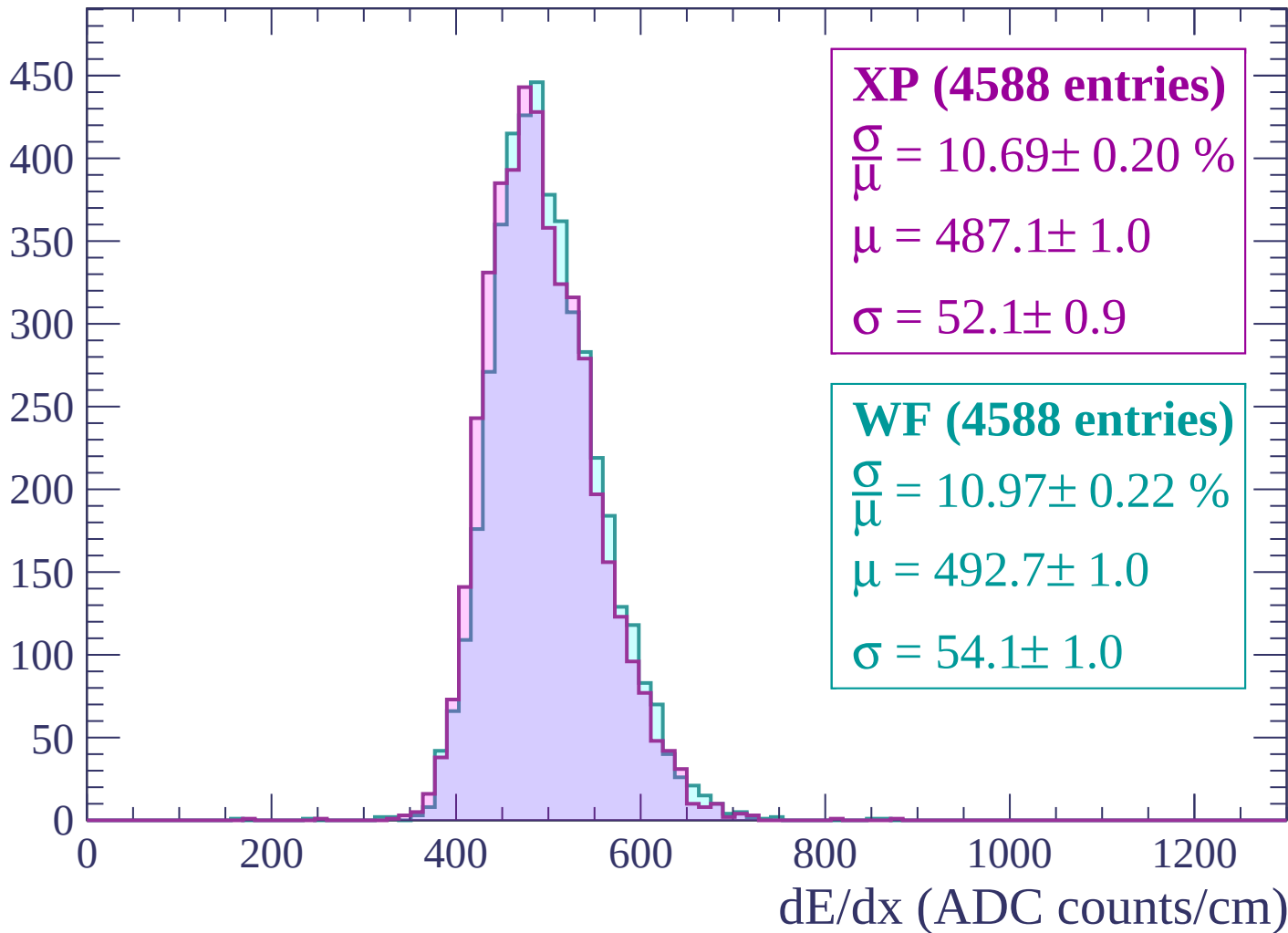
Energy loss | $1980 < p < 2040$

Count



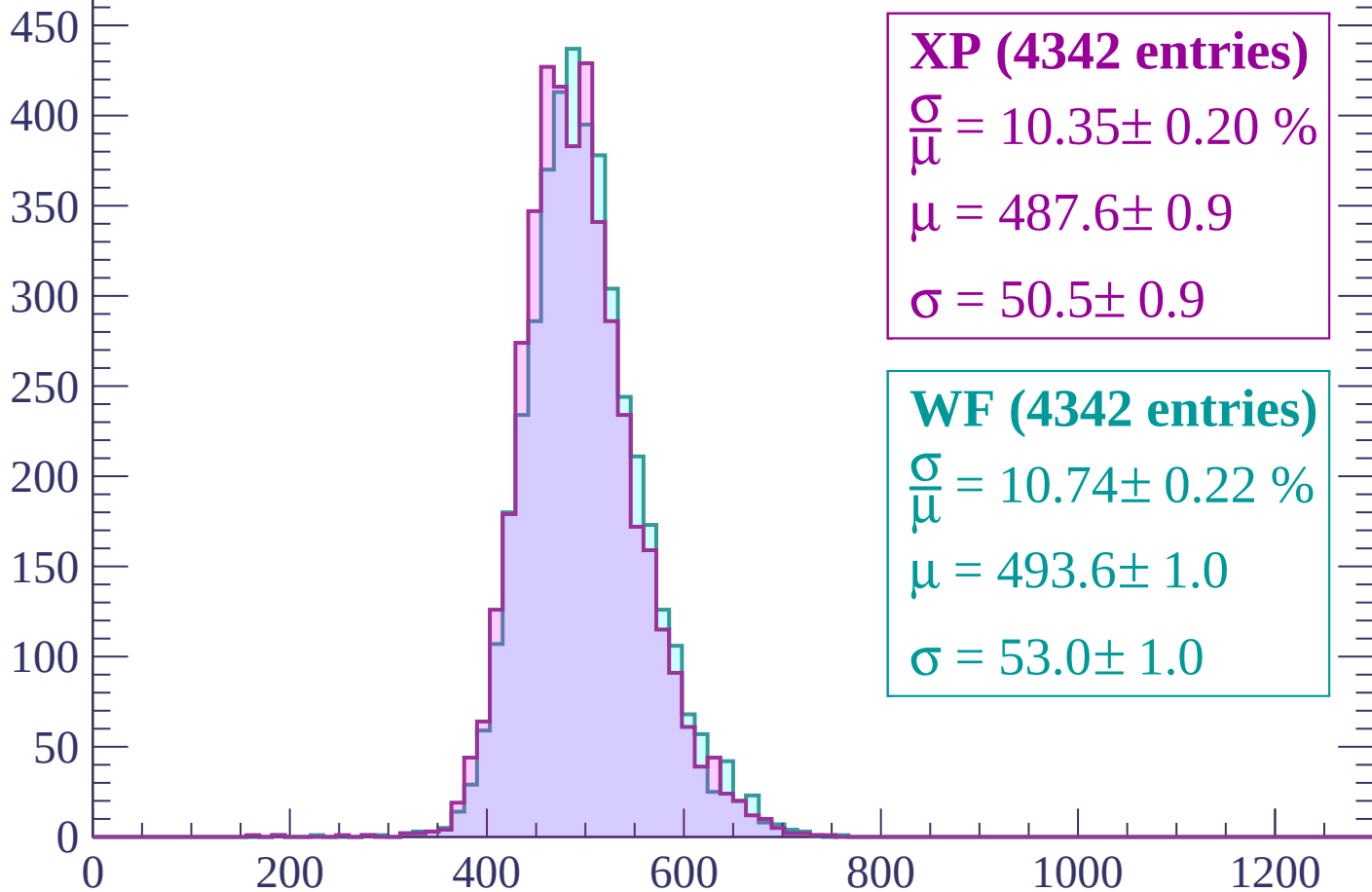
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



XP (4342 entries)

$\frac{\sigma}{\mu} = 10.35 \pm 0.20 \%$

$\mu = 487.6 \pm 0.9$

$\sigma = 50.5 \pm 0.9$

WF (4342 entries)

$\frac{\sigma}{\mu} = 10.74 \pm 0.22 \%$

$\mu = 493.6 \pm 1.0$

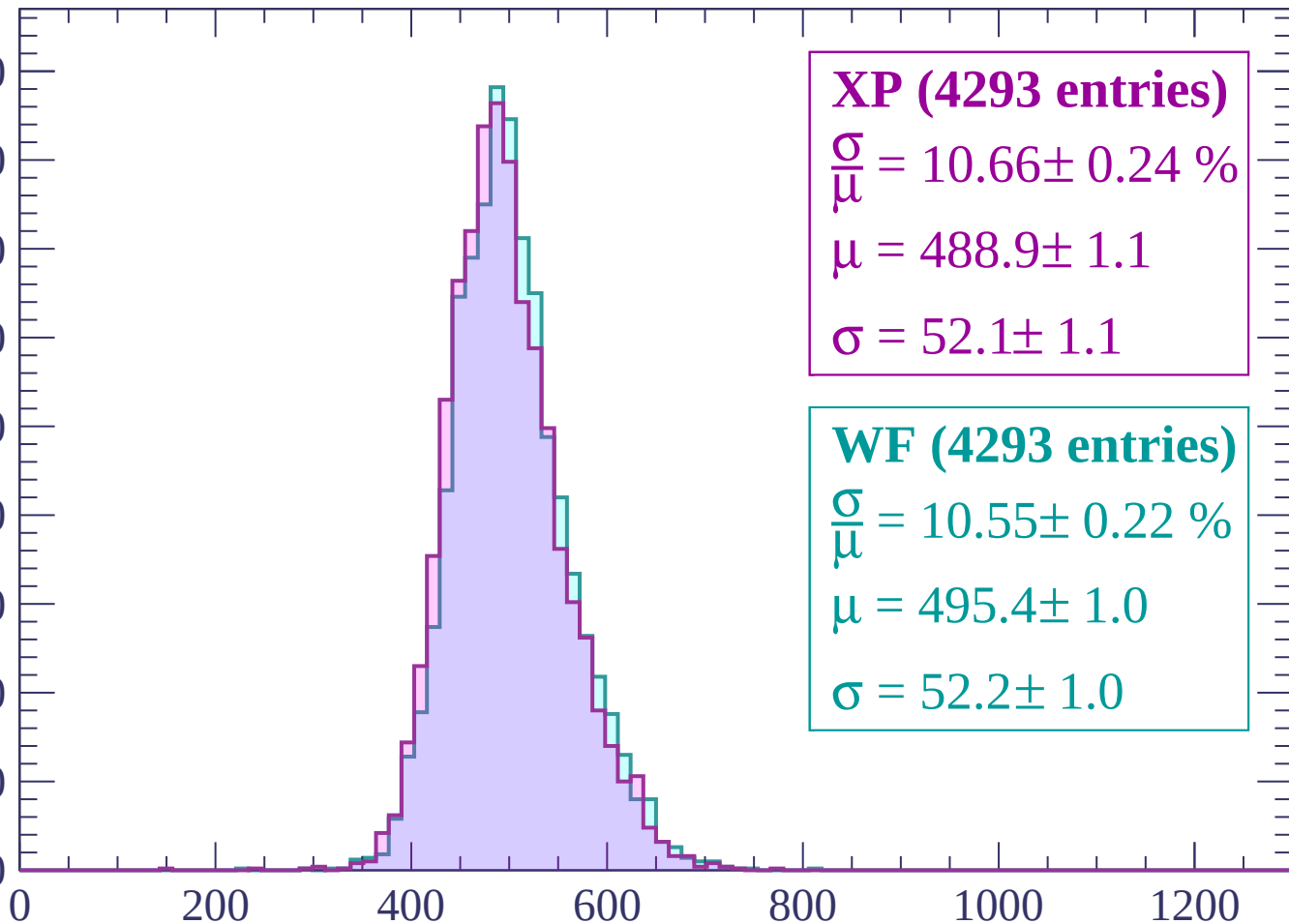
$\sigma = 53.0 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count

450
400
350
300
250
200
150
100
50
0



XP (4293 entries)

$\frac{\sigma}{\mu} = 10.66 \pm 0.24 \%$

$\mu = 488.9 \pm 1.1$

$\sigma = 52.1 \pm 1.1$

WF (4293 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.22 \%$

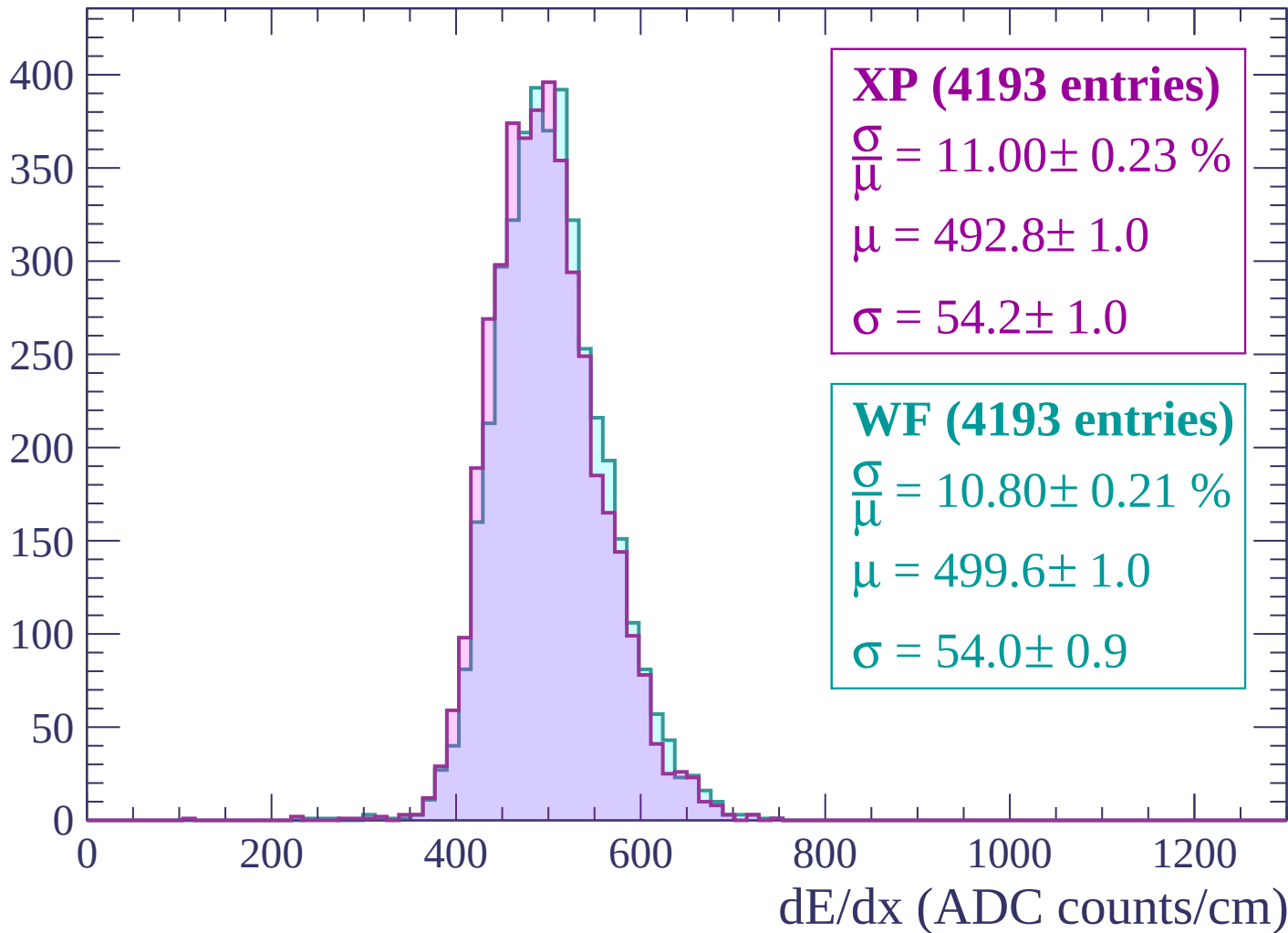
$\mu = 495.4 \pm 1.0$

$\sigma = 52.2 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

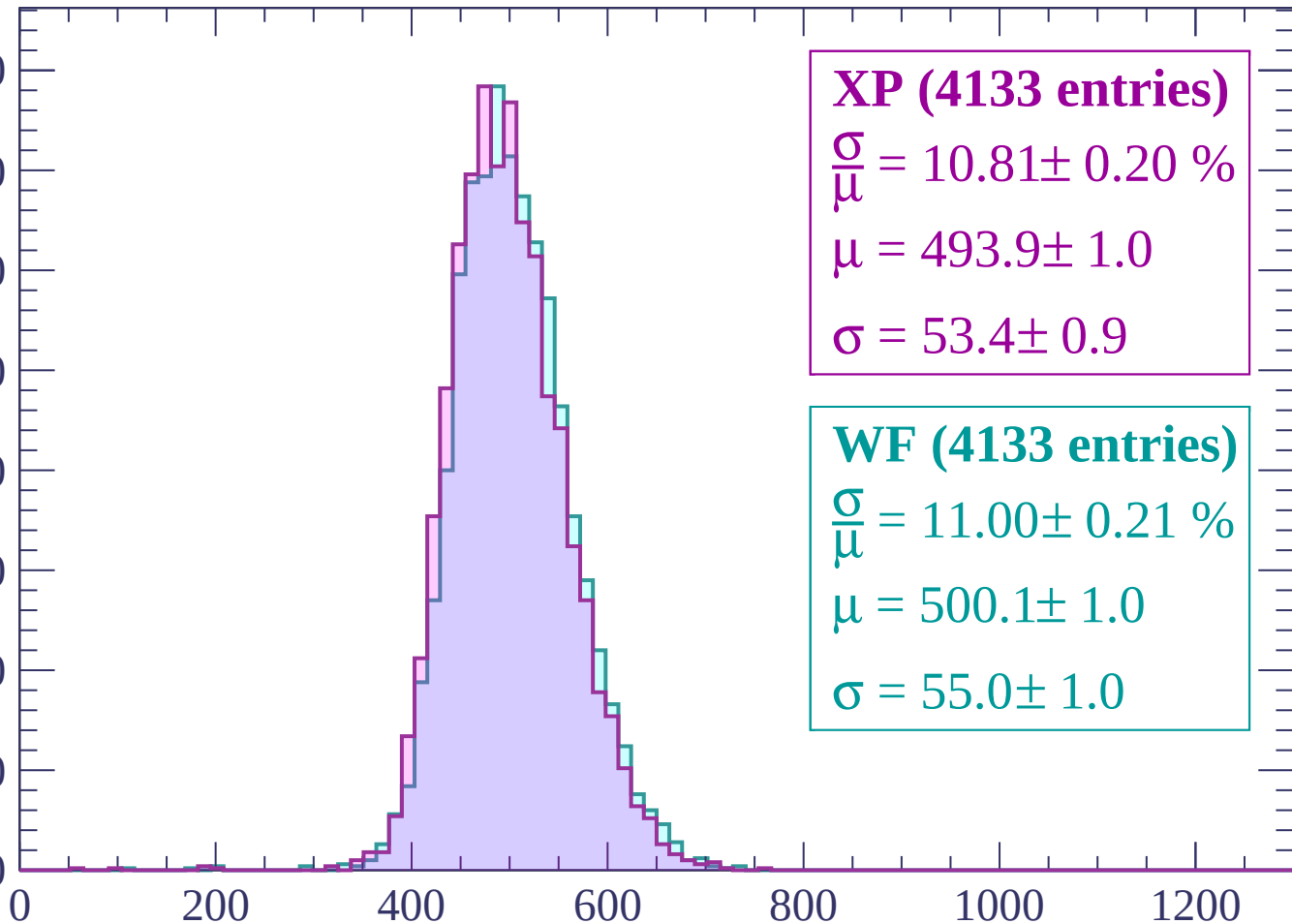
Count



Energy loss | $2280 < p < 2340$

Count

400
350
300
250
200
150
100
50
0



XP (4133 entries)

$\frac{\sigma}{\mu} = 10.81 \pm 0.20 \%$

$\mu = 493.9 \pm 1.0$

$\sigma = 53.4 \pm 0.9$

WF (4133 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.21 \%$

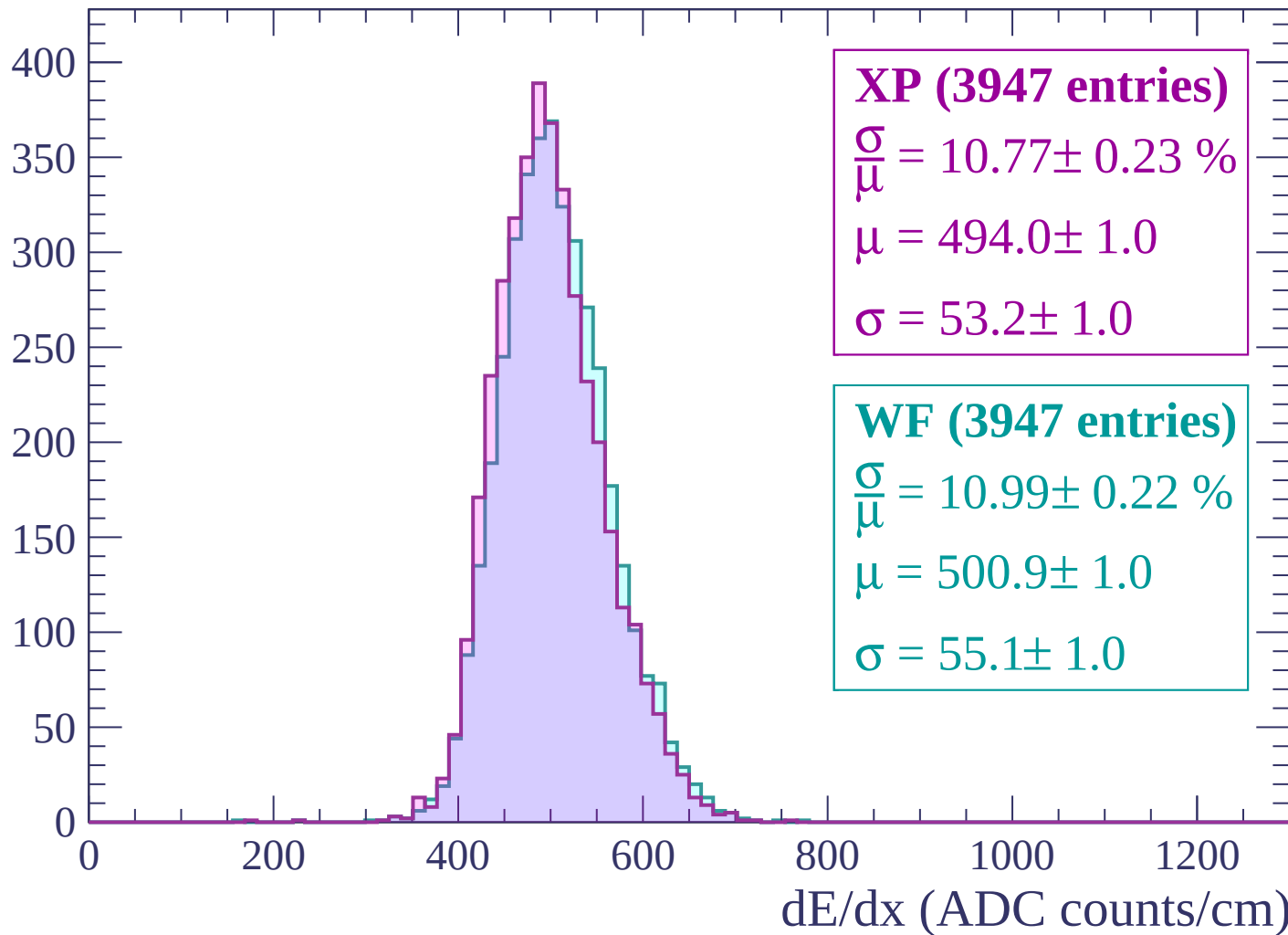
$\mu = 500.1 \pm 1.0$

$\sigma = 55.0 \pm 1.0$

dE/dx (ADC counts/cm)

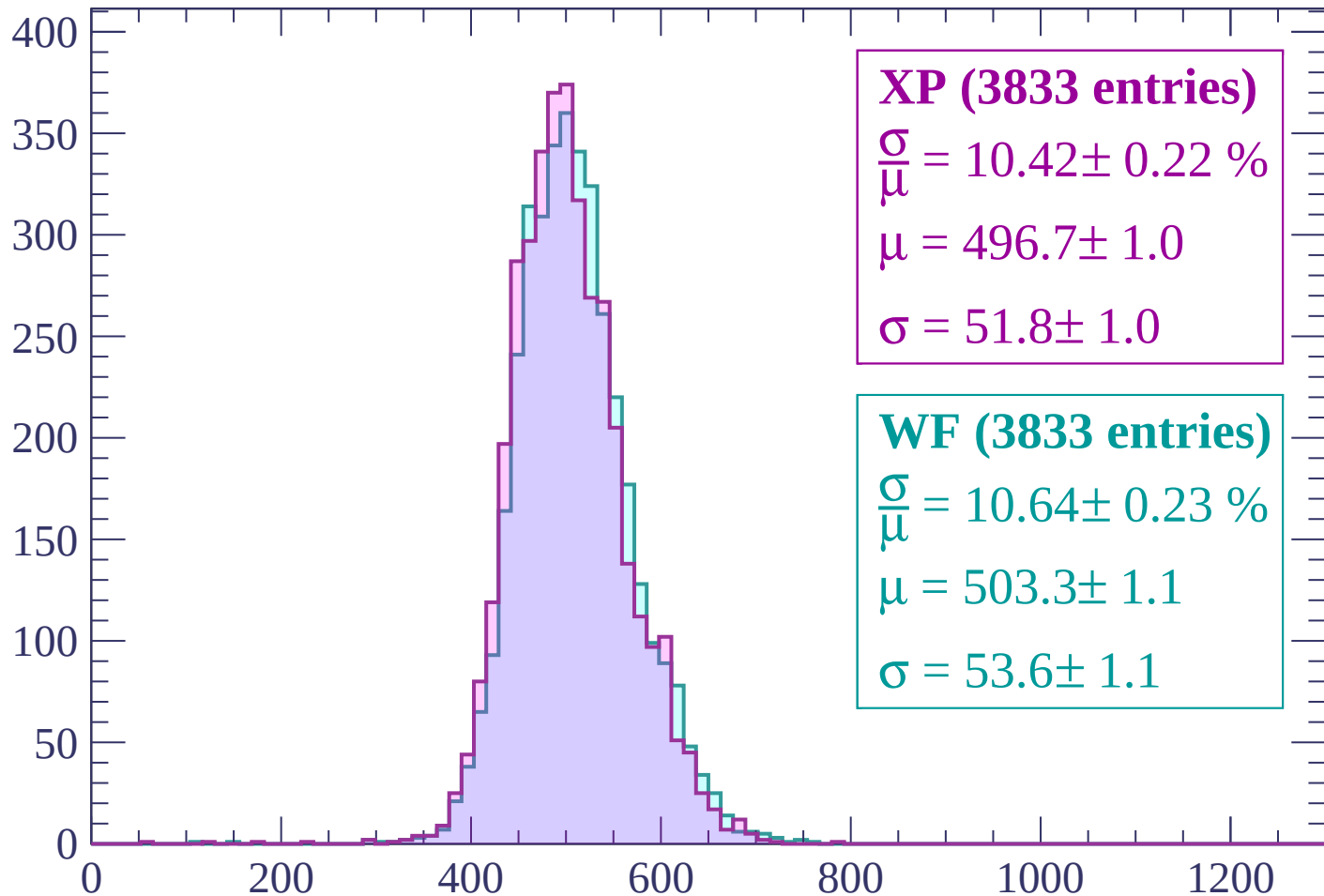
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (3833 entries)

$$\frac{\sigma}{\mu} = 10.42 \pm 0.22 \%$$

$$\mu = 496.7 \pm 1.0$$

$$\sigma = 51.8 \pm 1.0$$

WF (3833 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.23 \%$$

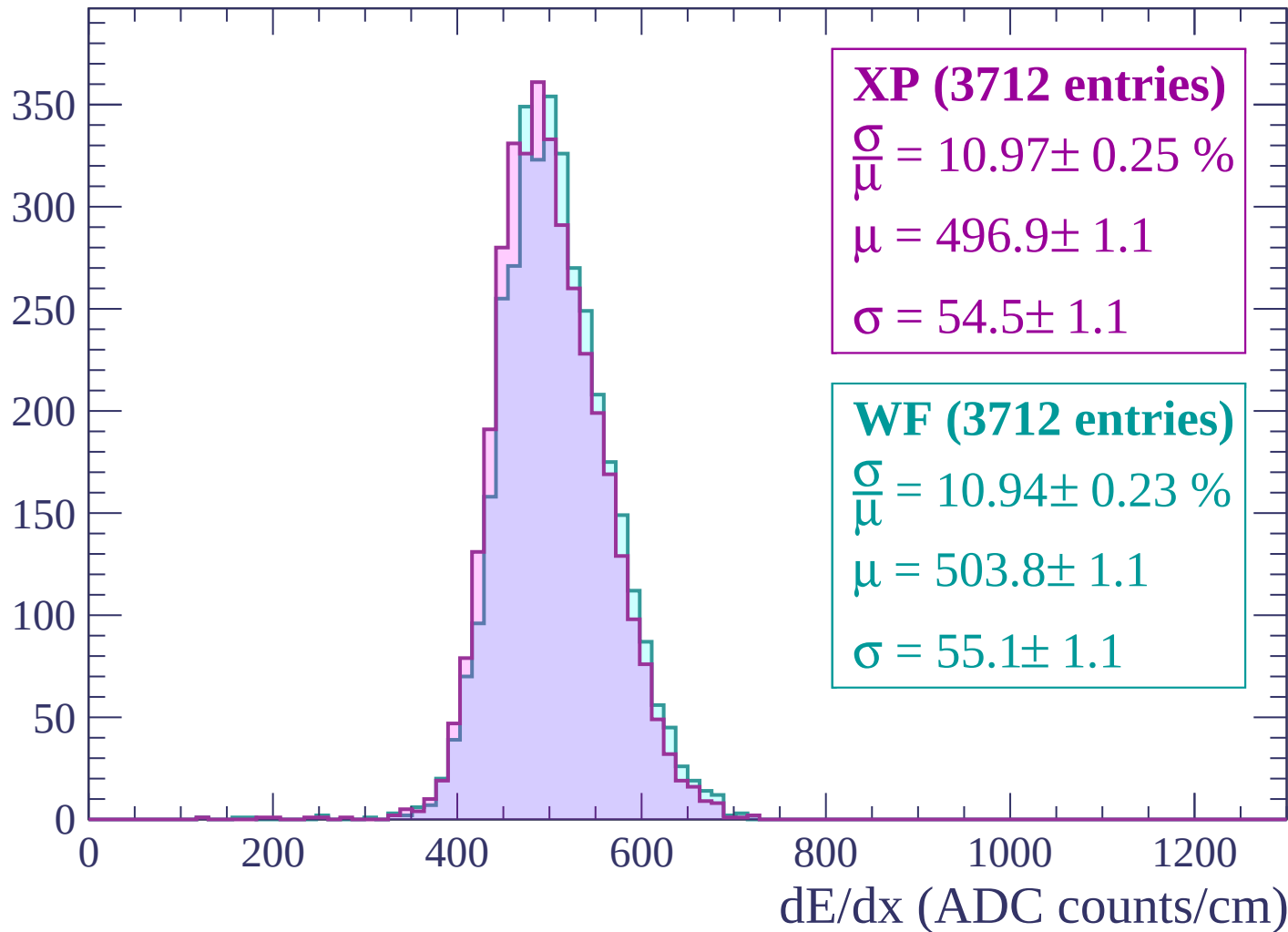
$$\mu = 503.3 \pm 1.1$$

$$\sigma = 53.6 \pm 1.1$$

dE/dx (ADC counts/cm)

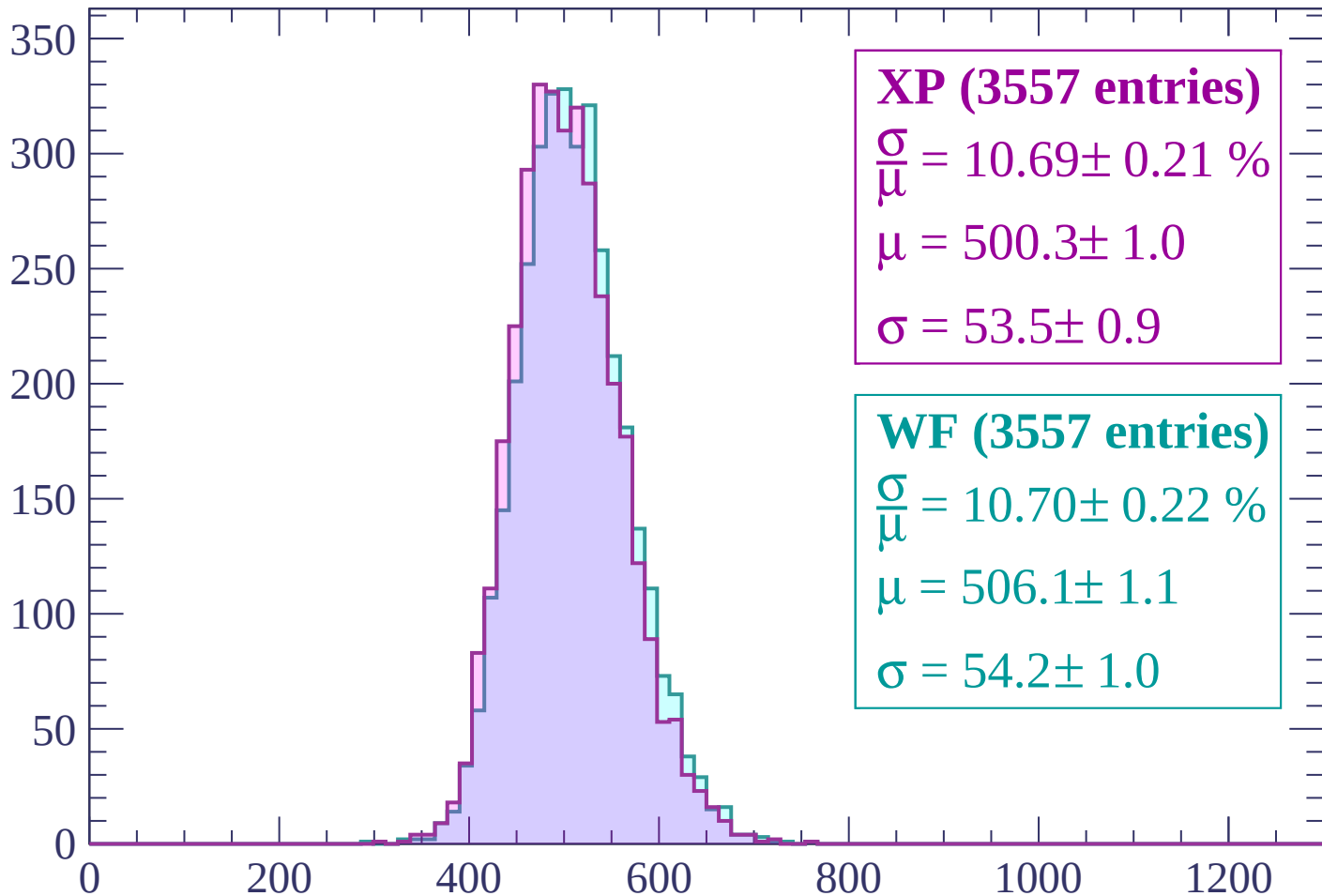
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (3557 entries)

$\frac{\sigma}{\mu} = 10.69 \pm 0.21 \%$

$\mu = 500.3 \pm 1.0$

$\sigma = 53.5 \pm 0.9$

WF (3557 entries)

$\frac{\sigma}{\mu} = 10.70 \pm 0.22 \%$

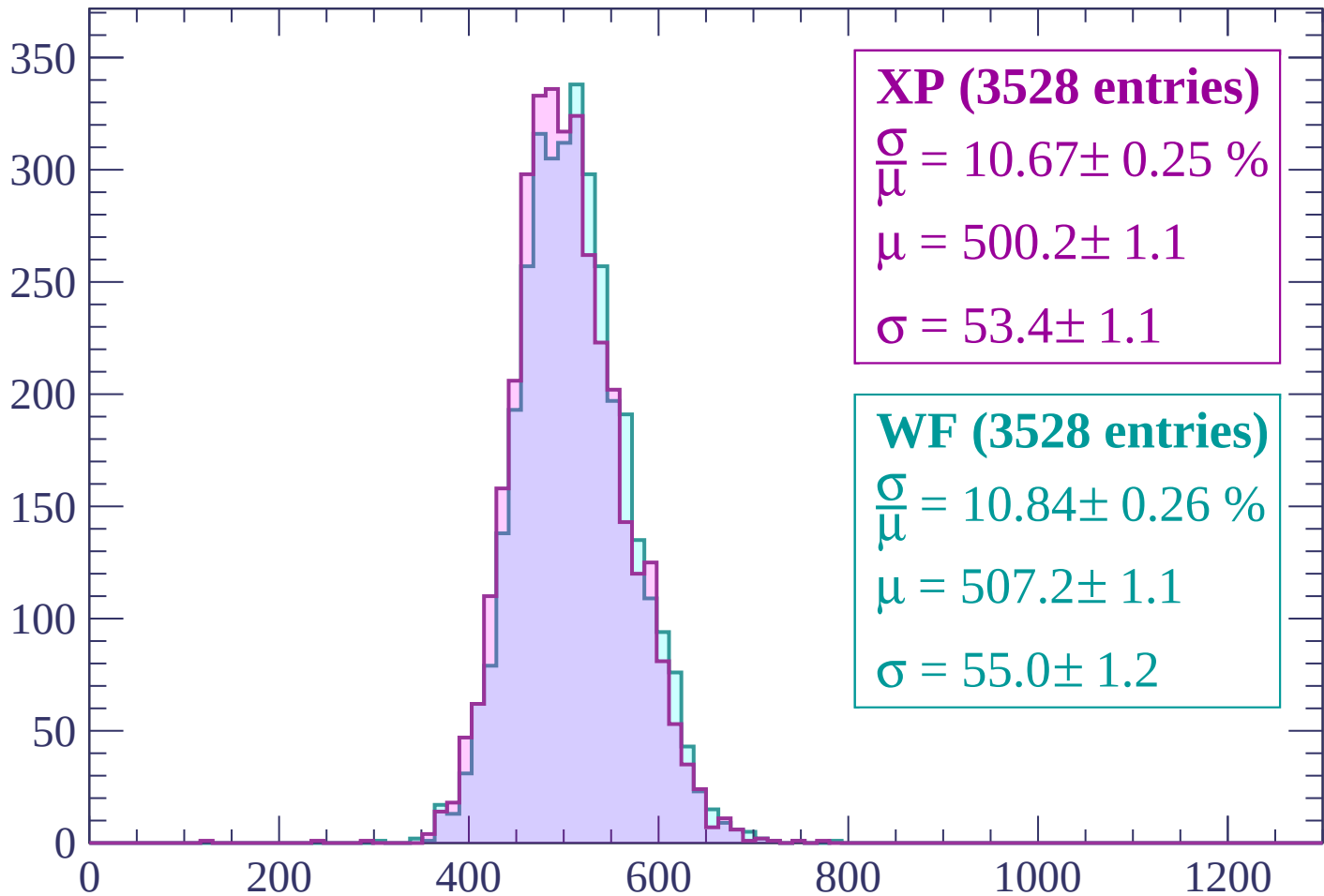
$\mu = 506.1 \pm 1.1$

$\sigma = 54.2 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (3528 entries)

$\frac{\sigma}{\mu} = 10.67 \pm 0.25 \%$

$\mu = 500.2 \pm 1.1$

$\sigma = 53.4 \pm 1.1$

WF (3528 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.26 \%$

$\mu = 507.2 \pm 1.1$

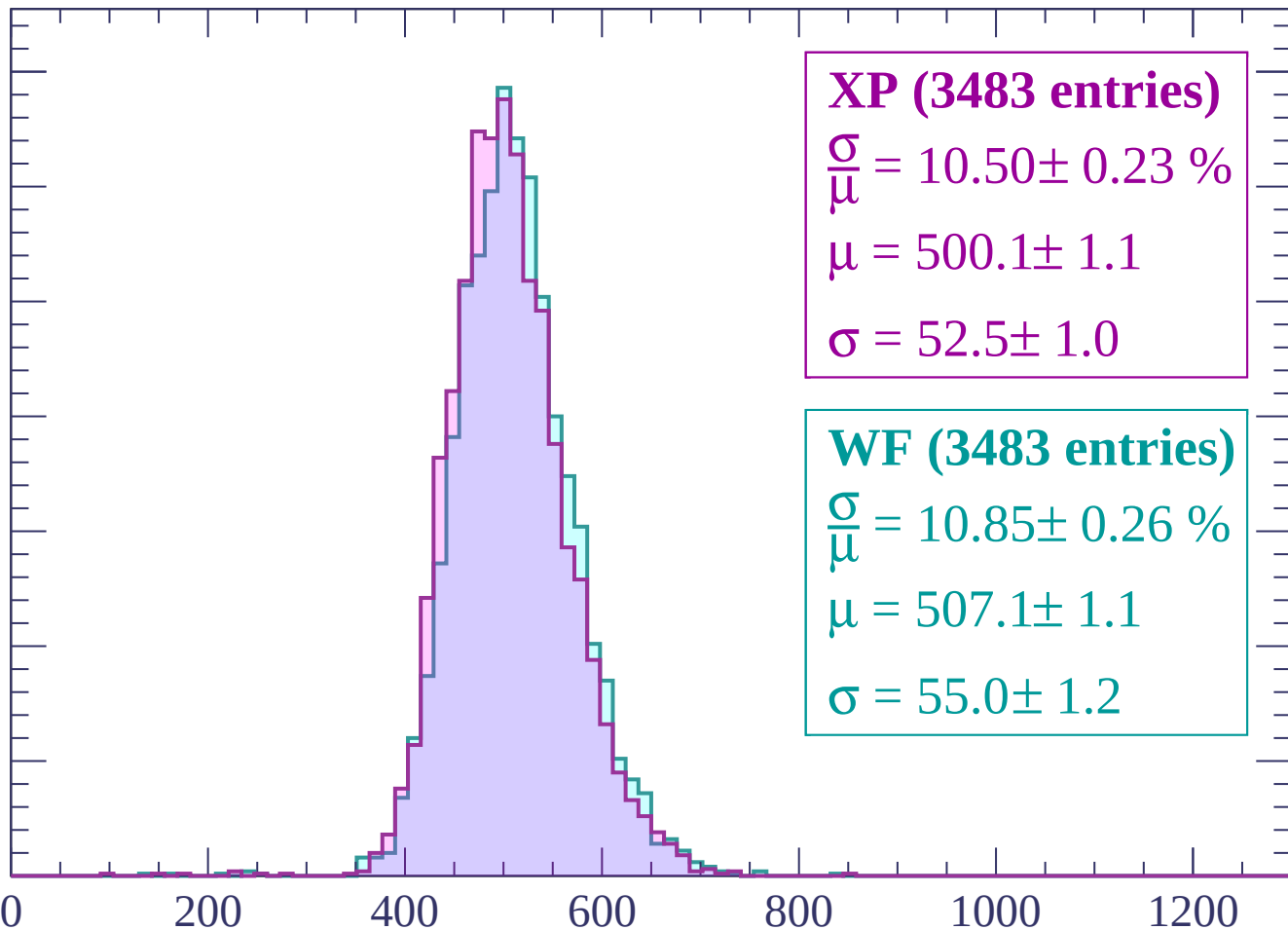
$\sigma = 55.0 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $2640 < p < 2700$

Count

350
300
250
200
150
100
50
0



XP (3483 entries)

$\frac{\sigma}{\mu} = 10.50 \pm 0.23 \%$

$\mu = 500.1 \pm 1.1$

$\sigma = 52.5 \pm 1.0$

WF (3483 entries)

$\frac{\sigma}{\mu} = 10.85 \pm 0.26 \%$

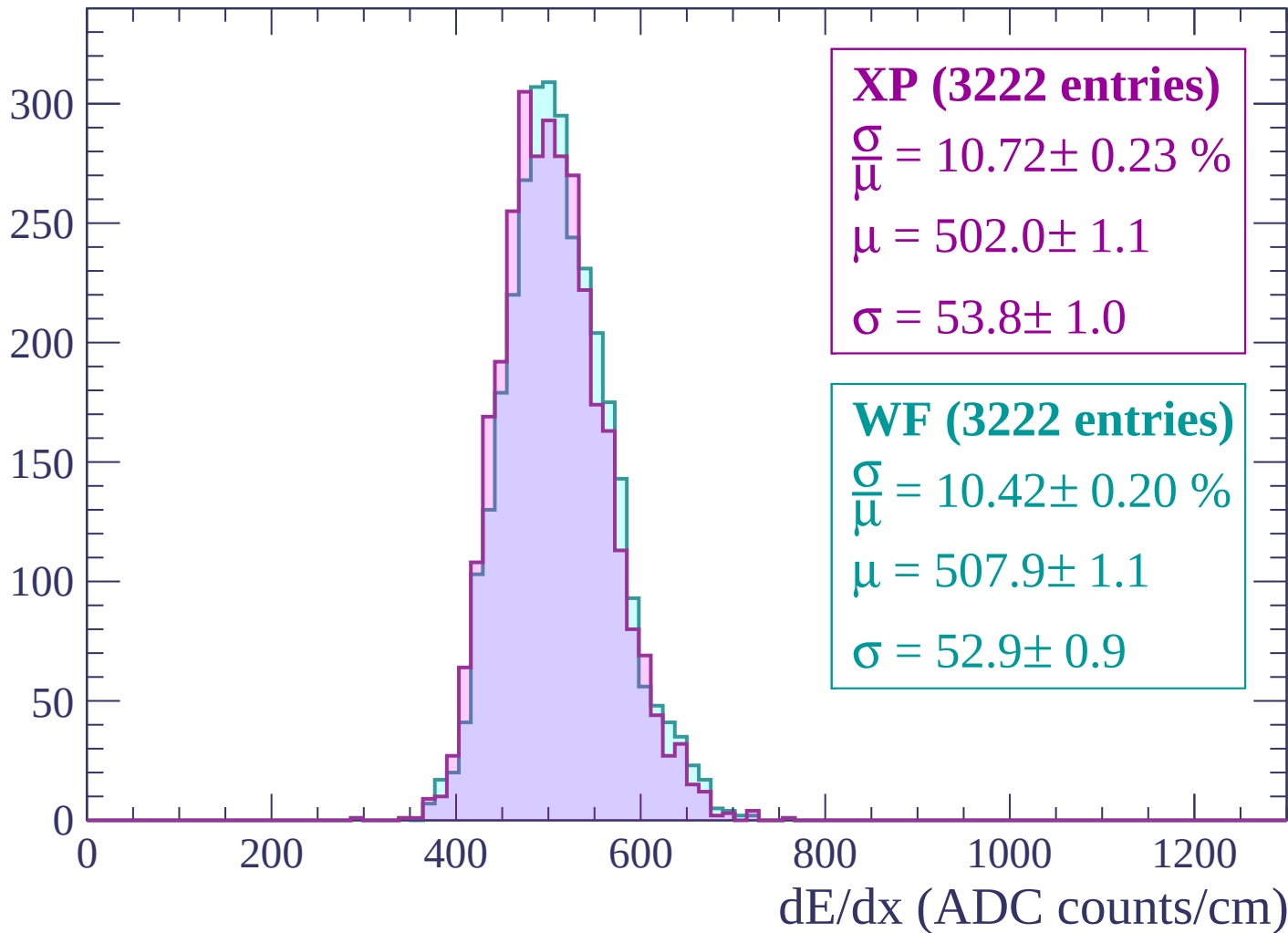
$\mu = 507.1 \pm 1.1$

$\sigma = 55.0 \pm 1.2$

dE/dx (ADC counts/cm)

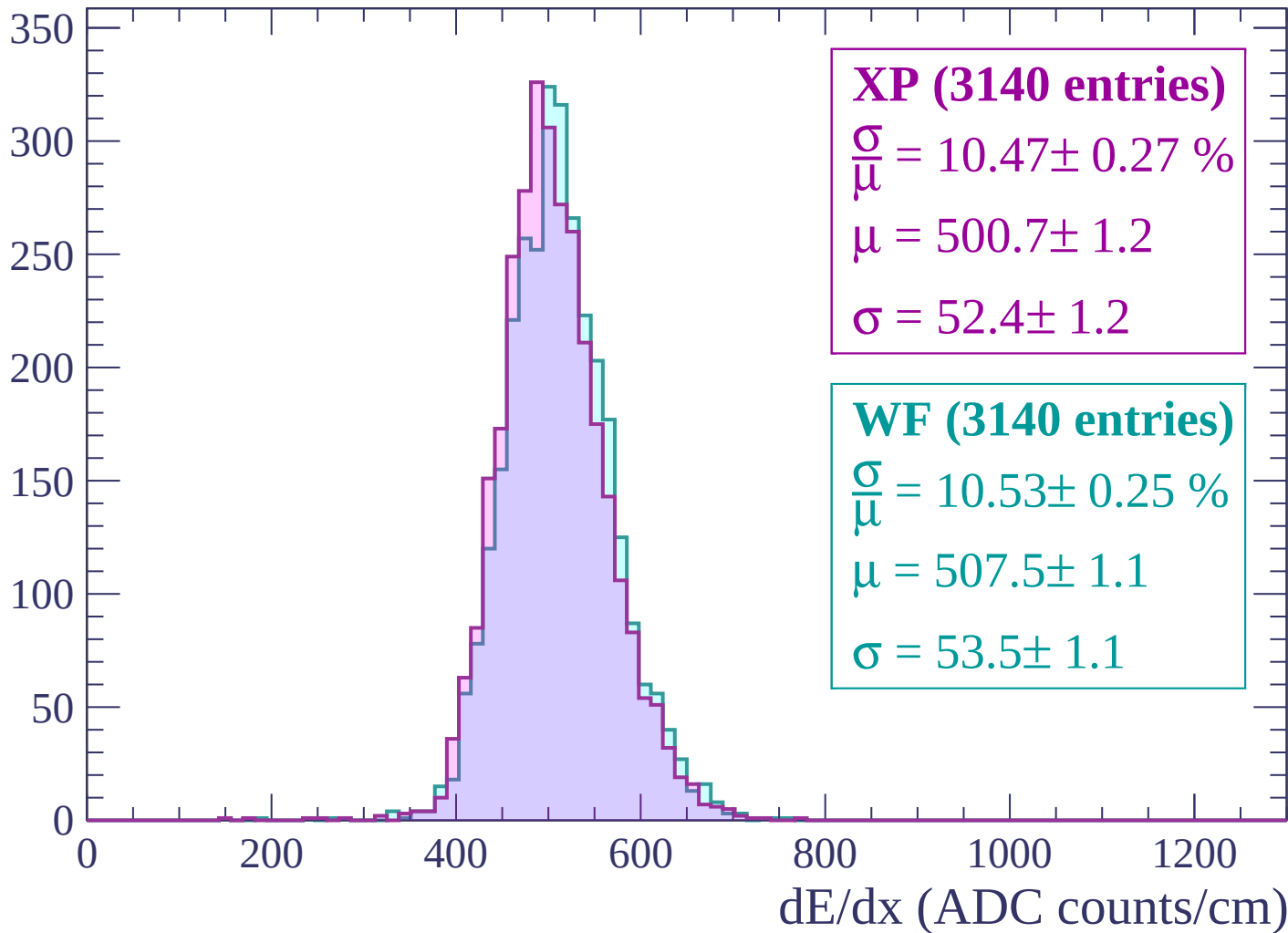
Energy loss | $2700 < p < 2760$

Count



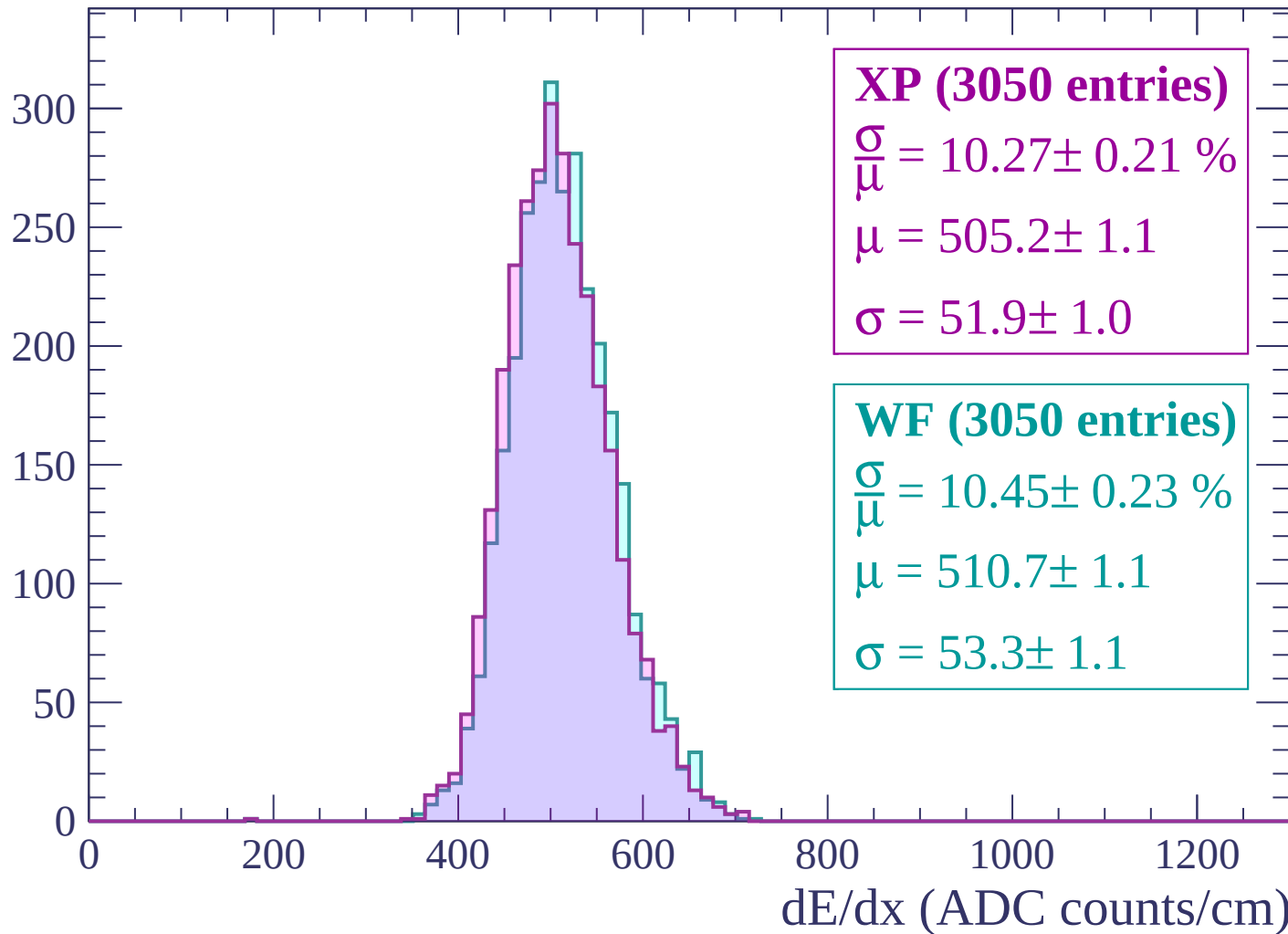
Energy loss | $2760 < p < 2820$

Count



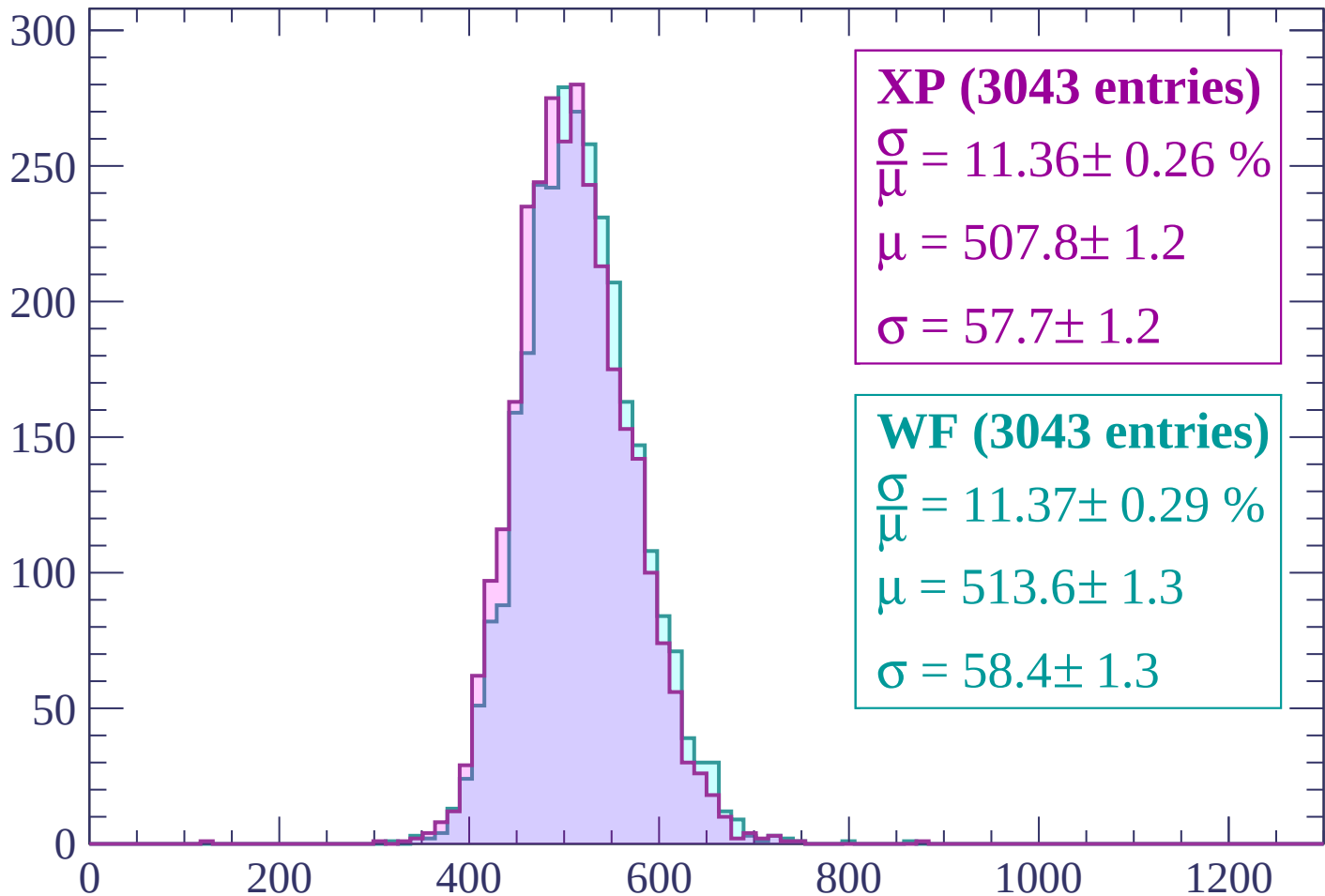
Energy loss | $2820 < p < 2880$

Count



Energy loss | $2880 < p < 2940$

Count



XP (3043 entries)

$$\frac{\sigma}{\mu} = 11.36 \pm 0.26 \%$$

$$\mu = 507.8 \pm 1.2$$

$$\sigma = 57.7 \pm 1.2$$

WF (3043 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 0.29 \%$$

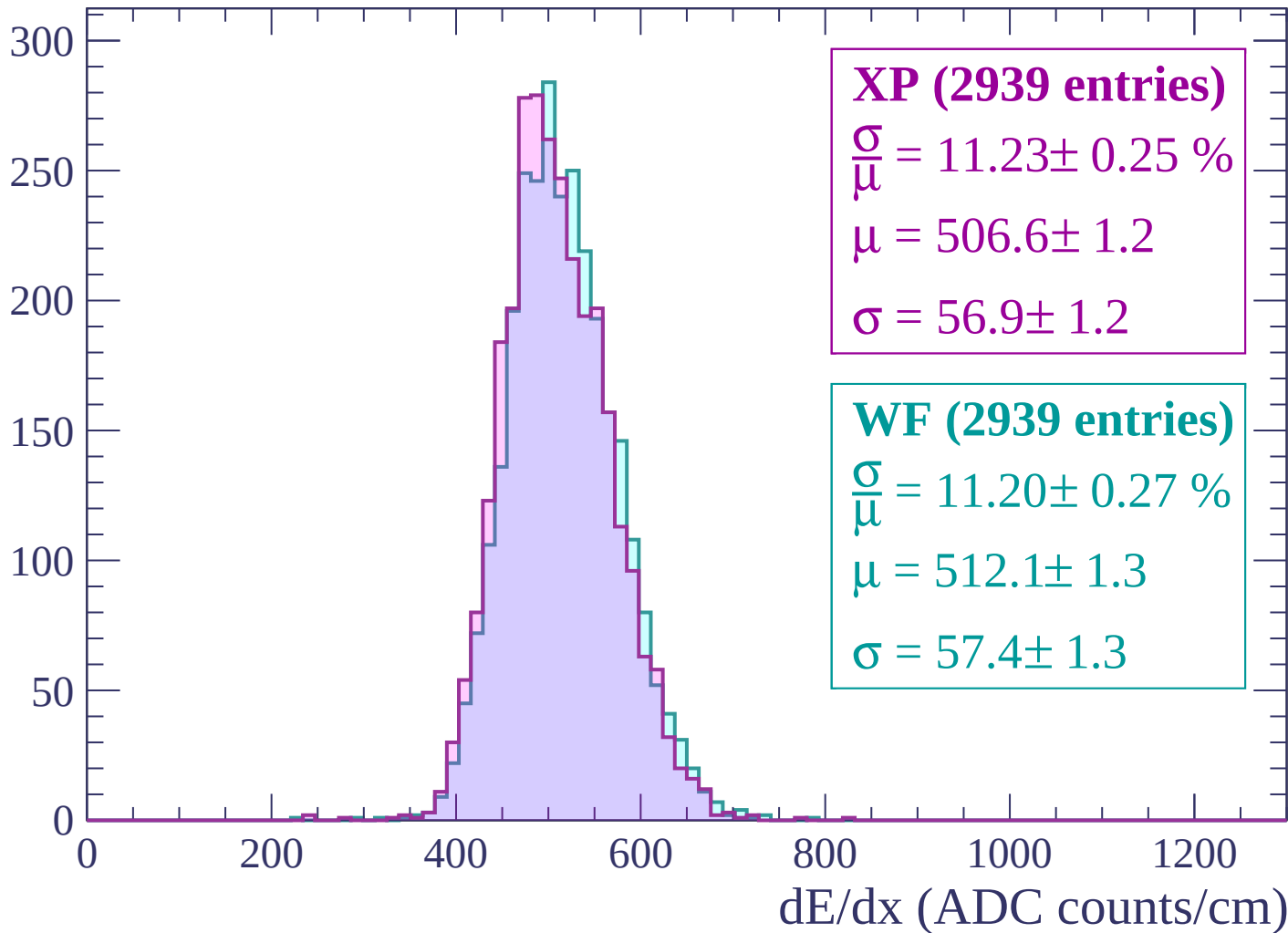
$$\mu = 513.6 \pm 1.3$$

$$\sigma = 58.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count

